

CH-1: EARTHWORK

Item No.	Description	Unit	Rate
1.1	EARTHWORK IN EXCAVATION		
	Earthwork in excavation for foundation trenches of walls, retaining walls, footings of column, steps, septic tank etc. including refilling (return filling) the quantity as necessary after completion of work, breaking clods in return filling, dressing, watering and ramming etc. and removal of surplus earth with all lead and lifts as directed and specified in the following classification of soils including bailing out water where necessary as directed and specified.		
	(A) Up to a depth of 2.00m below the existing ground level.		
	(a) In ordinary soil	Cum	108.82
	(b) In hard/ dense soil	Cum	138.12
	(c) In soft or laminated rock or medium shale	Cum	182.14
	(d) In medium rock of hard shale		
	(i) Using blasting materials	Cum	442.42
	(ii) Without using blasting materials	Cum	372.27
	(e) In hard rock		
	(i) Using blasting materials	Cum	540.27
	(ii) Without using blasting materials (where blasting is not possible/ not permissible)	Cum	579.64
	(B) Extra over item no.1.1(A) above for each 1m depth or part thereof beyond the initial depth of 2m	Cum	30.79
1.2	Shuttering and shoring the earth at different levels in foundations where necessary with required prop, struts, planks, timbers, nails etc. including dewatering if necessary, complete as directed and specified (measurement to be taken in the face area timbered)		
	(a) Upto a depth of 1.5 metre	Cum	75.38
	(b) Above 1.5 m upto 3.00 m depth	Cum	78.89
	(c) Above 3.00 m upto 4.50 m depth	Cum	80.62
	(d) Above 4.50 m upto 6.00 m depth	Cum	82.24
1.3	Earth/ sand filling in plinth in layers not more than 150mm thick including necessary carriage, watering, ramming etc. complete as directed and specified including payment of land compensation, forest royalty, sales tax and other duties and taxes as may be necessary.		
	(a) With earth or silt or sand obtained by head load with lead upto 90m and lift upto 2m	Cum	105.25
	(i) Extra for each 30m lead or part thereof beyond the initial lead of 90m (by head load)	Cum	13.92
	(ii) Extra for each 1.5m lift or part thereof beyond the initial lift of 2.00m (by head load)	Cum	6.96
	(b) With hill sand obtained by truck carriage loading and unloading, and carriage to the site of work.	Cum	333.77
	(c) With river sand or silt (predominantly non plastic) by truck carriage including loading and unloading.	Cum	345.61
	(d) With stone dust obtained by truck carriage including loading and unloading to the site of work.	Cum	428.51

	(e) Extra cost for lifting of sand for filling of sunk floor for each subsequent floors above the ground floor as directed and specified.	Cum	15.30
1.4	Raising low site around the building with approved soil obtained from outside by truck carriage including breaking clods, dressing etc. complete including paying necessary forest royelty, sales tax, land compensation, municipal gate fees, if any monopoly duty etc. (profile measurement to be taken and 12.5% deduction for shrinkage to be made from total quantity) etc. complete as directed and specified.		
	(a) Within Guwahati city only	Cum	276.81
	(b) Other than Guwahati city	Cum	210.20
1.5	Earth work in excavation in slushy/ marshy soil, decomposed organic matter mixed with brickbats, broken glass panes, broken earthen pots etc. removal of stumps and other deleterious matter in the proposed building site including constant water pumping for bailing out water as directed including return filling and ramming after completion of work with good quality earth (measurement will be on the basis of profile or section measurement).	Cum	236.24
1.6	Supplying and filling sand gravel in foundation trenches of column footings over slushy soil including local carriage of materials within 8 km pumping out of water, ramming etc. complete for strengthening the bearing capacity of the soil as directed and specified including payment of Forest Royalty and Sales Tax and other duties and taxes as may be necessary.	Cum	764.30
1.7	Extra lead per km truck carriage for sand gravel beyond the initial lead of 8 km.	Cum	7.04

CH-2: CEMENT CONCRETE WORKS

Item No.	Description	Unit	Rate
2.1	PLAIN CEMENT CONCRETE WORKS		
2.1.1	Plain cement concrete works with coarse aggregate of sizes 13mm to 32mm in foundation bed for footing steps, walls, brick works etc. as directed and specified including dewatering if necessary, and curing complete (shuttering where necessary shall be measured and paid separately).		
	(a) In prop 1cement: 3 sand :6 coarse aggregate by volume	cum	4292.86
	(b) In prop 1cement: 4 sand :8 coarse aggregate by volume	cum	3890.61
	(c) In prop 1cement: 5 sand :10 coarse aggregate by volume	cum	3569.75
	(d) In prop 1cement: 6 sand :12 coarse aggregate by volume	cum	3366.38
2.1.2	Plain cement concrete floor base in prop1:3:6 laid in alternate bays as specified with coarse agg. of size 13mm to 32mm including dewatering if necessary, and curing etc. complete.		
	a) 35mm thick	sqm	193.90
	b) 50mm thick	sqm	253.28
	c) 100mm thick	sqm	491.91

2.1.3	Providing and laying 25mm thick damp proof course with cement concrete in prop 1:1.5:3 with graded stone agg. of 10mm down nominal size including providing approved damp proof admixture in proportion as recommended by the manufacturer including curing etc. complete as directed.	sqm	181.99
2.1.4	Plain cement concrete apron in prop 1:4:8 with 20mm down graded coarse agg. Over soling including neat cement finish using cement slurry 2.75Kg. of cement per sq.m. of floor (soling to be measured separately).		
	(a) 35mm thick	sqm	192.13
	(b) 50mm thick	sqm	253.64
2.2	REINFORCED CEMENT CONCRETE WORKS		
2.2.1	Providing and laying plain/reinforced cement concrete works cement, coarse sand & 20mm down graded stone aggregate including dewatering if necessary, and curing complete but excluding cost of form work and reinforcement for reinforced cement concrete work (form work and reinforcement will be measured and paid separately)		
	(I) Using Mixer Machine		
	(A) In substructure up to plinth level		
	Foundation, footing, columns with base tie and plinth beam, pile cap, base slab, retaining walls, walls of septic tank, inspection pit and the like and other works not less than 100mm thick up to plinth level.		
	N) Without using admixture, plasticiser		
	a) M15 grade concrete or Prop. 1:2:4	cum	5496.65
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6063.65
	c) M25 grade concrete	cum	6603.01
	d) M30 grade concrete	cum	6672.15
	e) M35 grade concrete	cum	6801.22
	P) Using admixture, plasticiser etc for air entraining, water reducing purposes including addition of agent for changing normal setting time of concrete as required.		
	a) M15 grade concrete or Prop. 1:2:4	cum	5717.17
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6329.48
	c) M25 grade concrete	cum	6911.89
	d) M30 grade concrete	cum	6986.58
	e) M35 grade concrete	cum	7125.95
	(B) In super structure from plinth level up to 1st floor level.		
	i) Walls (thickness not less than 100mm) including attached pillars, buttresses, plinth and string course etc.		
	N) Without using admixture, plasticiser		
	a) M15 grade concrete or Prop. 1:2:4	cum	5668.82
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6241.39
	c) M25 grade concrete	cum	6786.00
	d) M30 grade concrete	cum	6855.86
	e) M35 grade concrete	cum	6986.19
	P) Using admixture, plasticiser etc for air entraining, water reducing purposes including addition of agent for changing normal setting time of concrete as required.		
	a) M15 grade concrete or Prop. 1:2:4	cum	5889.34
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6507.22

	c) M25 grade concrete	cum	7094.88
	d) M30 grade concrete	cum	7170.29
	e) M35 grade concrete	cum	7310.92
	ii) Columns, pillars, posts , struts, suspended floor, roof, landing, shelf and support, balcony, lintel, sill band, beam, girder, bressumer, cantilever, staircase (except spiral staircase and landing) including preparing the top surface and finishing of nosing.		
	N) Without using admixture, plasticiser		
	a) M15 grade concrete or Prop. 1:2:4	cum	5758.89
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6331.46
	c) M25 grade concrete	cum	6876.08
	d) M30 grade concrete	cum	6945.94
	e) M35 grade concrete	cum	7076.26
	P) Using admixture, plasticiser etc		
	a) M15 grade concrete or Prop. 1:2:4		5979.41
	b) M20 grade concrete or Prop. 1:1.5:3		6597.29
	c) M25 grade concrete		7184.96
	d) M30 grade concrete		7260.37
	e) M35 grade concrete		7400.99
	iii) Vertical and horizontal fins (thickness not more than 100mm) individually or forming box louvre and projected band.		
	N) Without using admixture, plasticiser		
	a) M15 grade concrete or Prop. 1:2:4	cum	6202.05
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6774.62
	c) M25 grade concrete	cum	7319.24
	d) M30 grade concrete	cum	7389.09
	e) M35 grade concrete	cum	7519.42
	P) Using admixture, plasticiser etc		
	a) M15 grade concrete or Prop. 1:2:4		6422.57
	b) M20 grade concrete or Prop. 1:1.5:3		7040.45
	c) M25 grade concrete		7628.12
	d) M30 grade concrete		7703.52
	e) M35 grade concrete		7844.15
	iv) On domes, vaults and shell roof		
	N) Without using admixture, plasticiser		
	a) M15 grade concrete or Prop. 1:2:4	cum	6328.84
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6906.96
	c) M25 grade concrete	cum	7456.88
	d) M30 grade concrete	cum	7527.39
	e) M35 grade concrete	cum	7658.99
	P) Using admixture, plasticiser etc		
	a) M15 grade concrete or Prop. 1:2:4	cum	6549.36
	b) M20 grade concrete or Prop. 1:1.5:3	cum	7172.79
	c) M25 grade concrete	cum	7765.76
	d) M30 grade concrete	cum	7841.82
	e) M35 grade concrete	cum	7983.72
	II) Using Batching Plant/ RMC/ Truck mounted batching plant including Transit Mixer and Concrete Pump		
	N) Without using admixture, plasticiser		
	a) M15 grade concrete or Prop. 1:2:4	cum	5812.96
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6363.40
	c) M25 grade concrete	cum	6886.98

	d) M30 grade concrete	cum	6954.11
	e) M35 grade concrete	cum	7079.41
	f) M40 grade concrete (PSC Grade)	cum	7173.39
	g) M45 grade concrete (PSC Grade)	cum	7504.54
	h) M50 grade concrete (PSC Grade)	cum	7737.24
	i) M55 grade concrete (PSC Grade)	cum	8104.20
	P) Using admixture, plasticiser etc		
	a) M15 grade concrete or Prop. 1:2:4	cum	6033.48
	b) M20 grade concrete or Prop. 1:1.5:3	cum	6629.23
	c) M25 grade concrete	cum	7195.86
	d) M30 grade concrete	cum	7268.54
	e) M35 grade concrete	cum	7404.14
	f) M40 grade concrete (PSC Grade)	cum	7505.85
	g) M45 grade concrete (PSC Grade)	cum	7864.24
	h) M50 grade concrete (PSC Grade)	cum	8116.09
	i) M55 grade concrete (PSC Grade)	cum	8513.23
2.2.2	Extra for using plasticiser in concrete if admixture, plasticiser etc are not included in the item rate		
	a) M15 grade concrete or Prop. 1:2:4	cum	220.52
	b) M20 grade concrete or Prop. 1:1.5:3	cum	265.83
	c) M25 grade concrete	cum	308.88
	d) M30 grade concrete	cum	314.43
	e) M35 grade concrete	cum	324.73
	f) M40 grade concrete (PSC Grade)	cum	332.46
	g) M45 grade concrete (PSC Grade)	cum	359.70
	h) M50 grade concrete (PSC Grade)	cum	409.03
	i) M55 grade concrete (PSC Grade)	cum	378.85
2.2.3	Extra over item Nos. 2.2.1 (I) (B), above for each subsequent floor or part thereof above the 1st floor level.	cum	306.19

CH-3: SHUTTERING WORKS

Item No.	Description	Unit	Rate
3.1	TIMBER SHUTTERING (FORMWORK)		
3.1.1	Providing form work of ordinary timber planking so as to give a rough finish including centering, shuttering, strutting and propping etc., height of propping and centering below supporting floor to ceiling not exceeding 4.0M and removal of the same for in-situ reinforced concrete and plain concrete work in:		
3.1.1.1	Foundation, footings, bases of columns, pile cap, raft and mass concrete works etc.		
	(a) Using 38mm thick plank	Sqm	276.07
	(b) Using 25mm thick plank	Sqm	252.24
3.1.1.2	Sides of tie beams, grade beams etc. at or below ground level.		
	(a) Using 38mm thick plank	Sqm	364.29
	(b) Using 25mm thick plank	Sqm	341.90
3.1.1.3	Columns, Pillars, Posts & Strut		
	(a) Square, Rectangular, polygonal in plan or any shape like Tee/L etc. having plane vertical face		
	(i) Using 38mm thick plank	Sqm	394.81
	(ii) Using 25mm thick plank	Sqm	364.01
	(b) Circular or curved in plan (Using 38mm thick plank)	Sqm	531.15

3.1.1.4	Sides and Soffits of Beams, beam haunchings, cantilever girders, bressumers, lintels and horizontal ties.		
	(a) For depth not exceeding 1.0M.		
	(i) Using 38mm thick plank	Sqm	212.37
	(ii) Using 25mm thick plank	Sqm	202.44
	(b) For depth exceeding 1.0M (Using 38mm thick plank).	Sqm	377.93
3.1.1.5	Flat Surfaces such as soffits of suspended floors, roofs, landings, cantilever slabs, chajjas, balconies and the like.		
	(a) Floors etc. upto 200mm in thickness.		
	(i) Using 38mm thick plank	Sqm	452.94
	(ii) Using 25mm thick plank	Sqm	429.93
	(b) Floors etc. above 200mm in thickness (Using 38mm thick plank).	Sqm	492.38
3.1.1.6	In Ribbed/ waffle slab including rib beams (measurements to be taken on flat surface) (Using 38mm thick plank)	Sqm	
3.1.1.7	Sloping or Battering surfaces such as sloped slabs etc. including folded plates.		
	(a) Where inclination to horizontal plane does not exceed 30° requiring shuttering only on the underside.		
	(i) Using 38mm thick plank	Sqm	512.08
	(ii) Using 25mm thick plank	Sqm	489.24
	(b) Where inclination to horizontal plane exceeds 30 degrees where shuttering may be provided both on under side & upper side, if required.		
	(i) Using 38mm thick plank	Sqm	785.13
	(ii) Using 25mm thick plank	Sqm	744.95
3.1.1.8	Vertical surface such as walls (any thickness), parapet walls, partitions, walls of septic tank, inspection pit and the like including attached pilasters, buttresses, plinth and string courses and the like.		
	(i) Using 38mm thick plank	Sqm	326.05
	(ii) Using 25mm thick plank	Sqm	303.19
3.1.1.9	Vertical ties, fins, sun breaker, louvers, bands, dropwalls etc.		
	(i) Using 38mm thick plank	Sqm	466.24
	(ii) Using 25mm thick plank	Sqm	441.68
3.1.1.10	Staircase with sloping or stepped soffits including risers and stringers but excluding Landing. (Using 38mm thick plank)	Sqm	496.80
3.1.1.11	Arches/ slabs having curved surface in one direction. (Using 38mm thick plank)	Sqm	2626.95
3.1.1.12	Domes, Vaults and shell roofs having curved surface in both directions (only area of the underside shall be measured for payment). (Using 38mm thick plank)	Sqm	5483.35
3.1.2	Extra for each subsequent 1.0M height or part thereof beyond the initial height of 4.0M as mentioned in item no. 3.1.1	Sqm	43.10
3.1.3	Providing form work of ordinary timber planking and removal of the same in edges of slabs and breaks in floors and walls etc. of in-situ reinforced concrete works upto first floor level.		
	(a) Thickness not exceeding 200mm.	R.M.	178.37
	(b) Thickness exceeding 200mm.	Sqm	823.21
3.1.4	Extra for each subsequent floor or part thereof above the First Floor level as mentioned in item no. 3.1.1 & 3.1.3(b).	Sqm	23.19

3.1.5	Extra for each subsequent floor or part thereof above the First Floor level as mentioned in item no. 3.1.3(a).	R.M.	4.64
3.2	PLYBOARD SHUTTERING (FORMWORK)		
3.2.1	Providing form work of 12mm thick Plywood Board so as to give a rough finish including centering, shattering, strutting and propping etc., height of propping and centering below supporting floor to ceiling not exceeding 4.0M and removal of the same for in situ reinforced concrete and plain concrete work in:		
3.2.1.1	Columns, Pillars, Posts & Strut of square/ rectangular/ polygonal in plan or any shape like Tee/L etc. having plane vertical face	Sqm	278.61
3.2.1.2	Sides and Soffits of Beams, beam haunchings, cantilever girders, bressumers, lintels and horizontal ties.		
	(a) For depth not exceeding 1.0M.	Sqm	231.00
	(b) For depth exceeding 1.0M.	Sqm	280.86
3.2.1.3	Flat Surfaces such as soffits of suspended floors, roofs, landings, cantilever slabs, chajjas, balconies and the like.		
	(a) Floors etc. upto 200mm in thickness.	Sqm	345.67
	(b) Floors etc. above 200mm in thickness.	Sqm	379.50
3.2.1.4	In Ribbed/ waffle slab including rib beams (measurements to be taken on flat surface)	Sqm	10222.84
3.2.1.5	Sloping or Battering surfaces such as sloped slabs etc. including folded plates.		
	(a) Where inclination to horizontal plane does not exceed 30o requiring shuttering only on the underside.	Sqm	383.09
	(b) Where inclination to horizontal plane exceeds 30o where shuttering may be provided both on under side & upper side, if required.	Sqm	573.00
3.2.1.6	Vertical surface such as walls (any thickness), parapet walls, partitions and the like including attached pilasters, buttresses, plinth and string courses and the like.	Sqm	225.65
3.2.1.7	Vertical ties, fins, sun breaker, louvers, bands, dropwalls etc.	Sqm	318.23
3.2.1.8	Staircase with sloping or stepped soffits including risers and stringers but excluding Landing.	Sqm	390.30
3.2.2	Extra for each subsequent 1.0M height or part thereof beyond the initial height of 4.0M as mentioned in item no. 3.2.1	Sqm	39.82
3.2.3	Providing form work of 12mm thick Plywood board and removal of the same in edges of slabs and breaks in floors and walls etc. of in-situ reinforced concrete works upto first floor level.		
	(a) Thickness not exceeding 200mm.	R.M.	171.87
	(b) Thickness exceeding 200mm.	Sqm	774.22
3.2.4	Extra for each subsequent floor or part thereof above the First Floor level as mentioned in item no. 3.2.1 & 3.2.3(b).	Sqm	23.19
3.2.5	Extra for each subsequent floor or part thereof above the First Floor level as mentioned in item no. 3.2.3(a).	R.M.	4.64
3.3	STEEL SHUTTERING:		

3.3.1	Providing formwork using 2mm pressed bend shuttering plate of mild sheet and tubular telescopic steel props including centering, shuttering, strutting and propping etc. complete and removal of the same for in situ reinforced concrete and plain concrete work in:		
3.3.1.1	Columns, Pillars, Posts & Strut of square/ rectangular/ polygonal in plan or any shape like Tee/L etc. having plane vertical face	Sqm	116.95
3.3.1.2	Sides and Soffits of Beams, beam haunchings, cantilever girders and bressumers.	Sqm	115.42
3.3.1.3	Flat Surfaces such as soffits of suspended floors, roofs, landings and the like.	Sqm	142.53
3.3.1.4	Sloping or Battering surfaces such as sloped slabs etc. including folded plates.		
	(a) Where inclination to horizontal plane does not exceed 30° requiring shuttering only on the underside.	Sqm	140.38
	(b) Where inclination to horizontal plane exceeds 30° where shuttering may be provided both on under side & upper side, if required.	Sqm	157.78
3.3.1.5	Vertical surface such as walls (any thickness), Lift shafts, partitions and the like including attached pilasters, buttresses, plinth and string courses and the like.	Sqm	143.23
3.3.2	Extra for each subsequent 1.0M height or part thereof beyond the initial height of 4.0M as mentioned in item no. 3.2.1	Sqm	23.42
3.3.3	Extra for each subsequent floor or part thereof above the First Floor level as mentioned in item no. 3.4.1	Sqm	25.69
3.4	L&T DOKA FORMWORK SYSTEM		
3.4.1	Providing formwork using L&T-Doka formwork system with 12mm thick Plywood Board sheathing supported by timber H-beams and steel props with tripods including centering, shuttering, strutting and propping etc. complete and removal of the same for in situ reinforced concrete and plain concrete work in:		
3.4.1.1	Columns, Pillars, Posts & Strut of square/ rectangular/ polygonal in plan or any shape like Tee/L etc. having plane vertical face	Sqm	1084.16
3.4.1.2	Sides and Soffits of Beams, beam haunchings, cantilever girders and bressumers.		
	(a) Using Flex system for height of propping and centering below supporting floor to ceiling upto 4.5m	Sqm	553.72
	(b) Using H.D.T. system for height of propping and centering below supporting floor to ceiling exceeding 4.5m and upto 6.0m.	Sqm	4702.83
3.4.1.3	Flat Surfaces such as soffits of suspended floors, roofs, landings and the like.		
	(a) Using Flex system for height of propping and centering below supporting floor to ceiling upto 4.5m	Sqm	539.15
	(b) Using H.D.T. system for height of propping and centering below supporting floor to ceiling exceeding 4.5m and upto 6.0m.	Sqm	4725.44
3.4.1.4	In Ribbed/ waffle slab including rib beams (measurements to be taken on flat surface)	Sqm	991.26

3.4.1.5	Sloping or Battering surfaces such as sloped slabs etc. including folded plates.		
	(a) Where inclination to horizontal plane does not exceed 30° requiring shuttering only on the underside.	Sqm	571.91
	(b) Where inclination to horizontal plane exceeds 30° where shuttering may be provided both on under side & upper side, if required.	Sqm	703.58
3.4.1.6	Vertical surface such as walls (any thickness), Lift shafts, partitions and the like including attached pilasters, buttresses, plinth and string courses and the like.	Sqm	1570.12
3.4.2	Extra for each subsequent floor or part thereof above the First Floor level as mentioned in item no. 3.4.1	Sqm	26.28
3.4.3	Providing scaffolding using L&T-Doka access scaffold system upto a height of 40m. supported by frames of timber H-beams, bracings including suitable couplers, pins, tower spindles and 2nd class timber plank platforms complete as specified and directed by the department. (measurement to be taken on elevation area of the building/ structure).	Sqm	50.22

Ch-4: MASONRY WORK

Item No.	Description	Unit	Rate
4.1.1	Providing soling in foundation and under floor with stone/ best quality picked jhama brick, sand packed and laid to level and in panel after preparing the subgrade as directed including all labour and materials and if necessary dewatering, complete.		
	(a) Brick on flat soling.	Sqm	368.71
	(b) Brick on edge soling	Sqm	532.75
	(c) Stone soling of thickness 150mm.	Sqm	217.83
4.1.2	Filling sunken floor with broken jhama brick bats and sand including compacting and supplying of materials complete as specified and directed for all levels.	Cum	4323.17
4.1.3	Extra cost for lifting of brickbats and sand for filling of sunk floor for each subsequent floors above the ground floor as directed and specified.	Cum	154.61
4.1.4	Brick work in cement mortar with 1st class brick including racking out joints and curing complete as directed.		
	(I) In Sub-structure upto plinth level including dewatering if necessary		
	(a) In proportion 1:3.	Cum	6045.24
	(b) In proportion 1:4.	Cum	5860.86
	(c) In proportion 1:5.	Cum	5630.36
	(d) In proportion 1:6.	Cum	5515.11
	(e) In proportion 1:1:6 (1 cement:1 lime: 6 sand)	Cum	5681.90
	(f) In proportion 1:1:9. (1 cement:1 lime: 9 sand)	Cum	5438.21
	(II) In superstructure above plinth level up to 1st floor level.		
	(a) In proportion 1:3.(1 cement:3 sand)	Cum	6211.75
	(b) In proportion 1:4.(1 cement:4 sand)	Cum	6027.36
	(c) In proportion 1:5.(1 cement:5 sand)	Cum	5796.87
	(d) In proportion 1:6.(1 cement:6 sand)	Cum	5681.61
	(e) In proportion 1:1:6.(1 cement: 1 Lime :6 sand)	Cum	5848.41
	(f) In proportion 1:1:9.(1 cement: 1 Lime :9 sand)	Cum	5604.71

4.1.5	Extra over item no. 4.1.5 above for brick work in each subsequent floor or part thereof above 1st floor level.	Cum	260.27
4.1.6	230mm thick brick wall in cement mortar with 1st class brick including racking out joints and curing complete as directed in structure		
	(I) In Sub-structure upto plinth level including dewatering if necessary		
	(a) In proportion 1:3.	Sqm	1390.26
	(b) In proportion 1:4.	Sqm	1347.85
	(c) In proportion 1:5.	Sqm	1294.84
	(d) In proportion 1:6. Analysis for 1m3	Sqm	1268.10
	(II) In superstructure above plinth level up to 1st floor level.		
	(a) In proportion 1:3.(1 cement:3 sand)	Sqm	1428.74
	(b) In proportion 1:4.(1 cement:4 sand)	Sqm	1386.33
	(c) In proportion 1:5.(1 cement:5 sand)	Sqm	1333.31
	(d) In proportion 1:6.(1 cement:6 sand)	Sqm	1306.58
4.1.7	Extra over item no. 4.1.6 (II) above for brick work in each subsequent floor or part thereof above 1st floor level.	Sqm	59.71
4.1.8	1st class brick nogged wall in cement mortar including racking out joints and curing complete as directed in super structure above plinth up to 1st floor level (protruding M.S rod/Tor steel of column to be embeded in cement mortar and will be measured and paid separately)		
	(A) 112mm thick brick wall		
	(a) In cement mortar in proportion 1:4. (1 cement:4 sand)	Sqm	653.01
	(b) In cement mortar in proportion 1:5. (1 cement:5 sand)	Sqm	635.49
	(c) In cement mortar in proportion 1:6. (1 cement : 6 sand)	Sqm	620.28
	(B) 75 mm thick wall		
	(a) In cement mortar in proportion 1:3 (1 cement:3 sand)	Sqm	486.60
	(b) In cement mortar in proportion 1:4 (1 cement: 4 sand)	Sqm	464.48
	(c) In cement mortar in proportion 1:5 (1 cement: 5 sand)	Sqm	452.03
4.1.9	112 mm thick reinforced brick wall in prop 1:5 reinforced with two lines of M.S rod/ Tor steel as specified embeded in 38 mm thick concrete course in prop.1:2:4 with coarse agg. 12mm down at every fourth courses embeded with protruding rod in column including curing etc.complete as directed (M.S rod/Tor steel shall be measured and paid separately) in super structure above plinth up to 1st floor level.	Sqm	730.00
4.1.10	75 mm thick reinforced brick wall in prop 1:5 reinforced with one line of M.S rod/ Tor steel as specified embeded in 28 mm thick concrete course in prop.1:2:4 with coarse agg. 12mm down at every third course embeded with protruding rod in column complete as directed (M.S rod / Tor steel shall be measured and paid separately) in super structure up to 1st floor level.	Sqm	520.68

4.1.11	Brick wall in cement lime mortar in prop. 1 cement: 1 lime putty: 6 fine sand including racking out joints and curing complete as directed in super structure above plinth up to 1st floor level (protruding M.S rod/Tor steel of column to be embeded in cement mortar and will be measured and paid separately)		
	a) 112mm thick brick wall	Sqm	645.17
	b) 75mm thick brick wall	Sqm	445.41
4.1.12	1st class brick nogged wall in lime mortar in prop. 1 lime : 2 fine sand including racking out joints and curing etc complete as directed (protruding M.S rod/Tor steel of column to be embedded in cement mortar and will be measured and paid separately) in super structure up to 1st floor level.		
	a) 112mm thick brick wall	Sqm	640.32
	b) 75mm thick brick wall	Sqm	452.69
4.1.13	R.B walling in prop.1:3 reinforced with H.B wire netting made of 20 Gauge G.I wire (diamond shape) in walls, railings, drop walls, parapet, fins, bands, mullions from ground floor level in superstructure up to 1st floor level including racking out joints and curing etc complete as directed (Protruding rods of columns to be tied with H.B. wire netting)		
	(a) 112 mm thick RB walls	Sqm	824.98
	(b) 75 mm thick RB walls	Sqm	621.67
4.1.14	Extra for brick walling for each subsequent floor or part thereof above the 1st floor level.		
	(a) 112mm thick brick/ R.B wall	Sqm	29.09
	(b) 75 mm thick brick/R.B wall	Sqm	21.43
4.2	STONE MASONRY WORK		
4.2.1	Stone masonry work in retaining wall, wing wall, abutment, foundation, steps, plinth etc. in cement mortar in prop 1:6 with levelling course of 1:6:12 with both faces hammer dressed including bonding, providing face stone, through stone and centering including racking of joints, curing and supplying and all carriage of stone as directed including payment of forest royalty and sales tax and carriage.		
	(a) Random Rubble Masonry	Cum	2574.55
	(b) Coarse Rubble Masonry	Cum	2672.81
4.3	Cellular Light Weight concrete Block work		
4.3.1	Cellular Concrete Block work using Cellular Light Weight concrete block of Size 500x200x100mm conforming to IS-2185 Pt-3 or pt-4 in cement morder of proportion 1: 5 including racking out joints and curing complete as directed instructure. (protruding rod is to be provided at every 3 course interval and will be measured and paid separately)		
	(a) In Sub-structure upto plinth level including dewatering if necessary	Cum	7540.66
	(b) In superstructure above plinth level up to 1st floor level.	Cum	7612.36
4.3.2	Extra over item no.4.3.1 (b) above for CC Block work in each subsequent floor or part thereof above 1st floor level.	Cum	130.14

4.3.3	200mm thick wall using Cellular Light Weight concrete block of Size 500x200x100mm conforming to IS-2185 Pt-3 or pt-4 in cement mortar of proportion 1: 5 including racking out joints and curing complete as directed in structure (protruding rod is to be provided at every 3 course interval and will be measured and paid separately)		
	(a) In Sub-structure up to plinth level including dewatering if necessary	Sqm	1553.46
	(b) In Super structure above plinth level upto 1st floor level	Sqm	1578.18
4.3.4	Extra over item no.4.3.3 (b) above for CC block work in each subsequent floor or part thereof above 1st floor level.	Sqm	59.71
4.3.5	100mm thick wall using Cellular Light Weight concrete block of Size 500x200x100mm conforming to IS-2185 Pt-3 or pt-4 in cement mortar of proportion 1: 5 including racking out joints and curing complete as directed in super structure above plinth up to 1st floor level (protruding rod is to be provided at every 3 course interval and will be measured and paid separately)	Sqm	784.41
4.3.6	Extra for CC block walling on item no 4.3.5 for each subsequent floor or part thereof above the 1st floor level.	Sqm	14.55

CHAPTER 5: FLOORING AND FLOOR FINISH

Item No.	Description	Unit	Rate
5.1	CEMENT CONCRETE FLOORING		
5.1.1	Cement concrete works in topping 1:1:2 (1cement :1sand :2 graded aggregates on R.C.C floor, tread and rise of steps finished with a floating coat of neat cement using cement slurry for bond @ 2.75 kg.of cement per sq. m. of floor to be laid in panels including curing complete as directed.		
	(a) 15 mm thick	Sq.m	226.67
	(b) 20 mm thick	Sq.m	270.39
5.1.2	25 mm thick cement concrete topping 1:2:4 (1cement : 2 coarse sand : 4 coarse aggregate of 12 mm nominal size) finished with a floating coat of neat cement finish to be laid in panels including curing complete as directed.	Sq.m	225.26
5.1.3	40 mm thick cement concrete floor consisting of 25 mm under layer of cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of 12.5 mm and down) and 15 mm thick wearing layer in cement concrete in prop. 1:1:2 (1cement :1 coarse sand : 2 coarse aggregate of size 10mm down) finished with a floating coat of neat cement finish using cement slurry for bond @ 2.75 kg. per square metre of floor area , wearing layer is to be laid in panels including curing etc. complete as directed.	Sq.m	405.53

5.1.4	65 mm thick cement concrete floor consisting of 50 mm under layer of cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of 25mm and down) and 15 mm thick wearing layer in cement concrete in prop. 1:1:2 (1cement : 1 coarse sand : 2 coarse aggregate of size 10mm down) laid in panels and finished with a floating coat of neat cement finish (using cement slurry for bond @ 2.75 kg. per square metre of floor area) including curing etc. complete as directed.	Sq.m	511.10
5.1.5	65 mm thick cement concrete floor consisting of 53 mm under layer of cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of 25mm and down) and 12 mm thick wearing coarse of cement sand plastering in prop. 1:2 laid in panels and finished with a floating coat of neat cement finish (using cement slurry for bond @ 2.75 kg. per square metre of floor area) including curing etc. complete as directed.	Sq.m	510.06
5.1.6	40 mm thick red oxide floor with under layer of 30 mm thick cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of size 20 mm and down) and top layer of 10mm thick plaster of cement red-oxide of iron per 50 kg of cement in prop. 1:3 (1 cement red oxide: 3 coarse sand) finished with a floating coat of neat cement red oxide mix of same proportion.	Sq.m	442.09
5.1.7	Providing and laying 50mm thick C.C. floor with metallic concrete hardener topping under layer of 35 mm thick cement concrete in prop.1:2:4 (1cement : 2 coarse sand : 4 coarse aggregate of 20 mm and down) and top layer of 15mm thick of metallic cement hardener consisting of mix 1:2 (1 cement hardener mix : 2 stone aggregate 6 mm and down) by volume with which metallic hardening compound of approved quality is mixed in the ratio of 4:1 (4 parts of cement : 1 part of metallic floor hardening compound of approved quality by weight) including finishing with curing complete as directed.	Sq.m	510.29
5.1.8	Providing and laying 65mm thick C.C. floor with metallic concrete hardener topping under layer of 50 mm thick cement concrete 1:2:4 (1cement:2 coarse sand:4 coarse aggregate of 20 mm and down) and top layer of 15mm thick of metallic cement hardener consisting of mix 1:2 (1 cement hardener mix : 2 stone aggregate 6 mm and down) by volume with which metallic hardening compound of approved quality is mixed in the ratio of 4:1 (4 parts of cement: 1 part of metallic floor hardening compound of approved quality by weight) including finishing with curing complete as directed.	Sq.m	621.47
5.1.9	Providing and fixing glass strips in joints of cement concrete flooring /Terrazo flooring in all level		
	(a) 25 mm wide		
	(i) 4 mm thick	R.m	27.88
	(ii) 6 mm thick	R.m	34.22
	(b) 40 mm wide		
	(i) 4 mm thick	R.m	35.23
	(ii) 6 mm thick	R.m	45.26

	(c) 50 mm wide		
	(i) 4 mm thick	R.m	40.56
	(ii) 6 mm thick	R.m	53.25
	(d) 65 mm wide		
	(i) 4 mm thick	R.m	48.20
	(ii) 6 mm thick	R.m	64.71
5.1.10	Cement plaster skirting with cement mortar in prop. 1:3 (1 cement : 3 coarse sand) finished with a floating coat of neat cement including rounding of junctions with floor.		
	(a) 15 mm thick	Sq.m	230.44
	(b) 20 mm thick	Sq.m	265.70
5.1.11	Red oxide plaster skirting with top layer of 5 mm thick cement plaster of cement mix using 3.5 kg. red oxide of iron per 50kg of cement in proportion 1:3 (1 cement: 3 coarse sand) finished with a floating coat of cement red oxide mix of same proportion.		
	(a) 15 mm thick with under layer of 10 mm thick cement plaster.	Sq.m	291.24
	(b) 20 mm thick with under layer of 15 mm thick cement plaster.	Sq.m	397.19
5.2	CEMENT CONCRETE TILE FLOORING		
5.2.1	Pre-cast cement concrete tiles 20 mm thick with aggregate of sizes up to 6 mm laid in floors, treads of steps and landing with 20 mm thick bed of cement mortar in prop. 1:6 (1 cement : 6 coarse sand) joined with neat cement slurry with pigment to match th		
	(a) Any shade or combination of shades using white cement.	Sq.m	841.63
	(b) Any shade or combination of shades using approx. 50% white cement 50% grey cement.	Sq.m	825.80
	(c) Any shade or combination of shades using grey cement.	Sq.m	809.97
5.2.2	Extra over item no. 5.2.1 for rubbing and polishing the cement concrete tiles in flooring, skirting, dado etc.	Sq.m	142.39
5.3	BRICK FLOORING		
5.3.1	Brick-on-edge floor with 1st class brick and grouted with cement mortar in prop.1:4 (1 cement : 4 sand) including cement pointing in proportion 1:1 (1 cement :1 sand) finishing and curing complete.	Sq.m	717.19
5.3.2	Brick-on-flat floor with 1st class brick and grouted with cement mortar in prop. 1:4 (1 cement : 4 sand) including 15 mm thick cement plastering in proportion 1:2 (1 cement : 2 sand) finishing with a floating coat of neat cement finish using cement slurry @ 2.75 kg. per sq. metre of floor area and curing complete.	Sq.m	672.06
5.4	WOODEN FLOORING		
5.4.1	Wooden planking in flooring including fixing to wooden base frame with necessary wood-screws complete (base of the wooden frame to be measured and paid separately.)		
	With 38 mm thick Indian Teak wood plank, tongue and groove jointed.	Sq.m	2887.70
5.4.2	Wooden planking in flooring including fixing to wooden frame with iron screws complete (base of the wooden frame to be measured and paid separately.)		
	(a) 38mm thick Hollock wood plank, butt jointed.	Sq.m	1886.99

	(b) 25mm thick Hollock wood plank, tongue, and groove jointed.	Sq.m	1325.78
	(c) 25mm thick Hollock wood plank, butt jointed.	Sq.m	1287.72
	(d) 38mm thick Sal wood plank, butt jointed	Sq.m	3465.00
	(e) 25 mm thick Sal wood plank, butt jointed.	Sq.m	2425.82
5.4.3	Supplying, fitting and fixing 38mm thick (finished size) wooden flooring with sal wood plank butt jointed, fitted on sal wood joist of size 100mm x 125mm spaced not more than 900mm apart fitted with 16mm dia bolt and nuts with iron clamp of 50mm x 5mm size inserting in the cement concrete floor base (floor base to be measured and paid separately) including providing 25mm thick damp proof course with water proofing compound as recommended by the manufacturer over C.C floor base including providing kiricide oiling two coats to the timber joists, underside of the flooring planks, two coats of anti-corrosive paint to the iron works, including providing ventilation gaps for the timber works of specified size (75mm x 75mm) covered with welded mesh fabrics at the face of the gap including dressing, levelling, finishing complete as specified and directed by the department at all levels (measurement shall be taken at the finished area of the works).	Sq.m	5114.32
5.5	TERRAZO FLOORING		
5.5.1	40 mm thick marble chips flooring rubbed and polished to granolithic finish with under layer of 32 mm thick cement concrete in prop. 1:2:4 (1 cement:2coarse sand:4 graded stone aggregate of 10 mm and down graded) and top layer 8 mm thick with marble chips of any colour or combination of colours and 1 mm to 4 mm nominal sizes laid in cement marble powder mix 3:1 (3 cement:1marble powder by weight) in proportion 4:7 (4cement marble powder mix :7 marble chips by volume) including providing cement slurry including curing etc. complete.		
	(a) Pigment of any shade with ordinary cement in top layer only.	Sq.m	745.38
	(b) Pigment of any light shade with white cement in top layer	Sq.m	758.14
	(c) Pigment of any shade with approx. 50% white cement and 50% grey cement in top layer.	Sq.m	752.88
	(d) White cement without any pigment on top layer.	Sq.m	738.05
	(e) Light shade pigment with ordinary cement on top layer	Sq.m	729.83
5.5.2	40 mm thick marble chips flooring rubbed and polished to granolithic finish with under layer of 30 mm thick cement concrete in prop. 1:2:4 (1 cement: 2 coarse sand:4 graded stone aggregate of 10 mm and down graded) and top layer 10 mm thick with marble chips of any colour or combination of colours of required sizes and 1 mm to 4 mm nominal size laid in cement marble powder mix 3:1 (3 cement:1marble powder by weight) in prop. 4:7 (4cement marble powder mix :7 marble chips by volume).		
	(a) Pigment of any shade with ordinary cement in top layer only.	Sq.m	802.60
	(b) Pigment of any light shade with white cement in top layer	Sq.m	777.69

	(c) Pigment of any shade with approx. 50% white cement and 50% grey cement in top layer.	Sq.m	842.07
	(d) White cement without any pigment on top layer.	Sq.m	757.60
5.5.3	40 mm thick marble chips flooring rubbed and polished to granolithic finish with under layer of 28 mm thick cement concrete in prop. 1:2:4 (1 cement: 2 coarse sand:4 graded stone aggregate of 10 mm and down graded) and top layer 12 mm thick with marble chips of any colour or combination of colours of required sizes 1 mm to 4 mm nominal sizes laid in cement marble powder mix 3:1 (3 cement : 1 marble powder by weight) in proportion 4:7 (4 cement marble powder mix :7 marble chips by volume).		
	(a) Pigment of any shade with ordinary cement in top layer only.	Sq.m	808.35
	(b) Pigment of any light shade with white cement in top layer	Sq.m	985.91
	(c) Pigment of any shade with approx. 50% white cement and 50% grey cement in top layer.	Sq.m	930.11
	(d) White cement without any pigment on top layer.	Sq.m	871.68
5.5.4	40 mm thick marble chips flooring rubbed and polished to granolithic finish with under layer of 25 mm thick cement concrete in prop. 1:2:4 (1 cement: 2 coarse sand:4 graded stone aggregate of 10 mm and down graded) and top layer 15 mm thick with marble chips of any colour or combination of colours of required sizes from 7 mm to 10 mm nominal sizes laid in cement marble powder mix 3:1 (3 cement:1marble powder by weight) in proportion 4:7 (4 cement marble powder mix :7 marble chips by volume)		
	(a) Pigment of any shade with ordinary cement in top layer only.	Sq.m	908.24
	(b) Pigment of any light shade with white cement in top layer	Sq.m	941.63
	(c) Pigment of any shade with approx. 50% white cement and 50% grey cement in top layer.	Sq.m	920.68
	(d) White cement without any pigment on top layer.	Sq.m	905.24
5.5.5	18 mm thick marble chips skirting or dado rubbed and polish to granolithic finish with under layer 12 mm thick in cement plaster in prop. 1:3 (1 cement :3 coarse sand) and top layer 6 mm thick with marble chips of any colour or combination of colours of 4 mm nominal size laid in cement marble powder mix 3:1(3 cement : 1 marble powder by weight in proportion of 4:7 (4 cement marble powder mix : 7 marble chips by volume)		
	(a) Pigment of any shade with ordinary cement in top layer only.	Sq.m	509.92
	(b) Pigment of any light shade with white cement in top layer	Sq.m	596.13
	(c) Pigment of any shade with approx. 50% white cement and 50% grey cement in top layer.	Sq.m	554.05
	(d) White cement without any pigment on top layer.	Sq.m	556.39
5.5.6	Providing and fixing aluminium strips of 40 mm wide and 1.5 mm thick in terrazo floors in item no. 5.5.1, 5.5.2, 5.5.3 and 5.5.4 .	R.m	65.08

5.5.7	Extra over item no.5.5.1,5.5.2, 5.5.3 and 5.5.4 for Terrazo floors laid as floor borders, margins and similar bands in different shade.		
	(a) Exceeding 75mm but not exceeding 300 mm in width.	Sq.m	17.43
	(b) In narrow band not exceeding 75 mm in width.	Sq.m	32.97
5.5.8	Extra over item nos. 5.5.1, 5.5.2, 5.5.3 and 5.5.4.for laying Terrazo flooring on stair case treads including the cost of forming noses.	Sq.m	23.08
5.5.9	Cutting and cleaning the existing mosaic floor with machine using acid and wax polishing the same as directed for all levels.	Sq.m	66.94
5.6	TERRAZO TILE FLOORING		
5.6.1	Pre-cast Terrazo tiles 20 mm thick with marble chips of sizes up to 6 mm laid in floors, treads of steps, landing with a bed of 10mm thick cement plaster in proportion 1:3 (1 cement : 3 coarse sand) jointed with neat white cement slurry mixed with pigment to match the shade of the tiles including rubbing and polishing complete.		
	(a) With precast tiles of any approved shade / shades using white cement.	Sq.m	1242.77
	(b) Using 50% white cement and 50% ordinary cement.	Sq.m	1227.06
	(c) Using ordinary cement	Sq.m	1208.80
5.6.2	Pre-cast Terrazzo tiles 20 mm thick with marble chips of sizes up to 6 mm laid in skirting and rises of steps with 10mm thick cement plaster in proportion 1:3 (1 cement : 3 coarse sand) jointed with neat white cement slurry including rubbing and polishing complete. with tiles of any approved shade/shades.		
	(a) Using white cement	Sq.m	1267.93
	(b) Using 50% white cement and 50% ordinary cement.	Sq.m	1252.22
	(c) Using ordinary cement	Sq.m	1236.51
5.6.3	Pre-cast Terrazo tiles 20 mm thick with marble chips of sizes up to 6 mm fixed on walls with 10 mm thick cement plaster in proportion 1:3 (1 cement : 3 coarse sand) jointed with neat white cement slurry including rubbing and polishing complete with tiles of any approved shade / shades (walls means both exterior and interior walls)		
	(a) Using white cement	Sq.m	1267.93
	(b) Using 50% white cement and 50% ordinary cement.	Sq.m	1252.22
	(c) Using ordinary cement	Sq.m	1236.51
5.6.4	Chequered Terrazo tiles of 22 mm thickness with marble chips of sizes up to 6 mm fixed on wall with 10 mm thick bed of cement plaster in proportion 1:3 (1 cement : 3 coarse sand) jointed with neat white cement slurry mixed with pigment to match the shades of tiles including rubbing and polishing complete with tiles of any approved shade/shades.		
	(a) Using white cement	Sq.m	1270.65
	(b) Using 50% white cement and 50% ordinary cement.	Sq.m	1227.06
	(c) Using ordinary cement	Sq.m	1221.17
5.7	TILES WORKS		
5.7.1	CERAMIC TILES		
	A) ON FLOOR		

5.7.1.1	Providing Ceramic Tiles of approved quality, size, shape and thickness not less than 8 mm on floors, skirtings, treads and risers of steps over cement mortar bed 15 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with fix-A Tile Choksey/ Sika/ Pedelite/ Rouf/ White cement slurry mixed with approved pigment to match the shade of granite slab, complete at all levels as specified and directed (coloured pigment should be in conformity with colour of slab and as approved and directed by the Department).		
	(a) Normal range Somany/ Orient/Nitco /Asian Make		
	(I) of size 300mmx300mm and above	Sq.m	680.00
	(II) of size 400mmx400mm and above	Sq.m	720.00
	(b) Executive range Somany/ Orient/Nitco/Asian Make		
	(I) of size 300mmx300mm and above	Sq.m	750.00
	(II) of size 400x400 mm and above	Sq.m	760.00
	(c) Delux range Somany/ Orient/Nitco/Qutone Make		
	(I) of size 300mmx300mm and above	Sq.m	970.00
	(II) of size 400mmx400mm and above	Sq.m	980.00
	(III) of size 600mmx600mm and above	Sq.m	990.00
	B) ON WALLS		
5.7.1.2	Providing polished ceramic wall tiles of approved quality, size, shape and thickness not less than 8mm on walls and skirtings over cement mortar bed 10 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey / Sika / Pedelite / Rouf) /white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed. (Cement plastering to be measured and paid separately). (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department). (Walls means both interior and exterior walls).		
	a) Normal range Somany/ Orient/Nitco/Asian Make		
	i) of size 200mmx200mm and above	Sq.m	610.00
	ii) of size 200mmx300mm and above	Sq.m	640.00
	b) Executive range Somany/ Orient/Nitco/Asian Make		
	i) of size 200mmx300mm and above	Sq.m	715.00
	ii) of size 250mmx350mm and above	Sq.m	855.00
	iii) of size 148.5mmx450mm and above	Sq.m	870.00
	c) Delux range Somany/ Orient/Nitco/Varmora Make		
	i) of size 300mmx450mm and above	Sq.m	931.53
	ii) of size 250mmx500mm and above	Sq.m	1225.00
	iii) of size 300mmx600mm and above	Sq.m	1545.91
	iv) of size 300mmx900mm and above	Sq.m	2217.81
5.7.2	VITRIFIED TILES		
	A) ON FLOOR		

5.7.2.1	Providing VITRIFIED floor tiles of approved quality of specified size, shape and thickness not less than 18mm on floors, skirtings, risers and treads of steps over 15 mm thick base of cement mortar in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey/Sika/Pedelite/Rouf) / white cement slurry mixed with approved pigment to match shade of tiles, mixed with approved pigment to match the shade of the tiles, complete at all levels as specified and directed. (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department)		
	a) Normal range Somany/ Orient/Nitco /Varmora/VITA/ Marbitto/ Make		
	i) Of size 400mmx400mm and above	Sq.m	1215.00
	ii) Of size 600mmx600mm and above	Sq.m	1300.00
	b) Executive range Somany/ Orient/Nitco /Varmora/VITA/ Marbitto/ Make		
	i) Of size 800mmx800mm	Sq.m	1450.00
	c) Delux range Somany/ Orient/Nitco /Varmora/VITA/ Marbitto/ Make		
	i) Of size 1000mmx1000mm and above	Sq.m	2000.00
	ii) Of size 1200mmx1600mm and above	Sq.m	2399.00
	B) ON WALLS		
5.7.2.2	Providing polished vitrified wall tiles of approved quality, size, shape and thickness not less than 8mm on walls and skirtings over cement mortar bed 10 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)/white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed. (coloured pigment should be in conformity with colour of tiles and as approved and directed by the department).		
	i) Of size 300mmx450mm and above	Sq.m	1049.33
	ii) Of size 150mmx605mm and above (Strips)	Sq.m	1250.00
	iii) Of size 300mmx600mm and above	Sq.m	1304.51
	iv) Of size 300mmx900mm and above	Sq.m	1970.01
5.7.3	CONCRETE TILES		
	(A) FLOOR TILES		
5.7.3.1	Providing polished SUNROCK-Hi CTS/ MULTIWYN tiles of approved quality, size, shape and thickness not less than 18mm on floors and skirtings over cement mortar bed 10 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey / Sika / Pedelite / Rouf) /white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed. (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department).		
	(a) SUNROCKHi CTS TILES		
	i) Well, Chocolate, Diamond Square, Ramp, Beach Sand (300x300) mm	Sq.m	884.47
	ii) Old Castle (455x455) mm	Sq.m	1023.33

	iii) Banana (370x140) mm	Sq.m	982.19
	(b) MULTIWYN TILES		
	Legato, Ribbera, Matrix, Esquire, Enigma, Flora, Little Flora, Nirvana, Purl, Contoura, Cobble, Canaletta, Nexus, Coral	Sq.m	1045.19
	(B) WALL TILES		
5.7.3.2	Providing Transporting to site, laying, fitting and fixing at all levels tiles for following purpose of SUNROCK-Hi-CTS tiles of high density cement concrete designer coloured tiles made of 100% pure white cement tested as per IS-1237-1980 of approved quality, size shape of thickness 5mm to 50mm to be anchored by adhesive cement and copper wire anchoring including cutting where necessary finished with flush pointing with polymer modified cementations tile joint filler of "ROOF RAINBOW TILE MATE-140" white cement slurry mixed with approved pigment to match shade of tiles complete as specified and directed. (Coloured pigment should be in conformity with color of tiles and as approved and directed by the department. (Cloud, Wave, Old Rock, English Stone)	Sq.m	1119.77
5.7.3.3	Providing polished MULTIWYN tiles of approved quality, size, shape and thickness not less than 8mm on walls and skirtings over cement mortar bed 10 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey / Sika / Pedelite / Rouf) /white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed. (Cement plastering to be measured and paid separately). (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department). (Walls means both interior and exterior walls). (Willow, Brick, Marvel, Sigma, Enamora, Pyramid, Crocotile, Bricket, Denver)	Sq.m	1045.19
	(C) PAVERS		
5.7.3.4	Providing polished tiles of approved quality, size, shape of specified thickness on pavers over a base of 100mm compacted sand gravel including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey / Sika / Pedelite / Rouf) /white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed. (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department).		
	SUN ROCK HIGH CTS / Devi Tiles (Pavers)		
	(a) 80 mm thick	Sq.m	1234.83
	(b) 60 mm thick	Sq.m	1122.76
	(c) 50 mm thick	Sq.m	1021.25
	(d) 40 mm thick	Sq.m	848.96
5.7.3.5	Providing polished Multiwyn/DEVI/VITA tiles of approved quality, size, shape of specified thickness on pavers over a base course including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey / Sika / Pedelite / Rouf) white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed.		

	(A) MULTIWYN Tiles (Pavers)		
	Patella, West wind, Harringbone, Flexi, Super	Sq.m	1045.19
	(B) DEVI Tiles (Pavers)		
	i) of size 300x300mm of 25 thick	Sq.m	773.90
	ii) of size 300x400mm of 30 thick	Sq.m	843.33
5.8	KOTA STONE SLAB / SAND STONE FLOORING		
5.8.1	Kota-stone slab flooring of 18 mm thick over 20mm thick base of cement plaster 1:6 (1 cement : 6 coarse sand) laid over and jointed with grey cement slurry mixed with pigment to match the shade of the slab including rubbing and polishing complete.		
	(1) 550mm x 550mm	Sq.m	1131.64
	(2) 600mm x 900mm and above	Sq.m	1374.76
5.8.2	Kota stone slab of 18 mm thick in riser of steps, skirting, dado and pillars laid with 10mm thick cement plaster 1:3 (1 cement : 3 coarse sand) and jointed with grey cement slurry mixed with pigment to match the shade of the slabs including rubbing and polishing complete.		
	(1) 550mm x 550mm	Sq.m	1128.55
	(2) 600mm x 900mm and above	Sq.m	1387.12
5.8.3	Fine dressed and rubbed stone slab flooring over 18 mm thick base of cement mortar 1:5 (1cement : 5 coarse sand) with joints 3 mm wide, sides buttered with cement mortar 1:2 (1 cement : 2 stone dust) admixed with pigments to match the shade of stone and pointing with the same mortar.		
	Red sand stone		
	(i) 25 mm thick	Sq.m	1316.48
	(ii) 40 mm thick	Sq.m	2218.72
5.8.4	Extra for nosing in steps of red or white, fine dressed and rubbed with sand stone.	R.m	145.85
5.9	MARBLE FLOORING		
5.9.1	18 mm thick Marble stone slab flooring of sizes 1200x1200 including treads and risers of steps, skirting, laid over 20 mm thick base of cement plaster 1:6 (1 cement : 6 coarse sand) laid and jointed with grey cement slurry including rubbing and polishing complete with marble slab of specified quality and shade at all levels.		
	(i) Dungri slab	Sq.m	3960.89
	(i) Marwat Slab	Sq.m	1891.07
	(III) Kumari Slab	Sq.m	1540.12
	(IV) Rajnagar Slab	Sq.m	1610.84
5.9.2	18 mm thick marble stone tiles in flooring including risers and treads of steps, skirting, dado, walls, pillars laid with 10 mm thick base of cement plaster in prop. 1:3 (1 cement : 3 coarse sand) and jointed with grey cement slurry including rubbing and polishing complete with marble tiles of specified quality and shade at all levels.		
	(a) With 18mm thick marble stone tiles of size 600mm x 600 mm		
	(i) Dungri slab	Sq.m	1438.70
	(ii) Marwat Marble	Sq.m	1323.95
	(iii) Kumari Marble	Sq.m	1066.40
	(iv) Rajnagar Marble	Sq.m	1151.81

	(b) With 18mm thick marble stone tiles of size 600mm x 450 mm, 600mm x 300mm		
	(i) Dungri slab	Sq.m	1251.89
	(ii) Marwat Marble	Sq.m	1109.10
	(iii) Kumari Marble	Sq.m	964.99
	(iv) Rajnagar Marble	Sq.m	1023.70
	(c) With 18mm thick marble stone tiles of size 300mm x 300mm		
	(i) Dungri slab	Sq.m	1181.16
	(ii) Marwat Marble	Sq.m	1037.05
	(iii) Kumari Marble	Sq.m	936.96
	(iv) Rajnagar Marble	Sq.m	966.32
5.9.3.	18 mm thick Makrana white marble stone tiles of size 600mm x 450mm in flooring including risers and treads of steps, skirting, nosing of treads, dado, walls, pillars laid with 10 mm thick base of cement plaster in prop. 1:3 (1 cement : 3 coarse sand) and jointed with grey cement slurry including rubbing and polishing complete with marble tiles of specified quality and shade at all levels.(For using Makrana slab, prior permission should be taken from The Chief Engineer, PWD(Bldg.) and rate should be approved before use)	Sq.m	3334.91
5.9.4.	Extra for Granite finish in marble slab / tiles at all levels.	Sq.m	227.69
5.10	GRANITE FLOORING		
5.10.1	Providing polished granite tiles of approved quality, size and shape and thickness not less than 8mm, on walls over 15mm thick, cement mortar bed in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary, finished with flush pointing on wall with ordinary cement slurry mixed with approved pigment to match shade of tiles, complete at all levels as specified and directed.	Sq.m	1157.36
5.10.2	Providing 18mm thick Granite slab on vertical face of pedestal laid over 10 mm thick base of cement mortar in prop. 1:3 (1 cement : 3 coarse sand) jointed with grey cement slurry mixed with pigment to match the shade of the slab including rubbing and polishing complete as directed with approved quality of shade.	Sq.m	2443.02
5.10.3	Providing polished granite tiles of approved quality, size, shape and thickness not less than 18 mm on floors, skirtings, risers and treads of steps over 15 mm thick base of cement mortar in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey/Sika/Pedelite/Rouf) / white cement slurry mixed with approved pigment to match shade of tiles, mixed with approved pigment to match the shade of the tiles, complete at all levels as specified and directed. (Cement plastering to be measured and paid separately). (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department)		
	(i) With Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)	Sq.m	2261.22
	(ii) With White cement	Sq.m	2229.16

5.10.4	Providing polished granite tiles of approved quality, size, shape and thickness not less than 8mm on walls and skirtings over cement mortar bed 15 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile (Choksey / Sika / Pedelite / Rouf) /white cement slurry mixed with approved pigment to match shade of tiles complete at all levels as specified and directed. (Cement plastering to be measured and paid separately). (Coloured pigment should be in conformity with colour of tiles and as approved and directed by the Department). (Walls means both interior and exterior walls).		
	(i) With Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)	Sq.m	1150.26
	(ii) With White cement	Sq.m	1168.92
5.10.5	Providing polished granite slab of approved quality, size, shape and thickness not less than 18 mm on floors, skirtings, treads and risers of steps over cement mortar bed 15 mm thick in prop. 1:3 (1 cement : 3 coarse sand) including cutting where necessary finished with flush pointing with Fix-A-Tile Choksey/Sika/Pedelite/Rouf) /white cement slurry mixed with approved pigment to match the shade of granite slab, complete at all levels as specified and directed. (Cement plastering to be measured and paid separately). (Coloured pigment should be in conformity with colour of slab and as approved and directed by the Department).		
	(i) Normal Quality		
	(i) With Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)	Sq.m	2246.07
	(ii) With White cement	Sq.m	2228.58
	(i) Executive Quality		
	(i) With Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)	Sq.m	2661.63
	(ii) With White cement	Sq.m	2644.14
	(iii) Deluxe Quality		
	(i) With Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)	Sq.m	3350.38
	(ii) With White cement	Sq.m	3332.89
5.11	ANTISKID TILES		
5.11.1	Providing Antiskid Tiles of approved quality size, shape not less than 8mm on floor, skirting over a cement mortar bed 15mm thick of 1:3 (1 cement: 3 coarse sand) approved make fix with Fix-A-Tile (Choksey/Sika/Pedelite/Rouf)/ white cement complete at all level as specified and directed.		
	(A) CERAMIC		
	i)Normal range (Sizes 300 mm x 300mm and above)		
	(a)Orient made: Galaxy Brown, Galaxy creama, Galaxy Pink, Galaxy Green, Pulsar Beige, Pulsar Blue, Pulsar Dove, Pulsar Pink, Burj Brown, Burj Green, Burj Slate	Sq.m	695.47
	ii)Executive range-1 (Sizes 400 mm x 400mm and above)		
	(a) Somany made- VC Shield (Group- 25) (406x406) mm Deluxe -VC	Sq.m	660.75
	iii)Executive range-2 (Sizes 600 mm x 600mm and above)		
	Somany made- VC Shield (Group- 25) (605x605) mm		

	a) Deluxe -VC	Sq.m	1015.62
	b) Deluxe -Rotto VC	Sq.m	1015.62
5.11.2	VITRIFIED		
	i)Normal range (Sizes 300 mm x 300mm and above)		
	Somany made- Durastone (Unpolished heavy duty Vitrified Tiles)		
	a) Premium	Sq.m	968.05
	b) Deluxe	Sq.m	984.76
	ii)Executive range-1 (Sizes 600 mm x 300mm and above)		
	a) Nitco made (Oak Wood)	Sq.m	1700.93
	iii)Executive range-2 (Sizes 600 mm x600mm and above)		
	i) Nitco made		
	a) Imperia Series- Gres Tiles	Sq.m	1230.34
	b) Imperia Series- Gres Lepato (1), Gres Lepato (2), Gres Lepato + Full Body	Sq.m	1286.92
	c) Tuscan Series- Gres Lepato	Sq.m	1493.92
	d) Tuscan Series- Gres Tiles	Sq.m	1313.92
	e) PVT- Hi Polish- PGVT	Sq.m	1604.50
	ii) Somany made- Duragres (Glazed Vitrified Tiles) (605x605) mm		
	a) Gress Matt	Sq.m	1419.35
	b) Gress Stone	Sq.m	1544.07
	c) Gress VC Shield	Sq.m	1544.07
5.12	WOODEN LAMINATED FLOORING/ENGINEERED WOOD FLOORING/CARPET TILES/LOOP PILE CARPETS/DECORATIVE GRASS/LANDSCAPING TURF		
5.12.1	Providing,fitting and fixing Loop pile carpet from Uniproducts (India) Ltd.of thickness not less than 4mm with maximum 7mm pile height with pile fibre of 100% BCF polypropelene of width 3.66m laid on existing concrete floor with approved transparent polythene and foam sheet to make the floor surface even and sound free complete as directed and specified		
	i) Melody Brand	Sq.m	1286.00
	ii) Aspire Brand	Sq.m	1149.70
	iii) Flute/Rainbow Brand	Sq.m	1190.46
	iv) Sonata/Splendour Brand	Sq.m	1489.80
	v) Trump Brand	Sq.m	1717.80
	vi) Harmony Brand	Sq.m	1762.38
	vii) Galaxy Brand	Sq.m	1601.89
	viii) New Royal Guard Brand	Sq.m	1543.30
	viii) Trump (Grey) Brand	Sq.m	1035.06
	ix) Basic Brand	Sq.m	726.81
	ix) Piano Brand	Sq.m	1068.18
5.12.2	Providing,fitting and fixing Synthetic Decorative Grass/Landscaping Turf of width 4.00m from Uniproducts (India) Ltd.of thickness not less than 10mm laid on existing concrete floor complete as directed and specified.		
	(A) Decorative Grass from Uniproducts(India) Ltd.		
	i) SNM-102	Sq.m	1495.48
	ii) SNM-152	Sq.m	1738.77

	(B) Landscaping Turf from Uniproducts(India) Ltd.		
	i) Lawn 66(18)	Sq.m	3044.38
	ii) Field 12	Sq.m	2801.09
	iii) Lawn 66(29)	Sq.m	3644.32
	iii) HT PA 13nd	Sq.m	4896.43
5.12.3	Providing,fitting and fixing wooden flooring from Uniproducts (India) Ltd.of thickness 14mm made of pure wood(Engineered Wood) of different types with a top layer of 3mm thick hard wood and Under layer of 2mm thickness made of polythene and foamsheet laid on existing concrete floor complete as directed and specified		
	Engineered Wood (New) from Uniproducts(India) Ltd.		
	i) Walnut Brand	Sq.m	8824.86
	ii) Merbau/Jatoba/Cherry Brand	Sq.m	8047.47
	iii) Oak Brand	Sq.m	6755.60
5.12.4	Providing,fitting and fixing Laminated abrasion resistant wooden flooring from Uniproducts (India) Ltd.of thickness not less than 8.2mm laid on existing concrete floor with underlayer of approved transparent polythene and foam sheet to make the floor surface even and sound free complete as directed and specified		
	Laminate Floorings-A from Uniproducts(India) Ltd.		
	i) Unitex AC3 Brand	Sq.m	1562.40
	ii) Unitex AC4 Brand	Sq.m	1576.10
5.12.5	Providing,fitting and fixing Carpet tiles of size 500x500mm with thickness not less than 4mm made of 100% Olefin Polypropylene material,with dual backing; Primary backing with PVC and Secondary with Reinforced Fibre glass from Uniproducts (India) Ltd.laid on existing concrete floor complete as directed and specified		
	Carpet Tiles from Uniproducts(India) Ltd.		
	i) Ultimate Brand	Sq.m	1962.94
	ii) Essential Plus Brand	Sq.m	1403.76
	iii) Coral Brand	Sq.m	1247.39
	iv) Tetris/Firkin Brand	Sq.m	2636.76
	v) Junction Brand	Sq.m	2327.23
	vi) Highway Brand	Sq.m	2080.12
	vii) Insight Brand	Sq.m	2226.61
	viii) Zebra/Hendriki/Tristis/Kuda Brand	Sq.m	1947.65
	ix) Pearl/Sapphire Brand	Sq.m	1675.07
	x) Ural Brand	Sq.m	3513.11
	xi) Runway Brand	Sq.m	3015.07

5.13.1	Application of MYK laticrete Spectral OCK Pro Premium Grout into joints per mm with joint gap (starting from 0.5mm) in between tiles which surface & joints are thoroughly cleaned, fully dried, dust, oil & other impurities & acids or alkali substance free, using standard Spectral LOCK Pro premium grout grouting mixing techniques of MYK Laticrete & fill the joints with full surety that full depth & width are fully packed with mixed materials, remove excess materials without leaving excess materials on the surface of tiles following the process of MYK Laticrete spectral LOCK pro Premium Grout, after 15-45 minutes (depending upon the site temperature of the application) Note : Lightly run the wet sponge on the surface of the tile & joint with sponge provided to remove any excess grout. then take a clean thick white towel/sponge with normal detergent soap and sponge to remove any haze from the tiles & other things as per specification of the manufacturing company.		
	a) For 400mm X 400mm size tiles		
	i) For 8mm thick Tiles	Sq.m	88.18
	ii) For 10mm thick Tiles	Sq.m	110.22
	iii) For 11mm thick Tiles	Sq.m	121.24
	iv) For 12mm thick Tiles	Sq.m	132.27
	b) For 600mm X 600mm size tiles		
	i) For 10mm thick Tiles	Sq.m	70.86
	ii) For 11mm thick Tiles	Sq.m	77.94
	iii) For 12mm thick Tiles	Sq.m	85.03
	c) For 800mm X 800mm size tiles		
	i) For 10mm thick Tiles	Sq.m	51.67
	d) For 1000mm X 1000mm size tiles		
	i) For 10mm thick Tiles	Sq.m	63.13
	ii) For 11mm thick Tiles	Sq.m	69.44
	iii) For 12mm thick Tiles	Sq.m	75.75
	e) For 1200mm X 1200mm size tiles		
	i) For 10mm thick Tiles	Sq.m	54.11
	ii) For 11mm thick Tiles	Sq.m	59.52
	iii) For 12mm thick Tiles	Sq.m	64.93
	f) For 1200mm X 600mm size tiles		
	i) For 10mm thick Tiles	Sq.m	144.29
	ii) For 11mm thick Tiles	Sq.m	158.72
	iii) For 12mm thick Tiles	Sq.m	173.15

5.13.2	Application of MYK laticrete LATAPOXY SP-100 Surface Grout per mm with joint gap using joint spacer to provide required gap into the joints in between tiles which surface & joints are thoroughly cleaned, fully dried, dust, oil & other impurities & acids or alkali substance free, using standard epoxy grouting mixing techniques of MYK LATAPOXY SP-100 & fill the joints with full surety that full depth & width are packed with mixed materials, remove excess materials as much as possible without leaving excess materials on the surface of tiles following the process of MYK Laticrete LATAPOX Sp-100, after 15-45 minutes (depending upon the site temperature of the application scrub the surface of the tile & joints with clean water & scrubber to remove any excess materials) Note : the take a clean thick towel & drag across the tile to remove any residue, after 24 hours do a second cleaning with normal shopdetergent to remove any haze from the tiles & other thing as per instruction of the manufacturing company.		
	a) For 400mm X 400mm size tiles		
	i) For 8mm thick Tiles	Sq.m	104.55
	ii) For 10mm thick Tiles	Sq.m	130.69
	iii) For 11mm thick Tiles	Sq.m	143.76
	iv) For 12mm thick Tiles	Sq.m	156.83
	b) For 600mm X 600mm size tiles		
	i) For 10mm thick Tiles	Sq.m	87.13
	ii) For 11mm thick Tiles	Sq.m	95.84
	iii) For 12mm thick Tiles	Sq.m	104.55
	c) For 800mm X 800mm size tiles		
	i) For 10mm thick Tiles	Sq.m	65.34
	ii) For 11mm thick Tiles	Sq.m	71.88
	iii) For 12mm thick Tiles	Sq.m	78.41
	d) For 1000mm X 1000mm size tiles		
	i) For 10mm thick Tiles	Sq.m	52.28
	ii) For 11mm thick Tiles	Sq.m	57.50
	iii) For 12mm thick Tiles	Sq.m	62.73
	e) For 1200mm X 1200mm size tiles		
	i) For 10mm thick Tiles	Sq.m	43.56
	ii) For 11mm thick Tiles	Sq.m	47.92
	iii) For 12mm thick Tiles	Sq.m	52.28
	f) For 1200mm X 600mm size tiles		
	i) For 10mm thick Tiles	Sq.m	65.35
	ii) For 11mm thick Tiles	Sq.m	71.88
	iii) For 12mm thick Tiles	Sq.m	79.19
CH - 6: PLASTERING AND HERITAGE FINISHES			
Item No.	Description	Unit	Rate
6	PLASTERING		
6.1	LIME PLASTERING		

6.1.1	15 mm thick lime plaster 1:1:1 (1 lime: 1 surkhi:1 sand) in single coat on fair side of brick/concrete walls for interior plastering up to second floor level including arises, internal round angles, chamfers and / or rounded angles not exceeding 80mm girth and finished even and smooth including curing complete as directed.	Sqm	160.86
6.1.2	15 mm thick lime plaster 1:1:1 (1 lime: 1 surkhi:1 sand) in single on half brick walls for interior plastering up to second floor level including arises, internal rounded angles, chamfers and/or rounded angles not exceeding 80mm girth and finished even and smooth including curing complete as directed.	Sqm	165.63
6.1.3	Extra over item no. 6.1.1 and 6.1.2 for each subsequent floor or part thereof beyond 2nd floor level.	Sqm	15.31
6.2	CEMENT PLASTERING		
6.2.1	10 mm thick Cement plaster in single coat on fair side of brick/concrete walls for interior plastering up to 1st floor level including arises or rounded angles not exceeding 80mm girth and finished even and smooth including curing complete as directed.		
	a) In cement mortar 1:3	Sqm	128.16
	b) In cement mortar 1:4	Sqm	116.85
	c) In cement mortar 1:5	Sqm	107.87
	d) In cement mortar 1:6	Sqm	101.88
6.2.2	15 mm thick Cement plaster in single coat on single or half brick wall for interior plastering up to 1st floor level including arises, internal rounded angles, not exceeding 80mm girth and finished even and smooth including curing complete as directed.		
	A) On rough side		
	a) In cement mortar 1:3	Sqm	167.05
	b) In cement mortar 1:4	Sqm	147.23
	c) In cement mortar 1:5	Sqm	136.63
	d) In cement mortar 1:6	Sqm	127.87
	B) On fair side		
	a) In cement mortar 1:3	Sqm	165.52
	b) In cement mortar 1:4	Sqm	145.70
	c) In cement mortar 1:5	Sqm	135.09
	d) In cement mortar 1:6	Sqm	126.34
6.2.3	20 mm thick Cement plaster in two coats on single or half brick walls for interior plastering up to 1st floor level including arises, internal rounded angles, chamfers and / or rounded angles not exceeding 80mm in girth and finished even and smooth including curing complete as directed.		
	A) On rough side (backing coat 15 mm and finishing coat 5mm thick)		
	a) In cement mortar 1:3	Sqm	210.14
	b) In cement mortar 1:4	Sqm	183.86
	c) In cement mortar 1:5	Sqm	169.57
	d) In cement mortar 1:6	Sqm	157.59
	B) On fair side (backing coat 10 mm and finishing coat 10mm thick)		
	a) In cement mortar 1:3	Sqm	243.74
	b) In cement mortar 1:4	Sqm	217.46

	c) In cement mortar 1:5	Sqm	203.17
	d) In cement mortar 1:6	Sqm	191.19
6.2.4	Extra over item no 6.2.1 to 6.2.3 for providing and mixing water proofing materials in cement mortar in prop as recommended by the manufacturer including curing complete as directed.		
	<i>i) 10 mm thick plaster</i>		
	a) In cement mortar 1:3	Sqm	6.93
	b) In cement mortar 1:4	Sqm	5.15
	c) In cement mortar 1:5	Sqm	4.21
	d) In cement mortar 1:6	Sqm	3.42
	<i>ii) 15 mm thick plaster</i>		
	a) In cement mortar 1:3	Sqm	10.39
	b) In cement mortar 1:4	Sqm	7.75
	c) In cement mortar 1:5	Sqm	6.31
	d) In cement mortar 1:6	Sqm	5.11
	<i>ii) 20 mm thick plaster</i>		
	a) In cement mortar 1:3	Sqm	13.89
	b) In cement mortar 1:4	Sqm	10.35
	c) In cement mortar 1:5	Sqm	8.45
	d) In cement mortar 1:6	Sqm	6.80
6.2.5	Extra over item no 6.2.1 to 6.2.3 for plastering on ceiling and soffits of stairs up to 1st floor level (instead of plastering on walls) including curing complete as directed.	Sqm	21.78
6.2.6	Extra over item no 6.2.1 to 6.2.3 for interior plastering above 1st floor level for every addl floor or part thereof including curing complete as directed.		
	a) On Single coat plaster	Sqm	3.87
	b) On two coat plaster	Sqm	4.64
6.2.7	Extra over item no 6.2.1 to 6.2.3 for exterior plastering over wall up to a height of 1st floor level above ground level including curing complete as directed.		
	a) On Single coat plaster	Sqm	3.83
	b) On two coat plaster	Sqm	4.98
6.2.8	Extra over item no 6.2.1 to 6.2.3 for exterior plastering over wall for every addl floor or part thereof beyond 1st floor level including curing complete as directed.		
	a) On Single coat plaster	Sqm	3.53
	b) On two coat plaster	Sqm	4.41
6.2.9	15 mm thick plain cement mortar bands in cement mortar 1:4 in width including curing complete as directed.		
	<i>A) Up to 300 mm width</i>		
	a) Flush band	Sqm	251.73
	b) Sunk band	Sqm	267.09
	c) Raised band	Sqm	292.64
	<i>B) Above 300 mm width</i>		
	a) Flush band	Sqm	190.37
	b) Sunk band	Sqm	205.71
	c) Raised band	Sqm	231.28
6.3	Pointing		

6.3.1	Pointing on brick work with cement mortar including curing complete as directed.(Measurement shall be taken on surface area of the wall).		
	A) Cement mortar prop. 1:3		
	a) Flush pointing	Sqm	74.85
	b) Ruled pointing	Sqm	77.41
	c) Cut or weather struck pointing	Sqm	77.41
	c) Raised and Cut pointing	Sqm	119.34
	B) Cement mortar prop. 1:4		
	a) Flush pointing	Sqm	71.49
	b) Ruled pointing	Sqm	73.98
	c) Cut or weather struck pointing	Sqm	73.98
	d) Raised and Cut pointing	Sqm	115.92
6.3.2	Pointing on brick flooring with cement mortar including curing complete as directed.(Measurement shall be taken on surface area of the floor).		
	A) On flat brick flooring		
	i) Cement mortar prop. 1:2		
	a) Flush pointing	Sqm	55.19
	b) Ruled pointing	Sqm	58.26
	ii) Cement mortar prop. 1:3		
	a) Flush pointing	Sqm	52.33
	b) Ruled pointing	Sqm	55.39
	iii) Cement mortar prop. 1:4		
	a) Flush pointing	Sqm	49.93
	b) Ruled pointing	Sqm	53.00
	iv) Cement mortar prop. 1:5		
	a) Flush pointing	Sqm	48.64
	b) Ruled pointing	Sqm	51.71
	B) On edge brick flooring		
	i) Cement mortar prop. 1:2		
	a) Flush pointing	Sqm	73.54
	b) Ruled pointing	Sqm	76.09
	ii) Cement mortar prop. 1:3		
	a) Flush pointing	Sqm	68.84
	b) Ruled pointing	Sqm	71.39
	iii) Cement mortar prop. 1:4		
	a) Flush pointing	Sqm	65.24
	b) Ruled pointing	Sqm	67.80
	iv) Cement mortar prop. 1:5		
	a) Flush pointing	Sqm	63.30
	b) Ruled pointing	Sqm	65.86
6.3.3	Pointing on stone masonry with cement mortar including curing complete as directed.(Measurement shall be taken on surface area of the wall)		
	A) On random rubble uncoursed stone masonry		
	i) Cement mortar prop. 1:3		
	a) Flush pointing	Sqm	80.21
	b) Ruled pointing	Sqm	83.28
	c) Raised and Cut pointing	Sqm	134.42
	ii) Cement mortar prop. 1:4		
	a) Flush pointing	Sqm	77.44
	b) Ruled pointing	Sqm	80.51

	c) Raised and Cut pointing	Sqm	131.65
	B) On square, rubble coursed or uncoursed stone masonry		
	i) Cement mortar prop. 1:3		
	a) Flush pointing	Sqm	75.61
	b) Ruled pointing	Sqm	78.16
	c) Raised and Cut pointing	Sqm	124.19
	ii) Cement mortar prop. 1:4		
	a) Flush pointing	Sqm	72.84
	b) Ruled pointing	Sqm	75.40
	c) Raised and Cut pointing	Sqm	121.43
6.3.4	Raised and cut pointing on Ashler masonry or concrete block walling in white cement mortar 1:3 (1 white cement : 3 marble dust) including curing complete as directed. (Measurements shall be taken on surface area of the wall)	Sqm	211.99
6.4	HERITAGE FINISHING		
6.4.1	Supplying and laying of permanent wall finish for interior and exterior walls with surface coating of Heritage Surface Texture of Bakelite Hylam Limited for all types of buildings at all levels as specified and directed by the department.		
	(a) Granules (Code : HGR)	Sqm	484.30
	(b) Granules with Top Coat (Code : HGRT)	Sqm	546.93
	(c) Fine Granules (Code : FG)	Sqm	337.98
	(d) Fine Granules with top coat (Code : FGHT)	Sqm	483.16
	(e) Flakes (Code : HFR)	Sqm	568.72
	(f) Flakes with Top coat (Code : HFRT)	Sqm	651.31
	(g) Fine Flakes (Code : FFH)	Sqm	496.67
	(h) Fine Flakes with Top coat (Code : FFHT)	Sqm	650.66
	(i) Granite Finish (Code : HFG)	Sqm	813.54
	(j) Sand Dune	Sqm	273.69
	(k) Silica Duke	Sqm	111.84
	(l) Lineal Finish	Sqm	322.73
	(m) Roto Finish	Sqm	302.91
	(n) Roller Coat (Application by Special Roller)	Sqm	231.37
	(o) Roller Coat (Application by brush)	Sqm	106.03
	(p) Spray Coat (Bubble Finish)	Sqm	382.85
	(q) Spray Coat (Compact Finish)	Sqm	698.69

Ch 7: CEILLING, LINING AND PARTITION

Item No.	Description	Unit	Rate
7.1	WATTLE/ SPILT BAMBOO CEILING		
7.1.1	Providing wattle /spilt bamboo ceiling mud plastered on top and 15mm cement plaster in proportion 1:4 in two coats -10 mm first coat and 5mm second coat on bottom, white wash three coats on cement plastered surface complete (timber ceiling joists to be measured and paid separately).	Sqm	183.83
7.1.2	Providing wattle/ spilt bamboo ceiling 15mm cement plastered in proportion 1:10 on top and hessian cloth at bottom fitted with 38mmx12mm 1st class hollock / bonsum / sundi timber beading chamfered and finished with 2 coats of enamel painting to the timber beads complete (timber ceiling joists to be measured and paid separately).	Sqm	211.87

7.1.3	Providing wattle/spilt bamboo ceiling 15mm cement plastered at top in proportion 1:10 and markin cloth at bottom fitted with 38mm x12mm 1st class hollock/ bonsum/ sundi timber beading chamfered and finished with 2 coats of chalk washing to the markin cloth and 2 coats of enamel painting to timber beads complete (timber ceiling joists to be measured and paid separately).	Sqm	215.83
7.1.4	Providing wattle/ spilt bamboo crete ceiling with 25 mm thick mud plaster on top and bottom in two coats - 15mm and 10mm, the final coat being trowel finished and white wash 3 coats below. (ceiling joists to be measured and paid separately).	Sqm	166.50
7.1.5	Providing single layer bamboo-mat ceiling with lattice-pattern jaffri or mesh size 40mmx 40mm made out of bamboo slips of size 5mmx2mm size at bottom stiffened with bamboo slip of 25 mm x 5 mm size. (ceilling joist to be measured and paid separately).	Sqm	161.08
7.1.6	Repairing and refixing tarja mat ceiling.	Sqm	38.77
7.1.7	Taking out and refixing dharamat ceiling after repairing as necessary including changing damage kamies, dhara, jaffri works, bethua bamboo ceiling joints, as and where necessary, including cleaning the top and bottom faces of the ceiling as directed complete,including tying with galvd wire for fixing in proper position.	Sqm	39.34
7.2	GYPBOARD CEILING		
7.2.1	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) M/F Suspended Ceiling which includes G.I. perimeter channels of size 0.55mm thick (having one flange of 20mm and another flange of 30mm and a web of 27mm) along with perimeter of ceiling, screw fixed to brick wall/ partition with the help of nylon sleeves and screws, at 610mm c/c. Then suspending G.I. intermediate channels of size 45mm (0.9mm thick with two equal flanges of 15mm each) from the soffit at 1220mm c/c with ceiling angle of width 25mmx10mmx0.55mm thick fixed to soffit with G.I.cleat and steel expansion fasteners.		
	Ceiling section of 0.55mm thickness having knurled web of 51.5mm and two equal flanges of 26mm each with lips of 10.5mm are then fixed to the intermediate channel with the help of connecting clip and in direction perpendicular to the intermediate channel at 457mm c/c, 9.5mm/ 12.5mm tapered edge Gypboard (conforming to IS:2095-1982) is then screw fixed to ceiling section with 25mm long drywall screws at 230mm c/c. Screw fixing is done mechanically either with screw driver or drilling machine with suitable attachment. Finally, the boards are to be jointed and finished so as to have a flush look which includes filling and finishing the tapered and square edges of the boards with jointing compound, joint paper tape and two coats of drywall topcoat suitable for Gypboard complete at all levels as specified and directed. (For light fittings, providing opening for doors, window, ventilators etc, cut out made with frame of perimeter channel supported suitably to be measured and paid separately where necessary)		

(i)	9.5mm thick	Sqm	1214.80
(ii)	12.5mm thick	Sqm	1105.50
7.2.2	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) suspended ceiling which includes G.I. perimeter channels of size 0.55mm thick (having one flange of 20mm and another flange of 30mm and a web of 27mm) along with perimeter of ceiling, screw fixed to brick wall/ partition with the help of nylon sleeves and screws, at 610mm c/c. Then suspending G.I. intermediate channels of size 45mm (0.9mm thick with two equal flanges of 15mm each) from the soffit at 1220mm c/c with ceiling angle of width 25mmx10mmx0.55mm thick fixed to soffit with G.I.cleat and steel expansion fasteners. Ceiling section of 0.55mm thickness having knurled web of 51.5mm and two equal flanges of 26mm each with lips of 10.5mm are then fixed to the intermediate channel with the help of connecting clip and in direction perpendicular to the intermediate channel at 298mm c/c.		
	Then 12.5mm thick square edge Fulltone/ Sundance/ Cratone/ Louvertone/ Textured. Gypboard panels of size 595mmx595mm (conforming to IS:2095-1982) is then screw fixed to ceiling section with 25mm long drywall screws at 290mm c/c. Screw fixing is done mechanically either with screw driver or drilling machine with suitable attachment. Finally, the boards are to be jointed and finished so as to have a flush look which includes filling and finishing the tapered and square edges of the boards with jointing compound, joint paper tape and two coats of drywall topcoat suitable for Gypboard complete at all levels as specified and directed. (For light fittings, grill diffusers etc. cut out to be made with the frame of perimeter channel supported suitably and measured and paid separately as and where necessary). (Glass wool to be measured and paid separately).		
(A)	M/F Frame Work with Design Type Textured Gypboard Tiles 9.5mm thick.	Sqm	1453.00
(B)	M/F Frame Work with Design Type Semi Perforated Gypboard (Sundance) Tiles (NRC value 0.20 with 50mm glass wool mat backing) 12.5mm thick.	Sqm	1462.80
(C)	M/F Frame Work with Acoustic Type Fissured Gypboard (Cratone) Tiles (NRC value 0.57 with 50mm glass wool mat backing) 12.5mm thick.	Sqm	1679.40
(D)	M/F Frame Work with Acoustic Type Fully Perforated Gypboard (Fulltone) Tiles (NRC value 0.75 with 50mm glass wool mat backing) 12.5mm thick.	Sqm	1661.00
(E)	M/F Frame Work with Acoustic Type Slit Gypboard (Louvertone) Tiles (NRC value 0.72 with 50mm glass wool mat backing) 12.5mm thick.	Sqm	1684.50

7.2.3	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) pre-coated G.I. angle size 24mmx24mmx80mm thick fixed to the brick wall/ partition at the perimeter of ceiling with nylon sleeves and screws, at 610mm c/c. Then suspending main 'T' section (bottom pre-coated) size 24mmx38mmx0.70mm and 3600mm long from the soffit in one direction with the help of metal Rawl Plug. Soffit cleat and G.I. Rod 4mm diameter with galvanised level clip 1220mm c/c G.I. cross	Sqm	
	(A) BPB Grid ceiling system with plain Gypboard Tiles 12.5mm thick.	Sqm	1294.70
	(B) BPB Grid ceiling system with plain Gypboard Tiles 9.5mm thick.	Sqm	1286.80
	(C) BPB Grid ceiling system with plain Gypboard Tiles 8.0 mm thick.	Sqm	1270.60
	(D) BPB Grid ceiling system with P.V.C. laminated plain Gypboard (Polyshield-P) Tiles 9.5mm thick.	Sqm	1599.20
	(E) BPB Grid ceiling system with P.V.C. laminated crack Gypboard (Polyshield-C) Tiles 9.5mm thick.	Sqm	1608.50
	(F) BPB Grid ceiling system with Design Type Textured Gypboard Tiles 9.5mm thick.	Sqm	1389.90
	(G) BPB Grid ceiling system with Design Wall Paper Gypboard (Paprex Panel) Tiles	Sqm	
	(i) 8 mm thick	Sqm	1487.60
	(ii) 9.5mm thick	Sqm	1453.60
	(H) BPB Grid ceiling system with Fabric Based Sound absorb Panel (595mmx595mm).	Sqm	2115.10
	(I) BPB Grid ceiling system with Design Type Semi Perforated Gypboard (Sundance) Tiles	Sqm	1399.70
7.2.4	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) wall lining which includes 12.5mm/ 9.5mm thick Gypboard (conforming to IS:2095-1982) screw fixed to the Ceiling section of size 0.55mm thick having a knurled web of 51.5mm and two equal flanges of 26mm each with lips of 10.5mm made of G.I. Steel placed at 600mm c/c in perimeter channels of size 0.55mm thick having one flange of 20mm and flange of 30mm and a web of 27mm made of G.I. Steel fixed to the floor & ceiling with nylon sleeves & wood screws at 600mm c/c.		
	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) wall lining which includes 12.5mm/ 9.5mm thick Finally the boards are to be jointed and finished so as to have a Flush look which includes filling and finishing with jointing Compound, joint paper tape, two coat drywall top coat suitable for Gypboard complete as specified and directed at all levels (Glass wool to be measured and paid separately) (For light finishing, skirting providing opening for doors, windows, ventilators etc. cut out made with the frame of perimeter channel supported suitably to be measured and paid suitably when necessary)		
	(A) M/F wall lining with plain Gypboard 9.5mm thick.	Sqm	1224.50
	(B) M/F wall lining with plain Gypboard 12.5mm thick.	Sqm	1218.20

7.2.5	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) wall lining which includes 12.5mm thick Louvertone/ Sundance/ Cratone/ Fulltone (slit) Gypboard ceiling panels of size 595mmx595mm (conforming to IS:2095-1982) screw fixed to the Ceiling section of size 0.55mm thick having a knurled web of 51.5mm and two equal flanges of 26mm each with lips of 10.5mm made of G.I. Steel placed at 298mm c/c in perimeter channels of size 0.55mm thick having one flange of 20mm and flange of 30mm and a web of 27mm made of G.I. Steel fixed to the floor & ceiling with nylon sleeves & wood screws at 600mm c/c.		
	Finally the boards are to be jointed and finished so as to have a Flush look which includes filling and finishing with jointing Compound, joint paper tape, two coat drywall top coat suitable for Gypboard complete as specified and directed at all levels (Glass wool to be measured and paid separately) (For light finishing, skirting providing opening for doors, windows, ventilators etc. cut out made with the frame of perimeter channel supported suitably to be measured and paid suitably when necessary).		
	(A) M/F wall lining with Design Type Semi Perforated (Sundance) Gypboard 12.5mm thick (NRC value=0.20 with 50mm glass wool mat backing).	Sqm	1332.80
	(B) M/F wall lining with Acoustic Type Fissured (Cratone) Gypboard 12.5mm thick (NRC value=0.57 with 50mm glass wool mat backing).	Sqm	1569.00
	M/F wall lining with Acoustic Type Fully Perforated (Fulltone) Gypboard 12.5mm thick (NRC value=0.75 with 50mm glass wool mat backing).	Sqm	1549.00
	(D) M/F wall lining with Acoustic Type Slit (Louvertone) Gypboard 12.5mm thick (NRC value=0.72 with 50mm glass wool mat backing).	Sqm	1574.60
7.2.6	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) 97mm thick metal stud partition (Single layer) including one layer of tapered edge 12.5mm thick Gypboard (conforming to IS:2095-1982) screw fixed to drywall screw of 25mm at 300mm c/c to either side of 75mm studs (0.55mm thick having one flange of 34mm and another flange of 36mm made of G.I. steel) with joints staggered to avoid leakage through joints and jointing the square and tapered edges of the boards and finishing so as to have a flush look which includes filling and finishing with jointing Compound, joint paper tape, two coat drywall top coat suitable for Gypboard complete as specified and directed at all levels (Glass wool to be measured and paid separately) (For light finishing, skirting providing opening for doors, windows, ventilators etc. cut out made with the frame of perimeter channel supported suitably to be measured and paid suitably when necessary).		
	(A) Partition with Plain Gypboard	Sqm	2006.60
	(B) Partition with Acoustic Type Fissured (Cratone) Gypboard (NRC value=0.57 with 50mm glass wool mat backing)	Sqm	2442.00

	(C) Partition with Acoustic Type Fully Perforated (Fulltone) Gypboard (NRC value=0.75 with 50mm glass wool mat backing)	Sqm	2876.70
	(D) Partition with Acoustic Type Slit (Lovertone) Gypboard (NRC value=0.72 with 50mm glass wool mat backing)	Sqm	3170.70
7.2.7	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) M/ F suspended ceiling which includes G.I. perimeter channels of size 0.55mm thick (having one flange of 20mm and another flange of 30mm and a web of 27mm) along with perimeter of ceiling, screw fixed to brick wall/ partition with the help of nylon sleeves and screws, at 610mm c/c. Then suspending G.I. intermediate channels of size 45mm (0.9mm thick with two equal flanges of 15mm each) from the soffit at 1220mm c/c with ceiling angle of width 5mmx10mmx0.55mm thick fixed to soffit with G.I.cleat and steel expansion fasteners. Ceiling section of 0.55mm thickness having knurled web of 51.5mm and two equal flanges of 26mm each with lips of 10.5mm are then fixed to the intermediate channel with the help of connecting clip and in direction perpendicular to the intermediate channel at 457mm c/c, 9.5mm/ 12.5mm Gypboard. False ceiling (Light Duty) with Metal Polyester Laminated Board on Back (MPL) (conforming to IS:2095-1982) is then screw fixed to ceiling section with 25mm long drywall screws at 230mm c/c. Screw fixing		
	is done mechanically either with screw driver or drilling machine with suitable attachment. Finally, the boards are to be jointed and finished so as to have a flush look which includes filling and finishing the tapered and square edges of the boards with jointing compound, joint paper tape and two coats of drywall topcoat suitable for Gypboard complete at all levels as specified and directed. (For light fittings, providing opening for doors, window, ventilators etc., cut out made with frame of perimeter channel supported suitably to be measured and paid separately as and where necessary).		
	i) 9.5mm MPL	Sqm	1487.60
	ii) 12.5mm MPL	Sqm	1502.70
7.2.8	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) M/ F suspended ceiling which includes G.I. perimeter channels of size 0.55mm thick (having one flange of 20mm and another flange of 30mm and a web of 27mm) along with perimeter of ceiling, screw fixed to brick wall/ partition with the help of nylon sleeves and screws, at 610mm c/c. Then suspending G.I. intermediate channels of size 45mm (0.9mm thick with two equal flanges of 15mm each) from the soffit at 1220mm c/c with ceiling angle of width 5mmx10mmx0.55mm thick fixed to soffit with G.I.cleat and steel expansion fasteners. Ceiling section of 0.55mm thickness having knurled web of 51.5mm and two equal flanges of 26mm each with lips of 10.5mm are then fixed to the intermediate channel with the help of connecting clip and in direction perpendicular to the intermediate channel at 457mm c/c, 9.5mm/ 12.5mm thick Gypboard.	Sqm	

	False ceiling (Light Duty) with Moisture Resistant Board (conforming to IS:2095-1982) is then screw fixed to ceiling section with 25mm long drywall screws at 230mm c/c. Screw fixing is done mechanically either with screw driver or drilling machine with suitable attachment. Finally, the boards are to be jointed and finished so as to have a flush look which includes filling and finishing the tapered and square edges of the boards with jointing compound, joint paper tape and two coats of drywall topcoat suitable for Gypboard complete at all levels as specified and directed. (For light fittings, providing opening for doors, window, ventilators etc., cut out made with frame of perimeter channel supported suitably to be measured and paid separately as and where necessary).		
	(i) 9.5mm MRB	Sq.m.	1427.00
	(ii) 12.5mm MRB	Sqm	1457.60
7.2.9	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) 50mm floor & ceiling channels (with flanges of 32mm each and 0.55mm thick) with the help of nylon sleeves and steel screws to the floor and R.C.C. respectively. 48mm studs (with flanges of 36mm and 34mm and 0.55mm thick) are then placed at every 595mm c/c in between floor and ceiling channels. The studs are then anchored to the wall at every 600mm c/c with help of L cleats fixed to studs with nut and bolts and to the wall with metal rawl plugs. Then 15mm thick Gypboard Silt Tone Sound Smart panels having 3mm thick non woven carpet (plain) are then fixed to the studs with a suitable beading. 50mm thick glass wool shall be placed at the back side of the panels to get the required absorption. (For skirting light fittings, grills diffusers etc. cut out to be made with the frame of perimeter channels supported suitably and measured and paid separately as and when necessary). (Glass wool to be measured and paid separately).	Sqm	2762.30
7.2.10	Extra for providing Twiga Insul glass wool mat backing with chicken mesh cover of Twiga Fibre Glass insulation as specified and directed at all levels.		
	(i) 25mm thick	Sqm	507.70
	(ii) 50mm thick	Sq.m.	702.20
7.2.11	Extra for providing cut outs for light/ A.C. diffusers, drops, access panel etc. in Gypboard ceiling portion, partition wall, lining etc. complete at all levels as specified and directed.(net area of cut out portion to be measured and paid only).	Sq.m.	605.00
7.2.12	Extra for providing two coats of water resistant sealant to upper surface of the board to prevent it from damage due to water leakage complete at all levels as specified and directed.	Sqm	44.00
7.2.13	Supplying, fitting, fixing and painting where necessary (one coat primer and two coats of paint) moulded timber skirting with best quality 1st class local wood timber planks of 75mm height and 20mm thick at bottom of Gypboard partition and wall lining etc. complete as specified and directed at all levels.	Sqm	1016.40

7.3	WOODEN WORK IN CEILING AND WALLS		
7.3.1	Providing and fixing wood plank ceiling with tongued and grooved joint leaving a gap of 10 mm as per drawing including necessary nails & wood screws complete (Timber frame work to be measured and paid separately).		
	(a) Teak wood plank (20mm thick)	Sqm	1457.72
	(b) Hollock (25mm thick)	Sqm	1157.48
	(c) Bonsum / Sundi wood plank (25mm thick)	Sqm	1130.19
	(d) Gamari timber plank (25mm thick)	Sqm	1157.48
7.3.2	Providing and fixing wood plank ceiling with butt and lap joints including necessary nails and wood screws complete (Timber frame work to be measured and paid separately).		
	(a) Hollock (25mm thick)	Sqm	1330.44
	(b) Bonsum/ Sundi timber plank (25mm thick)	Sqm	1320.13
	(c) Gamari timber plank (25mm thick)	Sqm	1320.13
	(d) 2nd class timber plank (25mm thick)	Sqm	1073.92
7.3.3	Providing and fixing plain, square edged cover fillet in ceiling and lining with wood screws including mitreing at junctions.		
	(a) Teak (40mmx12mm).	Rm	59.08
	(b) Hollock timber plank (40mmx12mm)	Rm	46.49
	(c) Extra for making chamfered edges of cover fillets.	Rm	12.37
7.3.4	Providing, fitting and fixing frame work for wall panelling with Hollock/ Bonsum/ Sundi battens of size 50mmx25mm fixed over tapering plugs made out of trapezoidal section having base 50mmx50mm at the bottom and 38mmx38mm at the top with depth of 50mm embedded in cement sand mortar in prop 1:3 and spaced 450mm to 500mm apart complete with counter sunk wood screw of 50mm length.	Sqm	357.22
7.3.5	Providing and fixing Teak wood lining with tongue and groove joint leaving a gap of 10mm as per drawing including fitting and fixing with countersunk screws with the frame work complete (frame work to be measured and paid separately).		
	i) 12 mm thick.	Sqm	1296.76
	ii) 20 mm thick.	Sqm	1867.51
	iii) 25 mm thick.	Sqm	2240.63
7.3.6	Providing and fixing 1st class local wood Hollock / Bonsum / Sundi / Gamari lining with tongue and groove joints leaving a gap of 10mm as per drawing including fitting and fixing with countersunk screws with the frame work complete (frame work will be measured and paid separately).		
	i) 12 mm thick.		
	a) Hollock	Sqm	1050.15
	b) Gamari	Sqm	1050.15
	ii) 20 mm thick.		
	a) Hollock	Sqm	1420.63
	b) Gamari	Sqm	1420.35
	iii) 25 mm thick.		
	a) Hollock	Sqm	1653.55
	b) Gamari	Sqm	1653.55
7.3.7	Providing and fixing 1st class local wood Hollock /Bonsum/ Gamari/ Sundi lining with butt joints including fitting and fixing with countersunk screws with the frame with complete (frame work to be measured and paid separately).		

	i) 12 mm thick.		
	a) Hollock	Sqm	800.62
	b) Bonsum / Sundi	Sqm	798.29
	c) Gamari	Sqm	798.29
	ii) 20 mm thick.		
	a) Hollock	Sqm	1121.27
	b) Bonsum / Sundi	Sqm	1121.27
	b) Gamari	Sqm	1121.27
	iii) 25 mm thick.		
	a) Hollock	Sqm	1324.16
	b) Bonsum / Sundi	Sqm	1318.33
	b) Gamari	Sqm	1318.33
7.3.8	Providing and fixing timber skirting with butt joining and screws (frame work and cover fillets to be measured and paid separately).		
	a) Teak wood / Sisu wood.		
	i) 12 mm thick	Sqm	908.85
	ii) 20 mm thick	Sqm	1438.21
	iii) 25 mm thick	Sqm	1776.35
	b) Gamari timber plank		
	i) 12 mm thick	Sqm	702.45
	ii) 20 mm thick	Sqm	1023.10
	iii) 25 mm thick	Sqm	1220.15
	c) Hollock/Bonsum/Sundi		
	i) 12 mm thick	Sqm	702.45
	ii) 20 mm thick	Sqm	1023.10
	iii) 25 mm thick Hollock	Sqm	1225.98
7.4	CEILING TILES		
7.4.1	Supplying, fitting and fixing ceiling tiles of 10mm thick made from Duratuff medium density fibre board in size 1200mm x 1200mm x 10mm with Sal wood timber frame (to be measured and paid separately)leaving an expansion gap of 3mm with necessary nails, screws etc. including 1st class local wood (Hollock /Bonsum / Sundi etc.)beading of size 50mm x 12mm complete as directed and specified.	Sqm	321.08
7.5	BISON PANEL CELING AND PARTION		
7.5.1	Providing, fitting and fixing in position false ceiling of 6mm thick Bison panel (cement bonded particle board) on timber frame of 1st class local wood (Hollock/ Bonsum/ Sundi) of 50mm x 50mm spaced at not more than 1.20m apart , fixed to the wall at the end with necessary nails, wood screws, hooks,bolts etc. including painting (one coat primer and two coats of painting) as necessary complete as directed by the Department for all levels.		
	(a) Both sides commercial	Sqm	786.20
	(b) One side laminated	Sqm	995.66
7.5.2	Providing, fitting and fixing in position Bison panel (cement bonded particle board) partition wall on one face of the timber frame of 1st class local wood (Hollock/ Bonsum/ Sundi)of 50mm x 80mm spaced at not more than 1.20m apart , fixed with necessary nails, wood screws, bolts etc. including painting (one coat primer and two coats painting) as necessary complete as directed by the Department for floors.		

	(A) Both sides commercial		
	i) 10mm thick Bison Panel	Sqm	1012.05
	ii) 12mm thick Bison Panel	Sqm	1178.58
	iii) 16mm thick Bison Panel	Sqm	1292.52
	iv) 20mm thick Bison Panel	Sqm	1362.42
	(B) One side laminated and one side commercial		
	i) 10mm thick Bison panel	Sqm	1297.47
	ii) 12mm thick Bison panel	Sqm	1328.25
	iii) 16mm thick Bison panel	Sqm	1397.88
	(C) Both side laminated		
	i) 10mm thick Bison Panel	Sqm	1271.28
	ii) 12mm thick Bison Panel	Sqm	1299.35
	iii) 16mm thick Bison Panel	Sqm	1383.83
7.5.3	Providing, fitting and fixing in position 6mm thick Bison panel (cement bonded particle board) partition wall on both faces of the timber frame of 1st class local wood (Hollock/ Bonsum/ Sundi) with horizontal and vertical batten of size 50mm x 80mm spaced at not more than 1.20m apart , with necessary nails, wood screws, bolts etc. including painting (one coat primer and two coats painting) as necessary complete as directed by the Department for all levels.		
	(a) Both sides commercial	Sqm	1360.44
	(b) One side laminated	Sqm	1372.01
7.6	PLYWOOD / HARDBOARD PARTION		
7.6.1	Providing, fitting and fixing in position plywood / hardboard partion walls with 6mm thick plyboard / hardboard (best) with horizontal and vertical Sal battens of size 50mm x 80mm spaced at not more than 1.20m apart with necessary nails, wood screws, bolts etc. complete as directed by the Department for all levels (Painting / Varnishing to be measured and paid separately).		
	A) 6mm thick Plyboard		
	a) On one face of the frame	Sqm	951.13
	b) On both faces of the frame	Sqm	1455.22
	B) 6mm thick Hardboard		
	a) On one face of the frame	Sqm	596.91
	b) On both faces of the frame	Sqm	767.02
7.7	DUROSAM CEILING TILES		
7.7.1	Providing, fitting and fixing in position ceiling tiles of Natural Fibre Thermosets Composite of jute, polymer and chemicals of Durosam brand or equivalent as directed by department in size 610mm x 610mm x 4mm thick with necessary nails, screws etc. complete as specified and directed at all levels (Framework to be measured and paid seperately.)	Sqm	1623.30
7.8	Pre laminated Particle Board		

7.8.1	Providing, fitting and fixing in position 62mm thick double skin partition with well seasoned Bonsum/ Sundi / Hollock frame in size 75mm x 38mm and internal frame of 50mm x 38mm and fixing on both sides 9mm thick flat pressed melamine faced pre-laminated graded wood particle boards bonded with phenol formaldehyde synthetic resin of approved shade as specified and directed at all levels. One side decorative (Bhutan board/Nepal Board/Bajaj board or equivalent board as directed by department)	Sqm	3812.63
7.8.2	Providing, fitting and fixing in position 62mm thick double skin partition with well seasoned Sal/ Bonsum/ Sundi / Hollock frame in size 75mm x 38mm and internal frame of 50mm x 38mm and fixing on both sides 8mm thick flat pressed melamine faced pre-laminated baggage board medium density fibre boards (MDF) bonded with phenol formaldehyde synthetic resin of approved shade as specified and directed at all levels. a) With Bajaj - exterior grade(one side decorative) (i) 8 mm tick	Sqm	4440.87
7.8.3	Providing, fitting and fixing in position 38/30mm thick single skin partition with well seasoned Bonsum/ Sundi / Hollock frame in size 75mm x 38mm / 75mm x 50mm and fixing 12mm / 18mm thick flat pressed melamine faced pre-laminated graded wood particle boards bonded with phenol formaldehyde synthetic resin of approved shade as specified and directed at all levels.		
	(I) Exterior grade - One side decorative		
	(a) 12mm thick	Sqm	1386.84
	(b) 19mm thick	Sqm	1695.41
	(II) Exterior grade - Both side decorative		
	(a) 12mm thick	Sqm	1519.28
	(b) 19mm thick	Sqm	1792.48
7.9	OTHER TYPES OF CEILING		
7.9.1	Providing, fitting and fixing hard board sheet ceiling with approved quality with necessary nails, wood screws including 1st class local wood 50mm x 12mm (hollock/ bonsum/ sundi) beading including painting two coats to timber beads complete as directed (ceiling joist to be measured and paid separately).		
	(a) 3 mm thick	Sqm	184.95
	(b) 4.5 mm thick	Sqm	231.26
	(c) 6 mm thick	Sqm	262.13
7.9.2	Providing, fitting and fixing in position ceiling tiles of unveneered teakwood particle board bonded with phenol formaldehyde synthetic resin equivalent to anchor nova teak super in size 600 mm x 600mmx12mm leaving an expansion gap of 10mm as per the drawing with necessary links and wood screws complete (timber frame work to be measured and paid separately).		
	(a) Plain	Sqm	615.72
	(b) Half perforated/ Full perforated	Sqm	622.09

7.9.3	Providing, fitting and fixing in position ceiling tiles of Insulex board(soft) in size 600mm x 600mm x 12mm leaving an expansion gap of 10mm as per the drawing with necessary nails and wood screws complete (timber frame work to be measured and paid separately).	Sqm	628.46
7.9.4	Providing 10mm thick plaster of paris (Gypsum, Anhydrous) ceiling up to a height of 5m above floor level over 1st class Hollock / Bonsum / Sundi wood strips 25mm x 6mm with 10mm gap in between with rabbit wire mesh fixed to wooden frame (frame work to be measured and paid separately).		
	(a) Flat surface	Sqm	799.70
	(b) Curved surface	Sqm	936.96
	(c) Extra for sunk or raised mouldings in plaster of paris (Gypsum, anhydrous) ceiling	Sqm	336.05
7.9.5	Providing and fixing expanded polystyrene plain tiles (thermocol) ceiling with fevicol @ 5 lits.per 10 m2 and brass screws etc.complete (frame work and cover fillets to be measured and paid separately).		
	(a) 15mm thick	Sqm	319.64
	(b) 20mm thick	Sqm	340.66
	(c) 25mm thick	Sqm	368.68
7.9.6	Providing and fixing decorative type veneered particle board lining with butt joint leaving a gap of 10mm as per drawing including fitting and fixing with countersunk screws with the frame work (frame work and cover fillets to be measured and paid separately).		
	(A) Decorative		
	i) 12 mm thick.	Sqm	1465.22
	ii) 20 mm thick.	Sqm	1694.50
	iii) 25 mm thick.	Sqm	2016.76
	(B) One Side Decorative		
	i) 12 mm thick.	Sqm	1188.27
	ii) 20 mm thick.	Sqm	1493.97
	iii) 25 mm thick.	Sqm	1672.30
7.9.7	Providing and fixing one side teak veneered particle board lining with butt joint leaving a gap of 10mm as per drawing including fitting and fixing with countersunk screws with the frame work complete as specified and directed at all levels(frame work to be measured and paid separately).		
	i) 12 mm thick.	Sqm	1061.97
	ii) 20 mm thick.	Sqm	1367.67
	iii) 25 mm thick.	Sqm	1546.00
7.10	Ceiling with Exposed Metal (G.I.) Frame work with E-Grid - 2430		

7.10.1	Providing and fixing, in position suspended false ceiling using E-Grid (exposed ceiling suspension 2430 system) of grid size 600mm x 600mm/1200mmx600mm with main runner 38 x 24mm, cross tee 29mm x 24mm & wall angle 24mm x 24mm . Using 4mm/6mm thick High Pressure Steam Cured Non Asbestos Fibre Cement Board as per IS 14862, to be placed in a grid of G.I. system duly suspended with 4mm/6mm dia G.I./M.S bar hanger fixed with steel dash fasteners or iron purlin from the roof with adjustment clip providing positions for light fixtures, air conditioning diffuser and ducts, sprinklers, smoke detectors etc. The main runners to be secured to structural soffit by hanger @ 1220mm maximum c/c. The last hanger at the end of each main runner should not be placed at more than 450mm from the adjacent wall. Flush fitting 1220mm cross tee to be interlocked between main runners @ 610mm c/c, so as to form 600mm x 600mm/1200mmx600mm grid, all complete as per the drawing & directions of engineer-in-charge.		
	Finally, if specified, boards are to be painted with 2 coats of acrylic emulsion paint of approved shade/make over a coat of cement primer before installing it on the grid.		
	(A) Standard Ceilings of Everest make or equivalent as directed by the departement		
	(i) Using 4 mm thick FC Board	Sqm	786.38
	(ii) Using 6 mm thick FC Board	Sqm	907.12
	(B) Designer Ceilings of Everest make or equivalent as directed by the departement		
	(i) Using 4 mm thick FC Board	Sqm	906.69
	(ii) Using 6 mm thick FC Board	Sqm	1079.86
7.11	ACOUSTICAL APPLICATION		
7.11.1	Providing and fixing Mineral Fiber Acoustic Ceiling Tiles in 600mm x 600mm of 13mm/15mm thickness to be placed in E-Grid Suspended Ceiling Grid 2430 System consisting of Main Runner 3600 mm long, Cross Tee 1200mm/600mm long and Wall Angle. The Wall Angle shall be fixed on PVC Dash Fasteners on the perimeter of the wall by steel screws with distance 300 mm c/c. Main Runner shall be placed along the shorter dimension of the area to be covered. The distance between the two Main Runners to be maintained at 1200mm. The Cross Tee 1200mm will be inserted in the pre-cut slots of Main Runner at a regular interval of 600 mm to form a modular grid of 1200mm x 600mm. Additional Cross Tees of 600 mm shall be placed perpendicular to the Cross Tee 1200 mm long to finally form a grid of 600mm x 600mm.E-Grid of module size 600mm x 600mm shall be supported by 4/6 mm dia G.I. wire from purlins/soffit Mineral Fiber Acoustic Ceiling Tiles of Everest Industries Limited should be placed in the E-Grid module to form a False Ceiling including the cuttings complete as directed.		
	A) Everest Designer Ceilings or equivalent as directed by the departement (Primered)		
	A) Using 13mm thick ceiling tiles	Sqm	1235.05
	B) Using 15mm thick ceiling tiles	Sqm	1753.07
7.12	FIBRE CEMENT BOARD WALL PANELLING		

7.12.1	Providing and fixing in position concealed G.I. sections for paneling using High Pressure Steam Cured), pre-primed, self-embossed, designer cement board or the non-primed, plain Fibre cement boards of 6mm thickness confirming IS 14862-2000 fixed on the supporting "W" profile (0.55mm thick) having a knurled web of 51.5mm and two flanges of 26mm each with lips of 10.55mm) placed @600mm c/c vertically and 11200mm c/c horizontally in perimeter channel (having one flange of 20mm and another flange of 30mm with thickness of 0.55mm and web of length 27mm). Perimetre channel is fixed on the floor and the ceiling with the nylon sleeves @ 300mm c/c with expandable fasteners. Board is fixed to the "W" profile with 25mm long self embedding type Fibre cement Screws @ 300mm c/c, all complete as per the drawing & directions of Engineer-in -charge. The boards may be finished so as to have a uniform gap/groove or Flush finish. The flush finish should be done with Elasto prefab joint sealant. Internal Jointing Compound and paper tape as per the procedure laid down by company.		
	Note: Alternatively, the uniform gap/groove may be covered with decorative molding or may be finished with Jointing filler including 2 coats of acrylic emulsion paint of approved shade/make, over a coat of cement primer before installing it on the grid as required.		
	I) Everest Plain FC board or equivalent	Sqm	1088.53
	II) Everest Designer FC board or equivalent	Sqm	1241.71
	III) Everest HDFC Board or equivalent	Sqm	1182.85
7.12.2	Providing and fixing double skinned partitions using G.I. sections High Pressure Steam Cured non asbestos Fibre Cement wall board conforming to IS 14862-2000 , fixed on frame work using 25mm fully threaded self embedding type Fibre Cement Screws @300mm c/c on either side of Steel G.I. 51mm studs (0.55mm thick having one flange of 40mm and another flange of 42mm) placed at 600mm c/c in 51mm floor and ceiling channel (0.55mm thick having equal flange of 32mm G.I. steel) with joints staggered to avoid through joints, all complete as per the drawing & direction of Engineer-in -Charge. The boards may be finished so as to have a uniform gap/groove or Flush finish. The flush finish should be done with Elasto prefab joint sealant. Everest Internal Jointing Compound and paper tape as per the procedure laid down by the company. Alternatively, the uniform gap/groove may be covered with decorative molding or may be finished with Jointing filler.		
	Finally, if specified the boards to be painted with 2 coats of acrylic emulsion paint of approved shade/make, over a coat of cement primer before installing it on the grid.		
	A) For Dry Area		
	i) Using 8mm thick FC Wall board of Everest make or equivalent	Sqm	1731.53

	ii) Using 10 mm thick FC Wall board of Everest make or equivalent	Sqm	1943.08
	iii) Using 12 mm thick FC Wall board of Everest make or equivalent	Sqm	2154.31
	B) For Wet Area		
	i) Using 9mm thick FC Wall board of Everest make or equivalent	Sqm	2123.99
	ii) Using 12 mm thick FC Wall board of Everest make or equivalent	Sqm	2537.10
7.12.3	Providing and fixing 50mm/ 75mm thick Non asbestos Everest Solid Wall Panels (Fibre reinforced aerated cement core sandwich panels) made out of light weight fibre reinforced aerated cement/concrete core composed of port land cement, flyash, binders etc., and 4mm thick Non asbestos fibre cement Boards (Everest Wall Boards) as per IS 14862:2000 on either side of the core, having a tongue and groove on longitudinal side of the modules. These Everest Solid Wall Panels are erected / installed using galvanised iron steel tracks of 1mm thick & of size 15x50x25mm as bottom track, 38x50x38mm as top track used for fixing to the floor and top of the wall respectively. Finally grooves of the panels should be finished with Everest recommended jointing compound.		
	i) Using 50 mm thick SW panel of Everest make or equivalent	Sqm	2255.57
	i) Using 75 mm thick SW panel of Everest make or equivalent	Sqm	2578.06
7.12.4	Providing and fixing in position High Pressure Steam Cured Non Asbestos Fibre Cement Everest Heavy Duty Wall Board (Density>1500 Kg/m3) confirming IS 14862 & Type A as Eves Board to be fixed on iron or timber Section as per the drawing & directions of engineer-in-charge. Finally, if specified, the boards are to be painted with 2 coats of acrylic emulsion paint of approved shade/make over a coat of cement primer after installing it on to the concealed metal framework.		
	i) Using 9mm thick board of Everest make or equivalent	Sqm	1475.35
	ii) Using 12mm thick board of Everest make or equivalent	Sqm	1681.90

CH-8 : ROOFING & ROOF TREATMENT

Item No	Description	Unit	Rate
8.1	ROOFING		
8.1.1	Providing corrugated galvenised Iron sheet roofing of TATA SHAKTEE / SAIL including fitting and fixing necessary galvenised J or L hooks, bolts and nuts 8 mm dia with bitumen washer 25 mm dia x 3 mm thick and 1.6 mm thick limpet washer complete excluding cost of roof truss, purlin etc. (Roof trusses and purlin etc.to be measured and paid separately).		
	(a) 0.45 mm thick	Sqm	450.45
	(b) 0.50 mm thick	Sqm	484.76
	(c) 0.55 mm thick	Sqm	512.95
	(d) 0.60 mm thick	Sqm	550.94

	(e) 0.63 mm thick	Sqm	576.68
8.1.2	Providing galvd iron ridging of TATA SHAKTEE / SAIL including supplying and fixing necessary galvd screws/washers etc. complete as directed.		
	(a) 0.45 mm thick		
	(i) 150 mm lapping	RM	134.92
	(ii) 230 mm lapping	RM	152.69
	(b) 0.50 mm thick		
	(i) 150 mm lapping	RM	143.63
	(ii) 230 mm lapping	RM	162.86
	(c) 0.55 mm thick		
	(i) 150 mm lapping	RM	150.84
	(ii) 230 mm lapping	RM	171.27
	(d) 0.60 mm thick		
	(i) 150 mm lapping	RM	169.59
	(ii) 230 mm lapping	RM	193.14
	(e) 0.63 mm thick		
	(i) 150 mm lapping	RM	176.08
	(ii) 230 mm lapping	RM	200.71
8.1.3	Labour for fitting and fixing Galvd corrugated iron roofing including supplying and fixing with new galvd screw, J hook, bolts nuts, washer etc. complete as directed.	Sqm	120.72
8.1.4	Labour for taking out and refitting Galvd ridging including supplying and fixing necessary galvd screw, bolts and nuts complete as directed.	RM	66.70
8.1.5	Providing 13 mm mesh hexagonal wire net made of one I.S. designation (1mm dia) wire above post plate in eaves with 1st class Hollock/Bonsum/ Sundi timber frame to fit exactly with the corrugation of roofing sheet including kiricide oiling two coats to timber works complete as directed.	Sqm	519.36
8.1.6	Providing 13 mm mesh hexagonal wire net made of one I.S. designation (1mm dia) wire above post plate in eaves with 1st class (Hollock/Bonsum) timber frame to fit exactly with the corrugation of roofing sheet including painting two coats with approved paint to timber works complete as directed.	Sqm	543.57
8.1.7	Providing 0.63 mm thick(24 gauge) galvd plain sheet in valley , lining, flashing etc. and fitted with gavd. Iron J or L hook bolts and nuts, G.I. limpet and bitumen washer complete, bent to shape and fixed in a positions as directed.	Sqm	427.61
8.1.8	Supplying, fitting and fixing wind ties with 40mm x6mm M.S. flat iron and necessary 10 mm dia G.I hook bolt, G.I flat washer and bitumen washer etc. complete including two coats painting with red oxide anticorrosive paint.	RM	201.58
8.1.9	Labour for taking out and refitting M.S. flat wind ties in position with new 10 mm dia J hook, G.I. flat washer, bitumen washer etc. including painting one coat with red oxide paint complete.	RM	63.54
8.1.10	Providing bracket made out of M.S. flat, bent to proper size and shape as per drawing and fitting with necessary bolts, nuts, sand washers etc. for fixing A.C / G.I. sheet gutters with purlins.	kg	208.03
8.1.11	Providing thatch roofing with sun grass (Ullo) in layer of 75 mm thick with half spilt bamboo fillet and tying with galvd. wire complete.		

	(a) 75 mm thick.	Sqm	277.14
	(b) 150 mm thick.	Sqm	531.14
8.1.12	Providing and laying thatching, capping to ridge and hip with new sun grass of 75 mm thick (fine work) stiffened with bamboo slips of size 30mm x 6mm and tied with galvd. tying wire complete.	RM	186.27
8.1.13	Re-thatching with new sun grass including dismantling and removing old thatching and also defective bamboo and replacing with new one and stacking the old thatch at safe place.		
	(a) 75 mm thick.	Sqm	288.52
	(b) 150 mm thick.	Sqm	558.65
8.1.14	Over thatching 75 mm thick with new sun grass, filling in holes and gaps in the roof etc.	Sqm	2819.04
8.1.15	Providing and laying bamboo slip jaffri of meshes not exceeding 63mm x 38mm including tying to the purlins and rafters with cane slip or galvd. tying wire.		
	(a) Fine work	Sqm	51.59
	(b) Rough work	Sq.m	42.07
8.1.16	Providing single layer of bamboo mat lining over roof frame.	Sqm	104.91
8.1.17	Providing bamboo maruli ties, purlins, rafters including tying with cane sling / Galvd. tying wire.		
	(a) With Bhaluka bamboo.	RM	34.49
	(b) With Jati bamboo.	RM	24.97
8.1.18	Providing bamboo roof truss with whole ruas bam boo and kamis, ruas placed at 150 mm apart and kamis 200mm to 230mm apart including tying all members with galvd tying wire.	Sqm	116.29
8.1.19	Providing bamboo jaffri over thatch roof with 200mm sq mesh bamboo slips of 35mm width	Sqm	33.52
8.1.20	Providing whole bamboo roof frame with old and new materials, new materials being at least 25% of the total required.	Sqm	34.93
8.1.21	Providing bamboo roof frame with kami of size 30mm x 6mm.	Sqm	42.07
8.1.22	Renewing / providing 5 layer of bamboo single roofing with matured muli bamboo tarja, woven closely round tiers of single kamis spaced 150mm apart on ground with 600mm and 900 mm single seperately. First 900mm long tiers are tied at lower ends of roof frames projected 450mm, towards the eaves, 750mm tiers tied with 150mm overlap of first tiers, 3 nos 600 mm tiers tied with 150mm overlap of second tiers, 2 nos 600mm tiers tied with 150mm overlap of third tiers and forth 600mm tiers continued up to ahead of ridge, the last but one will be 750mm and the last one will be 600mm long (this section gives 5 layers) Then with 900mm wire, diagonally woven double bamboo tarja mat over the ridge complete with all tying with galvd wire as directed and specified.	Sqm	738.25

8.1.23	Providing corrugated 1.20 mm thick transparent fibre sheet roofing including fitting and fixing necessary galvanised J or L hooks, bolts and nuts of 8 mm dia with bitumen washer 25 mm dia. X 3 mm thick and 1.6 mm thick limpet washer complete excluding cost of roof truss, purlins etc. (Roof trusses, purlin, etc. to be measured and paid separately).	Sqm	1319.67
8.1.24	Providing 1.2mm thick transparent fibre sheet ridging including supplying and fixing necessary galvanised screws / washers etc. complete as specified and directed.		
	(a) 1.20 mm thick - 150 mm lapping	m	244.22
	(b) 1.20 mm thick - 300 mm lapping	m	280.66
8.1.25	Providing coloured corrugated transparent fibre sheet roofing including fitting and fixing necessary galvanised J or L hooks, bolts and nuts of 8 mm dia. with bitumen washer 25 mm dia x 3 mm thick and 1.6 mm thick limpet washer complete excluding cost of roof truss, purlins etc. (Roof trusses, purlins etc. to be measured and paid separately)		
	(a) 1.00 mm thick	Sqm	1134.15
	(b) 1.20 mm thick	Sqm	1417.55
8.1.26	Providing coloured transparent fibre sheet ridging including supplying and fixing necessary galvanised screws / washer etc. complete as specified and directed.		
	(a) 1.20 mm thick - 150 mm lapping	RM	247.19
	(b) 1.20 mm thick - 300 mm lapping	RM	283.63
8.1.27	Providing Iron Plastic Laminated corrugated coloured sheet Roofing at all levels including fitting and fixing galvanised J or L hooks, bolts and nuts of 8 mm diameter with Bitumen washer 35 mm x 35 mm x 3 mm and G.I. Diamond washer 40 mm x 40 mm x 3 mm complete excluding cost of roof truss, purlins etc. (Roof trusses, Purlins etc. to be measured and paid separately).		
	(a) 0.50 mm thick	Sqm	462.17
	(b) 0.55 mm thick	Sqm	499.33
	(c) 0.60 mm thick	Sqm	537.37
	(d) 0.65 mm thick	Sqm	575.41
8.1.28	Providing Colour Coated (Iron Plastic Laminated Sheet) Ridging including supplying and fixing necessary galvanised screws / washer etc. complete as specified and directed.		
	(A) 0.50 mm thick		
	(i) 150 mm lapping	RM	408.71
	(ii) 230 mm lapping	RM	472.56
	(B) 0.55 mm thick		
	(i) 150 mm lapping	RM	446.76
	(ii) 230 mm lapping	RM	516.95
	(C) 0.60 mm thick		
	(i) 150 mm lapping	RM	486.12
	(ii) 230 mm lapping	RM	562.88
	(D) 0.65 mm thick		
	(i) 150 mm lapping	RM	524.17
	(ii) 230 mm lapping	RM	607.27

8.1.29	Providing circular corrugated / trouhed Everlast Aluminium roofing sheet of Aluminium Alloy procured from primary aluminium producer (On submission of certificate of alloy purchasing from primary producer) including fitting and fixing necessary J or L hooks bolts and nuts 8 mm dia with bitumen / EPDM washer 25mm dia x 3mm thick and 1.6mm limplet washer complete excluding cost of truss, purlin etc.(required certificate to be attested along with the bill)		
	(a) 0.40mm thick	Sqm	467.88
	(b) 0.45mm thick	Sqm	510.02
	(c) 0.50mm thick	Sqm	573.51
8.1.30	Providing aluminium ridging of Aluminium Alloy procured from primary aluminium producer (On submission of certificate of alloy purchasing from primary producer) including fitting and fixing necessary Aluminium screws, washers etc. HINDALCO make or equivalent Quality	Sqm	392.14
8.1.31	Providing plain Polycarbonate transparent sheet Ridging including supplying and fixing necessary self drilling, self tapping screws etc. complete as specified and directed.		
	(i) 150 mm lapping	RM	509.21
	(ii) 230 mm lapping	RM	584.30
8.1.32	Providing plain Polycarbonate transparent sheet roofing at all levels with self drilling, self tapping screws complete (Roof trusses, purlins etc. to be measured and paid separately.)		
	(i) 1.00 mm thick	Sqm	963.78
	(ii) 2.00 mm thick	Sqm	1421.60
	(iii) 3.00 mm thick	Sqm	2036.58
8.1.33	Providing Pre Painted Galvanized Iron Sheet Roofing (PPGI) at all levels including fitting and fixing with self drilling, self tapping screws complete. (Roof trusses, purlins etc. to be measured and paid separately.) TATA Blue scope/ Dyna roof / Durakolor /Wonder Roof or equivalent as directed by the Departement		
	(i) 0.40 mm thick	Sqm	687.38
	(ii) 0.45 mm thick	Sqm	725.42
	(iii) 0.50 mm thick	Sqm	802.77
	(iv) 0.55 mm thick	Sqm	869.97
	(v) 0.60 mm thick	Sqm	930.84
	(vi) 0.80 mm thick	Sqm	1164.16
8.1.34	Providing Pre Painted Galvanized Iron Sheet (PPGI) accessories (Ridges / Valley / Gutter /Flashing) at all levels including fitting and fixing self drilling, self tapping screws complete complete. TATA Blue scope/ Dyna roof / Durakolor /Wonder Roof or equivalent as directed by the Departement		
	(i) 0.40 mm thick	RM	333.83
	(ii) 0.45 mm thick	RM	355.49
	(iii) 0.50 mm thick	RM	398.81
	(iv) 0.55 mm thick	RM	437.04
	(v) 0.60 mm thick	RM	471.44
	(vi) 0.80 mm thick	RM	602.67

8.1.35	Providing Pre Painted Galvalume Sheet Roofing (PPGL) at all levels including fitting and fixing self drilling, self tapping screws complete (Roof trusses, purlins etc. to be measured and paid separately.)		
	(a) TATA Blue scope/ Dyna roof / Durakolor /Wonder Roof or equivalent as directed by the Departement		
	(i) 0.40 mm thick	Sqm	664.55
	(ii) 0.45 mm thick	Sqm	700.86
	(iii) 0.50 mm thick	Sqm	767.43
	(iv) 0.55 mm thick	Sqm	837.64
	(v) 0.60 mm thick	Sqm	895.74
	(vi) 0.80 mm thick	Sqm	1126.92
8.1.36	Providing Pre Painted Galvalume Sheet (PPGL) accessories (Ridges / Valley / Gutter / Flashing) at all levels including fitting and fixing self drilling, self tapping screws complete. TATA Blue scope/ Dyna roof / Durakolor /Wonder Roof or equivalent as directed by the Departement		
	(i) 0.40 mm thick	RM	351.52
	(ii) 0.45 mm thick	RM	376.56
	(iii) 0.50 mm thick	RM	416.06
	(iv) 0.55 mm thick	Sqm	476.49
	(v) 0.60 mm thick	RM	512.53
	(vi) 0.80 mm thick	RM	627.56
8.1.37	Providing Bare Galvalume Sheet Roofing at all levels including fitting and fixing self drilling, self tapping screws complete. (Roof trusses, purlins etc. to be measured and paid separately.) TATA Blue scope/ Dyna roof / Durakolor /Wonder Roof or equivalent as directed by the Departement		
	(i) 0.40 mm thick	Sqm	567.46
	(ii) 0.45 mm thick	Sqm	606.96
	(iii) 0.50 mm thick	Sqm	669.39
	(v) 0.60 mm thick	Sqm	780.96
	(vi) 0.80 mm thick	Sqm	1021.93
8.1.38	Providing Bare Galvalume Sheet roofing accessories (Ridges / Valley / Gutter /Flashing) at all levels including fitting and fixing self drilling, self tapping screws complete. TATA Blue scope/ Dyna roof / Durakolor /Wonder Roof or equivalent as directed by the Departement		
	(i) 0.40 mm thick	RM	274.14
	(ii) 0.45 mm thick	RM	296.85
	(iii) 0.50 mm thick	RM	330.91
	(iv) 0.60 mm thick	RM	465.90
	(v) 0.80 mm thick	RM	603.42
8.1.39	Providing Pre Painted Steel Tile Sheet at all levels including fitting and fixing self drilling, self tapping screws complete (Roof trusses, purlins etc. to be measured and paid separately.)		
	Dyna roof /Wonder Roof / Durakolar		
	(i) 0.40 mm thick	Sqm	750.68
	(ii) 0.50mm thick	Sqm	855.39

8.1.40	Providing Pre Painted Steel Tile Sheet accessories at all levels including fitting and fixing self drilling, self tapping screws complete.		
	Dyna roof/ Wonder Roof/ Durakolor		
	(i) 0.40 mm thick	Sqm	528.98
	(ii) 0.50 mm thick	Sqm	556.74
8.1.41	Providing colour coated DURASHINE/TATA BLUESCOPE zinc-aluminium profile sheet (total coated thickness=0.45mm) of nominal 1015mm effective cover width and nominal 28mm deep ribs with subtle square fluting in the five pan at nominal 203mm centre-to-centre including fasteners class 2 SDS, fitting and fixing complete. (Roof trusses, purlins etc. to be measured and paid separately.)		
	(A) DURASHINE/TATA BLUESCOPE ROOF PROFILE (0.45mm(TCT) x 1072mm)	Sqm	730.23
	(B) DURASHINE/TATA BLUESCOPE WALL PROFILE (0.45mm(TCT) x 1150mm)	Sqm	673.44
	(C) DURASHINE TILE PROFILE (0.50mm(TCT) x 505mm)	Sqm	1243.86
8.1.42	Providing colour coated DURASHINE/TATA BLUESCOPE zinc-aluminium roof, wall and tile profile sheet accessories (total coating thickness=0.45mm) including fasteners class 2 SDS, fitting and fixing complete. (Roof trusses, purlins etc. to be measured and paid separately.)		
	(i) Plain Ridge	Sqm	408.32
	(ii) Crimped Curved Ridge	Sqm	847.73
	(iii) Crimped End Curved Ridge	Sqm	847.73
	(iv) Gutter / Flashings	Sqm	520.51
	(v) Plain sheet	Sqm	576.75
8.1.43	Providing colour coated pre painted Aluminium (PPA) profiled sheet in AA 3105 Alloy of Balco/ Nalco/ Hindalco make including lapping, fasteners class 2 SDS / aluminium hooks & washers fitting and fixing complete. (Roof trusses, purlins etc. to be measured and paid separately.)		
	i) 0.50 mm thick (total coated thickness=0.56 mm) and 1297 mm wide having nominal 32 mm deep ribs with subtle square luting at nominal 200 mm C/C.	Sqm	679.27
	ii) 0.56 mm thick (total coated thickness=0.62 mm) and 1297 mm wide having nominal 32 mm deep ribs with subtle square luting at nominal 200 mm C/C.	Sqm	742.35
	iii) Tiled profile Mangaloran architecture PPA of 0.50 mm thick (total coated thickness=0.56 mm) and 1297 mm wide	Sqm	1171.31
8.1.44	Providing colour coated pre painted Aluminium (PPA) sheet in AA 3105 Alloy of Balco/ Nalco/ Hindalco make roof, wall and tile profile sheet accessories including laping complete.		
	i) PP aluminium plain ridg of 405mm wide	RM	382.63
	ii) PP aluminium Crimped curved ridge of 405mm wide	RM	817.89
	iii) PP aluminium Cutter / flashing of 405mm wide	RM	477.25
	CLAY ROOF TILES		
8.1.45	Providing and laying machine made clay roof tiles on sloping roof and chajjas over 20mm thick base of cement mortar 1:3 and jointed with grey cement slurry and joints flush with mortar 1:3 complete.	Sqm	958.53
8.2	ROOF TREATMENT		

8.2.1	Grading roof for water proofing treatment with		
	(a) Cement concrete 1:2:4 (cement: 2 coarse sand: 4 graded stone agg. 20 mm nominal size)	Cum	6333.98
	(b) Cement mortar	Cum	7399.55
	(i) 1:3 (1 cement: 3 coarse sand)		
	(ii) 1:4(1 cement: 4 coarse sand)	Cum	6181.68
8.2.2	Providing and laying six course water proof treatment with bitumen felt over roof consisting of first, third and fifth courses of blown or residual bitumen applied hot @ 1.45kg, 1.20kg and 1.45kg per sq.m of areas respectively, second and fourth courses of roofing with felt type 3 grade - I (hessian base self finished bitumen felt) and sixth and final course with stone grit 6mm and down per sq.m including preparation of surface, excluding grading complete.	Sqm	253.70
8.2.3	Providing and laying four course water proofing treatment with bitumen felt over roof consisting of first and third courses of blown or residual bitument applied hot @ 1.45kg per sq.m of area for each course second course of roofing with felt type 3 grade 1 (hessian base self finished bitumen) and fourth and final course with stone grit 6mm and down size, felt pea gravel spread @ 0.006m3 per sq.m including preparation of surface, excluding grouting etc. as directed and specified.	Sqm	175.99
8.2.4	Providing and application of Insultech safeguard coating for heat reflecting and waterproofing of approved colour as per direction of engineer in charge	Sqm	453.07
8.2.5	Providing and fixing of leak proof,fire retardant and Uv resistant skylighting galzing with Lexan polycarbonate sheet fixed with specially designed Domal management system confirming to IS:737 alloy & temper 6063 - T6 with a UTS of 180 to 200 mpa and weight of 1.03 kg/mtr. approx with EPDM rubber gasket of specific gravity as per ISO 2781,age resistant as per ASTM - D573 hardness as per 2240 & tensile strength as per ASTM 412 including the silicon sealent GE Bayer Weathersil NY as per drawing and directed by the Engineer in charge.		
	(a) Polycarbonate Sheet (Compact)		
	(i) 3mm thick	Sqm	5265.42
	(i) 4mm thick	Sqm	6271.78
	(b) Polycarbonate Sheet (Multiwall)		
	(i) 6mm thick	Sqm	4093.46
	(ii) 8mm thick	Sqm	4806.82
	(iii) 10 mm thick	Sqm	5456.50
	(c) Polycarbonate Sheet (76/16 Corrugated)		
	(i) 1.00mm thick	Sqm	1876.34
	(ii) 1.20mm thick	Sqm	2080.16
8.2.6	Providing laying and fixing of FINROOF Finolex PVC roofing sheet of approved thickness / colour and profile as per drawing and as directed by the engineer in charge		
	(a) 76/16 Greca Profile		
	(i) 1.0 mm thick	Sqm	735.92
	(ii) 1.3 mm thick	Sqm	774.13
	(b) 250/40 Industrial Profile		

	(i) 1.8 mm thick	Sqm	971.58
	(ii) 2.0 mm thick	Sqm	1041.65
	(iii) 2.5 mm thick	Sqm	1169.04
CH-9 : WOOD WORKS			
Item No	Description	Unit	Rate
9.1	CHOWKATHS		
9.1.1	Providing dressed wood work in post, walls,battens, post plate bressumers etc. includingsupplying and fixing with spikes, nails,bolt and nuts of 12mm dia(M.S) and required length etc.complete with kiricide oiling two coats to faces in contact with C.C or masonry (M.S flats, angle cleats and bolt and nuts required for flats,angle cleats wherever use shall be measured and paid separately)		
	(a) With sal wood	Cum	98776.83
	(b) With Hollock timber	Cum	56388.61
	(c) With Bonsum/ Sundi timber	Cum	57577.71
	(d) With Bola timber	Cum	49716.53
	(e) With 2nd class treated timber	Cum	39900.78
9.1.2	Providing wood work in frame (chowkaths) of doors, windows, clerestory windows and other similar works wrought, framed and fixed in position in contact with C.C or brick masonry wall including supplying, fitting and fixing with M.S. hold fast (40mmx3mmx250mm) as per design embeded in cement concrete block in proportion 1:2:4 and with two coats of kiricide oiling to the timber faces in contact with C.C and masonry as directed and specified.		
	(a) With sal wood	Cum	121163.03
	(b) With 1st class timber (Bonsum/ Sundi)	Cum	77474.00
9.2	ROOF TRUSS		
9.2.1	Providing undressed wood work in roof truss, rafter, purlin, tie and the like including hoisting and fixing in position with necessary spikes, nails including M.S straps with bolt and nuts etc.complete with kiricide oiling two coats to all timber members and two coats of anticorrosive paint to M.S straps (M.S flats, angle cleats, and bolt and nuts required for flat and angle cleats wherever used shall be measured and paid separately)		
	(a) With sal wood	Cum	87836.62
	(b) With Hollock/ Sundi/ Sam	Cum	51443.43
	(c) With 2nd class treated timber	Cum	35715.37
9.3	CEILING AND PARTITION		
9.3.1	Providing wood work in frame of false ceiling partitions etc.sawn, wrought, framed hoisted and fixed in positon with spikes, nails, M.S flat,angle/ cleats with bolt and nuts complete including kiricide oiling two coats to unexposed surfaces of the timber (M.S flats, angle cleats, and bolt and nuts required for flat, angle cleats wherever necessary shall be measured and paid separately)		
	(a) With sal wood	Cum	97125.37
	(b) With Gamari/ Bonsum/ Sundi	Cum	58905.75
9.4	BARGE BOARD		

9.4.1	Providing barge board of size 200mm x 20mm with 1st class local Hollock/ Bonsum timber including fitting and fixing with necessary wood screws etc. complete.	RM	349.75
9.5	PELMET		
9.5.1	Providing and fixing 12 mm thick and 150mm wide pelmet with 25 mm dia wooden curtain rod and bracket including fixing with 15mmx3mm M.S flat 100mm long plugs complete.		
	(a) First class Indian Teak wood/ Sisu wood	R.M.	498.76
	(b) First class local wood(Gamari)	R.M.	339.16
	(c) First class local wood(Hollock/ Bonsum)	R.M.	339.16
	(d) Veneered particle board (12 mm thick)	R.M.	484.60
	e) Extra for providing and fixing brass curtain rail with rollers, stoppers and brackets in pelmet instead of wooden curtain rods 25 mm dia with brackets (actual length of the rods to be measured)	R.M.	1260.10
	f) Extra for providing and fixing curtain rod of 1.25 mm thick wire brass plate with bracket instead of wooden curtain rod of 25mm dia brackets (actual length of the rods to be measured)		
	(i) 20 mm dia.	R.M.	1260.10
	(ii) 25 mm dia.	R.M.	1266.57
	g) Extra for providing and fixing curtain rod of 1.25mm thick chromium plated with brass plate with to chromium plated brass brackets fixed with CP brass screws etc. complete.(actual length of the rods to be measured)		
	(i) 20 mm dia.	R.M.	335.63
	(ii) 25 mm dia.	R.M.	335.63
9.5.2	Supplying, fitting and fixing pelmet to door and window frames with 12mm thick kit ply wood finished with matt finished sunmica on external faces and bottom and fixed with wooden brackets including providing 20mm dia aluminium hanger with necessary socket and clamps complete including inside painting as directed.		
	(a)With Sunmica	R.M.	2344.41
	(b)Without Sunmica	R.M.	1826.39
9.6	ARCHITRAVE / BEADING		
9.6.1	Providing and fixing 1st class timber moulded architrave of size 50x20mm to door and window frames with countersunk wood screw, wooden plugs and applying a coat of wood preservative on unexposed surface etc.		
	(a)With Teak wood/Sissu wood.	R.M.	166.37
	(b)With Bonsum/ Sundi wood.	R.M.	126.01
	(e)With Holok timber	R.M.	124.82
	(c)With Bola wood	R.M.	118.15
9.6.2	Providing and fixing 1st class timber moulded beading of size 25mm x 15mm around panel inserts of doors with countersunk wood screws including applying a coat of wood preservative on unexposed surface.		
	(a)With Teak wood/Sissu wood.	R.M.	115.30
	(b)With Bonsum/ Sundi wood.	R.M.	100.66
	(c)With Bola wood	R.M.	97.81
	(d)With Gamarai Wood.	R.M.	100.66
	(e)With Holok timber	R.M.	100.23

9.6.3	Providing and fixing 1st class timber beading for fixing clear sheet glass/ frosted glass including applying a coat of wood preservative on unexposed surface.		
	(a)With Teak wood/Sissu wood.	R.M.	65.85
	(b) With Bonsum/ Sundi wood.	R.M.	51.64
	(c)With Bola wood	R.M.	51.20
	(d)With Gamarai Wood.	R.M.	51.64
	(e) With Holok timber	R.M.	51.20
9.7	Railing/Balustrade		
9.7.1	Providing and fixing wooden 100mmx 60mm rounded hand rails in straight lengths complete.		
	(a) With Teak wood/Sissu wood.	R.M.	882.53
	(b) With 1st class local wood (Hollock)	R.M.	559.32
	(c) With 1st class Bola wood	R.M.	507.41
	(d) 1st class local wood (Gamari/Bonsum)	R.M.	568.57
9.7.2	Providing and fixing wooden 100mmx60 mm moulded hand rails in place of rounded hand rail.		
	(a) With 1st class Teak wood / Sissu wood	R.M.	1323.79
	(b) With 1st class local wood (Hollock)	R.M.	838.97
	(c) With 1st class Bola wood	R.M.	761.12
	(d) 1st class local wood (Gamari / bonsum)	R.M.	852.85
9.7.3	Extra for hand rails if sunk to receive iron core.	R.M.	57.92
9.7.4	Providing and fixing 25mmx25mm first class timber balustrade 25mmx25mm square, wrought fixed with ends housed (square or splayed) to hand rails and trends or strips complete.		
	(a) With 1st class Teak wood / Sissu wood	R.M.	246.75
	(b) With 1st class local wood (Gamari/Hollock/Bonsum)	R.M.	220.72
	(c) With 1st class Bola wood	R.M.	216.54
9.8	VENTILATING LOUVRE		
9.8.1	Providing 1st class local wood (Hollock/Bonsum)ventilating louvre in 50mm x 38mm size wooden frame with 150mm x 20mm slate made of 1st class local wood including fitting and fixing with nails and painting with two coats of kiricide oiling complete.		
	(i) 1st class local wood (Gamari/Bonsum/Holok)	Sq.m.	1689.07
9.8.2	Providing first clas local wood (Hollock/ Bonsum ventilating louvre of wooden frame (minimum size of frame members 150mm x 75mm) with clear glass/ frosted glass including fitting and fixing and painting two coats of kiricide oil complete as directed for all levels.	Sq.m.	2251.75
9.8.3	Providing Sal wood ventialating louvre of wooden frame (minimum size frame members 150mmx75mm) with clear glass/frosted glass including fitting and fixing and painting with two coats of kiricide oiling complete as directed for all levels.	Sq.m.	3675.18
9.9	DOORS / WINDOWS / VENTILATORS / CLERESTORY WINDOWS		

9.9.1	Providing fitting and fixing 1/3 panelled 2/3 glazed wooden shutters for doors/windows including oxidised iron butt hinges (100mm x75 mm x3.55 mm) with necessary screws including glass panes (other fittings to be measured and paid separately)		
	A) With Teak wood / Sissu wood.		
	(i) 40 mm thick.	Sq.m.	4333.57
	(ii) 35 mm thick.	Sq.m.	4033.34
	(iii) 30 mm thick.	Sq.m.	3561.77
	B) With 1st class local wood (Gamari/Hollock/Bonsum)		
	(i) 40 mm thick.	Sq.m.	2989.95
	(ii) 35 mm thick.	Sq.m.	2823.88
	(iii) 30 mm thick.	Sq.m.	2566.27
	C) With 1st class Local wood (Bola wood)		
	(i) 40 mm thick.	Sq.m.	2718.16
	(ii) 35 mm thick.	Sq.m.	2577.90
	(iii) 30 mm thick.	Sq.m.	2350.21
9.9.2	Providing, fitting and fixing full panelled doors/ windows including oxidised M.S butt hinges (100mm x75 mm x3.55mm) with necessary screws (other fittings to be measured and paid separately)		
	A) With Teak wood / Sissu wood.		
	(i) 40 mm thick.	Sq.m.	5026.46
	(ii) 35 mm thick.	Sq.m.	4623.51
	(iii) 30 mm thick.	Sq.m.	3618.01
	B) With 1st class local wood (Gamari/Hollock/Bonsum)		
	(i) 40 mm thick.	Sq.m.	3316.46
	(ii) 35 mm thick.	Sq.m.	3150.88
	(iii) 30 mm thick.	Sq.m.	2532.56
	C) With 1st class Local wood (Bola wood)		
	(i) 40 mm thick.	Sq.m.	3265.22
	(ii) 35 mm thick.	Sq.m.	3103.58
	(iii) 30 mm thick.	Sq.m.	2500.03
9.9.3	Providing, fitting and fixing factory made wooden panelled doors with panel inserts of 75mm to 50mm width and 12mm to 20mm thickness,joined together with continuous tongue and groove joints glued together with Movicol or other approved adhesive leaving a vertical groove of 7.5mm to 15mm between the inserts,the edges of panel inserts being feather tongue into styles and rails including providing three pairs of oxidised M.S butt hinges (100mm x75 mm x3.55mm) and furnishing with moulded timber beading of size 25mmx15mm around the panels complete as per drawing (moulded timber bead and all fittings except hinges to be paid separately)		
	a) With Teak wood / Sissu wood.		
	(i) 40 mm thick with panel inserts of 20mm thickness.	Sq.m.	5377.75
	(ii) 35 mm thick with panel inserts of 20mm thickness.	Sq.m.	4593.09
	(iii) 30 mm thick with panel inserts of 20mm thickness.	Sq.m.	4390.23
	b) With 1st class local wood (Hollock / Gamari/Sundi/ Bonsum)		

	(i) 40 mm thick with panel inserts of 20mm thickness.	Sq.m.	3740.34
	(ii) 35 mm thick with panel inserts of 20mm thickness.	Sq.m.	3303.89
	(iii) 30 mm thick with panel inserts of 20mm thickness.	Sq.m.	3191.05
	c) With 1st class local wood (Bola/Sundi)		
	(i) 40 mm thick with panel inserts of 20mm thickness.	Sq.m.	3692.10
	(ii) 30 mm thick with panel inserts of 20mm thickness.	Sq.m.	3265.91
	(iii) 30 mm thick with panel inserts of 20mm thickness.	Sq.m.	3155.72
9.9.4	Providing fitting and fixing 1st class local wood battened, framed and braced door and window shutters including oxidised iron butt hinges (100mm x58 mm x1.90 mm) 6 nos with necessary screws.		
	(a) Using Bonsum/Holok wood	Sq.m.	2152.51
	(b) Using Gamri wood	Sq.m.	2152.51
9.9.5	Providing fitting and fixing 1st class local wood battened and framed door and window shutters including oxidised iron butt hinges (100mmx 58mmx 1.90mm) 6 nos with necessary screws.	Sq.m.	
	(a) Using Bonsum/Holok wood	Sq.m.	1931.47
	(b) Using Gamari wood	Sq.m.	1931.47
9.9.6	(a) Providing fitting and fixing 1st class local wood battened, ledged and braced door and window shutters including oxidised iron butt hinges (100mmx 58mmx1.90mm) 6 nos with necessary screws.		
	(a) Using Bonsum/Holok wood		
	(i) 25mm thick	Sq.m.	3303.39
	(ii) 20mm thick	Sq.m.	2871.47
	(b) Using Gamari wood		
	(i) 25mm thick	Sq.m.	3303.39
	(ii) 20mm thick	Sq.m.	2871.47
9.9.7	Providing, fitting and fixing fully glazed window with sash bars including oxidised M.S butt hinges (75mmx 60mmx3.15mm) 2nos.on each leaf and 3 nos. on single leaf (glass panes to be measured and paid separately)		
	A) With Teak wood / Sissu wood.		
	(i) 40mm thick	Sq.m.	3013.90
	(ii) 35mm thick	Sq.m.	2604.24
	(iii) 30mm thick	Sq.m.	2351.00
	B) With Holock/Gamari/Bonsum wood.		
	(i) 40mm thick	Sq.m.	1967.52
	(ii) 35mm thick	Sq.m.	1830.81
	(iii) 30mm thick	Sq.m.	1689.94
	C) With 1st class local wood		
	(i) 40mm thick	Sq.m.	1941.52
	(ii) 35mm thick	Sq.m.	1808.02
	(iii) 30mm thick	Sq.m.	1670.47
9.9.8	Providing, fitting and fixing fully glazed window without intermediate sash bars including oxidised M.S butt hinges (75mm x 60mm x 3.15mm) 2nos.on each leaf and 3 nos. on single leaf (glass panes to be measured and paid separately)	Sq.m.	3170.31

9.9.9	Providing, fitting and fixing fully glazed clerestory window and fan light including fitting and fixing 2 M.S pivot hinges (75mm x 20mm x 3mm), eye hook and nylon chord complete (glass panes shall be measured and paid separately)		
	A) With Teak wood / Sissu wood.		
	(i) 40mm thick	Sq.m.	2484.49
	(ii) 35mm thick	Sq.m.	2286.07
	(iii) 30mm thick	Sq.m.	2279.95
	B) With Hollock/Gamari/Bonsum wood.		
	(i) 40mm thick	Sq.m.	1758.06
	(ii) 35mm thick	Sq.m.	1647.69
	(iii) 30mm thick	Sq.m.	1528.12
	C) With 1st class local wood		
	(i) 40mm thick	Sq.m.	1736.65
	(ii) 35mm thick	Sq.m.	1628.88
	(iii) 30mm thick	Sq.m.	1512.12
9.9.10	Providing, fitting and fixing fully glazed wooden door shutters with necessary butt hinges, screws etc. complete as per approved drawing as directed and specified at all levels.(fittings e.g. tower bolt, door handle, aldop, glass panes etc. should be measured and paid seperately.)		
	(a) With Teak or Sissu Wood		
	(I) 40mm thick	Sq.m.	2533.85
	(II) 35mm thick	Sq.m.	2341.13
	(b) With Gamari Wood.		
	(I) 40mm thick	Sq.m.	1846.28
	(II) 35mm thick	Sq.m.	1739.05
	(c) With 1st class local wood		
	(I) 40mm thick	Sq.m.	1712.33
	(II) 35mm thick	Sq.m.	1621.76
9.10	VENEERED PARTICLE BOARD DOOR SHUTTERS		
9.10.1	Providing, fitting and fixing factory made exterior type fully panelled single leaf door shutters of approved brand conforming to relevent I.S code having kiln seasoned and vacuum pressure treated Hollock / Bonsum / Sundi rail and styles with pannel of Anchor brand 12mm thick veneered Teak wood particle board bonded with phenol - formaldehyde synthetic resin adhesive conforming to relevent I.S code including 4 nos of oxidised iron butt hinges (100mm x75 mm x3.55mm) to each leaf complete.		
	(i) 40mm thick	Sq.m.	2289.07
	(ii) 35mm thick	Sq.m.	2189.00
	(iii) 30mm thick	Sq.m.	2088.33
9.10.2	Extra for double leaf door shutters, 10% cost for single leaf shutters as in item nos 9.10.1		
9.11	CUPBOARD SHUTTERS		
9.11.1	Providing and fixing 25mm thick wooden shutters for cup board including black enamelled M.S butt hinges (75mmx 40mmx1.12mm) 6 nos. with necessary wood screws (fully panelled)		
	(i) Using wooden plank		
	a)With Teak wood / Sissu wood.(fully panelled)	Sq.m.	2650.88

	b) With Gamari/ Sundi/ Bonsum wood.	Sq.m.	2028.73
	c) With 1st class local timber.	Sq.m.	2028.73
	d) With 1st class local timber (Hollock)	Sq.m.	2010.26
	(ii) Using Particle/Block bord		
	a) With Teak wood particle board (12mm thick)	Sq.m.	2658.24
	b) With Teak wood particle board (19mm/20mm thick)	Sq.m.	2964.21
	c) With block board (both side commercial 16mm thick)	Sq.m.	1819.36
	d) With block board (both side commercial 19mm/20mm thick)	Sq.m.	2985.88
	e) With block board (one side commercial and other side teak 12mm thick)	Sq.m.	2093.46
	f) With block board (one side Teak and other sidecommercial 19mm/20mm thick)	Sq.m.	2985.88
	(g) With Anchor brand veneered, plyboard bonded with phenol formaldehyde synthetic resin 12 mm thick.	Sq.m.	2326.77
9.12	BLOCK BOARD SHUTTERS		
9.12.1	Providing and fixing flush door shutters solid core construction with frame of 1st class hard wood with cross band and face veneered ply wood face panels conforming to relevent I.S code including oxidised iron butt hinges (100mm x 75mm x 3.5mm) 6 nos with necessary wood screws.		
	a) Decorative type face panel and block board core.		
	(i) 40 mm thick .	Sq.m.	3190.73
	(ii) 35 mm thick .	Sq.m.	2990.57
	(iii) 30 mm thick .	Sq.m.	2812.24
	(iv) 25 mm thick .	Sq.m.	2615.72
	b) Non decorative type and block board core.		
	(i) 40 mm thick .	Sq.m.	2493.20
	(ii) 35 mm thick .	Sq.m.	2293.03
	(iii) 30 mm thick .	Sq.m.	2114.71
	(iv) 25 mm thick .	Sq.m.	1918.18
	c) One face decorative and other face nondecorative type and block board core.		
	(i) 40 mm thick .	Sq.m.	2857.13
	(ii) 35 mm thick .	Sq.m.	2656.96
	(iii) 30 mm thick .	Sq.m.	2439.99
	(iv) 25 mm thick .	Sq.m.	2253.77
9.13	WIRE GAUGE SHUTTERS		
9.13.1	Providing, fitting and fixing Wire gauge shutters using galvd. M.S. wire of I.S designation 85G with wire of dia 0.56mm for doors, windows and clerestory windows including oxidised M.S butt hinges (100mm x 58mm x1.90mm)		
	A) With Teak wood / Sissu wood.		
	(i) 40 mm thick .	Sq.m.	3445.84
	(ii) 35 mm thick .	Sq.m.	3154.94
	(iii) 30 mm thick .	Sq.m.	2860.22
	B) With Gamari/ Bonsum/ Sundi wood.		
	(i) 40 mm thick .	Sq.m.	2408.02
	(ii) 35 mm thick .	Sq.m.	2246.21
	(iii) 30 mm thick .	Sq.m.	2082.28
	C) With 1st class local wood(Hollock)		
	(i) 40 mm thick .	Sq.m.	2377.44
	(ii) 35 mm thick .	Sq.m.	2219.44
	(iii) 30 mm thick .	Sq.m.	2059.36

	D) With 1st class local wood. (Bola)		
	(i) 40 mm thick .	Sq.m.	2377.44
	(ii) 35 mm thick .	Sq.m.	2219.44
	(iii) 30 mm thick .	Sq.m.	2059.36
9.13.2	Providing, fitting and fixing Expanded metal 20mm x 60mm strands 3.25mm wide and 1.6mm thick for windows, openings etc. including fixing 50mm 20mm beading of timber with necessary nails, wood screws etc. complete		
	a) With 1st class local timber bead	Sq.m.	459.34
	b) With 2nd class local timber bead	Sq.m.	407.03
9.13.3	Providing, fitting and fixing fly-proof galvd. M.S. wire gauge of I.S. designation 85 G with wire of dia. of 0.56mm to windows and clerestory windows including timber beading of size 50mmx 20mm complete.		
	a) With 1st class local timber bead	Sq.m.	521.56
	b) With 2nd class local timber bead	Sq.m.	469.25
9.14.1	Supplying, fitting and fixing 30mm thick flush door(single leaf finished) made of phenol bonded Duratuff exterior grade (both side) plain medium density fibre board using 4 nos of hinges 125mm long ensuring that the hinges are not fixed less than 150mm from the edges with 8 nos. parallel sunk fully threaded screws 38mm long, should be used for fixing the hinges to the flush door after drilling a pilot hole 2.50mm dia 40mm long . Screws should not be hammered . Lock, tower bolts etc. necessary.		
	(a) 30mm thick.		
	i) Both side teak.	Sq.m.	2159.12
	ii) One side teak and other side commercial.	Sq.m.	1849.60
	(b) 40mm thick.		
	i) Both side teak.	Sq.m.	2378.21
	ii) One side teak and other side commercial.	Sq.m.	2211.59
9.15	MISCELLNEOUS		
9.15.1	Extra for using bright finished Nickel plated M.S piano hinges in place of oxidised M.S.butt hinges.	No	11.78
9.15.2	Extra for providing lapping with 1st class Teak wood battens of 25mm minimum depth on all edges of the shutters.	Sq.m.	424.48
9.15.3	Extra for wood work for each subsequent floor above 1st floor level.	cu.m.	153.10
9.15.4	Extra for fitting and fixing doors/windows for each subsequent floor above 1st floor level.	Sq.m.	8.83
9.15.5	Extra for providing lapping with 1st class Teak wood battens of 25mm minimum depth on all edges of the shutters.	Sq.m.	468.65
9.16.1	Providing, fitting and fixing Bison Panel board (Cement Bonded Particle Board) door shutters fixed to 100mmx38mm wooden frames with necessary hinges, screws, bolts etc. including painting of door shutters (one coat primer and two coats painting) complete as specified and directed by the department for all floors (Door fittings to be measured and paid separately)		
	(a) Using 10mm thick board		
	(i) Both sides commercial	Sq.m.	1960.30
	(ii) One side laminated	Sq.m.	1971.74
	(iii) Both sides laminated	Sq.m.	1924.58
	(a) Using 16mm thick board		

	(i) Both sides commercial	Sq.m.	1934.13
	(ii) One side laminated	Sq.m.	2011.86
	(iii) Both sides laminated	Sq.m.	2208.63
9.17.1	Providing, fitting and fixing 38mm Melamine faced pre-laminated cellular core Flush Door with 9mm/12mm thick high density exterior grade laminated particle board fixed to the internal wooden frame of Sal/Hollock wood with Polyurestene central core, fixed to the chowkath with necessary hinges, screws, bolts etc. as specified and directed at all levels.(Door fixtures to be measured and paid separately.)		
	(a)With Bhutan Board-Exterior Grade (One side Decorative)		
	(i) 9 mm thick	Sq.m.	1858.03
	(ii) 12mm thick	Sq.m.	2091.04
	(b)With Nepal Board-Exterior Grade (One side Decorative)		
	(i) 9 mm thick	Sq.m.	1607.13
	(ii) 12mm thick	Sq.m.	1728.14
9.18.1	Providing, fitting and fixing factory made Santro Flush door shutters having core & cross bands fabricated from seasoned 1st class timber, 25mm core battens, 2.5mm to 3.5mm uniformly thick cross bands, styles & rails made from hardwood timber of width not less than 50mm, fixed to the door frames (chowkath) with necessary hinges, screws, bolts etc. as specified and directed at all levels (Door fixtures and fittings to be measured and paid seperately.)		
	(a) 30 mm thick	Sq.m.	1988.33
	(b) 35mm thick	Sq.m.	2123.11
	(c) 40 mm thick	Sq.m.	2282.67
9.19.	DUROSAM DOORS, WINDOWS & CHOWKHATS		
9.19.1	Providing, fitting and fixing Natural Fibre Thermosets Composite of jute, polymer and chemicals, Durosam make door and window frames (chowkhats) fixed to the columns / stiffeners / walls with necessary holdfast / sleeves etc. as specified and directed at all levels.		
	(a) 75mm x 50mm Solid Frame.	R.M.	1256.89
	(b) 50mm x 25mm Solid Frame.	R.M.	731.77
	(c)75mm x 50mm Grooved Frame.	R.M.	1030.98
9.19.2	Providing, fitting and fixing Natural Fibre Thermosets Composite of jute, polymer and chemicals, Durosam make door/ window shutters fixed to the door/ window frames (chowkath) with necessary hinges, screws, bolts etc. as specified and directed at all levels (Door/ window fixtures and fittings to be measured and paid seperately.)		
	(a) Solid Core moulded/ Fabricated/ Panelled/ Glazed Doors	Sq.m.	6251.82
	(b) Flush Doors	Sq.m.	3357.58
	(c)Solid Core moulded/ Fabricated/ Panelled/ Glazed Windows	Sq.m.	4088.72

9.20.1	Providing fitting and fixing compressed ready to fit Godrej make wood door shutter (Godrej Entranza Moulded Panel Door) for various thickness Conforming to IS 15380-2003,printed,shutter comprising of High Density Fiberboard skins (HDF) of 3.5 mm (hardened finely processed natural wood fibre) with panel design compressed on both faces,sandwitching, termite and borer resistant chemical treated and kiln seasoned wooden stiles and rails of 50mm-60mm width,lock block and high density flax board core / chemical treated and kiln seasoned timber core infil. The panel would be of different design viz. Two panel,Four panel,Six panel (both side) Shutter should be bonded with Phenol Formaldehyde or adhesive / conforming to BWP grade of IS 848.Shutter should have trim ability option of 12.5 mm from all sides. The door should have flexibility for painting or polishing to match any interiors.		
	(i) 32 mm thick with panel inserts of 20mm thickness.	Sq.m.	2826.48
	(ii) 35 mm thick with panel inserts of 20mm thickness.	Sq.m.	2918.15
	(iii) 45 mm thick with panel inserts of 20mm thickness.	Sq.m.	3223.87

Ch-10: GRILL

Item No.	Description	Unit	Rate
10	GRILLS		
10.1	Providing, fitting and fixing M.S. round bars with M.S. flat stiffeners at required spacing in wooden frames of windows and clerestory windows	Kg	76.22
10.2	Providing, fitting and fixing M.S. grill of required pattern for windows/ clerestory windows/ openingwith M.S. flats at required spacing in frame all round, squire or round M.S. bars with round headed bolts and nuts or screws.		
	I) Plain grill		
	(a) Fixed to wooden frames	Kg	84.55
	(b) Fixed to steel frames	Kg	84.16
	(c) Fixed to Brickwork/P.C.C/R.C.C.	Kg	88.74
	II) Ornamental grill		
	(a) Fixed to wooden frames	Kg	101.34
	(b) Fixed to steel frames	Kg	100.58
	(c) Fixed to Brickwork/P.C.C/R.C.C.	Kg	106.20
10.3	Supplying, fitting and fixing anodised aluminium decorated grill with anodised aluminium rectangular frame of 'F' and 'H' section fitted with walls, columns or stiffeners by special type of screws as per specification by the Department for all levels. (Aluminium Alloy procured from primary aluminium producer (On submission of certificate of alloy purchasing from primary producer)	Sqm	2098.22
10.4	Extra for providing security grill welded to casement window frames With 16mm dia M.S round placed horizontally at a distance of 10 cm c/c @ 15kg/ 16kg per sq.meter as per approved drawing.	Sqm	1150.47

CHAPTER-11: ALUMINIUM DOORS, WINDOWS AND OTHER WORKS

Item No.	Description	Unit	Rate
	WINDOWS AND VENTILATORS		
11.1.1	Providing, fitting and fixing anodised aluminium windows and ventilators of standard sections of approved brand without horizontal glazing bars joints mitred and welded (manufactured to relevant IS specifications) fitted with 6mm x 3.15mm lugs 100mm long embedded in CC blocks 150mm x 100mm x 100mm in 1:3:6 mix including providing and fixing handles, frictions stays, joining cleat, bolting device, locking arrangement, spring catch as required complete as specified and directed for all levels. (Cement concrete to be measured and paid separately).		
	a) Side hung and Top hung windows and ventilators		
	i) 6 mm clear glass	Sqm	5980.92
	ii) 5mm frosted glass	Sqm	6020.77
	iii) 6mm coloured glass	Sqm	6136.02
	iv) 5 mm clear glass	Sqm	5852.89
	v) 5 mm coloured glass	Sqm	6007.99
	(b) Fixed windows and ventilators		
	i) 6mm clear glass	Sqm	2700.05
	ii) 5mm frosted glass	Sqm	2739.91
	iii) 6mm coloured glass	Sqm	2855.14
	iv) 5 mm clear glass	Sqm	2572.03
	v) 5 mm coloured glass	Sqm	2668.85
11.1.2	Providing, fitting and fixing anodised aluminium sliding windows and ventilators of standard sections of approved brand without horizontal glazing bars, joints mitred and welded (manufactured to relevant IS specifications) and providing and fixing handles, angle cleat, rubber gasket, roller, bolting device, locking arrangements, screws etc as required complete as specified and directed for all levels.		
	(a) 2 Track sliding window		
	(i) 6mm clear glass	Sqm	4766.85
	(ii) 5mm frosted glass	Sqm	4302.54
	(iii) 6mm coloured glass	Sqm	4414.41
	(iv) 5 mm clear glass	Sqm	4627.17
	(v) 5 mm coloured glass	Sqm	5224.19
	(b) 3 Track sliding window		
	(i) 6mm clear glass	Sqm	4839.56
	(ii) 5mm frosted glass	Sqm	4878.26
	(iii) 6mm coloured glass	Sqm	4990.13
	(iv) 5 mm clear glass	Sqm	4712.14
	(v) 5 mm coloured glass	Sqm	4865.83

11.1.3	Providing, fitting and fixing coloured anodised aluminium windows and ventilators of standard sections of approved brand without horizontal glazing bars joints mitred and welded (manufactured to relevant IS specifications) fitted with 6mm x 3.15mm lugs 100mm long embedded in CC blocks 150mm x 100mm x 100mm in 1:3:6 mix including providing and fixing handles, frictions stays, joining cleat, bolting device, locking arrangement, spring catch as required complete as specified and directed for all levels.(Cement concrete to be measured and paid separately).		
	a) Side hung and Top hung windows and ventilators		
	(i)6mm clear glass	Sqm	6972.55
	(ii)5mm frosted glass	Sqm	6082.45
	(iii)6mm coloured glass	Sqm	6194.32
	(iv) 5 mm clear glass	Sqm	5919.46
	(v) 5 mm coloured glass	Sqm	6070.02
	(b) Fixed windows and ventilators		
	(i)6mm clear glass	Sqm	2712.60
	(ii)5mm frosted glass	Sqm	2751.30
	(iii)6mm coloured glass	Sqm	2863.17
	(iv) 5 mm clear glass	Sqm	2588.31
	(v) 5 mm coloured glass	Sqm	2738.87
11.1.4	Providing, fitting and fixing coloured anodised aluminium sliding windows and ventilators of standard sections of approved brand without horizontal glazing bars,joints mitred and welded (manufactured to relevant IS specifications) and providing and fixing handles, angle cleat, rubber gasket, roller, bolting device, locking arrangements, screws etc as required complete as specified and directed for all levels.		
	(a) 2 Track sliding window		
	(i)6mm clear glass	Sqm	4949.42
	(ii)5mm frosted glass	Sqm	4988.12
	(iii)6mm coloured glass	Sqm	5099.98
	(iv) 5 mm clear glass	Sqm	4825.12
	(v) 5 mm coloured glass	Sqm	5085.13
	(b) 3 Track sliding window		
	(i)6mm clear glass	Sqm	5046.09
	(ii)5mm frosted glass	Sqm	5084.78
	(iii)6mm coloured glass	Sqm	5296.66
	(iv) 5 mm clear glass	Sqm	4921.79
	(v) 5 mm coloured glass	Sqm	5072.35
11.1.5	a) Providing, fitting and fixing anodised aluminium fixed windows and ventilators of standard sections of approved brand, joints mitred welded (manufactured to relevant I.S. specification) fitted with glass panes of required thickness and 12mm prelaminated boards and providing and fixing handles, angle cleat, rubber gasket,roller bolting device, screws as required complete as specified and directed for all levels.		
	(a) In Normal anodised Aluminium section		
	(i) With 6mm thick glass panes	Sqm	4967.67
	(ii) With 5mm thick glass panes	Sqm	4653.47
	(b) In Coloured anodised Aluminium section		

	(i) With 6mm thick glass panes	Sqm	5245.61
	(ii) With 5mm thick glass panes	Sqm	4557.82
11.2	DOORS		
11.2.1	Providing, fitting and fixing anodised aluminium framed glazed doors with anodised aluminium frame made of 100mm x 45mm x 2.5mm section of approved brand with door style of size 88mm x 45mm x 2.5mm, top rails 50mm x 45mm x 2mm lock rails 100mm x 45mm x 2mm and bottom rails 100mm x 45mm x 2.5mm fitted with glazing clip, special type rubber gasket complete including hydraulic floor spring, pivot, tower bolt in each leaf, aluminium door handle, lock, angles, cleat etc complete as specified and directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	10422.37
	(ii) 5mm frosted glass	Sqm	10461.07
	(iii) 6mm coloured glass	Sqm	10572.94
	(iv) Prelaminated Board	Sqm	10506.21
	(v) 5 mm clear glass	Sqm	10298.08
	(vi) 5 mm coloured glass	Sqm	10448.64
11.2.2	Providing, fitting and fixing anodised aluminium framed glazed doors with anodised aluminium frame made of 63mm x 38mm x 2.5mm section of approved brand with door style of size 50mm x 45mm x 2.0mm, top rails 50mm x 45mm x 2.5mm, middle rail 100mm x 45mm x 2mm and bottom rails 100mm x 45mm x 2.5mm fitted with glazing clip, special type rubber gasket complete including hydraulic floor spring, pivot, tower bolt in each leaf, aluminium door handle, lock, angles, cleat etc complete as specified and directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	9906.66
	(ii) 5mm frosted glass	Sqm	9945.36
	(iii) 6mm coloured reflectory glass	Sqm	11585.84
	(iv) Prelaminated Board	Sqm	9990.51
	(v) 5 mm clear glass	Sqm	9782.37
	(vi) 5 mm coloured glass	Sqm	9932.93
11.2.3	Providing, fitting and fixing Coloured anodised aluminium framed glazed doors with anodised aluminium frame made of 100mm x 45mm x 2.5mm section with door style of size 88mm x 45mm x 2.5mm, top rails 50mm x 45mm x 2mm lock rails 100mm x 45mm x 2mm and bottom rails 100mm x 45mm x 2.5mm fitted with glazing clip, special type rubber gasket complete including hydraulic floor spring, pivot, tower bolt in each leaf, aluminium door handle, lock, angles, cleat etc complete as specified and directed by the department at all levels.		
	(i) 6mm clear glass	sqm	10561.74
	(ii) 5mm frosted glass	sqm	10600.44
	(iii) 6mm coloured glass	sqm	10712.30
	(iv) Prelaminated Board	sqm	10645.58
	(v) 5 mm clear glass	sqm	10623.47
	(vi) 5 mm coloured glass	sqm	10588.01

11.2.4	Providing, fitting and fixing coloured anodised aluminium framed glazed doors with anodised aluminium frame made of 63mm x 38mm x 2.5mm section of approved brand with door style of size 50mm x 45mm x 2.0mm, top rails 50mm x 45mm x 2.5mm, middle rail 100mm x 45mm x 2mm and bottom rails 100mm x 45mm x 2.5mm fitted with glazing clip, special type rubber gasket complete including hydraulic floor spring, pivot, tower bolt in each leaf, aluminium door handle, lock, angles, cleat etc complete as specified and directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	10221.86
	(ii) 5mm frosted glass	Sqm	10260.56
	(iii) 6mm coloured glass	Sqm	10372.42
	(iv) Prelaminated Board	Sqm	10305.70
	(v) 5 mm clear glass	Sqm	10097.56
	(vi) 5 mm coloured glass	Sqm	10248.13
11.2.5	Providing, fitting and fixing coloured anodised aluminium sliding glazed doors with anodised aluminium frames made of 100mm x 45mm x 2mm section of approved brand with door style of size 81mm x 45mm x 2mm, top rails 55mm x 45mm x 2mm, lock rails 81mm x 45mm x 2mm, fitted with glazing clip, special type rubber gasket complete including roller, tower bolt, aluminium door handle, lock, angles, cleat etc. complete as specified and directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	7497.23
	(ii) 5mm frosted glass	Sqm	7528.25
	(iii) 6mm coloured glass	Sqm	7617.90
	(iv) Prelaminated Board	Sqm	7564.43
	(v) 5 mm clear glass	Sqm	7397.62
	(vi) 5 mm coloured glass	Sqm	7518.28
11.2.6	Providing, fitting and fixing anodised aluminium sliding glazed doors with anodised aluminium frames made of 100mm x 45mm x 2mm section of approved brand with door style of size 81mm x 45mm x 2mm, top rails 55mm x 45mm x 2mm, lock rails 81mm x 45mm x 2mm, fitted with glazing clip, special type rubber gasket complete including roller, tower bolt, aluminium door handle, lock, angles, cleat etc. complete as specified and directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	7116.53
	(ii) 5mm frosted glass	Sqm	7147.54
	(iii) 6mm coloured glass	Sqm	7237.19
	(iv) Prelaminated Board	Sqm	7582.41
	(v) 5 mm clear glass	Sqm	7016.91
	(vi) 5 mm coloured glass	Sqm	7137.58

11.2.7	Providing, fitting and fixing coloured anodised aluminium framed double leaf glazed doors with anodised aluminium frame made of 100mm x 45mm x 2mm sections of approved brand with door styles of size 81mm x 45mm x 2mm, top rails 50mm x 45mm x 2mm, lock rails 81 mm x 45mm x 2mm, bottom rails 100mm x 45mm x 2mm fitted with glazing clip, special type rubber gasket complete including hydraulic floor spring, pivot, tower bolt in each leaf, aluminium door handle, lock, angles, cleat etc complete as specified and directed by the Department at all levels.		
	(i) 6mm clear glass	Sqm	6647.53
	(ii) 5mm frosted glass	Sqm	6678.02
	(iii) 6mm coloured glass	Sqm	6766.16
	(iv) Prelaminated Board	Sqm	6713.59
	(v) 5 mm clear glass	Sqm	6549.59
	(vi) 5 mm coloured glass	Sqm	6668.23
11.2.8	Providing, fitting and fixing anodised aluminium framed double leaf glazed doors with anodised aluminium frame made of 100mm x 45mm x 2mm section of approved brands with door styles of size 81mm x 45mm x 2mm, top rails 50mm x 45mm x 2mm, lock rails 81 mm x 45mm x 2mm, bottom rails 100mm x 45mm x 2mm fitted with glazing clip, special type rubber gasket complete including hydraulic floor spring, pivot, tower bolt in each leaf, aluminium door handle, lock, angles, cleat etc complete as specified and directed by the Department at all levels.		
	(i) 6mm clear glass	Sqm	6416.13
	(ii) 5mm frosted glass	Sqm	6446.62
	(iii) 6mm coloured glass	Sqm	6534.77
	(iv) Prelaminated Board	Sqm	6482.19
	(v) 5 mm clear glass	Sqm	6258.98
	(vi) 5 mm coloured glass	Sqm	6377.62
11.2.9	Providing and fixing anodised aluminium framed double leaf doors of approved brand with styles size 45mm x 48mm x 3.2mm, top rails 50mm x 45mm x 2.5mm, lock rails 48 mm x 45mm x 2.5mm, bottom rails 100mm x 45mm x 2.5mm fitted with tapper clips of size 20mm x15mm x1.2mm complete with automatic floor mounted door closure, butterfly type handles, special cylindrical lock, concealed type aluminium tower bolt, neoprene dryset rubber gasket, door felt gasket, etc. complete as directed by the Department at all levels.		
	(i) 5mm thick frosted glass and 12mm thick prelaminated particle board	Sqm	7616.42
	(ii) 6mm clear glass	Sqm	7428.10
	(iii) 5mm plain frosted glass	Sqm	7513.88
	(iv) 6mm coloured glass	Sqm	7606.44
	(v) Prelaminated Board	Sqm	7497.47
	(vi) 5 mm clear glass	Sqm	7316.56
	(vii) 5 mm coloured glass	Sqm	7449.83

11.2.10	Providing and fixing anodised aluminium framed double leaf doors of approved brand with styles size 45mm x 48mm x 3.2mm, top rails 50mm x 45mm x 2.5mm, lock rails 48 mm x 45mm x 2.5mm, bottom rails 100mm x 45mm x 2.5mm fitted with taper clips of size 20mm x15mm x1.2mm complete with automatic floor mounted door closure, butterfly type handles, special cylindrical lock, concealed type aluminium tower bolt, neoprene dryset rubber gasket, door felt gasket, etc. complete as directed by the Department at all levels.		
	(a) Double leaf doors		
	(i) 5mm thick frosted glass and 12mm thick prelaminated particle board	Sqm	7660.40
	(ii) 6mm clear glass	Sqm	7609.01
	(iii) 5mm plain frosted glass	Sqm	7641.02
	(iv) 6mm coloured glass	Sqm	7733.59
	(v) Prelaminated Board	Sqm	7661.89
	(vi) 5 mm clear glass	Sqm	7610.60
	(vii) 5 mm coloured glass	Sqm	7721.20
11.2.11	Providing and fixing anodised aluminium framed single leaf door of approved brand with styles size 45mm x 48mm x 3.2mm, top rails 50mm x 45mm x 2.5mm, lock rails 48 mm x 45mm x 2.5mm, bottom rails 100mm x 45mm x 2.5mm fitted with taper clips of size 20mmx15mmx1.2mm complete with automatic hydraulic floor mounted door closure, butterfly type handles, special cylindrical locks, concealed type tower bolt, neoprene dryset rubber gasket, etc. complete as directed by the Department at all levels.		
	(i) 5mm thick frosted glass and 12mm thick prelaminated board	Sqm	7260.59
	(ii) 6mm clear glass	Sqm	7050.29
	(iii) 5mm plain frosted glass	Sqm	7085.79
	(iv) 6mm coloured glass	Sqm	7188.41
	(v) Prelaminated Board	Sqm	7127.20
	(vi) 5 mm clear glass	Sqm	6936.27
	(vii) 5 mm coloured glass	Sqm	7074.39
11.2.12	Providing and fixing coloured anodised aluminium framed single leaf door of approved brand with styles size 45mm x 48mm x 3.2mm, top rails 50mm x 45mm x 2.5mm, lock rails 48 mm x 45mm x 2.5mm, bottom rails 100mm x 45mm x 2.5mm fitted with taper clips of size 20mm x 15mm x 1.2mm complete with automatic hydraulic floor mounted door closure, butterfly type handles, special cylindrical locks, concealed type tower bolt, neoprene dryset rubber gasket, etc. complete as directed by the Department at all levels.		
	(i) 5mm thick frosted glass and 12mm thick prelaminated board	Sqm	7426.85
	(ii) 6mm clear glass	Sqm	7216.56
	(iii) 5mm plain frosted glass	Sqm	7252.06
	(iv) 6mm coloured glass	Sqm	7354.67
	(v) Prelaminated Board	Sqm	7293.47
	(vi) 5 mm clear glass	Sqm	7102.54
	(vii) 5 mm coloured glass	Sqm	7203.13
11.3	MISCELLANEOUS		

11.3.1	Suppling, fitting and fixing 20mm dia aluminium rod with aluminium bracket at both ends of the pelmet over door and windows as directed and specified at all levels.	M	115.44
11.3.2	Providing and fixing aluminium step edging with anodised aluminium angle to stairs at all levels.		
	(i) With angle of size 38mm x12 mm x 3mm	M	250.81
	(ii) With angles of size 57mm x12 mm x 3mm	M	285.94
11.3.3	Extra over item 11.3.2 for cutting R.C.C./brick steps for fixing of aluminium strip edging and making good to match the surroundings.	M	48.55
11.4	PARTITION WALLS		
11.4.1	Providing, fitting and fixing anodised aluminium partition walls of standard sections of approved brand, joints mitred and welded (manufactured to relevant I.S. specification) and providing and fitting angle cleat, rubber gasket, bolting device screws etc. at all levels.		
	(i)Partly 6mm thick glass panes and 12mm prelaminated ply board	Sqm	3429.01
	(ii) 6mm clear glass	Sqm	3593.12
	(iii) 5mm frosted glass	Sqm	3242.28
	(iii) 6mm coloured glass	Sqm	3345.88
	(iv) Prelaminated Board	Sqm	3225.66
	(v) Partly 5mm thick glass panes and 12mm prelaminated ply board	Sqm	3297.45
	(vi) 5 mm clear glass	Sqm	3091.34
	(vii) 5 mm coloured glass	Sqm	3230.77
11.4.2	Providing, fitting and fixing coloured anodised aluminium partition walls of standard sections of approved brand, joints mitred and welded (manufactured to relevant I.S. specification) and providing and fitting angle cleat, rubber gasket, bolting device screws etc. at all levels.		
	(i)Partly 6mm thick glass panes and 12mm prelaminated ply board	Sqm	3586.86
	(ii) 6mm clear glass	Sqm	3364.29
	(iii) 5mm frosted glass	Sqm	3400.12
	(iv) 6mm coloured glass	Sqm	3503.72
	(iv) Prelaminated Board	Sqm	3383.50
	(vi) Partly 5mm thick glass panes and 12mm prelaminated ply board	Sqm	3530.55
	(viii) 5 mm clear glass	Sqm	3249.18
	(viii) 5 mm coloured glass	Sqm	3388.61
11.4.3	Providing, fitting and fixing anodised aluminium hollow section of approved brand (Indal section 9210) of size 63.50mm x 38.10mm x 2.5mm thick for aluminium frame partition wall, with a maximum spacing of frame 1.20m for horizontal/vertical member including aluminium cleat 32mm x 50mm x 4mm thick and fixing of pre-laminated board/ Nepal board/ Bhutan board with taper cleat 20mm x 15mm x 1.2mm thick as directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	3581.54
	(ii) 5mm frosted glass	Sqm	3592.15
	(iii) 6mm coloured glass	Sqm	3619.62
	(iv) Prelaminated Board	Sqm	3685.73
	(v) 5 mm clear glass	Sqm	3551.02

	(vi) 5 mm coloured glass	Sqm	3589.10
11.4.4	Providing, fitting and fixing coloured anodised aluminium hollow section of approved brand (Indal section 9210) of size 63.50mm x 38.10mm x 2.5mm thick for aluminium frame partition wall, with a maximum spacing of frame 1.20m for horizontal/vertical member including aluminium cleat 32mm x 50mm x 4mm thick and fixing of pre-laminated board/ Nepal board/ Bhutan board with tapper cleat 20mm x 15mm x 1.2mm thick as directed by the department at all levels.		
	(a) 2 Track sliding window		
	(i) 6mm thick clear glass	Sqm	3668.96
	(ii) 5mm thick frosted glass	Sqm	3642.40
	(iii) 6mm thick coloured glass	Sqm	3705.93
	(iv) Prelaminated Board	Sqm	3772.04
	(v) 5 mm thick clear glass	Sqm	3638.44
	(vi) 5 mm thick coloured glass	Sqm	3675.41
11.5	PARTITION WALLS WITH DOORS		
11.5.1	Providing, fitting and fixing anodised aluminium framed glazed doors with partition walls partly glass and partly prelaminated board with anodised aluminium frame made of 100mm x 43mm x 2mm sections of approved brand with doors style of size 81mm x 43mm x 2mm, top rails 46mm x 43mm x 2mm, lock rails 81mm x 43mm x 2mm, fitting with glazing clip, special rubber type gasket complete including hydraulic floor spring pivot aluminium door handle lock angles cleat etc. complete as directed by the department at all levels.		
	(i) 6mm thick glass panes and 12mm prelaminated board	Sqm	3689.31
	(ii) 6mm clear glass	Sqm	3611.06
	(iii) 5mm frosted glass	Sqm	3638.42
	(iv) 6mm coloured glass	Sqm	3715.36
	(v) Prelaminated Board	Sqm	3668.19
	(vi) 5mm thick glass panes and 12mm prelaminated board	Sqm	3624.59
	(vii) 5 mm clear glass	Sqm	3521.02
	(viii) 5 mm coloured glass	Sqm	3627.48
11.5.2	Providing, fitting and fixing coloured anodised aluminium framed glazed doors with partition walls partly glass and partly prelaminated board with anodised aluminium frame made of 100mm x 43mm x 2mm sections of approved brand with doors style of size 81mm x 43mm x 2mm, top rails 46mm x 43mm x 2mm, lock rails 81mm x 43mm x 2mm, fitting with glazing clip, special rubber type gasket complete including hydraulic floor spring pivot aluminium door handle lock angles cleat etc. complete as directed by the department at all levels.		
	(i) 6mm thick glass panes and 12mm prelaminated board	Sqm	3857.22
	(ii) 6mm clear glass	Sqm	3778.97
	(iii) 5mm frosted glass	Sqm	3806.33
	(iv) 6mm coloured glass	Sqm	3885.43
	(v) Prelaminated Board	Sqm	3838.25
	(vi) 5 mm thick glass panes and 12mm prelaminated board	Sqm	3846.47
	(vii) 5 mm clear glass	Sqm	3691.09
	(viii) 5 mm coloured glass	Sqm	3797.54

11.5.3	Providing, fitting and fixing anodised aluminium framed sliding glazed doors with partition walls with anodised aluminium frames made of 100mm x 43mm x 2mm section of approved brand with door style of size 81mm x 43mm x 2mm, top rails 46mm x 43mm x 2mm, lock rails 81mm x 43mm x 2mm, paints and 12mm prelaminated board fitting with glazing clip, special type rubber gasket complete including roller, door handle, lock angles, cleat etc. complete as directed by the department at all levels.		
	(i) Partly 6mm thick coloured reflectory glass and partly with prelaminated board	Sqm	5882.03
	(ii) 6mm clear glass	Sqm	4488.98
	(iii) 5mm frosted glass	Sqm	4491.48
	(iv) 6mm coloured glass	Sqm	4646.02
	(v) Prelaminated Board	Sqm	4576.42
	(vi) 5 mm clear glass	Sqm	4359.33
	(vii) 5 mm coloured glass	Sqm	4516.37
11.5.4	Providing, fitting and fixing coloured anodised aluminium framed sliding glazed doors with partition walls with anodised aluminium frames made of 100mm x 43mm x 2mm section of approved brand with door style of size 81mm x 43mm x 2mm, top rails 46mm x 43mm x 2mm, lock rails 81mm x 43mm x 2mm, paints and 12mm prelaminated board fitting with glazing clip, special type rubber gasket complete including roller, door handle, lock angles, cleat etc. complete as directed by the department at all levels.		
	(i) Partly 6mm thick coloured reflectory glass and partly with prelaminated board	Sqm	6103.98
	(ii) 6mm clear glass	Sqm	4710.92
	(iii) 5mm frosted glass	Sqm	4751.28
	(iv) 6mm coloured glass	Sqm	4867.96
	(v) Prelaminated Board	Sqm	4798.37
	(vi) 5 mm clear glass	Sqm	4581.28
	(vii) 5 mm coloured glass	Sqm	4738.32
11.5.5	Providing, fitting and fixing anodised aluminium framed double leaf glazed doors with partition walls partly glazed and partly prelaminated board with anodised aluminium frame made of 100mm x 43mm x 2mm sections of approved brand with door styles of size 81mm x 43mm x 2mm, top rails 46mm x 43mm x 2mm, lock rails 81 mm x 43mm x 2mm fitting with glazing clips, special type rubber gasket including hydraulic floor spring at all levels.		
	(i) 6mm thick glass panes and 12mm prelaminated board	Sqm	5674.16
	(ii) 6mm clear glass	Sqm	5552.81
	(iii) 5mm frosted glass	Sqm	5592.42
	(iv) 6mm coloured glass	Sqm	5706.93
	(v) Prelaminated Board	Sqm	5638.63
	(vi) 5 mm thick glass panes and 12mm prelaminated board	Sqm	5582.85
	(vii) 5 mm clear glass	Sqm	5425.58
	(viii) 5 mm coloured glass	Sqm	5579.70

11.5.6	Providing, fitting and fixing coloured anodised aluminium framed double leaf glazed doors with partition walls partly glazed and partly prelaminated board with anodised aluminium frame made of 100mm x 43mm x 2mm sections of approved brand with door styles of size 81mm x 43mm x 2mm, top rails 46mm x 43mm x 2mm, lock rails 81mm x 43mm x 2mm fitting with glazing clips, special type rubber gasket including hydraulic floor spring at all levels.		
	(i) 6mm thick glass panes and 12mm prelaminated board	Sqm	5893.12
	(ii) 6mm clear glass	Sqm	5771.78
	(iii) 5mm frosted glass	Sqm	5811.39
	(iv) 6mm coloured glass	Sqm	5925.89
	(v) Prelaminated Board	Sqm	5857.60
	(vi) 5 mm thick glass panes and 12mm prelaminated board	Sqm	5801.82
	(vii) 5 mm clear glass	Sqm	5644.55
	(viii) 5 mm coloured glass	Sqm	5798.66
11.5.7	Providing, fitting and fixing anodised aluminium hollow section of approved brand (Indal section 9210) of size 63.50mm x 38.10mm x 2.5mm thick for aluminium frame partition wall, with a maximum spacing of frame 1.20m for horizontal/vertical member including aluminium cleat 32mm x 50mm x 4mm thick and fixing of glass panes and pre-laminated board with taper cleat 20mm x 15mm x 1.2mm thick fixing glass panes with rubber gasket, screws, etc. as directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	3718.46
	(ii) 5mm frosted glass	Sqm	3749.17
	(iii) 6mm coloured glass	Sqm	3837.94
	(v) Prelaminated Board	Sqm	3865.87
	(v) 12mm thick pre-laminated board and 5mm thick frosted glass panes	Sqm	3939.00
	(vi) 5 mm clear glass	Sqm	3619.83
	(vii) 5 mm coloured glass	Sqm	3739.30
11.5.8	Providing, fitting and fixing coloured anodised aluminium hollow section of approved brand (Indal section 9210) of size 63.50mm x 38.10mm x 2.5mm thick for aluminium frame partition wall, with a maximum spacing of frame 1.20m for horizontal/vertical member including aluminium cleat 32mm x 50mm x 4mm thick and fixing of glass panes and pre-laminated board with taper cleat 20mm x 15mm x 1.2mm thick fixing glass panes with rubber gasket, screws, etc. as directed by the department at all levels.		
	(i) 6mm clear glass	Sqm	3915.39
	(ii) 5mm plain frosted glass	Sqm	3936.79
	(iii) 6mm coloured glass	Sqm	3998.62
	(iv) Prelaminated Board	Sqm	3961.74
	(v) 12mm thick pre-laminated board and 5mm thick frosted glass panes	Sqm	3936.79
	(vi) 5 mm clear glass	Sqm	3860.61
	(vii) 5 mm coloured glass	Sqm	3943.83
11.6	CURTAIN WALL, CURTAIN GLAZING, STRUCTURAL GLAZING		

11.6.1	Providing, fitting and fixing aluminium curtain wall on the exterior face of building fitting to concrete member by drilling hole fixing by brass screw fabricated with anodised aluminium section of approved brand of size 101mm x 44.50mm by providing aluminium jointing angles including providing 6mm thick coloured reflectory type glass fitted with aluminium at all levels.	Sqm	6963.74
11.6.2	Providing, fitting and fixing aluminium curtain wall on the exterior face of building fitting to concrete member by drilling hole fixing by brass screw fabricated with coloured anodised aluminium section of approved brand of size 101mm x 44.50mm by providing aluminium jointing angles including providing 6mm thick coloured reflectory type glass fitted with aluminium at all levels.		
	(i) Partly 6mm thick coloured reflectory glass and partly with prelaminated board	Sqm	7419.93
	(ii) 6mm clear glass	Sqm	5523.88
	(iii) 5mm frosted glass	Sqm	5561.51
	(iv) 6mm coloured glass	Sqm	5581.58
	(v) Prelaminated Board	Sqm	5932.14
11.6.3	Providing, fitting and fixing aluminium curtain wall on the exterior face of building fitting to concrete member by drilling hole fixing by brass screw fabricated with coloured anodised aluminium section of approved brand 101mm x 55mm x 2mm by providing jointing angles including providing coloured reflectory type glass 6mm thick fitted with aluminium at all levels.	Sqm	5549.24
11.6.4	Providing, fitting and fixing aluminium curtain wall on the exterior face of building fitting to concrete member by drilling hole fixing by brass screw fabricated with anodised aluminium section of approved brand 101mm x 55mm x 2mm by providing jointing angles including providing coloured reflectory type glass 6mm thick fitted with aluminium at all levels.	Sqm	5643.86
11.6.5	Providing and fixing of aluminium curtain glazing made out of Anodised section of approved shade section of 6 nos 1.2m each @ 1.819 Kg/m, 6 nos 1.2m each @ 0.515 Kg/m, 6 nos 1.2m each @ 0.317 Kg/m of Jindal or equivalent complete with rubber gasket and all other accessories, including fixing of 5mm tinted glass as and where necessary.		
	I) Using Plain Anodised section	Sqm	7978.54
	II) Using Coloured Anodised section	Sqm	8337.78
11.6.6	Providing , fabrication and fixing of structural glazing made out of Anodised aluminium mother frame 6 nos 1.2m each @ 1.819 Kg/m and subframe of 16 nos 0.60 m each @ 0.31 Kg/m of HINDALCO/ NALCO/ BALCO/JINDAL/OTHER equivalent primary product and 5 mm reflective glass of approved shade. The glass to be pasted using structural silicon (DC 995) and weatherseal silicon to be used (DC 789), the structure complete with all accessories as and where necessary.	Sqm	9394.97

11.6.7	Providing , fabrication and fixing of structural glazing made out of coloured Anodised aluminium mother frame 6 nos 1.2m each @ 1.819 Kg/m and subframe of 16 nos 0.60 m each @ 0.31 Kg/m of Jindal or equivalent and 5 mm reflective glass of approved shade. The glass to be pasted using structural selicon (DC 995) and weatherseal silicon to be used (DC 789), the structure complete with all accessories as and where necessary.	Sqm	
11.7.1	Supplying,fitting and fixing of Automatic sliding door and operator, modular design including internal cover with operator dimensions (H X T) not exceeding 120mm x 175mm and of length required to suit the opening size given below: Clear opening of size 2400mm X 2400mm: compact linear sliding door automatic, with self cleaning rollers, TUV type tested as per DIN 18650.The Automatic sliding door consists of 16 bit Microprocessor Control unit, monitoring of closing force, automatic reverse on contact with obstruction, operator commissioning without any additional instrumentation and self-teaching. Operating Modes: Continuous open, automatic operation, closing time, night lock, multi-sensor detector(radar motion detector with direction detecion and self-monitoring, double 3D safeguard light curtain (light barrier) for convenient safeguarding of the opening width) with Electro magnetic locking, integrated in the drive unit, with manual emergency unlocking of GODREJ/GEZE make.The operator will be provided with short circuit proof power supply unit 220 / 240 V, 50 / 60 hz .		
	(a) Door Automatic ST10S	Each	98881.98
	(b) Door Automatic ST10D	Each	112615.59
	(c) Door Automatic ST20 S	Each	137336.08
	(d) Door Automatic ST20 D	Each	168923.38
11.7.2	Providing, fitting and fixing anodised aluminuim framed glazed doors with anodised aluminium frame made of 100mm x 45mm x 2.5mm section with door style of size 88mm x 45mm x 2.5mm, top rails 50mm x 45mm x 2mm and bottom rails 100mm x 45mm x 2.5mm fitted with glazing clip, special type rubber gasket complete etc complete as specified and directed by the department at all levels.		
	(a) with 12 mm thick toughened glass	Sqm	10891.60
	(b) with 10 mm thick toughened glass	Sqm	11098.68
	(c) with 8 mm thick toughened glass	Sqm	10650.63
	(d) with 12 mm thick coloured glass	Sqm	10846.71
	(e) with 10 mm thick coloured glass	Sqm	10891.60
	(f) with 8 mm thick coloured glass	Sqm	10650.63
CH-12 PVC, uPVC, FRP DOORS & DOOR FRAMES			
Item No.	Description	Unit	Rate
	P.V.C. DOORS & DOOR FRAMES		
12.1	Providing, fitting and fixing 40mm x 48mm wood reinforced PVC chowkhats fixed to the columns / stiffeners / walls with necessary holdfast / sleeves etc. as specified and directed at all levels.	Rm	186.12

12.2	Providing, fitting and fixing 24mm/ 29mm thick polymer reinforced moulded PVC door shutters fixed to the chowkath with necessary hinges, screws, bolts etc. as specified and directed at all levels (Door fixtures and fittings to be measured and paid seperately)		
	a) 24mm thick	Sqm	999.44
	b) 29mm thick	Sqm	1511.53
12.3	Providing, fitting and fixing PVC door frame of Duroplast make DPF-5060 make suitable for DPS-3720 made from extruded PVC multichamber hollow section having dimensions of 50mmx60mm (Toletance + 1mm). Horizontal and vertical sections shall be joined by steel screws after inserting M.S. tubular rectangular galvanised L-shaped brackets of size 100mmx100mm or plastic brackets of size 75mmx75mm with metred cut at the corners. Hinge side of the frame shall be reinforced with M.S.Tubular section of size 40mmx20mm. The frame will be fixed to the column/ stiffner/ wall with the help of 3 nos 75mm long and 4mm dia steel screws and expandable fastners on each side of the door frame complete in all respects as sepcified and directed at all levels.	Rm	700.79
12.4	Providing, fitting and fixing PVC Door shutters of Duroplast DPS 3720 make made fromm extruded PVC stiles and rails made from rigid PVC multicavity hollow chamber section of size 90mmx37mm (Toletance+ 1mm). The corners of the stiles and rails shall be metred out and fixed to each other with the help of solid plastic L-shaped brackets or M.S. rectangular tubular galvanised two nos of size 200mmx100mm and 100mmx100mm at each corner by self tapping steel screws. PVC multicavity hollow section of size 105mmx37mm shall be fixed in the middle by aluminium cleat and steel screws as lock rail. The section frame shall then be filled with PVC panels of approved colour of size 100mmx20mm. Then two numbers of bright steel rods of 6mm diameter shall be inserted horizontally through stiles and panels with two numbers of checknuts and washers all complete as specified and directed at all levels. (Door fixtures and fittings to be measured and paid seperately)	Sqm	1511.53
12.5	Providing, fitting and fixing PVC Door shutters of 37mm (single panel seamless finish) of Syntex make, at all levels made of extruded sections in the configuration of 'A'.The overall dimension is 37mm x 90mm, with usual process variation having a wall thickness of 2mm with a variation of ± 0.03 mm provided with all plastic reinforcement to avoid rusting. The shutter frame is mitred and joined at the corners by means of stainless steel screws.The corners of the shutter frame are being reinforced by polymeric reinforcement of the size 125mm x 250mm and the lock rail is reinforced by 45 x18 x255 size reinforcement.	Sqm	1947.01

	The lock rail is made of extruded hollow PVC section in the configuratino of 'H' having wall thickness of 2mm with a variation of +0.03mm and effective dimension of 37mm x 105mm. The infill is made of multi-chambered hollow PVC extruded panel having an effective dimension of 762mm x 20mm or less as per requirement with usual process variation having a wall thickness of 1mm +0.03mm. The entire door shutter should be provided with S.S. screws at appropriate places for rust free quality. (Door fixtures and fittings to be measured and paid separately.)		
12.6	Providing, fitting and fixing PVC Door shutters of 37mm (single panel seamless finish) of Syntex make, at all levels made of extruded sections in the configuration of 'A'. The overall dimension is 37mm x 90mm, with usual process variation having a wall thickness of 2mm with a variation of ± 0.03mm provided with all plastic reinforcement to avoid rusting. The shutter frame is mitred and joined at the corners by means of stainless steel screws. The corners of the shutter frame are being reinforced by polymeric reinforcement of the size 125mm x 250mm and the lock rail is reinforced by 45 x 18 x 255 size reinforcement. The lock rail is made of extruded hollow PVC section in the configuratino of 'H' having wall thickness of 2mm with a variation of ± 0.03mm and effective dimension of 37mm x 105mm.	Rm	209.70
	The infill is made of multi-chambered hollow PVC extruded panel having an effective dimension of 762mm x 20mm or less as per requirement with usual process variation having a wall thickness of 1mm +0.03mm. The entire door shutter should be provided with S.S. screws at appropriate places for rust free quality. (Door fixtures and fittings to be measured and paid separately.)		
12.7	FRP PANNELED DOOR Supplying & fixing of factory made 30mm thick Fibreglass Reinforced Plastic (FRP) panneled door shutter of required colour of FIBREWAYS made with fire retardant grade unsatured polyster resin, moulded to 3mm thick FRP laminate for forming hollow rails and styles, with wooden frame and suitable blocks of seasoned wood inside at required places for fixing of fittings, cast monolithically with 5mm thick FRP laminate for panels and conforming to IS:14856-2000 including fixing to frames complete as per direction of Engineer-in charge.	Sqm	2647.85
12.8	FRP FLUSH DOOR Supplying & fixing of factory made 30mm thick Fibreglass Reinforced Plastic (FRP) panneled door shutter of required colour of FIBREWAYS made with fire retardant grade unsatured polyster resin, moulded to 3mm thick FRP laminate for forming hollow rails and styles, with wooden frame and suitable blocks of seasoned wood inside at required places for fixing of fittings, cast monolithically with 5mm thick FRP laminate for panels and conforming to IS:14856-2000 including fixing to frames complete as per direction of Engineer-in charge.	Sqm	3047.82

12.9	FRP FRAME providing & fixing of factory made Fibreglass Reinforced Plastic (FRP) door frame of FIBREWAYS having three legged of cross section 90mmx45mm having single rebet of 32 mmx15mm to receive shutter of 30 mm thickness .The laminate door frame moulded with fire retardant grade unsaturated polyster resin and chopped mat .Door frame laminate and shall be filled with suitable wooden block in all the three legs the frame shall be covered with covered glass from all sides. MS stay shall be provided at the bottom to steady the frame.	RM	575.50
12.10	FRP SHEET MACHINE MADE supplying & fixing of factory Machine made Fibreglass Reinforced Plastic (FRP) Translucent Roofing u/v stabilized sheets of ROOFFIT in desired thickness ,size profile -(plain,corrugated,trapezoidal) & colour manufactured as per IS 12866 on continuous machine auto controlled production line with auto batching,bubble free mixer,impregnation ,heating,refractor system using continuous glass fibre roving of 2400 tex reinforced powder grade material as per IS:11551 & sheet grade unsaturated polyester resin as per IS:6746 duly mixed with u/v protection & extra clarity in continuous hot cured production translucent pigmented or gelcoated opaque sheet complete as per specification,drawing or direction of Engineer in charge.	Sqm	1121.04
12.11	FRP CHAJJA providing & fixing of factory made Fibreglass Reinforced Plastic (FRP) chajja 4 mm thick of FIBREWAYS in required colour size and design made by Resin Transfer Moulding (RTM) Machine Technology, resulting in void free compact laminate in single piece ,having smooth gradual slope curvature for easy drainage of water and duly reinforced by two numbers vertically and one number horizontally 50x2mm thick M.S. plate with 12 mm in built hole for grouting on the existing wall along with the 50 mm flange duly inserted and sealed in the wall complete in one single piece casted monlythically including all necessary fittinga. ?The FRP chajja should be manufactured using unsaturated polyster resin as per IS 6746 duly reinforced with fibre glasss chopped strand mat (CSM)as per IS: 11551 complete with protective Gelcoat U/V coating on top for completen resistance from the extreme of temperature,weather & sunlight.	Rm	4795.82
	uPVC Doors & Windows		
	uPVC Windows		
12.12	Fixed Windows		

12.12.1	Supplying and installing uPVC- Fixed Windows made out of Lead free green profile having "GREENLINE" mark of BIS standard uPVC multichambered sections of wall thickness 2.4mm with corners fusion welded, fully reinforced with galvanized steel 1.5/2mm including glazing bead , grooving bead, drain cap, fisher screws, packing pieces, necessary stainless steel screws etc. complete as directed. The windows must be installed complete with all kinds of ironmongery including EPDM gaskets, bridging wedges and glass packers and with suitable water draining system. Application of Silicon sealant from inside / outside of dowercorning / GE or equivalent. With 5mm clear glass		
	(a) Fixed window using uPVC sections of size 58mm x 52mm x 2.4mm thick wall for frame and 58mm x 68mm having 2.4mm thick wall for glass sash having reinforcement of 1.5mm thick in both frame & sash. (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	6272.22
	(b) ProVent fixe window Using uPVC sections of size 37mm x 43mm x 2.2mm thick wall for frame and 37mm x 56mm x 2.2mm thick wall for sash having 1.5mm thick reinforcement for both frame & sash. (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	5431.37
12.12.2	Supplying and installing uPVC- Sliding Windows made out of Lead free green profile having "GREENLINE" mark of BIS standard uPVC multichambered sections of wall thickness 2.4mm with corners fusion welded, fully reinforced with Galvanized steel 1.5/2mm including interlock profile, glazing bead, grooving bead, brush seal, aluminum sliding track, stainless steel rollers, sealing wedge block, Espag, int handle, pop up handle, keeper, drain cap, fisher screws, packing pieces with all necessary stainlessness screws etc. complete as directed. The windows must be installed complete with all kinds of ironmongery including EPDM gaskets, bridging wedges and glass packers and with suitable water draining system. Application of silicon sealant from inside / outside of Dowcorning / GE or equivalent. With 5mm clear glass		
	(a) 2 Track Sliding Window with Flyscreen using uPVC section of sizes 88mm x 50 mm x 2.4mm thick wall for frame with inbuilt flyscreen track, 37mm x 61mm x 2.4mm thick wall for glass sash, 25mm x 49mm x 2.4mm thick wall for flyscreen sash having 1.5mm thick reinforcement, Interlock: 43mm x 23mm (Glass Panel) (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)		
	(i) 2-Panel with 5mm clear float glass	sqm	8854.83
	(ii) 4-Panel with 5mm clear float glass	sqm	10116.09
	(b) 2 Track Sliding Window without Flyscreen using uPVC section of size 58mm x 53mm x 2.4mm thick wall for frame, 37mm x 61mm x 2.4mm thick wall for glass sash having 1.5mm thick reinforcement for both frame & sash, Interlock: 43mm x 23mm (Glass Panel) (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)		
	(i) 2-Panel with 5mm clear float glass	sqm	8194.16

	(ii) 4-Panel with 5mm clear float glass	sqm	9515.49
12.13	Casement Windows		
12.13.1	Supplying and installing uPVC- Casement Windows made out of Lead free green profile having "GREENLINE" mark of BIS standard uPVC sections of wall thickness 2.4mm, multichambered with corners fusion welded, fully reinforced with Galvanized steel 1.5/2mm including glazing bead, folding gasket, glazing gasket, friction stay, Espag handle, Espag, keeper, drain cap, run up block, fisher screws, packing pieces with all stainless steel screws complete as directed. The windows must be installed complete with all kinds of ironmongery including EPDM gaskets,bridging wedges and glass packers and with suitable water draining system. Fittings ROTO / GQ or Equivalent. Application of Silicon sealant from Inside / Outside of Dowcorning / GE or equivalent.		
	a) Single leaf Side hung Window using uPVC section of size 58mm x 57mm x 2.4mm thick wall for frame, 58mm x 68mm x 2.4mm thick wall for glass sash. With 5mm clear float glass (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	10716.71
	b) Double leaf Side hung Window with a fixed Mullion (2 Panels) using uPVC profile size of 58mm x 57mm x 2.4mm thick wall for frame, 58mm x 68mm x 2.4mm thick wall for glass sash, 58mm x 74mm x 2.4mm thick for mullion. With 5mm clear float glass	sqm	10716.71
	(c) Double leaf Side hung Window with a Dummy Mullion - French Window (2 Panels) using uPVC section of size 58mm x 57mm x 2.4mm thick wall for frame, 58mm x 68mm x 2.4mm thick wall for glass sash With 5mm clear float glass (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	10716.71
	(d) Single leaf Top hung Window using uPVC section of size 58mm x 57mm x 2.4mm thick wall for frame, 58mm x 68mm x 2.4mm thick wall for glass sash With 5mm clear float glass (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	9815.80
	(e) Double leaf Top hung Window using uPVC section of size of 58mm x 57mm x 2.4mm thick wall for frame, 58mm x 68mm x 2.4mm thick wall for glass sash, 58mm x 74mm x 2.4mm thick wall for mullion. With 5mm clear float glass (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	11077.07
	(f) ProVent openable Top hung Window using uPVC section of size 37mm x 43mm x 2.2mm thick wall for frame, 37mm x 56mm x 2.2mm thick wall for glass sash With 5mm clear float glass (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	8013.98
12.14	Sliding Doors		

12.14.1	Supplying and installing uPVC- sliding door made out of Lead free green profile having "GREENLINE" mark of BIS standard uPVC multichambered sections of wall thickness 2.8mm with corners fusion welded, fully reinforced with Galvanized steel 2mm including interlock profile, glazing bead, grooving bead, brush seal, locking arrangement, aluminum sliding track, stainless steel rollers, sealing wedge block, Espag, int handle, pop up handle, keeper, drain cap, fisher screws, packing pieces with all necessary stainlessness screws etc. complete as directed. The windows must be installed complete with all kinds of ironmongery including EPDM gaskets, bridging wedges and glass packers and with suitable water draining system. Fittings ROTO / GQ or Equivalent. Application of silicon sealant from inside / outside of Dowcorning / GE or equivalent.		
	(I) 2 Track Sliding door with Flyscreen using uPVC section of size 88mm x 50mm x 2.8mm thick wall for frame, 37mm x 75mm x 2.8mm thick wall for glass sash, 25mm x 49mm x 2.8mm thick wall for flyscreen sash, Interlock: 43mm x 23mm (Glass Panel) (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)		
	(a) 2-Panel		
	i) Using 5mm clear float glass	sqm	11857.86
	ii) Using 5mm clear toughened glass	sqm	10956.95
	(b) 4-Panel		
	i) Using 5mm clear float glass	sqm	12638.64
	ii) Using 5mm clear toughened glass	sqm	12638.64
	(II) 2 Track Sliding door without Flyscreen using uPVC section of size 58mm x 53mm x 2.8mm thick wall for frame, 37mm x 75mm x 2.8mm thick wall for glass sash, Interlock: 43mm x 23mm (Glass Panel) (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)		
	(a) 2-Panel		
	i) Using 5mm clear glass	sqm	10956.95
	ii) Using 5mm clear toughened glass	sqm	8794.77
	(b) 4-Panel		
	i) Using 5mm clear glass	sqm	10776.77
	ii) Using 5mm clear toughened glass	sqm	11077.07
12.15	Casement Doors		
12.15.1	Supplying and installing uPVC- Casement Door made out of Lead free green profile having "GREENLINE" mark of BIS standard uPVC sections of wall thickness 2.4mm, multichambered with corners fusion welded, fully reinforced with Galvanized steel 1.5/2mm including glazing bead, folding gasket, glazing gasket, locking arrangement, Espag handle, Espag, keeper, drain cap, run up block, fisher screws, packing pieces with all stainless steel screws complete as directed. The windows must be installed complete with all kinds of ironmongery including EPDM gaskets, bridging wedges and glass packers and with suitable water draining system. Fittings ROTO / GQ or Equivalent. Application of Silicon sealant from Inside / Outside of Dowcorning / GE or equivalent.		

	(a) Single leaf Side hung Door using uPVC section of size 58mm x 69mm x 2.4mm thick wall for frame, 58mm x 109mm x 2.4mm thick wall for glass sash, Aluminium threshold at bottom (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)		
	i) Using 5mm clear glass	sqm	15281.32
	ii) Using 5mm clear toughened glass	sqm	9875.86
	(b) Double leaf Side hung Door with a Dummy Mullion - French Door using uPVC section of size 58mm x 69mm x 2.4mm thick wall for frame, 58mm x 109mm x 2.4mm thick wall for glass sash, 58mm x 61mm x 2.4mm thick wall for dummy mullion, Aluminium threshold at bottom. (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)		
	i) Using 5mm clear glass	sqm	19125.20
12.14	Lift and Slide Patio Door		
12.14.1	Supplying and installing uPVC- lift or slide patio door made out of Lead free green profile having "GREENLINE" mark of BIS standard uPVC multichambered sections of wall thickness 2.8mm with corners fusion welded, fully reinforced with Galvanized steel 2mm including interlock profile, glazing bead, grooving bead, brush seal, locking arrangement, aluminum sliding track, stainless steel rollers, sealing wedge block, fisher screws, packing pieces with all necessary stainlessness screws etc. complete as directed. The windows must be installed complete with all kinds of ironmongery including EPDM gaskets, bridging wedges and glass packers and with suitable water draining system. Fittings ROTO / GQ or Equivalent. Application of silicon sealant from inside / outside of Dowcorning / GE or equivalent. uPVC section of size 174mm x 70mm x 2.8mm thick wall for frame, 62mm x 100mm x 2.8mm thick wall for glass sash. (WINSTA KOMMERLING/ FINESTA/ ENCRAFT)	sqm	26933.09

CH-13: PAINTING

Item No.	Description	Unit	Rate
13.1	WHITE / CHALK WASHING		
13.1.1	SURFACE PREPARATION		
13.1.1	(a) Removing white or colour wash by steel wire brushing and/ or scraping, sand papering and preparing the wall surface smooth including necessary repairs to scratches complete.	Sqm	13.57
	b) Extra over item no (a) above for scraping on ceiling and on sloping roof.	Sqm	1.36
	c) Washing and cleaning wall surface with soap, soda and clean water	Sqm	3.72
	d) Extra over item no (c) above for washing and cleaning ceiling and sloping roof.	Sqm	0.37
13.1.2	WHITE WASHING		

	(a) White washing with lime on wall surfaces (two coats) to give an even shade including thoroughly brooming the surface to remove all dirt, dust, mortar drops and other foreign matter.	Sqm	17.56
	(b) White washing with lime on old wall surfaces (one coat) to give an even shade including thoroughly brooming the surface to remove all dirt, dust, mortar drops and other foreign matter.	Sqm	9.06
	(c) Extra over item no. (a) above for every subsequent coat of white washing with lime on wall surfaces.	Sqm	8.51
	(d) Extra over item no. (a) above for white washing with lime on ceiling and on sloping roof.	Sqm	0.94
	e) Extra over item no. (b) above for white washing with lime on ceiling and on sloping roof.	Sqm	0.70
13.1.3	CHALK WASHING		
	(a) White washing with chalk on cloth or hessian ceiling (two coats) to give an even shade including thoroughly brooming the surface to remove all dirt, dust, mortar drops and other foreign matter.	Sqm	12.77
	b) Extra over item no. (a) above for every subsequent coat of white washing with chalk washing on cloth or hessian cloth ceiling.	Sqm	5.76
13.2.1	COLOUR WASHING / CEMENT WASHING / DISTEMPERING		
	a) Colour washing with lime on wall surface (two coats) over and including a priming coat of white washing to give an even shade after thoroughly brooming the surface to remove all dirt, dust, mortar drops and other foreign matter.	Sqm	27.93
	b) Extra over item no. (a) above for every subsequent coat of colour wash with lime on wall surfaces.	Sqm	10.53
	c) Cement washing with portland cement slurry on wall surface (one coats) to give a smooth bodied opaque finish including thoroughly brooming the surface to remove all dirt, dust, mortar drops and other foreign matter.	Sqm	7.74
13.2.2	DISTEMPERING		
13.2.2	a) Removing dry or oil bound distemper by washing and scraping and sand papering the wall surface smooth including necessary repairs to scratches complete.	Sqm	7.64
	b) Extra over item no (a) above for removing dry or oil bound distemper on ceiling and in sloping roof.	Sqm	0.76
	c) Applying one coat of distemper primer of approved brand and manufacture on wall surface after thoroughly brushing the surfaces free from mortar droppings and other foreign matter and including preparing the surface even and sand papered smooth.	Sqm	35.17
	d) Applying one coat of alkali resistance primer of approved brand and manufacture on wall surface after thoroughly brushing the surfaces free from mortar droppings and other foreign matter and including preparing the surface even and sand papered smooth.	Sqm	35.17
	e) Extra over item no. (c) and (d) above for applying one coat of distemper primer/alkali resistant primer over ceiling and on sloping roof.	Sqm	1.60

	f) Distempering with dry distemper of approved brand and manufacture(two coats)and of required shade on new wall surface to give an even shade, over and including priming coat of white primer after thoroughly brushing the surface free from mortar dropings and other foreign matter and including preparing surface even and sand papered smooth.	Sqm	78.37
	g) Distempering with dry distemper of approved brand and manufacture (twocoats) and of required shade on new wall surface to give an even shade,after thoroughly brushing the surface free from mortar dropings and other foreign matter and including preparing surface even and sand papered smooth.	Sqm	51.46
	h). Distempering with dry distemper of approved brand and manufacture(one coat) and of required shade on old wall surface to give an even shade,after thoroughly brushing the surface clean of all grease, dirt, loose pieces of scales including preparing surface even and sand papered smooth.	Sqm	29.14
	i). Extra over item no (f), (g) and (h) above for distempering every subsequent coat of distempering with dry distemper of approved brand and manufacture.	Sqm	29.14
	j). Extra over item no. (f) above for distempering with dry distemper on ceiling and or sloping roof.	Sqm	4.93
	k) Extra over item no. (g) above for distempering with dry distemper on ceiling and or sloping roof.	Sqm	3.23
	l) Extra over item no. (h) above for distempering with dry distemper on ceiling and or sloping roof.	Sqm	1.69
	m) Distempering two coats with oil bound distemper of approved brand and manufacture and of required shade on new wall surface to give an even shade, over and including a priming coat with distemper primer of approved brand and manufacture after thoroughly brooming the surface free from mortar dropings and other foreign matter and including preparing surface even and sand papered smooth.		
	(i).Priming coat with distemper primer	Sqm	34.56
	(ii). Top coat	Sqm	48.68
	(n) Distempering two coats with oil bound distemper of approved brand and manufacture and of required shade on new wall surface to give an even shade, over and including a priming coat with alkali resistant primer of approved brand and manufacture after thoroughly brooming the surface free from mortar dropings and other foreign matter and also including preparing the surface even and sand papered smooth.		
	(i).Priming coat with alkali resistant primer	Sqm	34.56
	(ii). Top coat	Sqm	48.68
	(o) Distempering (one coat) with oil bound distemper of approved brand and manufacture and of required shade on old wall surface to give an even shade, after thoroughly brooming the surface free from dirt, loose scales and other foreign matter and also including preparing the surface even and sand papered smooth.	Sqm	27.81

	p) Extra over item no (m), (n) and (o) above for every subsequent coat of distempering with oil bound distemper of approved brand and manufacture.	Sqm	25.88
	q) Extra over item no (m), (n) above for distempering with oil bound distemper of approved brand and manufacture on celing and sloping roof.	Sqm	4.93
	r) Extra over item no (o) above for distempering with oil bound distemper of approved brand and manufacture on celing and sloping roof.	Sqm	3.23
13.2.3	CEMENT PAINT & PRIMER		
	a) Applying one coat of cement primer of approved brand and manufacture on new wall surface after thoroughly brooming the surfaces free from mortar droppings and other foreign matter and including preparing the surface even and sand papered smooth.	Sqm	40.66
	b) Finishing wall with water proofing cement paint of approved brand and manufacture and of required shade on new wall surface (two coats) to give an even shade after thoroughly brooming the surfaces to remove all dirt and remains of loose powdered materials.	Sqm	46.75
	c) Finishing wall with water proofing cement paint of approved brand and manufacture and of required shade on old wall surface (one coats) to give an even shade after thoroughly brooming the surfaces to remove all dirt and remains of loose or softened paint.	Sqm	26.24
	d) Extra over item no. (b) and (c) above for every subsequent coat of water proofing cement paint of approved brand and manufactured.	Sqm	21.24
	e) Extra over item no (b) above for finishing with cement paint on celing and sloping roof.	Sqm	4.93
	f) Extra over item no (c) above for finishing with cement paint on celing and sloping roof.	Sqm	3.23
	(g) Extra over item no (d) above for finishing with cement paint on celing and sloping roof.	Sqm	1.69
13.3	PLASTER OF PARIS & WALL PUTTY		
13.3.1	Providing surface preparation for walls, celling etc.using average 1.50 mm thick plaster of paris including clearing, rubbing with sand paper, filling gaps, depression etc. where necessary, with chalk powder and and plaster of paris paste, making the surface even and smooth complete at all levels as specified and directed.	Sqm	60.63
13.3.2	Providing two coats of acrylic water based wall putty/ wall filler of Asian Paints/ Berger/ Dulux over concrete/ cement plaster/ brick walls and ceilings both internal and external after removing all the loosely adhering material from the wall/ ceiling surface with the help of emery stone, putty blade or wire brush, moistening the wall with the help of sufficient quantity of clear water as specified and directed by the department.	Sqm	52.10

13.3.3	Applying water miscible dry powder based 1.5mm thick Vinyl wall putty of Seacem Paint (I) Pvt. Ltd. based on white cement and reinforced with vinyl resin specially designed self curing use as plaster over sand cement plastering over brick and cement concrete wall, ceilings and other surfaces for both interior and exterior surfaces to provide a firm and durable base for subsequent painting.	Sqm	52.60
13.3.4	Providing two coats of Birla White Wall Care Putty (Water Resistant White Cement based putty for concrete/ mortar walls and ceiling both internal and external) after removing all loosely adhering material from the wall surface with the help of emery stone, putty blade or wire brush and moistening the wall with sufficient quantity of clean water as specified and directed by the department. (Total thickness of two coats is maximum 1.5mm)		
	(i) Thickness = 1.5 mm thick	Sqm	66.60
	(ii) Thickness = 1.0mm thick	Sqm	50.20
13.3.5	Providing two coats of J.K Wall Putty (White Cement based putty for concrete/ mortar walls and ceiling both internal and external) after removing all loosely adhering material from the wall surface with the help of emery stone, putty blade or wire brush and moistening the wall with sufficient quantity of clean water as specified and directed by the department. (Total thickness of two coats is maximum 1.5mm)	Sqm	50.30
13.4	WALL PAINTING		
13.4.1	Wall painting (two coats) with acrylic emulsion paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on new surface to give an even shade after thoroughly brushing the surfaces free from mortar droppings and other foreign matter and sand papered smooth.		
	(a) Acrylic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	71.17
13.4.2	Extra over item no 13.4.1 above for every subsequent coat of wall painting with acrylic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).		
	(i).Acrylic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	49.82
13.4.3	Wall painting (one coat) with acrylic emulsion paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on old wall surface to give an even shade after thoroughly brushing the surfaces free from mortar droppings and other foreign matter and sand papered smooth.		
	(a). Acrylic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	41.67
13.4.4	Extra over item no 13.4.3 above for every subsequent coat of wall painting with acrylic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).		
	(i).Acrylic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	29.17

13.4.5	Extra over item no 13.4.1 above for painting with acrylic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on ceiling and on sloping roof.		
	(i).Acrylic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	4.95
13.4.6	Extra over item no 13.4.3 above for painting with acrylic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on ceiling and on sloping roof.		
	(i).Acrylic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	3.23
13.4.7	Wall painting (two coats) with Plastic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on new surface to give an even shade after thoroughly brushing the surfaces free from mortar droppings and other foreign matter and sand papered smooth.		
	(a).Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	76.99
13.4.8	Extra over item no 13.4.7 above for every subsequent coat of wall painting with Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).		
	(i).Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	53.89
13.4.9	Wall painting (One coat) with Plastic Emulsion Paint approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on old wall surface to give an even shade after thoroughly brushing the surfaces free from mortar droppings and other foreign matter and sand papered smooth.		
	(a).Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	45.26
13.4.10	Extra over item no 13.4.9 above for every subsequent coat of wall painting with Plastic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).		
	(i).Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	31.68
13.4.11	Extra over item no 13.4.7 above for painting with Plastic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on ceiling and on sloping roof.		
	(i).Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	4.93

13.4.12	Extra over item no 13.4.9 above for painting with Plastic emulsion paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) on ceiling and on sloping roof.		
	(i).Plastic Emulsion Paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac)	Sqm	3.23
13.5	EXTERIOR PAINTS / COATINGS		
13.5.1	Applying one self-priming coat of acrylic emulsion coating of Unicem "Rakshak" brand by thinning 1ltrs of paints with 1ltrs of water over freshly plastered wall surface after thoroughly brooming the surfaces free from mortar droppings and other foreign matter and including preparing the surface even and sand papered smooth as specified and directed by the department.		
	a) Unicem Shakti	Sqm	36.81
	b) Unicem Rakshak	Sqm	52.66
	c) Asian paint/ Berger paint/ ICI Paint / J & N paint/ Nerolac	Sqm	55.29
13.5.2	Finishing wall with two coats of acrylic emulsion exterior coatings of Unicem "Rakshak" brand of required shade by thinning 1ltr. of paint with 750ml of water to give an even shade after thoroughly brooming the surfaces to remove all dirt and remains of loose powdered materials as specified and directed by the department.		
	a) Unicem Shakti	Sqm	46.78
	b) Unicem Rakshak	Sqm	85.77
	c) Asian paint/ Berger paint/ ICI Paint / J & N paint/ Nerolac	Sqm	90.06
13.5.3	Extra over item no.13.5.2 above for every subsequent coat of acrylic emulsion exterior coatings of approved brand of required shade.	Sqm	
	a) Unicem Shakti	Sqm	27.86
	b) Unicem Rakshak	Sqm	42.38
	c) Asian paint/ Berger paint/ ICI Paint / J & N paint/ Nerolac	Sqm	46.62
13.5.4	Extra over item no.13.5.2 above for applying/ finishing on sloping roof etc.	Sqm	
	a) Unicem paint	Sqm	6.89
	b) Asian paint/ Berger paint/ ICI Paint / J & N paint/ Nerolac	Sqm	7.58
13.6	PAINTING ON WOOD AND WOOD WOOD BASED SURFACES		
13.6.1	Removing old paint or polish from wood and wood base surfaces with paint remover of approved brand and manufacture and making the surface even.	Sqm	49.30
13.6.2	Removing old paint or polish from wood and wood base surfaces with caustic soda solution and making the surface even	Sqm	36.40
13.6.3	Applying priming coat over new wood and wood based surfaces over 100mm in girth/width after and including preparing the surface by thoroughly cleaning oil, grease, dirt and other foreign matter, sand papering and knotting.		
	(a) With ready mixed paint, Wood primer (pink)	Sqm	36.90
	(b) With ready mixed paint, wood primer (white).	Sqm	36.90
	(c) With ready mixed paint, Alluminium primer	Sqm	30.16

13.6.4	Applying priming coat over new wood based surfaces up to 100mm in girth/width after and including preparing the surface by thoroughly cleaning oil, grease, dirt and other foreign matter, sand papering and knotting.		
	(a) With ready mixed paint, wood primer (pink)	Rm	3.95
	(b) With ready mixed paint, wood primer (white).	Rm	3.95
	(c) With ready mixed paint, Alluminium primer.	Rm	3.23
13.6.5	Painting two coats (excluding priming coat) on new wood and wood based surface with enamel paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) to give an even shade including cleaning the surfaces of all dirt, dust and other foreign matter sand papering and stopping.		
	(i) Surfaces over 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	61.95
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	73.00
	(ii) Surfaces up to 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	7.00
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	8.25
13.6.6	Painting one coat (excluding priming coat) on old wood and wood based surfaces with enamel paint to give an even shade including cleaning the surface of all dirt, dust and other foreign matter sand papering and stopping.		
	(i) Surfaces over 100mm in width or girth.		
	a). General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	41.26
	b). High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	47.78
	(ii) Surfaces up to 100mm in width or girth.		
	a). General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	4.42
	b). High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	5.13
13.6.7	Extra over item no13.6.5and 13.6..6 for every subsequent coat of paint.		
	(i) Surfaces over 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	36.90
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	43.47
	(ii) Surfaces up to 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	4.17
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	4.92
13.7	PAINTING ON STEEL AND OTHER METAL SURFACE		

13.7.1	Synthetic enamel paint, aluminium paint / bituminous paint on steel and other metal surfaces. i) Removing old paint from steel and other metal surface and making the surface even.		
	a) With hand scraping	Sqm	42.07
	b) With paint remover	Sqm	44.82
13.7.2	Applying primary coat over new steel and other metal surface over 100m in width or girth after preparing the surface by thoroughly cleaning oil,grease, dirt and other foreign matter and scoured with wire brushes, fine steels, wood scrapers and sand paper.		
	a) With ready mixed "red-lead/ red oxide" primer.	Sqm	33.96
13.7.3	Applying primary coat over new steel and other metal surface upto 100m in width or girth after preparing the surface by thoroughly cleaning oil,grease, dirt and other foreign matter and scoured with wire brushes, fine steels, wood scrapers and sand paper.		
	a) With ready mixed "red-lead/ red oxide" primer.	Rm	3.80
13.7.4	Painting two coats (excluding priming coat) on new steel and other metal surface with enamel paint of approved brand and manufacture(Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) to give an even shade including cleaning the surface of all dirt, dust and other foreign matter.		
	(i) Surfaces over 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	61.95
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	73.00
	(ii) Surfaces up to 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	7.00
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	8.25
13.7.5	Painting one coat (excluding priming coat) on previously painted steel and other metal surface with enamel paint of approved brand and manufacture(Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) to give an even shade including cleaning the surface of all dirt, dust and other foreign matter.		
	(i) Surfaces over 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	41.26
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	47.89
	(ii) Surfaces up to 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	4.66
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	5.41
13.7.6	Extra over item no.13.7.4 and 13.7.5 for every subsequent coat of paint.		
	(i).Surfaces over 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	36.90

	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Sqm	43.11
	(ii) Surfaces up to 100mm in width or girth.		
	a) General purpose (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	4.17
	b) High gloss (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac).	Rm	4.92
13.7.7	Painting two coats (excluding priming coat) on new steel and other metal surface with alluminium paint of approved brand and manufacture(Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) and brushing to give an even shade including cleaning the surface of all dirt, dust and other foreign matter.		
	(a) Surfaces over 100mm in width or girth.	Sqm	60.44
	(b) Surfaces up to 100mm in width or girth.	Rm	6.83
13.7.8	Painting one coats (excluding priming coat) on new steel and other metal surface with alluminium paint of approved brand and manufacture(Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) and brushing to give an even shade including cleaning the surface of all dirt, dust and other foreign matter.		
	(a) Surfaces over 100mm in width or girth.	Sqm	38.84
	(b) Surfaces up to 100mm in width or girth.	Rm	4.39
13.7.9	Painting three coats (excluding priming coat) on new steel and other metal surface with ready mixed bituminous paint of approved brand and manufacture (Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) after applying black anticorrosive paint to give an even shade including cleaning the surface of all dirt, dust and other foreign matter.	Sqm	47.76
13.7.10	Painting one coats (excluding priming coat) on new steel and other metal surface with ready mixed bituminous paint of approved brand and manufacture(Asian paint/ Berger paint/ ICI paint/ J & N paint/ Nerolac) after applying black anticorrosive paint to give an even shade including cleaning the surface of all dirt, dust and other foreign matter.	Sqm	24.41
13.8	PAINTING ON ROOF SURFACES		
13.8.1	Applying mordent solution of approved brand and manufacture on new C.I sheet for preparation of surfaces for painting @5.00 litre/100 sq. metre of surface and rubbing the solution with brush including thoroughly cleaning the surface of oil, grease, dirt and other foreign matter.	Sqm	26.51
13.8.2	Painting two coats on new G.C.I sheets with ready mixed anticorrosive paint of approved brand and shade suitable for painting over G.C.I sheets to give an even shade including the cleaning the rust patches with emery paper, dust ,dirt, grease and other foreign matter with water complete.	Sqm	55.87
13.8.3	Painting one coats on old G.C.I sheets with ready mixed anticorrosive paint of approved brand and shade suitable for painting over G.C.I sheets to give an even shade including the cleaning the rust patches with emery paper, dust ,dirt, grease and other foreign matter with water complete.	Sqm	42.27

13.8.4	Extra over item no.13.8.3 for every subsequent coat of paint.		19.12
	SPECIAL WATHER PROOF COLOURING		
13.9.1	Finishing old / new wall with water proofing weather coat smooth anti-fungal exterior painting of approved make of Berger paint of required shade after cleaning and clearing the surface etc. including scaffolding complete as directed at all levels (two coats)	Sqm	127.80
13.9.2	Finishing old/ new wall with water proofing weather coat smooth anti-fungal exterior painting of Apex brand of Asian Paints and of required shade after cleaning and clearing the surface etc. including scaffolding complete as directed at all levels (two coats)	Sqm	127.58
13.9.3	Finishing old/ new wall with water proofing weather coat smooth anti-fungal premium textured exterior finish of Sandex Matt brand of Snowcem India Ltd. of required shade after clering and clearing the surfaces etc.including scaffolding complete as directed at all levels (two coats).	Sqm	142.93
13.9.4	Finishing old/ new wall with water proofing weather coat smooth anti-fungal premium acrylic smooth exterior finish of Snowcryl XT brand of Snowcem India Ltd. of required shade after clering and clearing the surfaces etc. including scaffolding complete as directed at all levels (two coats).	Sqm	129.21
13.9.5	Finishing wall with water proofing weather coat smooth anti-fungal exterior painting of Ace brand of Asian Paints and of required shade on old/new wall surface after cleaning and clearing the surface etc. including scaffolding complete as directed at all levels (two coats)	Sqm	101.78
13.9.6	Finishing old/new wall with water proofing weather coat smooth anti-fungal exterior painting of approved make of Dulux (ICI) paint of required shade after cleaning and clearing the surface etc. including scaffolding complete as directed at all levels (two coats)	Sqm	127.80
13.9.7	Supplying and applying interior/exterior walls/ floors/ roofs one coat of Epoxy Primer of Jenson & Nicholson/ Nerolac brand as priming coat after cleaning and clearing the surfaces as specified and directed and then applying two coats of Epoxy paints of Jenson & Nicholson/ Nerolac brand of required shade on walls/ floors/ roof surfaces as specified and directed complete including scaffolding at all levels.	Sqm	224.27
13.9.8	Finishing old/ new wall with water proofing weather coat smooth anti-fungal exterior painting of Nerolac Paints Limited of required shade on old/ new wall surface to give an even shade after cleaning and clearing the surface etc. including scaffolding complete as directed at all levels (two coats).	Sqm	127.80
13.10.1	Providing and applying Spectrum aggregate textured wall coating in the branded name of SPECTRUM with its all tools, plants and machineries and scaffolding etc. as per manufacturers guideline and specification of the product. a) Spectrum Antonio		
	a) Spectrum Antonio	Sqm	652.52
	b) Spectrum Granito natural	Sqm	769.21
	c) Spectrum Naturaplast	Sqm	800.27

d) Spectrum Permastone	Sqm	799.10
e) Spectrum Granula	Sqm	707.21
f) Spectrum Perlina (Regular)	Sqm	748.60
g) Spectrum resana	Sqm	254.56
h) Spectrum Star Spray	Sqm	399.56
i) Spectrum Quartz King	Sqm	204.11
j) Spectrum Spray Coat	Sqm	637.99
k) Spectrum Random Stat	Sqm	719.14
l) Spectrum Super fine	Sqm	306.16
m) Spectrum Permasan	Sqm	716.48
n) Spectrum Ornaplast	Sqm	642.14
o) Spectrum Graniplast	Sqm	713.30
p) Spectrum Classic	Sqm	212.58
q) Spectrum Perlina (fine)	Sqm	593.43
r) Spectrum Rustik (fine)	Sqm	593.43
s) Spectrum Rustik (Regular)	Sqm	787.39

CH-14: VARNISHING, POLISHING, OILING

Item No.	Description	Unit	Rate
14.1	VARNISHING		
14.1.1	Varnishing two coats (excluding priming coat) on new wood and wood based surfaces under coating with flatting varnish and finishing coat with copal varnish or spar varnish to give an even surface including cleaning the surface of all dirt,dust and filling the knots,crack etc with specially made wood putty and sand papering so as to produce a smooth,dry and matt surface.		
	(i) Surfaces over 100mm in width or girth.	Sqm	83.47
	(ii) Surfaces upto 100mm in width or girth.	Rm	7.71
14.1.2	Varnishing one coat (excluding priming coat) on previously varnished wood and wood based surfaces with copal/ spar varnish to give an even surface including cleaning the surface of all dirt, dust and sand papering so as to produce a papering so as to produce a smooth,dry and matt surface.		
	(i) Surfaces over 100mm in width or girth.	Sqm	41.96
	(ii) Surfaces upto 100mm in width or girth.	Sqm	4.08
14.2	POLISHING		
14.2.1	Polishing with french sprit polishing on new wood and wood based surfaces to give an even surface including cleaning the surface of all dirt, dust sand papered smooth and including a coat of wood filter	Sqm	115.21
14.2.2	Polishing with french sprit polishing on previously polished wood and wood based surfaces to give an even surface including cleaning the surface of all dirt,dust, sand papered smooth and including a coat of wood filter	Sqm	62.39
14.2.3	Applying wax-polish on new wood and wood based surfaces with bees wax polish in prop. (2 bees wax: 1.5 linseed oil:1 terpentine oil:0.5 varnish by weight to give an even surface including cleaning the surface of all dirt,dust, sand papered smooth .	Sqm	62.88

14.2.4	Applying wax-polish on previously wax polish wood and wood based surfaces with bees wax polish in prop. (2 bees wax:1.5 linseed oil:1 turpentine oil: 0.5 varnish by weight) to give an even surface including cleaning the surface of all dirt,dust, sand papered smooth .	Sqm	35.37
14.2.5	Applying wax-polish on new concrete/masonry floor with wax polish of approved brand and manufacture to give an even surface including cleaning the surface of all dirt,dust and other foreign matter.	Sqm	20.12
14.2.6	Applying wax-polish on previously polished concrete/ masonry floor with wax polish of approved brand and manufacture to give an even surface including cleaning the surface of all dirt,dust and other foreign matter .	Sqm	19.69
14.3	OILING		
14.3.1	Applying two coats of oil with raw linseed oil on new wood and wood based surface including cleaning the surface of all dirt,dust and sand papered so as to produce a smooth, dry and neat surface.	Sqm	25.46
14.3.2	Applying one coat of oil with raw linseed oil on previously oiled wood and wood based surfaces to give an even surface including cleaning the surface of all dirt,dust and sand papering so as to produce a smooth, dry and neat surface.	Sqm	15.90
14.3.3	Applying two coats of wood preservative on wood and wood based surfaces including preparing the surface by thoroughly cleaning oil, grease dirt and other foreign matter, sand papering and knotting.		
	(a) With kericide oil	Sqm	24.59
	(b) With Creosote oil	Sqm	25.35
14.3.4	Applying one coat of wood preservative on previously created wood and wood based surfaces, including preparing the surface by thoroughly cleaning oil, grease dirt and other foreign matter, sand papering and knotting.		
	(a) With kericide oil	Sqm	19.43
	(b) With Creosote oil	Sqm	20.04
14.3.5	Painting two coats (excluding priming coat) on new wood and wood based surface with coal tar, new wood and wood based surface with coal tar, black paint to give an even surface including cleaning all dirt, dust and other foreign matter.	Sqm	44.15
14.3.6	Painting one coat (excluding priming coat) on previously painted wood and wood based surfaces with coaltar,black paint to give an even surface including cleaning all dirt, dust and other foreign matter.	Sqm	24.79
14.3.7	Applying two coats of wood preservative oil,colourless (Burger Products) on new wood and wood based surfaces, after including preparing the surface by thoroughly cleaning oil,grease,dirt and other foreign matter, sand papering and knotting.	Sqm	105.17

14.3.8	Applying one coats of wood preservative oil,colourless (Burger Products) on new wood and wood based surfaces, after including perparing wood based surfaces, after including perparing the surface by throughly cleaning oil,grease,dirt and other foreign matter, sand papering and knotting.	Sqm	60.38
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CH-15: DOOR AND WINDOW FITTINGS

Item No.	Description	Unit	Rate
15.1	ELECTRO COLOURED IRON FITTINGS		
15.1.1	Supplying, fitting, fixing electrocoloured iron fittings of approved make, resonably smooth, free from sharp edges and corners, flaws and other defects and with counter sunk holes for wood screws including necessary screws etc. complete.		
	a) Sliding door bolts		
	i) 300mm x 16mm	Each	169.88
	ii) 250mm x 16mm	Each	152.04
	b) Tower bolts		
	i) 300mm x 10mm	Each	164.68
	ii) 250mm x 10mm	Each	139.71
	iii) 200mm x 10mm	Each	121.86
	iv) 150mm x 10mm	Each	92.12
	v) 100mm x 10mm	Each	64.78
	c) Cast iron stove door handles		
	i) 75mm	Each	29.35
	ii) 100mm	Each	35.30
	iii) 150mm	Each	41.24
	d) MS door handles		
	i) 75mm	Each	32.91
	ii) 100mm	Each	38.86
	iii) 150mm	Each	43.62
	e) MS gate shutter hook & eye		
	i) 75mm	Each	8.53
	ii) 100mm	Each	10.32
	iii) 150mm	Each	13.29
	iv) 250mm	Each	21.02
	v) 300mm	Each	37.67
	f) MS wire hasp and staples		
	i) 90mm / 75mm	Each	70.85
	ii) 115mm	Each	76.80
	iii) 150mm	Each	100.58
	g) MS butt hinge		
	I) Heavy type		
	i) 50mm x 40mm x 2.5mm	Pair	63.47
	ii) 75mm x 60mm x 3.15	Pair	82.51
	II) Medium type		
	i) 50mm x 37mm x 1.5mm	Pair	44.45
	ii) 75mm x 47mm x 1.60mm	Pair	58.72
	iii) 100mm x 58mm x 1.90mm	Pair	76.56
	III) Light type		
	i) 50mm x 30mm x 0.90mm	Pair	36.12

	ii) 75mm x 40mm x 1.12mm	Pair	44.45
	iii) 100mm x 50mm x 1.25mm	Pair	61.10
	h) MS parliamentary hinge		
	i) 75mm (100mm x20mm x2.24mm)	Pair	183.61
	ii) 100mm (125mm x 27mm x 2.80mm)	Pair	232.37
	ii) 125mm (125mm x 27mm x 2.80mm)	Pair	296.61
	i) MS double acting spring hinge		
	a) 100mm	Pair	332.29
	b) 125mm	Pair	367.97
	c) 150mm	Pair	504.76
	j) MS continuous hinge		
	35mm x 1.00m	RM	206.33
	k) MS door latches		
	i) 300mm x 20mm x 6mm	Each	128.09
	ii) 250mm x 10mm x 6mm	Each	107.87
	l) MS mortice night latches	Each	1064.67
	m) MS rim latches		
	i) 100mm	Each	740.82
	ii) 125mm	Each	788.40
	iii) 150mm	Each	835.98
15.2	ALUMINIUM FITTINGS		
15.2.1	Supplying, fitting, fixing anodised aluminium fittings of approved make, reasonably smooth, free from sharp edges and corners, flaws and other defects and with counter sunk holes for screws including necessary aluminium screws etc. complete. (anodised to bright natural matt & satin finished)		
	a) Sliding door bolts		
	i) 300mm x 16mm	Each	272.17
	ii) 250mm x 16mm	Each	260.27
	b) Tower bolts		
	i) 300mm x 12mm	Each	140.90
	ii) 250mm x 12mm	Each	121.86
	iii) 200mm x 12mm	Each	104.02
	iv) 150mm x 10mm	Each	88.56
	v) 100mm x 10mm	Each	68.34
	c) Door handles		
	i) 75mm	Each	36.49
	ii) 100mm	Each	38.86
	iii) 150mm	Each	42.43
	d) Gate shutter hook & eye		
	i) 100mm	Each	22.22
	ii) 125mm	Each	24.59
	iii) 150mm	Each	26.97
	iv) 250mm	Each	44.81
	v) 300mm	Each	60.27
	e) Butt hinge		
	i) 50mm x 35mm x 3.2mm	Pair	65.86
	ii) 75mm x 45mm x 3.20mm	Pair	77.75
	iii) 75mm x 65mm x 4mm	Pair	87.26
	iv) 100mm x 63mm x 3.2mm	Pair	99.15
	v) 100mm x 75mm x 4mm	Pair	107.48
	f) Parliamentary hinge		

	i) 75mm (100mm x31mm x6.5mm)	Pair	161.01
	ii) 100mm (50mm x 22.5mm x 6.5mm)	Pair	195.51
	g) Continuous piano hinge		
	40mm x .8m	RM	182.54
	h) Mortice night latches		1307.99
	i) Rim latches		
	i) 100mm	Each	871.66
	ii) 125mm	Each	954.92
	iii) 150mm	Each	1050.08
	j) Fan light pivot	Each	56.85
	k) Fan light catcher	Each	82.23
	l) Hydraulic door closure	Each	1077.24
	m) Mortice lock vertical type		
	i) 65mm(Godrej With handle(2 lever)	set	1076.05
	ii) 75mm(Godrej with handle(4 lever)	set	1288.96
	iii) 100mm(Godrej with handle (6 lever)	set	1652.92
	iv) Aldrop with In built Lock-6 lever kadi tala(Godrej Make)	set	1026.08
	v) pin cylinder Mortise Lock Oliver/Matiz/Victoria -Godrej make		2213.14
	n) Drawer locks		
	i) 40mm	Each	184.49
	ii) 50mm	Each	204.72
	iii) 65mm	Each	246.34
	iv) 75mm	Each	252.29
	o) Cupboard locks		
	i) 40mm	Each	177.36
	ii) 50mm	Each	192.82
	iii) 65mm	Each	210.66
	iv) 75mm	Each	246.34
	p) Finger plate	Each	324.88
	q) Kicking plate	Each	327.79
	r) Curtain rod		
	i) 20mm x 2mm	RM	117.91
	ii) 25mm x 2mm	RM	147.65
	s) Wardrobe knob	Each	41.26
	t) Door stopper with rubber stopper	Each	85.81
15.3	BRASS FITTINGS		
15.3.1	Supplying, fitting, fixing brass fittings of approved make, resonably smooth, free from sharp edges and corners, flaws and other defects and with counter sunk holes for screws including necessary c.p. brass screws etc. complete.(bright natural matt & satin finished).		
	a) Sliding door bolts		
	i) 300mm x 16mm	Each	824.08
	ii) 250mm x 16mm	Each	764.59
	b) Tower bolts		
	i) 300mm x 10mm	Each	376.41
	ii) 250mm x 10mm	Each	271.74
	iii) 200mm x 10mm	Each	236.04
	iv) 150mm x 10mm	Each	212.25
	v) 100mm x 10mm	Each	146.84
	c) Door handles		
	i) 75mm	Each	59.08

	ii) 100mm	Each	65.03
	iii) 150mm	Each	82.87
	D) Gate shutter hook & eye		
	i) 100mm	Each	44.81
	ii) 150mm	Each	55.51
	iii) 250mm	Each	62.65
	iv) 300mm	Each	72.16
	e) Butt hinge		
	i) 125mm x 50mm x 4mm	Pair	448.02
	iii) 100mm x 50mm x 4mm	Pair	329.08
	iii) 100mm x 40mm x 2.5mm	Pair	293.40
	iv) 75mm x 50mm x 4mm	Pair	274.37
	v) 75mm x 40mm x 2.5mm	Pair	195.87
	vi) 50mm x 40mm x 2mm	Pair	82.87
	f) Parliamentary hinge		
	i) 75mm (75mm x 20mm x 3.2mm)	Pair	472.64
	ii) 100mm (100mm x 27mm x 5mm)	Pair	660.57
	ii) 150mm (100mm x 27mm x 5mm)	Pair	815.19
	g) Hasp and staple		
	i) 90mm	Pair	199.30
	ii) 115mm	Pair	211.19
	iii) 150mm	Pair	246.88
	h) Spring hinge		
	i) 100mm	Pair	951.34
	ii) 125mm	Pair	1040.55
	iii) 150mm	Pair	1165.44
	i) Flush bolts		
	i) 250mm x 20mm x 4mm	Each	369.54
	ii) 150mm x 20mm x 4mm	Each	226.81
	iii) 100mm x 20mm x 4mm	Each	161.39
	j) Door latches		
	i) 300mm x 16mm x 5mm	Each	248.22
	ii) 250mm x 16mm x 5mm	Each	236.33
	k) Night latches		
	i) Mortice night latches (Godrej make)	Each	980.91
	ii) Rim latches (Godrej make)	Each	1254.65
	l) Fan light pivot	Each	53.70
	m) Fan light catcher	Each	103.65
	n) Mortice lock vertical type		
	i) 65mm	set	791.76
	ii) 75mm	set	1362.69
	o) Mortice cylindrical type (Godrej make)-polished Brass	Each	1323.44
	p) Drawer locks		
	i) 40mm	Each	285.07
	ii) 50mm	Each	332.64
	iii) 65mm	Each	386.17
	iv) 75mm	Each	439.69
	q) Cupboard locks		
	i) 40mm	Each	232.07
	ii) 50mm	Each	279.64
	iii) 65mm	Each	333.17
	iv) 75mm	Each	386.69
	r) Finger plate	Each	446.21

	s) Kicking plate	Each	256.44
	t) Curtain rod		
	i) 20mm x 2mm	RM	239.24
	ii) 25mm x 2mm	RM	310.61
	u) Wardrobe knob	Each	72.18
	v) Door stopper with rubber stopper	Each	176.85
	w) Ball catcher	Each	50.76
	x) Magnet catcher	Each	50.76
	y) Peeping plate	Each	167.18
15.4.1	Supplying fitting, fixing Stainless steel fittings of approved make reasonably smooth, free from sharp edges and corners, flaws and other defects and counter sunk holes for screws including necessary c.p. brass screws etc. complete (bright natural matt & satin finished)		
	a) Sliding door bolts		
	i) 300mm x 16mm	Each	354.71
	ii) 250mm x 16mm	Each	324.97
	iii) 200mm x 16mm	Each	295.24
	b) Tower bolts		
	i) 300mm x 10mm	Each	191.81
	ii) 250mm x 10mm	Each	163.26
	iii) 200mm x 10mm	Each	134.70
	iv) 150mm x 10mm	Each	102.36
	v) 100mm x 10mm	Each	73.82
	c) Door Handle		
	i) 75mm	Each	60.04
	ii) 100mm	Each	77.88
	iii) 150mm	Each	113.56
	d) Butt hinge		
	i) 125mm x 50mm x 2.5mm	Each	107.61
	ii) 100mm x 50mm x 2.5mm	Each	89.77
	iii) 75mm x 50mm x 2.5mm	Each	68.13
	iv) 75mm x 40mm x 1.8mm	Each	46.72
	v) 50mm x 40mm x 1.8mm	Each	32.16
15.4	DOOR CLOSER & FLOOR SPRING		
15.4.1	Providing,fitting and fixing Door closer of various type UL listed and fire rated good quality depending on the door specification.Types like single speed,Double Speed,Heavy Duty etc are available and as directed & specified by the department.		
	A) Godrej		
	a) Single Speed Door Closer(size166.5x 48mmx 67mm)-Model C071	Each	1163.92
	b) Double Speed Door Closer(size 140mm x 39mm x 60mm)-Model Number C 752	Each	1447.02
	c) Heavy Duty Door Closer(size 180 mm x 65mm x 64mm)	Each	1938.25
	d) Sleek Door Closer(215mm x 51mm x 45mm)	Each	1700.37
	e) Concealed Door Closer (230mm x 76.5mm x 30.5mm)	Each	3990.03
	B) Fire Rated Door Closer-EN rated Godrej Make		
	a) Single speed Door Closer _ Model LKYGDC680	Each	3014.69
	b) Double speed Fire Rated Door Closer Model LKYGDC834	Each	4304.05
	c) Fire rated Track Arm Door Closer Model LKYGDC610	Each	4910.65

15.4.2	Providing,fitting and fixing Floor Spring of various weight & type UL listed and fire rated good quality depending on the door specification.Types like single speed,Double Speed,Heavy Duty etc are available and as directed & specified by the department		
	A) Architectural Range		
	a) Compact Floor spring B710 (size 274mm x 97mm x 40mm)	Each	4709.90
	b) Heavy duty Floor Spring D100 (size 308mm x 108mm x40.5mm)	Each	5604.36
	B) Godrej		
	a) Fire rated floor spring B 803(size 307mm x 109mm x 40mm)-80 kg door/ B 804(size 307mm x 109mm x 40mm)-100 kg door (Godrej make)	Each	7829.80
15.5	DRAPERY ROD		
15.5.1	Supplying, fitting and fixing drapery rods along with all accessories in doors, windows and other openings complete as specified and directed by the departmentb at all levels.		
	(A) MAC Drapery rod		
	i) Gold line and silver line	RM	1151.17
	ii) Starline and pearline	RM	1033.42
	iii) Club line	RM	1745.88
	iv) Print line	RM	1508.00
	v) Wood line	RM	1543.68
	vi) Silk line	RM	1745.88
	vii) Nature line	RM	2221.66
	viii) Studio line	RM	1389.06
	ix) Soft line	RM	1022.70
	x) Fine line	RM	1056.01
	xi) Speer line	RM	794.34
	xii) Sleek line	RM	413.71
	(B) VISTA Drapery rod		
	i) Vista sherwood	RM	1032.23
	ii) Florence III vista	RM	1435.44
	iii) Flerona (Rotiron)	RM	1091.69
	iv) Flored colour gold (pulling system)	RM	1239.18
	v) Flored color color (Pulling system)	RM	1239.18
	vi) Soffio M-IV	RM	913.29
	vii) Soffio M-II	RM	675.39
	(C) PVC coated Drapery rod	RM	592.13
	(D) Mettalic Drapery rod	RM	1151.17
	(D) Mettalic Drapery rod		
	(E) Globus Collection -23mm		
	i) Bellini	RM	2554.70
	ii) Renoir Pyramid	RM	2371.53
	iii) Renoir Tulip	RM	2601.09
	iv) Renoir Orchid	RM	3515.76
	v) Lorenzo	RM	2417.91
	vi) Monet Oyestar	RM	2784.26
	vii) Monet Tulip	RM	3515.76
	(F) Globus Collection -19mm		

	i) Angelo	RM	2096.76
	ii) Salvador	RM	2096.76
	iii) Van Gough	RM	1778.00
	iv) Picaso	RM	1594.82
15.6	BLIND		
15.6.1	Supplying, fitting and fixing MAC blinds horizontal and vertical complete as specified and directed by the department at all levels.		
	(A) Vertical blinds		
	(a) Regular		
	i) 100 mm	Sqm	1861.25
	ii) 89 mm	Sqm	1914.78
	iii) 75 mm	Sqm	2009.94
	iv) 50 mm	Sqm	2150.29
	(b) Blackout		
	i) 100 mm	Sqm	1914.78
	ii) 89 mm	Sqm	2043.25
	iii) 75 mm	Sqm	2151.48
	iv) 50 mm	Sqm	2340.61
	(B) Horizontal blinds		
	i) 15mm micro blind	Sqm	1745.88
	ii) 25 mm slim line blind	Sqm	1805.35
	iii) 25 mm perforated blind	Sqm	2090.82
	iv) 25 mm printed blind	Sqm	2697.44
	v) 25 mm dust guard blind	Sqm	4541.06
	vi) 25 mm pearl finish blind	Sqm	5373.66
	vii) 25 mm special effect blind	Sqm	2865.14
	viii) 25 mm timber blind	Sqm	5549.69
15.6.2	Extra for providing 175 mm x 175 mm size Vision panel in door shutters including providing and fixing 4 mm thick clear glass and teak wood beading 12mm x 12mm , all along the glass edges on both sides including necessary screws etc. and painting / polishing of wooden beading complete as directed.		
	i) Square	Sqm	330.46
	ii) Round	Sqm	372.08

CH-16: GLASS FITTINGS

Item No.	Description	Unit	Rate
16	GLASS FITTING		
16.1	Providing and fixing clear sheet glass bedded in putty and fixed with/without wooden bidding including necessary cutting and glass to be required sized(payment for wooden bead shall be made separately)		
	(1) Area of glass panes not exceeding 0.15m2		
	(c) 3.00mm thick.	Sqm	486.11
	(2) Area of glass panes exceeding 0.15m2 but not exceeding 0.52m2.		
	(a) 3.00mm thick.	Sqm	447.30
	(b) 4.00mm thick.	Sqm	562.01
	(c) 5.00mm thick.	Sqm	642.09
	(3) Area of glass panes exceeding 0.52m2.		

	(a) 5.00mm thick.	Sqm	691.60
16.2	Providing and fixing plain frosted glass of ordinary glazing quality beded in putty and fixed with/without wooden bidding including necessary cutting and glass to be required sized(payment for woodenbead shall be made separately)		
	(1) Area of glass panes not exceeding 0.15m2		
	(a) 3.00mm thick.	Sqm	600.17
	(b) 3.50mm thick.	Sqm	676.25
	(2) Area of glass panes exceeding 0.15m2 but not exceeding 0.52m2.		
	(a) 3.50mm thick.	Sqm	726.19
	(b) 4.00mm thick.	Sqm	671.34
	(c) 5.00mm thick.	Sqm	811.36
	(3) Area of glass panes exceeding 0.52m2.		
	(a) 5.00mm thick.	Sqm	858.64
16.3	Providing and fixing pin-headed glass 3mm thick beded in putty and fixed with/without wooden bidding including necessary cutting and glass to be required sized(payment for wooden bead shall be made separately)		
	(i) Area of glass panes not exceeding 0.15m2	Sqm	428.49
	(ii) Area of glass panes exceeding 0.15m2 but not exceeding 0.52m2.	Sqm	444.67
16.4	Providing and fixing plain with special metal, sachputty and glazing clip in steel window / doors / cleastory windows including necessary cutting to required sized for all floors.		
	(1) Clear sheet glass panes		
	(iii) 3.00mm thick.	Sqm	433.90
	(iv) 4.00mm thick.	Sqm	554.31
	(v) 5.00mm thick.	Sqm	632.84
	(b) Frosted Glass panes.		
	(i) 3.00mm thick.	Sqm	433.90
	(ii) 4.00mm thick.	Sqm	554.31

CHAPTER 17: FENCING & GATES

Item no	Description	Unit	Rate
17.1	Spilt jati bamboo Zenana fencing up to 1800 mm high with well matured Bhaluka bamboo post1500 mm to 1800mm apart driven at least 600 mm under ground and three lines of double bamboo kami tied with wire or cane (spilt bamboo woven in 4 nos of bamboo kamies)	Sqm	188.50
17.2	Jati bamboo compound Zeora fencing 1050mm to 1200mm high kamies of sizes 30mmx6mm placed not more than 100mm apart and driven at least 100 mm under groundincluding Bhaluka bamboo post 1500 to 1800mm apart driven at least 450mm under ground and tied with three lines of double longitudinal bamboo kamies and half bamboo on top (Lattice pattern).		
	a) Single	Sqm	115.24
	b) Double	Sqm	154.92
17.3	Bhaluka bamboo dressed post 75mm to 100mm dia driven at least 600mm under ground and tied with wire or cane (in zenana fencing)	Each	85.11

17.4	Bhaluka bamboo dressed post 75mm to 100mm dia driven at least 450mm under ground and tied with wire or cane (in zeora fencing)	Each	56.74
17.5	Providing barbed wire on the existing barbed wire fencing with seven strands of barbed wire 12 S.W. G.x 2pliesx 4 points at 75mm apart including supplying, fitting and fixing with galvanised staples 38mm x 38mm x 3mm sizes with existing posts of the fencing, including removing rotten barbed wire as directed, complete with strengthening of post and tightening of the old barbed wire and joinining old serviceable pieces on remaining lines as and where necessary and as directed.	M	28.08
17.6	Supplying , fitting and fixing barbed wire fencing with seven strands of barbed wire 12 S.W.G.x 2pliesx 4 points at 75mm apart including supplying fitting and fixing sawn sal wood posts of 100mmx100mm size struts 1800mm long at 1500mm apart driven at least 600mm under ground with providing 100mmx75mm size struts 1800mm long in corner, straining and end post as necessary including coaltaring two coats in posts and struts and supplying necessary nails, staples etc.complete as directed and specified (the distribution of the barbed wire should be placed at equal intervals for the intermediate except bottom and top which is to be placed at 100mm and 230mm clearance respectively).	M	1096.24
17.7	Supplying , fitting and fixing galvanised barbed wire fencing with seven strands of barbed wire 12 S. W. G .x 2plies x 4 points at 75mm apart with angle iron intermediate fencing posts not more than 1800mm apart and straining and end post etc. as necessary to be fitted with straining eye bolts and staples etc.complete as directed, specification of the angle iron post as given below.	M	630.70
	a) Intermediate post 35mm x35mm x5mm x1800mm long bottom torched and duly slotted. b) Straining post 65mm x 65mm x 6mm x 1820mm long with 2 stays of 35mmx35mmx1520mm long and 3 base plates 200mm x 200mm x 5mm, duly slotted to take eye bolts etc. c) Corner posts 65mm x 65mm x 6mm x1820mm long with 2 stays of 35mm x 35mm x 5mm x 1 670 mm and 3 base plates 200mm x 200mm x 5mm,duly slotted to take eye bolts etc. d) Straining or end posts 65mmx 65mm x 6mmx 1820mm long with 1 stays of 35mm x 35mm x 5mmx 1820 mm long with base plates 200mmx 200mm x 5mm, duly slotted to take eye bolts etc.		
	(All the angle iron posts to be embeded 600mm in C.C.pillars of size 300mmx300mmx600mm in prop.1: 3: 6 and one coat of anticorrosive ready mixed painting to all iron works) including excavation of foundation trenches for C.C. blocks and refilling and removal of surplus soil after completion of works.		

17.8	Supplying, fitting and fixing goat proof wire fencing 1070mm high with horizontal wire spacing at 130 mm, 130mm, 150mm, 150mm, 180mm and 180mm from bottom onwards and vertical wires at 100mm apart top and bottom wires should be of 8 S.W G. and intermediate and vertical wire should be of 10S.W.G.with angle iron posts specified in item no 17.7 above including supplying, fitting and fixing with necessary straining eye bolts and staples etc. and painting one coat to all iron posts etc. with anti-corrosive black paint etc. complete as directed and specified(intermediate and straining post should be placed not more than 4000mm and not less than 3000 mm apart and all posts embedded in C.C. pillars of size 300mm x 300mm x 600mm in prop 1:3:6).	M	530.42
17.9	Supplying, fitting and fixing goat proof wire fencing 1200mm high with horizontal wire spacing at 100mm, 100mm, 100mm, 100mm, 100mm, 100mm,150mm, 150mm, 150mm and 150mm from bottom onwards and vertical wires at 130mm apart, first 3 horizontal wires from top barbed wire apart, first 3 horizontal wires from top barbed wire 12 S.W G.x 2 plies x 4 points x 100 mm apart bottom wire 10 S.W.G. fitted with angle iron posts specified in item no.17.7 above including supplying, fitting and fixing with necessary hooks, bolts and staples etc. and painting one coat to all iron posts etc. with anti-corrosive black paint etc. complete as directed and specified(intermediate and straining post should be placed not more than 4000 mm and not less than 3000 mm apart and all posts embedded in C.C. pillars of size 300mm x 300mm x 600mm in prop 1:3:6).	M	530.42
17.10	Supplying, fitting and fixing chain link fencing of 10 S.W.G, G.I. Wire 50mm x50mm mesh fitted to M.S angle iron post at 2000mm apart of specification detailed below including supplying, fitting and fixing hooks, eye bolts with nuts, staples and bolts including three horizontal lines of G.I wire 10 S.W.G at top,middle and bottom and with one vertical line to each fencing post for keeping the fencing tight and erect complete, including one coat of primer painting to M.S angle iron post in P.C.C base etc. with following.	Sqm	656.82
	a) Corner and end posts with M.S angle of size 50mmx50mmx 5mm and length not less than 1800mm only duly slotted and bottom torqued with two numbers as stay posts..		
	b). Intermediate post with M.S angle of size 40mmx40mmx5mm and length not less than 1800mm only duly slotted and bottom torqued and splayed.		
	c).Stay post with M.S angle of size 35 mmx35mmx5mm and length not less than 1800mm duly slotted and bottom torqued and splayed. All M.S angle iron posts are to be fitted to P.C.C base embedded not less than 600mm (P.C.C. work shall be measured and paid separately).		

17.11	Supplying, fitting and fixing double leaf heavy duty iron gate, frame made from 50mm x 6mm M.S. flat iron with another horizontal flat at the middle, 20mm dia M.S bars at 200mm apart from bottom flat to top flat and 20 mm dia M.S. bars at 200mm apart from bottom flat to middle flat including drilling, welding etc. complete. All vertical bars are to be protruded at least 100mm above the middle and top flat is to be flattened and pointed as directed. Necessary locking arrangement on both faces, arrangement for temporary closing the gate and 2 nos.(minimum) strong iron hinges of M.S bars and flat of same sizes on each leaf to be embedded in C.C brick pillars as necessary including a red oxide painting to all iron works as directed and specified. (Separate payment will be made for posts).	Qtl	7436.57
17.12	Supplying, fitting and fixing double leaf heavy duty iron gate, frame made from 40mm x 40mm x 6mm M.S. angle and with M.S. flat of 40mm x 6mm size as per approved design.Necessary locking arrangement for temporary closing the gate and 2nos. (minimum) strong iron hinges of M.S bars and flat of same sizes on each leaf to be embedded in C.C brick pillars as necessary including a red oxide painting to all iron works as directed and specified.(Separate payment will be made for posts).	Qtl	6808.23
17.13	Providing rails for plying of rollers of gate made made from 50mm x 6mm size M.S. flat and to be fitted with 3 nos of hinged hold fasts to be embedded in concrete prop. 1:2:4 of 10mm wide and 75 mm deep over brick flat soling 250mm wide including curing etc. complete.	RM	446.97
17.14	Providing roller to the gate including welding etc.complete		
	a) With 25mmx50mm (dia) roller	Each	251.84
	b) With 12.50mmx50mm (dia) roller	Each	216.16

CH-18: STEEL AND IRON WORKS

Item No	Description	Unit	Rate
18.1	REINFORCEMENT		
18.1.1	Supplying, fitting and fixing in position reinforcement bars conforming to relevant I.S. Code for R.C.C. work/ R.B. walling including straightening, cleaning, cutting and bending to proper shapes and length as per details, supplying and binding with 20G annealed black wire and placing in position with proper blocks, supports, chairs, spacers etc. complete. (No extra measurement for lap, hook, chair, anchor etc. will be entertained in the measurement as they are included in the rate) (Upto 1st floor level)		
	a) From Primary Producer: TATA/SAIL/Essex Steel/ Jindal steel/Shyam steel/RINL		
	(i) TMT Corrosion resistant steel (CRS) reinforcement bar	Qtl	7158.90
	(ii)Super Ductile (SD) TMT reinforcement bars	Qtl	6902.86
	b) Other ISI approved TMT reinforcement bar (SAI/BISCON/THERMAX) (For Assam Type Bldg., drain works,,retaining wall&boundary wall etc.	Qtl	6026.82
	c) ISI approved Mild steel reinforcement bar(M.S.Bar)	Qtl	6767.66

18.1.2	Extra over item 18.1.1(a,b,c) for each subsequent floor or part thereof above first floor level.	Qtl	37.75
18.2	BEAMS AND JOISTS		
18.2.1	Structural steel work in single sections including cutting, hoisting, fixing in position and applying a priming coat of red lead paint (up to 4.6m height) from plinth level including drilling holes, supplying, fitting and fixing with bolts and nuts or welding, if necessary as directed.		
	a) In beams and joists fitted independently without connecting plates.	Qtl	6946.00
	b) In flat up to 25mm thickness	Qtl	6620.00
	c) In M.S. angles, channels, Tees etc.	Qtl	7074.00
18.2.2	Steel work riveted and/ or bolted/ riveted in built up sections, frame work including cutting, hoisting, fixing in position and applying a priming coat of red- lead paint up to 4.6m height above plinth including drilling holes, supplying, fitting and fixing with bolts and nuts or welding, if necessary as directed.		
	a) In beams and joists with connecting plates/ angle cleats	Qtl	6729.00
	b) In single stanchions (columns) composed of R.S.J. or channels with caps, bases, splices, angle brackets etc.	Qtl	7618.00
	c) In trusses and trussed purlins up to 25m span and 15.0m overall height.	Qtl	7316.00
18.2.3	Steel work welded in built up sections, framed work including cutting, hoisting and fixing in positions and applying priming coat of red-lead paint including drilling holes, supplying, fitting and fixing with bolts and nuts or welding, if necessary as directed.		
	a) In beams and joists with connecting plates/ angled cleat.	Qtl	7079.00
	b) In trusses and trussed purlins up to 25m span and 15.0m overall height.	Qtl	7320.00
18.3	ROOF TRUSS		
18.3.1	Providing fitting, hoisting and fixing of roof trusses including purlins fabricated using MS circular hollow section conforming to relevant I.S. code, as per approved design and drawings including providing M.S. cleats, base plates, bolts and nuts and one coat of red oxide Zinc Chromate primer and two coats of approved enamel paints complete including fitting necessary cleats etc. for fixing ceiling joists as per design and drawing as directed.		
	a) TATA make Circular Hollow section- CHS of Grade YST-310 conforming to I.S. Specification : 1161-1998	Qtl	8839.00
	b) Using other ISI marked approved circular hollow section	Qtl	8410.00
18.3.2	Providing, fitting, hoisting and fixing of roof trusses fabricated out of combination of any two or all of the following sections (i) Square section (ii) Rectangular Hollow section of specification RSH/SHS and (iii) M.S. circular hollow section of Grade 210 including purlins, bottom runners and providing M.S. cleat, M.S. angle base plank, bolts and nuts with red oxide primer including fitting bottom runner and cleat etc. for fixing ceiling joist as per design complete as directed as per specifications.		
	a) Using TATA make RHS/SHS of Grade 310 conforming to relevant I.S. code 4923 : 1997	Qtl	9334.00

	b) Using other ISI marked approved brand tube	Qtl	9055.00
18.4	COLLAPSIBLE / ROLLING SHUTTERS		
18.4.1	Providing and fixing in position collapsible M.S. shutters with vertical channels 20mmx10mmx2mm braced with flat iron diagonals 20mmx5mm size with top and bottom rails of T-iron 40mmx40mmx6mm with 38mm dia steel pulleys complete with bolts and nuts, locking arrangements, stoppers, handles including applying a priming coat of red lead paint.	Sq.m.	4245.50
18.4.2	Providing and fixing rolling shutters of approved make made of 80mm wide M.S. laths interlocked together through their entire length and jointed together at the ends by end-locks mounted on specially designed pipe shaft with bracket plates, guide channels and arrangements for inside and outside locking with push-pull operation complete including cost of wire, spring and hood cover for the shutters.	Sq.m.	3069.00
18.5	WINDOWS / VENTILATORS		
18.5.1	Providing, fitting and fixing steel windows/ ventilators (openable) of standard rolled steel sections as per relevant I.S. Code-1038 joints mitred and welded with 15mmx3mmx100mm lugs embedded in cement concrete block of M-10 grade including providing and fixing of projecting hinges (not more than 65mm and not less than 15mm wide) bolting device and steel handles, peg stays of 300mm long etc. complete as per drawing including applying a priming coat of red-lead paint.	Sq.m.	2632.90
18.5.2	Providing, fitting and fixing steel windows/ ventilators (fixed) of standard rolled steel sections as per relevant I.S. Code, joints mitred and welded with 15mmx3mmx100mm lugs embedded in cement concrete block of M-10 grade including applying a coat of red-lead primer.	Sq.m.	1554.40
18.5.3	Providing, fitting and fixing steel windows of standard rolled steel sections as per relevant I.S. Code, joints mitred and welded with 15mmx13mmx100mm lugs embedded in cement concrete block of M-10 grade including providing and fixing of projecting hinges (not more than 65mm and not less than 15mm wide) bolting device, steel handles, pegs, stays of 300mm long etc. complete including providing 12mmx12mm sq. bars duly welded to steel frame at not more than 12cm c/c applying a priming coat of red-lead paint etc. complete as per drawing.		
	a) Openable	Sq.m.	3087.70
	b) Fixed	Sq.m.	2119.60
18.5.4	Providing, fitting and fixing steel ventilators (openable) of standard rolled steel sections as per relevant I.S. Code, joints mitred and welded with 15mmx13mmx100mm lugs embedded in cement concrete block of M-10 grade including providing and fixing of projecting hinges (not more than 65mm and not less than 15mm wide) bolting device, handles, pegs, stays of 300mm long etc. complete including providing 2nos. of 12mmx12mm sq. bars (horizontal) duly welded to steel frame and applying a priming coat of red oxide paint etc. complete as per drawing.	Sq.m.	4461.90
FLATS/ BOLTS AND NUTS			

18.6.1	Supplying fitting and fixing M.S.angles, M.S. flat, M.S. bolts and nuts including painting with red lead paint one coat complete as directed.		
	a) M. S. Angles.....	Qtl	10344.45
	b) M. S. Flats.....	Qtl	9604.99
	c) M. S. bolts & nuts.....	Qtl	9982.23
DOOR FRAMES			
18.7.1	Supplying, fitting and fixing door frame made of M.S. equal angle of size 40mmx40mmx5mm fabricated, welded, mitred and joined as per relevant I.S. Codes including fitting 8 nos. of M.S. hold fast, providing closing/ holding device for 2 nos. tower bolts of door for each side and 1 no. for fitting and fixing Aldrop (for single leaf) to the frame for fixing the door to the frame, applying primer to all steel surfaces fixing the frame to the wall/ R.C.C. members by embedding the hold fast securely in C.C. block of M-15 grade complete at all levels as specified and directed. (Door hinges to be paid as per relevant item of the schedule of rates).	M	354.60
18.7.2	Providing, fitting and fixing powder coated pressed steel door frame made of 1.25mm/ 1.6mm thick sheet of various sizes with necessary hinge jamb, lock jamb, head and steel base ties at the bottom of the frame. The whole frame shall be welded or rigidly fixed together by mechanical means with base ties of pressed mild steel angle of size 20x20x1.25mm thick. Frame to be filled up with 1:4:8 P.C.C. after fitting the frame with brick/ concrete works etc.complete as directed by the department for all levels. (P.C.C. to be measured and paid for seperately).		
	(a) 1.25mm thick sheet		
	(i) Size 90mm x 60mm (Single rebate)	M	713.20
	(ii) Size 105mm x 60mm (Single rebate)	M	713.20
	(iii) Size 125mm x 60mm (Single rebate)	M	725.20
	(iv) Size 125mm x 60mm (Double rebate)	M	853.30
	(v) Size 165mm x 60mm (Double rebate)	M	942.70
	(b) 1.60mm thick sheet		
	(i) Size 90mm x 60mm (Single rebate)	M	785.60
	(ii) Size 105mm x 60mm (Single rebate)	M	815.90
	(iii) Size 125mm x 60mm (Single rebate)	M	849.70
	(iv) Size 125mm x 60mm (Double rebate)	M	948.80
	(v) Size 165mm x 60mm (Double rebate)	M	1009.20
CATTLE TRAP			
18.8.1	Supplying fitting and fixing M.S. cattle trap fabricated, welded, mitred and jointed as per relevant IS code with ISMB 250 joist spaced not more than 1800mm apart and ISA 65 x 65 x 6 spaced not more than 200mm apart and fixing the same over RCC member / Brick wall complete as specified and directed.	Sq.m.	5381.77
18.9	Steel Railing		

18.9.1	Providing and fixing Stainless Steel (Grade304) railing made of Hollow tubes, challels, plates etc., including welding, grinding, buffing polishing and making curvature (wherever required) and fitting the same with necessary stainless steel nuts and bolts complete i.e. fixing the railing with necessary accessories & stainless steel dash fasteners, stailless steel bolts etc. of required size on the top of the floor or the side of waist slabe with suitable arrangement as per approval of Engineer in-charge (For payment, only weight of stainless steel member is to be considered. No extra payment for fixing accessories such as nuts, bolts, fasteners etc. will be considered.	Kg	582.31
18.9.2	Providing and fixing M.S. Tube hand rail of approve size by welding etc. to steel ladder railing, belcony railing, staircase railing and similar works, including applying priming coat of approve steel primer complete as specified and directed by department.	Kg	91.27

CH-19: PLINTH PROTECTION WORKS

Item No	Description	Unit	Rate
19.1	Providing plinth protection 50mm thick to cement concrete 1 : 3 : 6 with coarse agg.of 20mm nominal size including finishing the surface with 10mm thick cement plaster in proportion 1:3 with a floating coat of neat cement finished.	Sqm	342.80
19.2	Providing plinth protection with bricks flat laid in cement mortar 1: 4 and finished with 15mm thick cement plaster in prop. 1:2 with a floating coat of neat cement finish.	Sqm	665.72
19.3	Providing plinth protection with bricks laid on edge and grouted with cement mortar 1:4 and finishing with cement pointing in prop. 1:1 complete.	Sqm	710.43
19.4	Providing drain with cement concrete (1:3:6) 10cm thick sides 10cm thick bed over brick flat soling including 15mm thick cement plastering in prop. 1:3 finished with a floating coat of cement slurry as directed with necessary shattering for sides and earth work in excavation of foundation trenches and refilling the sides after completion of work etc. as specified.		
	i) 300mm wide and average 150mm deep with bed slope 1 in 150 with initial depth of 100mm.	Rm	888.38
	ii) 300mm wide and average 250mm deep with bed slope 1 in 150 with initial depth of 100mm.	Rm	1093.04
19.5	Providing drain with brick work in cement mortar in proportion 1:5 with half brick thick side walls and 100mm thick C.C (1:3:6) base over one brick flat soling including 15mm thick cement plastering in prop. 1:3 finished with a floating coat of cement slurry as directed with necessary shattering for sides and earth work in excavation of foundation trenches and refilling the sides after completion of work etc. as specified.		
	i) 300mm wide and average 150mm deep with bed slope 1 in 150 with initial depth of 100mm.	Rm	796.21
	ii) 300mm wide and average 250mm deep with bed slope 1 in 150 with initial depth of 100mm.	Rm	975.10

CH-20: LANDSCAPING

Item No.	Description	Unit	Rate
	LAWN MAKING BY DIBBLING		
20.1	Supplying and dibbling of selected lawn grass [Doob grass (Cynodon dactylon), cushion grass (Axonopus affinis)] after preparing the land by ploughing / working to a depth of 40 -45 cm, removing of all unwanted debris (rubbles, pebbles, plant roots etc.), mixing of organic manure(10kg - 15kg/sq.m), levelling the surface and initial maintenance by proper and periodic rolling, mowing and irrigation etc.(as specified) for a period of six months from the date of dibbling including the application of recommended dose of fertilizers(N:P:K) mixture as specified and directed by the department.		
	(a) By Doob grass (Cynodon dactylon)	Sqm	332.13
	(b) By Carpet grass (Axonopus affinis)	Sqm	335.04
	LAWN MAKING BY TURFING		
20.2	Supplying and laying of selected grass turves [size 60 cm x 30 cm] after preparing the land by removing of all unwanted debris (rubbles, pebbles, plant roots etc.), mixing of organic manure, levelling the surface and initial maintenance by proper and periodic rolling, mowing and irrigation etc.(as specified) for a period of six months from the date of laying of grass including the application of recommended dose of fertilizers(N:P:K) mixture as specified and directed by the department.		
	(a) By Bahpotia grass and Doob grass (Cynodon dactylon) :	Sqm	434.46
	(b) BY CARPET GRASS (Axonopus affinis) :	Sqm	496.28
	POT PLANTING		
20.3	Supplying and planting of terrestrial orchids with required size of pot including filling the pot to the top surface with appropriate soil media made of garden soil, river sand, cow-dung, leaf mould, fertilizer, pesticides as per specification and necessary maintenance for a period of 4 months from the date of planting complete as specified and directed . (Mixing of different ingredients of soil media to be made in presence of Deptt. official / Specialist concerned)		
	(A) With Aeirides		
	(I) Using earthen pot of 30 cm dia.	each	426.30
	(II) Using earthen pot of 60 cm dia.	each	455.56
	(III) Using concrete pot of 60 cm dia.	each	536.98
	(B) With Dendrobium (Hybrid)		
	(I) Using earthen pot of 30 cm dia.	each	660.02
	(II) Using earthen pot of 60 cm dia.	each	686.95
	(III) Using concrete pot of 60 cm dia.	each	871.46
	(C) With Mokara		
	(I) Using earthen pot of 30 cm dia.	each	950.79
	(II) Using earthen pot of 60 cm dia.	each	977.73
	(III) Using concrete pot of 60 cm dia.	each	1162.23
	(D) With Vanda (hybrid)		
	(I) Using earthen pot of 30 cm dia.	each	2388.12

	(II) Using earthen pot of 60 cm dia.	each	2430.36
	(III) Using concrete pot of 60 cm dia.	each	2614.86
20.4	Supplying and planting of house plants in required size of pot including filling the pot to the top surface with appropriate soil media of garden soil, river sand, cow-dung, leaf mould, fertilizer, pesticides as per specification and necessary maintenance for a period of 6 months from the date of planting complete as specified and directed . (Mixing of different ingredients of soil media to be made in presence of Deptt. official / Specialist concerned)		
	(A) With Fancy leaf caladium/Coleus		
	(i) Using earthen pot of 30 cm dia.	each	262.98
	(ii) Using earthen pot of 60 cm dia.	each	340.85
	(iii) Using concrete pot of 60 cm dia.	each	423.49
	(B) With Indian rubber plant / Prayer plant		
	(i) Using earthen pot of 30 cm dia.	each	262.98
	(ii) Using earthen pot of 60 cm dia.	each	340.85
	(iii) Using concrete pot of 60 cm dia.	each	423.49
	(C) With Fragrant dracaena / Silver net leaf / Philodendron / Devil's back bone /Peperomea/Croton		
	(i) Using earthen pot of 30 cm dia.	each	262.98
	(ii) Using earthen pot of 60 cm dia.	each	340.85
	(iii) Using concrete pot of 60 cm dia.	each	423.49
	(D) With Peacock/spider/Umbrella		
	(i) Using earthen pot of 30 cm dia.	each	286.24
	(ii) Using earthen pot of 60 cm dia.	each	364.11
	(iii) Using concrete pot of 60 cm dia.	each	446.75
	(E) With Maiden hair fern/Aloe/Boston fern/Swiss cheese plant/Forget me not		
	(i) Using earthen pot of 30 cm dia.	each	297.87
	(ii) Using earthen pot of 60 cm dia.	each	375.74
	(iii) Using concrete pot of 60 cm dia.	each	458.38
	(F) With Marry Christmas begonia / Polypodium / Aluminium plant		
	(I) Using earthen pot of 30 cm dia.	each	303.69
	(II) Using earthen pot of 60 cm dia.	each	381.55
	(III) Using concrete pot of 60 cm dia.	each	464.19
	(G) With English Ivy		
	(I) Using earthen pot of 30 cm dia.	each	286.24
	(II) Using earthen pot of 60 cm dia.	each	364.11
	(III) Using concrete pot of 60 cm dia.	each	446.75
	(H) With Iron Cross Begonia		
	(I) Using earthen pot of 30 cm dia.	each	558.16
	(II) Using earthen pot of 60 cm dia.	each	636.02
	(III) Using concrete pot of 60 cm dia.	each	718.66
	(I) With Anthurium		
	(J) With Selloum		
	(K) With Birdnest fern/Good luck plant		

20.5	Supplying and planting of palms with required size of pot including filling the pot to the top surface with appropriate soil media made of garden soil, river sand, cow-dung, leaf mould, fertilizer, pesticides as per specification and necessary maintenance for a period of 4 months from the date of planting complete as specified and directed . (Mixing of different ingredients of soil media to be made in presence of Deptt. official / Specialist concerned)		
	(A) With Bottle palm		
	(I) Using earthen pot of 30 cm dia.	each	244.43
	(II) Using earthen pot of 60 cm dia.	each	335.03
	(III) Using concrete pot of 60 cm dia.	each	417.67
	(B) With Erica palm (Butterfly palm)/ Kentia palm		
	(I) Using earthen pot of 30 cm dia.	each	271.00
	(II) Using earthen pot of 60 cm dia.	each	348.86
	(III) Using concrete pot of 60 cm dia.	each	431.50
	(C) With Chinese Palm		
	(I) Using earthen pot of 30 cm dia.	each	294.26
	(II) Using earthen pot of 60 cm dia.	each	372.13
	(III) Using concrete pot of 60 cm dia.	each	454.77
	(D) With Fan Palm /Finger palm		
	(I) Using earthen pot of 30 cm dia.	each	500.00
	(II) Using earthen pot of 60 cm dia.	each	577.87
	(III) Using concrete pot of 60 cm dia.	each	660.51
20.6	Supplying and planting of seasonal flower in required size of pot including filling the pot to the top surface with appropriate soil media made of garden soil, river sand, cow-dung, leaf mould, fertilizer, pesticides as per specification and necessary maintenance for a period of 4 months from the date of planting complete as specified and directed . (Mixing of different ingredients of soil media to be made in presence of Deptt. official / Specialist concerned)		
	(I) Using earthen pot of 30 cm dia.	each	184.38
	(II) Using earthen pot of 60 cm dia.	each	274.98
	(III) Using concrete pot of 60 cm dia.	each	357.62
20.7	Supplying and planting of cactus in required size of pot including filling the pot to the top surface with appropriate soil media made of garden soil, sand, cow-dung, fertilizer, pesticides as per specification and necessary maintenance for a period of 6 months from the date of planting complete as specified and directed . (Mixing of different ingredients of soil media to be made in presence of Deptt. official / Specialist concerned)		
	(A) With Peanut Cactus / Rat tail Cactus		
	(I) Using earthen pot of 30 cm dia.	each	369.74
	(II) Using earthen pot of 60 cm dia.	each	455.44
	(III) Using concrete pot of 60 cm dia.	each	538.08
	(B) With Prickly pear		
	(I) Using earthen pot of 30 cm dia.	each	393.00
	(II) Using earthen pot of 60 cm dia.	each	478.70
	(III) Using concrete pot of 60 cm dia.	each	510.41
	(C) With Parodia globular		
	(I) Using earthen pot of 30 cm dia.	each	270.88
	(II) Using earthen pot of 60 cm dia.	each	513.59

	(III) Using concrete pot of 60 cm dia.	each	596.23
	BED PLANTING		
20.8	Supplying and planting of Dahlia / Chrysanthemum / Gladiolus on the ground (bed planting, bed size 45 cm x 45 cm x 45 cm) including the preparation of the soil, providing organic manure and fertilizer, pesticides as per specification and necessary maintenance for a period of 4 months from the date of planting complete as specified and directed .	each	136.04
20.9	Supplying and planting of selected rose plants on the ground (as specimen bush) including the preparation of the pit (45 cm x 45 cm x 45 cm), removing unwanted debris, filling the pit with manure and fertilizers as per specification and necessary maintenance for a period of 1 year from the date of planting complete as specified and directed .		
	(A) With Pompon Rose	each	476.22
	(B) With Hybrid tea rose / Miniature rose	each	464.59
	HEDGE PLANTING / EXTERNAL HEDGE		
20.10	Supplying and planting of suitable plant materials for preparation protective / external hedge (50 cm wide) including the preparation of the ground by mixing organic manure and fertilizers as per specification and necessary maintenance for a period of one year from the date of planting complete as specified and directed .		
	(A) ADOPTING SINGLE ROW PLANTING METHOD		
	(i) With Duranta	metre	352.72
	(ii) With Kamini	metre	396.80
	(iii) With Chinese Bamboo	metre	401.45
	(iv) With Thuja / Ixora	metre	494.49
	(B) ADOPTING CROSS WISE DOUBLE ROW PLANTING METHOD		
	(i) With Duranta	metre	483.35
	(ii) With Kamini	metre	596.97
	(iii) With Chinese Bamboo	metre	606.27
	(iv) With Thuja / Ixora	metre	792.37
	EDGE PLANTING / INTERNAL HEDGE		
20.11	Supplying and planting of suitable plant materials for edge making / Internal hedge (20 cm wide) including the preparation of the ground by mixing organic manure and fertilizers as per specification and necessary maintenance for a period of one year from the date of planting complete as specified and directed .		
	(A) ADOPTING SINGLE ROW PLANTING METHOD		
	(i) With Duranta	metre	362.03
	(ii) With Coleus	metre	400.65
	(iii) With Brindavan / Latifolius	metre	436.34
	(iv) With Acalypha wilkesiana	metre	517.76
	(v) With Acer palmatum	metre	558.46
	(B) ADOPTING CROSS WISE DOUBLE ROW PLANTING METHOD		
	(i) With Duranta	metre	501.96
	(ii) With Coleus	metre	579.22
	(iii) With Brindavan / Latifolius	metre	676.06
	(iv) With Acalypha wilkesiana	metre	838.89

	(v) With Acer palmatum	metre	920.31
	TREE PLANTING		
20.12	Supplying and planting of ornamental trees (1'-0" height sapling) including pit making (60 cm x 60 cm x 60 cm), filling the pit with appropriate soil media, manure and fertilizers as per specification and necessary maintenance for a period of one year from the date of planting complete as specified and directed .		
	SHADE TREE		
	(A) Neem / Mahaneem / Beef wood / Bakul / Acacia / Siris / Devils tree / Kadam / Palash / Pink Cassia / peepal / Silver oak / Simalu / Ajar	each	299.34
	(B) Kanchan / Nahar / Arjunu	each	305.16
	(C) Sonaru	each	310.97
	(D) Bottle Brush / Nil Gulmohar / Debadaru / Weeping Willow	each	340.05
	(E) Ravenala	each	340.05
	(F) Paine	each	362.21
	(G) Christmas tree / Kuki	each	378.55
	Flowering Tree		
	(A) Gulmohar / Ashok	each	305.16
	(B) Kanchan	each	328.42
	SHRUB PLANTING		
20.13	Supplying and planting of ornamental shrub (below 1'-0" height) including pit making (60 cm x 60 cm x 60 cm), filling the pit with appropriate soil media, manure and fertilizers as per specification and necessary maintenance for a period of one year from the date of planting complete as specified and directed .		
	(A) Radhachura / Krishnachura / Rat ki rani / Ixora / Jetuka	each	368.02
	(B) Allmanda / Khorika zai / May flower / Kamini / Kathanda / Tecoma / Acalypha / Bougainvillea / Dracaena / Togor / Nilakantha / Mussaenda	each	373.84
	(C) Korobi / Azalea	each	379.66
	(D) Thuja	each	379.66
	(E) Poinsettia / Forget me not	each	391.29
	AQUATIC PLANT		
20.14	Supplying and planting of aquatic plants (water Lily and Lotus) after preparation of under water soil including organic manuring, supplying the required quantity of water and maintenance for a period of one year from the date of completion complete as specified and directed.	Sqm	516.78
20.15	Preparation of a carpet bedding by raising the backside of the bed to a height of 45 cm (average) after mixing organic manure, fertilizer including supplying and planting of grass and specimen plants (Brindavan) to engrave the desired letters, necessary maintenance for a period of 6 months from the date of planting complete as directed and specified by the department.	Sqm	686.22

20.16	Construction of a pergola (Width = 2.0 m, Height of the side posts = 1.8 m, Height at the apex = 2.8 m) with angle iron of size 50 mm x 50 mm x 5 mm (placing at 2000 mm apart) embedding the iron posts in P.C.C. base (not less than 600 mm) covering at the top and sides by chain link fencing, painting one coat of primer and two coats of paint including supplying and planting of flowering climbers as enclosed from both sides (3 nos. per metre in one side) of the pergola to facilitate climbing by pit making (30 cm x 30 cm x 30 cm) filling the pit with the mixture of soil and organic manure and necessary maintenance by trimming etc complete as specified and directed by the department. (P.C.C. and soling to be measured and paid separately)	metre	5699.85
20.17	Supplying and placing of orchid house of size 3.00 m x 1.5 m x 2.00 m with tub stand, make of 15 mm dia. G.I. Pipe with M.S. rod, M.S. flat etc. including roof frame, painting two coats, covering the roof with 75 % heat absorbing black nets and three sides with wire net complete as directed and specified.	each	23587.77
20.18	Edging with bricks laid dry lengthwise including excavation, refilling, consolidating with hand packing and spreading neatly surplus earth within a lead of 50 m.	each	38.93
20.19	Half brick circular tree guard internal diameter 1.25 m and height 1.2 m above ground and 0.02 m below ground bottom two coarses laid dry and top three coarses in cement mortar 1:6 and the intermediate coarses being in dry honey comb masonry as per design complete.	each	2090.61
20.20	Making tree guard 53 cm dia and 1.3 m high as per design from empty coal tar drums including providing and fixing 2 nos. M.S. sheet rings (50 x 0.50) mm with rivets complete in all respects including painting inside and outside of tree guard .	each	135.44
20.21	Supplying and installation of Field Turf Turket PVC sports surface (vinyl sports floor covering) over fibreglass mat with PVC foam backing including anti bacterial coating and protected with dirt resistant treatment for ease of maintenance complete as specified and directed.		
	(A) F-25 (filament ht. 25mm)	Sqm	3582.50
	(B) PGF (filament ht. 32mm)	Sqm	6678.00
	(C) L-35 (filament ht. 35mm)	Sqm	4722.90
	(D) FTS (filament ht. 40mm)	Sqm	4815.00
	(E) FTT (filament ht. 60mm)	Sqm	7649.00
	(F) FTOS 1S (filament ht. 65mm)	Sqm	7649.00

CH-21 : PILING WORKS

Item No.	Description	UNIT	Rate
21.1	MANUALLY DRIVEN R.C.C. PILE		

21.1.1	Providing cast in-situ manually driven reinforced cement concrete piles of using required grade of cement concrete of required bore-diameter, bulb diameter, lengths and with reinforcement as per specifications and drawings. Reinforcement to be measured and paid separately as per item no. 21.5.1		
	(A) SINGLE UNDER - REAMED PILE		
	I) Concrete grade M20		
	a) Pile Dia 250 mm with 625 mm bulb diameter		
	i) Upto 5 M	Rm	506.40
	ii) Above 5M	Rm	513.07
	b) Pile dia 300 mm with 750 mm bulb diameter		
	i) Upto 5 M	Rm	688.34
	ii) Above 5M	Rm	717.79
	c) Pile dia 350 mm with 875 mm bulb diameter		
	i) Upto 10 M	Rm	912.21
	ii) Above 10 M	Rm	944.68
	d) Pile dia 400 mm with 1000 mm bulb diameter		
	i) Upto 10 M	Rm	1252.06
	ii) Above 10 M	Rm	1299.48
	e) Pile dia 450 mm with 1125 mm bulb diameter		
	i) Upto 10 M	Rm	1613.82
	ii) Above 10 M	Rm	1668.05
	f) Pile Dia 500 mm with 1125 mm bulb diameter		
	i) Upto 10 M	Rm	2029.11
	ii) Above 10 M	Rm	2087.37
	(B) DOUBLE UNDER - REAMED PILE		
	a) Pile dia 300 mm with 750 mm bulb diameter		
	i) Upto 5 M	Rm	807.27
	ii) Above 5M	Rm	866.99
	(b) Pile dia 350 mm with 875 mm bulb diameter		
	i) Upto 10 M	Rm	1097.95
	ii) Above 10 M	Rm	1121.74
	c) Pile dia 400 mm with 1000 mm bulb diameter		
	i) Upto 10 M	Rm	1483.69
	ii) Above 10 M	Rm	1504.10
	d) Pile dia 450 mm with 1125 mm bulb diameter		
	i) Upto 10 M	Rm	1940.76
	ii) Above 10 M	Rm	1953.80
	e) Pile Dia 500 mm with 1125 mm bulb diameter		
	i) Upto 10 M	Rm	2474.23
	ii) Above 10 M	Rm	2473.55
	i) Upto 10 M	Rm	
	ii) Above 10 M	Rm	
	C) UNIFORM PILE		
	a) Pile Dia 250 mm		
	i) Upto 5 M	Rm	413.24
	ii) Above 5M	Rm	442.28
	b) Pile Dia 300 mm		
	i) Upto 5 M	Rm	613.70
	ii) Above 5M	Rm	636.86
	c) Pile Dia 350 mm		
	i) Upto 10 M	Rm	866.52

	ii) Above 10 M	Rm	947.35
	d) Pile Dia 400 mm		
	i) Upto 10 M	Rm	1131.90
	ii) Above 10 M	Rm	1237.30
	e) Pile Dia 450 mm		
	i) Upto 10 M	Rm	1431.94
	ii) Above 10 M	Rm	1566.24
	f) Pile Dia 500 mm		
	i) Upto 10 M	Rm	1689.59
	ii) Above 10 M	Rm	1875.03
21.2	MACHINE DRIVEN PILES		
21.2.1	Providing cast in-situ machine driven reinforced cement concrete piles of required bore-diameter, bulb diameter, lengths and with reinforcement as per specifications and drawings including using of Bentonite solution. (Reinforcement to be measured and paid separately as per item no. 21.4.1)		
	(A) SINGLE UNDER - REAMED PILE		
	(I) Concret garade M20		
	i) Pile dia 300 mm with 750 mm bulb diameter	Rm	787.92
	ii) Pile dia 350 mm with 875 mm bulb diameter	Rm	1038.92
	iii) Pile dia 400 mm with 1000 mm bulb diameter	Rm	1362.82
	iv) Pile dia 450 mm with 1125 mm bulb diameter	Rm	1728.86
	v) Pile dia 500 mm with 1250 mm bulb diameter	Rm	2132.62
	vi) Pile dia 550 mm with 1375 mm bulb diameter	Rm	2564.82
	vii) Pile dia 600 mm with 1500 mm bulb diameter	Rm	2944.57
	(II) Concrete Grade M25		
	i) Pile dia 300 mm with 750 mm bulb diameter	Rm	813.25
	ii) Pile dia 350 mm with 875 mm bulb diameter	Rm	1112.88
	iii) Pile dia 400 mm with 1000 mm bulb diameter	Rm	1460.60
	iv) Pile dia 450 mm with 1125 mm bulb diameter	Rm	1854.07
	v) Pile dia 500 mm with 1250 mm bulb diameter	Rm	2288.96
	vi) Pile dia 550 mm with 1375 mm bulb diameter	Rm	2756.15
	vii) Pile dia 600 mm with 1500 mm bulb diameter	Rm	3174.85
	(B) DOUBLE UNDER - REAMED PILE		
	(I) Concret garade M20		
	i) Pile dia 300 mm with 750 mm bulb diameter	Rm	813.44
	ii) Pile dia 350 mm with 875 mm bulb diameter	Rm	1124.45
	iii) Pile dia 400 mm with 1000 mm bulb diameter	Rm	1489.20
	iv) Pile dia 450 mm with 1125 mm bulb diameter	Rm	1907.57
	v) Pile dia 500 mm with 1250 mm bulb diameter	Rm	2375.56
	vi) Pile dia 550 mm with 1375 mm bulb diameter	Rm	2881.45
	vii) Pile dia 600 mm with 1500 mm bulb diameter	Rm	3336.48
	(II) Concrete Grade M25		
	ii) Pile dia 350 mm with 875 mm bulb diameter	Rm	1204.53
	iii) Pile dia 400 mm with 1000 mm bulb diameter	Rm	1596.08
	iv) Pile dia 450 mm with 1125 mm bulb diameter	Rm	2045.73
	v) Pile dia 500 mm with 1250 mm bulb diameter	Rm	2549.68
	vi) Pile dia 550 mm with 1375 mm bulb diameter	Rm	3096.48
	vii) Pile dia 600 mm with 1500 mm bulb diameter	Rm	3783.89
	C) UNIFORM PILE		
	(I) Concret garade M20		
	i) Pile dia 300 mm	Rm	704.28

	ii) Pile dia 350 mm	Rm	953.04
	iii) Pile dia 400 mm	Rm	1236.74
	iv) Pile dia 450 mm	Rm	1549.85
	v) Pile dia 500 mm	Rm	1890.35
	vi) Pile dia 550 mm	Rm	2247.36
	vii) Pile dia 600 mm	Rm	2552.83
	(II) Concrete Grade M25		
	i) Pile dia 300 mm	Rm	754.18
	ii) Pile dia 350 mm	Rm	1020.97
	iii) Pile dia 400 mm	Rm	1325.44
	iv) Pile dia 450 mm	Rm	1662.13
	v) Pile dia 500 mm	Rm	2029.00
	vi) Pile dia 550 mm	Rm	2415.09
	vii) Pile dia 600 mm	Rm	2752.41
	(III) Concrete Grade M30		
	i) Pile dia 500 mm	Rm	2046.74
	ii) Pile dia 550 mm	Rm	2436.63
	iii) Pile dia 600 mm	Rm	2777.99
21.3	Stone Pile		
21.3.1	Stone Pile Providing stone piles of required diameter and length in ordinary / dense soil etc. using 25 mm down graded stone aggregate or river gravel including cost of boring including watering, ramming etc. as required, Cost of T&P complete as directed and specified by the department.		
	i) Pile Dia 250 mm	Rm	223.35
	ii) Pile Dia 300 mm	Rm	294.45
	iii) Pile Dia 350 mm	Rm	378.73
	iv) Pile Dia 400 mm	Rm	439.23
	v) Pile Dia 450 mm	Rm	555.82
	vi) Pile Dia 500 mm	Rm	686.22
	vii) Pile Dia 550 mm	Rm	988.04
	viii) Pile Dia 600 mm	Rm	1755.87
21.4	MISCELLANEOUS		
21.4.1	Extra for using admixture & plasticiser		
	(a) RCC grade M20		
	i) Pile Dia 250 mm	Rm	39.75
	ii) Pile Dia 300 mm	Rm	58.45
	iii) Pile Dia 350 mm	Rm	81.22
	iv) Pile Dia 400 mm	Rm	108.23
	v) Pile Dia 450 mm	Rm	139.76
	vi) Pile Dia 500 mm	Rm	175.98
	vii) Pile Dia 600 mm	Rm	263.25
	(b) RCC grade M25		
	i) Pile Dia 250 mm	Rm	46.19
	ii) Pile Dia 300 mm	Rm	67.92
	iii) Pile Dia 350 mm	Rm	94.38
	iv) Pile Dia 400 mm	Rm	125.77
	v) Pile Dia 450 mm	Rm	162.41
	vi) Pile Dia 500 mm	Rm	204.50
	vii) Pile Dia 600 mm	Rm	305.91
	(c) RCC grade M30		
	i) Pile Dia 500 mm	Rm	208.16

	ii) Pile Dia 550 mm	Rm	120.08
	Pile Dia 600 mm	Rm	311.38
21.4.2	Extra for boring specified diameter pile in high sub-soil water level requiring use of drilling mud and extra cement		
	(i) 250 mm dia	Rm	96.78
	(ii) 300 mm dia	Rm	124.10
	(ii) 350 mm dia	Rm	156.90
	(iii) 400 mm dia	Rm	184.23
	(iv) 450 mm dia	Rm	227.95
	(v) 500 mm dia	Rm	282.61
	(vi) 600 mm dia	Rm	337.27
21.4.3	Empty boring for specified diameter piles from ground level to cut-off level.		
	a) Pile Dia 250 mm	Rm	120.84
	b) Pile Dia 300 mm	Rm	171.72
	c) Pile Dia 350 mm	Rm	209.35
	d) Pile Dia 400 mm	Rm	236.38
	e) Pile Dia 450 mm	Rm	302.10
	f) Pile Dia 500 mm	Rm	346.62
	g) Pile Dia 550 mm	Rm	385.31
	h) Pile Dia 600 mm	Rm	421.35
21.5	REINFORCEMENT		
21.5.1	Supplying, fitting and fixing in position reinforcement bars conforming to relevant IS code, for reinforced cement concrete works in piles including straightening, cleaning, cutting and bending to proper shapes and length as per specifications and drawings. The rings should be welded with vertical reinforcements as per specified IS codes of practice. (Inclusive of lap length, hook, chair, bend, anchor...etc)		
	(A) TMT bar made of Primary producer (From Primary producer)	Qtl	7210.68
	(B) TMT bar of other ISI marked (From Secondary producer)	Qtl	6395.53
	(C) ISI approved M.S. Rod	Qtl	7074.82
21.6	LOAD TEST ON PILES		
	ROUTINE TESTS		
21.6.1	Routine test for test loading to 1.5 times the specified safe working load including construction of any test pile cap and dismantling the same after completion of test. (Rate for Safe working Load)	Ton	947.87
21.6.2	Initial test for test loading to 2 times the specified safe working load including construction of any test pile cap and dismantling the same after completion of test. (Rate for Safe working Load)	Ton	1143.05

21.6.3	Initial testing of pile using Low strain / Sonic integrity Test / sonic Echo Test method in accordance with is 14893 including surface preparation of pile top by removing soil, mud, dust & chipping lean concrete lump etc. and use of computerised equipment and high skill trained personal for conducting the test & submission of results, all complete as per direction of Engineer-in-charge. Note:- The inclusion of the above item in the schedule of work should be judiciously decided by the technical sanctioning authority, kiping in view the quality control, type of soil strata & imprortance of the project	Each Test	706.60
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CHAPTER 22 : WATERPROOFING, CONCRETE ADMIXTURES & CHEMICAL TREATMENT,REPAIR,REHABILITATION & RE-STRENGTHENING OF STRUCTURES

Item No.	Description	Unit	Rate
22	WATERPROOFING, REPAIR & RESTRENGTHENING OF STRUCTURES		
22.1	SURFACE PREPARATION		
22.1.1	Preparing the RCC surface in roof slab, chajjas, terraces, sunken slabs, septic tank, OH tank, raft slab, walls etc. by thoroughly cleaning with wire brush etc. followed by power washing for making the surface absolutely free from any foreign particles like dust, oil, grease etc. and allowing surface to dry complete as directed at all levels.	Sqm	26.78
22.1.1.1	Extra over Item No. 22.1.1, for removing and chipping of brick bat corba, tar felt or any other existing protective layer over roof slab etc.	Sqm	24.74
22.1.2	Doing patch work in roof slab/ Raft slab wherever necessary by laying small portions of Polymer modified mortar - Cement sand mortar (in prop 1:4) mixed with 100% acrylic Polymer Master Crete M-81 of Choksey Chemicals/Dr. Sealkit Sealcrete (Asian Chemicals) / Rheomix 115 of BASF@ 4% by weight of cement as specified & directed by the Department complete after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately).	Sqm	130.27
22.1.3	Doing patch work in roof slab/ Raft slab wherever necessary by laying small portions of two component Polymer Modified Mortar Sika Top 122 as specified & directed by the Department complete after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately).	Sqm	268.51
22.1.4	Doing patch work in existing roof slab wherever necessary by laying small portions of Prepacked Polymer modified cementitious mortar - Renderoc S2 of Fosroc Chemicals/Emaco S88CT of BASF as specified & directed by the Department complete after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately).	Sqm	1343.19

22.1.5	Doing patch work in existing roof slab wherever necessary by laying cement sand mortar (prop.1:4) admixed with mortar modifier cum bonding agent Nitobond SBR of Fosroc Chemicals/Rheomix 141 of BASF/Dr. Sealkit Master Bond Latex (Asian Chemicals) @ 3Ltrs per bag of cement after applying with brush a priming coat of Nitobond SBR/Rheomix 141 of BASF/Dr. Sealkit Master Bond Latex (Asian Chemicals) @ 3.5sq.m. per litre as specified & directed by the Department complete after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately).	Sqm	692.72
22.1.6	Preparing RCC surfaces in expansion joints by thoroughly cleaning with wire brush etc. followed by power washing to make the surface absolutely free from any foreign particles like dust, oil, grease etc. and allowing surface to dry complete as directed at all levels.	Rm	8.40
22.2	CONCRETE & MORTAR ADMIXTURES		
22.2.1	Extra for providing and mixing normal Plastisizer Master Plast PL-1 of Choksey Chemicals /Dr. Sealkit Normal Plast (Asian Chemicals)/ RHEOMAC 707 of BASF @ 0.3% by weight of cement in concreting RCC structures as specified and directed by the Department complete at all levels.		
	(i) In 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	116.34
	(ii) In 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	145.41
	(iii) In 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	221.82
22.2.2	Extra for providing and mixing Integral waterproofing compounds like Master Proof IWP-1 of Choksey Chemicals /Dr. Sealkit No.1 (Asian Chemicals)/ RHEOMAC 707 of BASF @ 0.2% by weight of cement for waterproofing structures like roof slabs, basement, sunken slabs, chajjas, tunnels, water tank etc. as specified and directed by the Department complete at all levels.		
	A. In concrete of proportion:		
	(i) 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	91.01
	(ii) 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	113.77
	(iii) 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	173.49
	B In Plasters/ mortars of proportion:		
	(i) 1 : 3 (1cement : 3coarse sand)	Cum	95.88
	(ii) 1 : 4 (1cement : 4coarse sand)	Cum	71.38
	(iii) 1 : 5 (1cement : 5coarse sand)	Cum	61.09
22.2.3	Extra for providing and mixing normal Plastisizer Conplast P211/Dr. Sealkit LW Super (Asian Chemicals) @ 80ml per bag of cement in concrete/ mortars as specified and directed by the Department complete at all levels.		
	A. In concrete of proportion:		
	(i) In 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	86.56

	(ii) In 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	108.11
	(iii) In 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	165.70
	B In Plasters/ mortars of proportion:		
	(i) 1 : 3 (1cement : 3coarse sand)	Cum	138.67
	(ii) 1 : 4 (1cement : 4coarse sand)	Cum	102.63
	(iii) 1 : 5 (1cement : 5coarse sand)	Cum	88.33
22.2.4	Extra for providing and mixing Integral waterproofing liquid admixture & plasticiser Conplast X421 IC of Fosroc Chemicals /Rheomac 707 of BASF/Dr. Sealkit Normal Plast (Asian Chemicals) @ 100ml per bag of cement in concrete for waterproofing structures like roof slabs, basement, sunken slabs, chajjas, tunnels, water tank etc. as specified and directed by the Department complete at all levels.		
	A In concrete of proportion:		
	(i) 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	75.75
	(ii) 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	94.98
	(iii) 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	145.41
	B In Plasters/ mortars of proportion:		
	(i) 1 : 3 (1cement : 3coarse sand)	Cum	121.68
	(ii) 1 : 4 (1cement : 4coarse sand)	Cum	90.59
	(iii) 1 : 5 (1cement : 5coarse sand)	Cum	77.17
22.2.5	Extra for providing and mixing normal Plastisizer Plastiment BV 40 of Sika India Ltd. /Dr. Sealkit Superplast WR 40 (Asian Chemicals)/Rheobuild 1100i of BASF@ 0.5%-0.8% by weight of cement in concrete for waterproofing structures like roof slabs, basement, sunken slabs, chajjas, tunnels, water tank etc. as specified and directed by the Department complete at all levels.		
	(i) In 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	282.64
	(ii) In 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	353.30
	(iii) In 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	538.79
22.2.6	Extra for providing and mixing Plastisizer Sikament 170 of Sika India Ltd. /Rheobuild 620 of BASF@ 1.0%-1.5% by weight of cement in concrete for waterproofing structures like roof slabs, basement, sunken slabs, chajjas, tunnels, water tank etc. as specified and directed by the Department complete at all levels.		
	(i) In 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	569.80
	(ii) In 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	712.26
	(iii) In 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	1086.17

22.2.7	Extra for providing and mixing waterproofing Plastisizer Plastocrete Plus of Sika India Ltd /Dr. Sealkit LW Super (Asian Chemicals)/ RHEOMAC 730 of BASF @ 200gms per bag of cement in concrete for waterproofing as specified and directed by the Department complete at all levels.		
	A In concrete of proportion:		
	(i) 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	358.77
	(ii) 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	448.45
	(iii) 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	683.89
	B In Plasters/ mortars of proportion:		
	(i) 1 : 3 (1cement : 3coarse sand)	Cum	571.79
	(ii) 1 : 4 (1cement : 4coarse sand)	Cum	426.04
	(iii) 1 : 5 (1cement : 5coarse sand)	Cum	364.36
22.2.8	Extra for fibre reinforcing concrete/ mortars by providing and mixing Recron3S of Reliance Industries Ltd. @ 125gms per bag of cement as specified and directed by the Department complete at all levels.		
	A In concrete of proportion:		
	(i) 1 : 2 : 4 (1cement : 2coarse sand : 4aggregate) or M-15 grade.	Cum	204.52
	(ii) 1 : 1.5 : 3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade.	Cum	249.82
	(iii) 1 : 1 : 2 (1cement : 1coarse sand : 2aggregate) or M-25 grade.	Cum	368.72
	B In Plasters/ mortars of proportion:		
	(i) 1 : 3 (1cement : 3coarse sand)	Cum	312.11
	(ii) 1 : 4 (1cement : 4coarse sand)	Cum	238.49
	(iii) 1 : 5 (1cement : 5coarse sand)	Cum	207.35
22.3	WATER PROOFING OF CHAJJAS, ROOF SLABS, TERRACES, SUNKEN SLABS, SEPTIC TANK, OH TANK, RAFT SLAB IN BASEMENT, WALLS ETC.		
22.3.1	(A) Supplying, mixing, laying and applying the following waterproofing systems over Raft slabs, Roof slabs, Terraces etc. including curing complete as specified and directed by the Department at all levels.: (i) 1mm thick single coat of Master Crete M-81 of Choksey Chemicals /Dr. Sealkit Sealcrete (Asian Chemicals)/ mixed with cement in the ratio 1:2 as a prime coat. (ii) 3mm thick Polymer modified mortar - Cement sand mortar (in prop 1:3) mixed with Master Crete M-81/Dr. Sealkit Sealcrete (Asian Chemicals) @ 15% by weight of cement. (iii) 1mm thick coat of Master Crete M-81 /Dr. Sealkit Sealcrete (Asian Chemicals) mixed with cement in the ratio 1:2 as final coat.	Sqm	1208.50
	(B) Using RHEOMIX 131 OF BASF	Sqm	1291.75

22.3.2	Supplying, providing, mixing and laying over waterproofing layers/ coatings with concrete screed of average 30mm thick or as per drawings/ requirements in proportion 1:2:4 (1cement : 2fine aggregate : 4coarse aggregate) (by volume) with 10mm and down well graded aggregates and admixed with a normal plasticizer like Master Plast PL-1 of Choksey Chemicals /Dr. Sealkit Normal Plast (Asian Chemicals)/Rheomac 707 of BASF @ 0.3% by weight of cement and thereafter scoring the top surface of the concrete screed @ 200mm c/c as specified and directed by the Department.	Sqm	161.98
22.3.3	Providing, supplying, mixing, laying 8mm thick cement Plastering in prop.1:6 over R.C.C. surfaces mixed with integral waterproofing compounds like Master Proof IWP-1 of Choksey Chemicals /Dr. Sealkit No.1 (Asian Chemicals) /Rheomac 707 of BASF@ 0.2% by weight of cement conforming to IS:2465 as specified and directed by the Department complete.	Sqm	51.32
22.3.4	Providing, supplying, mixing and champhering the edges along joint of wall/ parapet wall and raft/ roof slab having chamfered dimension of 150mm x 150mm with cement sand mortar (prop.1:4) mixed with Master Proof IWP-1 of Choksey Chemicals /Dr. Sealkit No. 1 of (Asian Chemicals)/ RHEOMAC 707 of BASF@ 0.2% by weight of cement over a primer coat on the area to be champhered with a combination of 100% acrylic Polymer Master Crete M-81/Dr. Sealkit Sealcrete (Asian Chemicals)/ RHEOMAC 115 of BASF of Choksey Chemicals and cement in the ratio 1:2 to be applied by brush as specified & directed by the Department complete.	Rm	272.64
22.3.5	Providing and laying modified bituminous water proofing membrane system in roof, Polymod-AP of choksey Chemicals, 3mm thick APP modified waterproofing membrane by torch on method using LPG torch flame after applying a coat of Bituminous primer "Primer Polymod-SB" @ 0.2kg/sq.m. complete as specified and directed by the Department followed by : Supplying, providing, mixing and laying of concrete screed of average 75mm thick in proportion 1:2:4 (1cement : 2fine aggregate : 4coarse aggregate) (by volume) with 12mm and down well graded aggregates and admixed with a normal plasticizer like Master Plast PL-1 of Choksey Chemicals /Dr. Sealkit Normal Plast (Asian Chemicals) @ 0.3% by weight of cement after making the concrete fibre reinforced for reducing the effect of thermal expansion by mixing concrete with Recron 3S @ 1.25kg per bag of cement and/ as specified and directed by the Department.	Sqm	882.68

22.3.6	(A) Providing and applying over roof slab/ raft slab, polymer modified bituminous waterproofing membrane Proofex SM of Fosroc chemicals on an unprimed substrate using specially designed spray application equipment to be applied in a single coat, using wet on wet passes to give a nominal dry film thickness of 2mm followed by applying ProofexGeo Textile having weight of 200gm per sq.m immediately after the application of Proofex SM membrane when the surface is still wet but not tacky and as specified and directed by the Department.	Sqm	594.54
	(B) Providing and application of Sonosheild HLM 5000 of BASF India Ltd, the Liquid, cold-applied elastomeric Asphalt modified Polyurethane waterproofing membrane system of 1.2mm thick having average elongation of 600% and average tensile strength of 1 MPA. The product shall conform ASTM C 836 & National standard of Canada 37.58 - M86 developed by CGSB.	Sqm	1222.24
22.3.7	Supplying, providing, mixing and laying over waterproofing layers/ coatings with cement mortar screed of average 25mm thick or as per drawings/ requirements in proportion 1:4 (1cement : 4sand) admixed with integral waterproofing admixture Conplast X421IC of Fosroc Chemicals /Rheomac 707 of BASF/Dr. Sealkit Normal Plast of (Asian Chemicals)@ 125ml per bag of cement complete as specified and directed by the Department.	Sqm	219.00
22.3.8	Providing, supplying, mixing and coving the edges along joint of parapet wall & roof slab/ chajja & exterior wall/ walls & raft slabs etc. with cement sand mortar (prop.1:4) mixed with mortar modifier cum bonding agent Nitobond SBR of Fosroc Chemicals/Rheomix 141 of BASF/Dr. Sealkit Masterbond Latex (Asian Chemicals) @ 2-3Ltrs per bag of cement as specified & directed by the Department complete.		292.45
22.3.9	Providing and applying over roof/ raft slab, two coats of waterproofing slurry prepared with cement, fine sand in prop 1:1 admixed with modified acrylic polymer like Cemflex of Sika India Ltd./Dr. Sealkit Sealcrete (Asian Chemicals),/ RHEOMIX 131 OF BASF by brush @ 0.25% of cement-sand mixture (by weight) after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately) as specified & directed by the Department complete.	Sqm	218.45
22.3.10	Providing and applying over sunken slab etc., two coats of waterproofing slurry prepared with neat cement, water and modified Styrene Butadiene Rubber based polymer like Sika Latex of Sika India Ltd./Dr. Sealkit Masterbond Latex (Asian Chemicals)/ RHEOMIX 141 by brush after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately) as specified & directed by the Department complete.	Sqm	69.51

22.3.11	Supplying, providing, mixing and laying of concrete screed of average 20mm thick or as per drawings/ requirements in proportion 1:2:4 (1cement : 2fine aggregate : 4coarse aggregate) (by volume) with 10mm and down well graded aggregates and admixed with a waterproofing plasticizer like Plastocrete Plus of Sika India Ltd./Dr. Sealkit LW Super (Asian Chemicals) /Rheomac 730 of BASF@ 0.2% by weight of cement and thereafter scoring the top surface of the concrete screed @ 200mm c/c as specified and directed by the Department.	Sqm	106.23
22.3.12	15mm thick plastering of the interior surface of concrete water tanks with cement mortar in prop 1:4 admixed with modified Styrene Butadiene Rubber based polymer like Sika Latex of Sika India Ltd./Dr. Sealkit Masterbond Latex (Asian Chemicals)/Rheomix 141 of BASF @ 200gms per bag of cement as specified and directed by the department complete.	Sqm	103.35
22.3.13	Providing, supplying, mixing and champhering the edges along joint of parapet wall & roof slab/ chajja & exterior wall/ walls & raft slabs etc. having champhered dimension of 150mm x 150mm with cement sand mortar (prop.1:4) mixed with Plastocrete Plus of Sika India Ltd./Dr. Sealkit LW Super (Asian Chemicals) /Rheomac 707 of BASF@ 0.2% by weight of cement over a primer coat on the area to be champhered with a combination of modified Styrene Butadiene Rubber based polymer like Sika Latex of Sika India Ltd./Rheomix 141 of BASF and cement in the ratio 1:8 to be applied by brush as specified & directed by the Department complete.	Sqm	158.18
22.3.14	Providing and laying Polymeric waterproofing membrane sheets Hyperplas Standard of IWL India Ltd. @3kg/sq.m. having 90micron HMHDPE film as centre core with polymeric mix of softening point 118oC and penetration of 25-40 covered on both side and laminated with 20micron HMHDPE, layed over first coat of water based emulsion primer @ 100gms/sq.m. and using hot blown bitumen 90/15 or 85/25 @ 1.2kg/sq.m. followed by repeation of the bitumen coat and then final coat of bituminous aluminium @ 100gms/sq.m. complete as specified and directed by the Department followed by :	Sqm	510.98
22.3.15	Providing and applying of one coat of Techoxy of Choksey Chemicals as Primer coat on RCC roof surface followed by 2 (two) coats of Tar-O-Thane with necessary tools inclusive of labour charges in RCC buildings at all levels.	Sqm	437.38
22.3.16	Providing and applying two coats of coal tar epoxy at an interval of 12 - hours each including all necessary tools and application in R.C.C. buildings at all levels.	Sqm	148.24
22.4	WATER PROOF COATINGS:		

22.4.1	Applying a two coats of Futura Sealer Coat of Choksey Chemicals over concrete surface/ external walls after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately) followed by applying over sealer coat two coats of Futura-5 at a spread rate of 45sq.ft/litre after proper thinning with 10-20% of water complete as specified & directed by the Department.	Sqm	306.50
22.4.2	Applying one coat of epoxy coating Techoxy of Choksey Chemicals followed by two coats of Polyurethane waterproof coating Techcoat 61P at an interval of 12hours each for waterproofing of external walls, roofs, terraces, chajjas, etc. as specified & directed by the Department complete.	Sqm	476.72
22.4.3	Applying 2 coats of 1mm thick acrylic polymer modified cementitious waterproofing coating Brushbond of Fosroc Chemicals/Dr. Sealkit Flexbond AP 121 C (Asian Chemicals)/ MASTERSEAL 550 of BASF for waterproofing structures like roof slabs, basement, sunken slabs, chajjas, tunnels, water tank etc. after proper surface preparation and soaking the surface with water prior to application complete as specified and directed by the Department at all levels.	Sqm	326.41
22.4.4	Applying 2 coats of 1mm thick elastomeric cementitious waterproofing coating Brushbond RFX of Fosroc Chemicals /Dr. Sealkit Flexbond AP 122 C (Asian Chemicals) / MASTERSEAL 550EL of BASF for waterproofing structures like roof slabs, basement, sunken slabs, chajjas, tunnels, water tank etc. after proper surface preparation and soaking the surface with water prior to application complete as specified and directed by the Department at all levels.	Sqm	375.97
22.4.5	Applying two coats of aliphatic acrylate based exterior coating Dekguard Std of Fosroc Chemicals/Masterseal 300H of BASF over one coat of Silane siloxine based Dekguard Std Primer/Masterseal 380 of BASF Primer after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately) as specified & directed by the Department complete.	Sqm	245.85
22.4.6	Providing and applying over roof slab/ raft slab, two coats of waterproofing slurry like SikaTop Seal 107 of Sika India Ltd./Dr. Sealkit Flexbond AP 121 C (Asian Chemicals) ./ MASTERSEAL 550 @ 1.5kg per sq.m. per coat, by brush after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately) as specified & directed by the Department complete.	Sqm	390.64
22.4.7	Providing and applying of waterproofing coating like Inertol Proxitar of Sika India Ltd. by brush for waterproofing of septic tanks, sewage treatment plants etc. after proper surface preparation (surface preparation including thorough cleaning of surface to be measured and paid separately) as specified & directed by the Department complete.	Sqm	197.66

22.4.8	Applying over exterior walls/ masonry walls etc. two coats of Silaoxane silane based water repellant like Sikagard 703W/Masterseal 380 of BASF @ 250ml per sq.m. wet on wet after properly diluting with water as specified & directed by the Department complete.	Sqm	153.92
22.5	GROUTING IN R.C.C. WALLS, SLABS & RAFT SLABS:		
22.5.1	Providing Grouting in R.C.C. walls by drilling holes @ 1M c/c upto half of the wall depth in zigzag manner or as per drawings using a drill machine and fixing a nozzle of 18 to 20mm dia. for injecting a grout slurry of cement, water and expanding grout admixture Master grout CPG1 of Choksey Chemicals /Dr. Sealkit NSPG (Asian Chemicals)/ FLOWCABLE 50 OF BASF @ 0.5% by weight of cement till the refusal of injected grout material from adjacent nozzle (grouting operation pressure shall be in between 2 to 4kg per sq.cm.). Followed by cutting the exposed nozzle so as to make the wall free from the grouting pipes and to seal the gap with polymer modified mortar as specified and directed by the Department complete at all levels.	Sqm	327.35
22.5.2	Providing Grouting in R.C.C. walls & slabs by drilling holes @ 300mm c/c upto half of the wall depth in zigzag manner or as per drawings using a drill machine and fixing a nozzle of 20mm dia. for injecting a grout/ grout slurry till the refusal of injected grout material from adjacent nozzle (grouting operation pressure shall be in between 2 to 4kg per sq.cm.). Followed by cutting the exposed nozzle so as to make the wall free from the grouting pipes and to seal the gap with polymer modified mortar as specified and directed by the Department complete at all levels.		
22.5.2.1	Grout slurry of cement, water and grout admixture Cebex 100 of Fosroc Chemicals /Flowcable 50 of BASF/Dr. Sealkit NSPG (Asian Chemicals) @ 225gms per bag of cement	Sqm	361.54
22.5.2.2	Grout of two part low viscous epoxy grout Conbextra EP10 of Fosroc Chemicals/Concresive 1315 of BASF	Sqm	351.94
22.5.3	Providing Grouting by drilling holes @ 1M c/c upto half of the slab/ wall depth in zigzag manner or as per drawings using a drill machine and fixing a nozzle of 18 to 20mm dia. for injecting a grout slurry of cement admixed with expanding grout admixture Intraplast EP of Sika India Ltd./Dr. Sealkit NSPG (Asian Chemicals) / FLOWCABLE 50 OF BASF@ 1.0% by weight of cement and water resistant bonding agent Sika Latex, till the refusal of injected grout material from adjacent nozzle (grouting operation pressure shall be in between 3 to 4kg per sq.cm.). Followed by cutting the exposed nozzle so as to make the wall free from the grouting pipes and to seal the gap with polymer modified mortar as specified and directed by the Department complete at all levels.	Sqm	345.05
22.6	TREATMENT OF CONSTRUCTION JOINTS		

22.6.1	Providing treatment of constructional joints in R.C.C. walls and slabs by supplying, mixing and applying a 2-component epoxy resin based bonding agent for old to new concrete like Master Bond - EP/ Concrecive 1414 of BASF/Dr. Sealkit Bond OTN (Asian Chemicals) (with an open time of 6 hours at 30 deg Celsius) as per detailed drawings or as specified and directed by the Department complete at all levels.		
	(i) Slab/ Wall thickness upto 200mm	Rm	218.03
	(ii) Slab/ Wall thickness greater than 200mm	Sqm	1501.63
22.6.2	Providing treatment of constructional joints in R.C.C. walls and slabs by supplying, mixing and applying a epoxy resin based bonding agent for old to new concrete like Nitobond - EP /Dr. Sealkit Bond OTN (Asian Chemicals)/ CONCRECIVE 1414 as per detailed drawings or as specified and directed by the Department complete at all levels.		
	(i) Slab/ Wall thickness upto 200mm	Rm	86.06
	(ii) Slab/ Wall thickness greater than 200mm	Sqm	833.30
22.6.3	Providing treatment of constructional joints in R.C.C. walls and slabs by supplying, mixing and applying a single component modified styrene butadiene emulsion bonding agent for old to new concrete like Nitobond - SBR /Rheomix 141 of BASF/Dr. Sealkit MasterBond Latex (Asian Chemicals) as per detailed drawings or as specified and directed by the Department complete at all levels.		
	(i) Slab/ Wall thickness upto 200mm	Rm	25.10
	(ii) Slab/ Wall thickness greater than 200mm	Sqm	250.82
22.6.4	Providing treatment of constructional joints in R.C.C. walls and slabs by spraying surface retarder like Rugasol 2 liquid on old concrete surface followed by cleaning with water jets to obtain exposed aggregates prior to laying of new concrete and applying a 2-component epoxy resin based bonding agent for old to new concrete like Sikadur 32LP /Dr. Sealkit Bond OTN (Asian Chemicals) (with an open time of 6 hours at 30 deg Celsius) as specified and directed by the Department complete at all levels.		
	(i) Slab/ Wall thickness upto 200mm	Rm	59.59
	(ii) Slab/ Wall thickness greater than 200mm	Sqm	589.16
22.7	TREATMENT OF EXPANSION JOINTS		
22.7.1	Providing treatment of expansion joints by the following process as specified and directed by the department. (a) Preparation of groove by nosing the groove in required sizes with Polymer modified mortar - Cement sand mortar (in prop 1:3) mixed with Master Crete M-81 of Choksey Chemicals /Dr. Sealkit Sealcrete (Asian Chemicals) /Rheomix 115 of BASF@ 15% by weight of cement over a bonding coat of Master Crete M-81/Rheomix 131 of BASF mixed with cement in the ratio 1:2 as specified & directed by the Department complete after proper surface preparation including cleaning/ chipping removing all sorts of loose particles and foreign materials and cleaning surface thoroughly (surface preparation including thorough cleaning of surface to be measured and paid separately).	Rm	194.81

	(b) Providing and applying two coat of Techseal RDL 942 Primer in both surface of the expansion joint followed by applying Tech seal 940 Polysulphide sealant on vertical joint surface after proper fixing of the necessary backup materials as specified and directed by the Department complete.	Rm	1047.52
	(C) Providing and applying MASTERFLEX 700i of BASF India Ltd, two-component, high performance polysulfide formulation having weathering resistance to ultraviolet rays. The product shall exhibit shore A hardness of 25 and have movement accommodation factor of 25%. The sealant must comply with the performance specifications as laid in BS:4254 and ASTM C 920. All the joints must be primed using compatible primer for the substrate from the MASTERFLEX PRIMER range. Sealant application shall be carried out, strictly in accordance with Manufacturer's recommendations.	Rm	1172.62
	(d) (Upto 40 mm:) Providing and applying two coat of Masterflex Primer in both surface of the expansion joint followed by applying of one component Polyurathene based joint sealant MASTERFLEX 472/474 of BASF India Ltd, which is having specific gravity of 1.30gm/cm ³ , elongation at break will be > 600% and recoverability will be > 80% joint surface after proper fixing of the necessary backup materials as specified and directed by the Department complete.	Rm	1126.16
22.7.2	Providing treatment of expansion joints by the following process as specified and directed by the department. (a) Preparation of groove of required sizes by nosing the groove with cement sand mortar (prop.1:4) mixed with mortar modifier cum bonding agent Nitobond SBR of Fosroc Chemicals /Rheomix 141 of BASF/Dr. Sealkit Master Bond Latex (Asian Chemicals) @ 2-3Ltrs per bag of cement as specified & directed by the Department complete after proper surface preparation including cleaning/ chipping removing all sorts of loose particles and foreign materials and cleaning surface thoroughly (surface preparation including thorough cleaning of surface to be measured and paid separately).	Rm	204.99
	(b) Providing and applying two coat of Primer 4 in both surface of the expansion joint followed by applying polysulphide sealant Thioflex 600 on vertical joint surface after proper fixing of the necessary backup materials as specified and directed by the Department complete.	Rm	1119.56

22.7.3	Providing treatment of expansion joints by the following process as specified and directed by the department. (a) Preparation of groove by nosing the groove in required sizes with Cement sand mortar (in prop 1:4) mixed with Sika Latex of Sika India Ltd. /Dr. Sealkit Master Bond Latex (Asian Chemicals)@ 4% by weight of cement as specified & directed by the Department complete after proper surface preparation including cleaning/ chipping removing all sorts of loose particles and foreign materials and cleaning surface thoroughly (surface preparation including thorough cleaning of surface to be measured and paid separately).	Rm	142.16
	(b) Providing and applying two coat of Sika Primer P in both surface of the expansion joint followed by applying Sikaflex Construction Polyurethane based sealant on vertical joint surface after proper fixing of the necessary backup materials as specified and directed by the Department complete.	Rm	560.24
22.8	CURING COMPOUNDS:		
22.8.1	Supplying and applying of Mastercure WB1 / WB2 of Choksey Chemicals (india) Ltd. /Dr. Sealkit Cure CC 102/ CC101 (Asian Chemicals) /MASTERKURE 106i OF BASF on the newly placed concrete as soon as possible after the concrete is free from visible surface water by spray method as specified and directed by the department.	Sqm	39.76
22.8.2	Supplying and applying of Concure WB of Fosroc (india) Ltd./Masterkure 107i of BASF on the newly placed concrete as soon as possible after the concrete is free from visible surface water as specified and directed by the department.	Sqm	52.88
22.9	METALLIC/ NON-METALLIC/ EPOXY BASED FLOOR HARDNER/ TOPPINGS		
22.9.1	Extra for Providing and applying Lithium silicate based concrete floor hardner Nitoflor LH of Fosroc (india) Ltd. using high volume sprayer or hand applying using lambs wool roller as specified and directed by the department.	Sqm	94.87
22.9.2	Applying a floating coat of concrete floor hardner Nitoflor Hardtop or Nitoflor Hardtop standard of Fosroc (india) Ltd. over freshly prepared concrete floors followed by sprinkling of Nitoflor Hardtop or Nitoflor Hardtop standard and finishing surface using power trowel as specified and directed by the department (Concrete floor & base courses to be measured and paid separately).		
	(a) Heavy duty	Sqm	261.60
	(b) Medium duty	Sqm	211.65
	(c) Light duty	Sqm	161.70
22.9.3	Extra for Providing Non-metallic, ready to use concrete floor hardner, Techfloor HT 200 of Choksey India Ltd. as specified and directed by the department.		
22.9.3.1	Cement Grey Shade		
	(a) Heavy duty	Sqm	311.72
	(b) Medium duty	Sqm	247.45
	(c) Light duty	Sqm	183.18
22.9.3.2	Dark Shade		
	(a) Heavy duty	Sqm	494.42
	(b) Medium duty	Sqm	384.03

	(c) Light duty	Sqm	273.64
22.9.3.3	Light Shade		
	(a) Heavy duty	Sqm	603.84
	(b) Medium duty	Sqm	453.57
	(c) Light duty	Sqm	303.30
22.9.4	Extra for Providing Non-metallic, ready to use concrete floor hardner, Chapdur of Sika India Ltd./ MASTERTOP 100 of BASF as specified and directed by the department.		
	(a) Heavy duty	Sqm	558.68
	(b) Medium duty	Sqm	416.32
	(c) Light duty	Sqm	273.96
22.9.5	Extra for Providing and applying 1mm self smoothing epoxy floor topping Nitoflor SL1000 of Fosroc (india) Ltd./ MASTERTOP 1210 PLUS OF BASF using 1mm serrated trowel and rolled with a spiked nylon roller after proper surface preparation and application of a priming coat of Nitoprime 25(M) as specified and directed by the department (surface preparation to be measured and paid separately).	Sqm	743.86
22.9.6	Extra for Providing and applying 1.5mm to 3mm self smoothing epoxy floor topping Nitoflor SL2000 of Fosroc (india) Ltd./ MASTERTOP 1230 PLUS of BASF using serrated trowel and rolled with a spiked nylon roller after proper surface preparation and application of a priming coat of Nitoprime 25(M) as specified and directed by the department (surface preparation to be measured and paid separately).	Sqm	1219.89
22.9.7	Extra for Providing and applying low build polysulphide-epoxy floor coatings, Techfloor LBC-140 of Choksey (india) Ltd. using brush or lambs wool roller after proper surface preparation and applying techfloor primer coat complete as specified and directed by the department (surface preparation to be measured and paid separately).	Sqm	442.62
22.9.8	Extra for Providing and applying high build polysulphide-epoxy floor coatings, Techfloor HBC-150 of Choksey (india) Ltd. using medium pile roller or a squeegee after proper surface preparation complete as specified and directed by the department (surface preparation to be measured and paid separately).	Sqm	1189.89
22.9.9	Extra for Providing and applying self leveling polysulphide-epoxy floor topping Techfloor SL-2 of Choksey (india) Ltd. /Dr. Sealkit EPU Floor Self L (Asian Chemicals) using steel trowel and rolled after proper surface preparation as specified and directed by the department (surface preparation to be measured and paid separately).	Sqm	1295.60
22.9.10	Providing and applying normal setting, self levelling epoxy modified cementitious floor topping Sikafloor 81 EpoCem of Sika (india) Ltd. using trowel followed by applying self smoothing epoxy floor coating Sikafloor 261(S) of Sika (india) Ltd. /Dr. Sealkit EPU Floor Self L (Asian Chemicals) / MASTERTOP 542 ECC of BASF using serrated trowel and rolled with a spiked nylon roller after proper surface preparation and application of a priming coat of Sikafloor 80 Primer as specified and directed by the department (surface preparation to be measured and paid separately).	Sqm	2024.16

22.1	REPAIR AND REHABILITATION OF STRUCTURES:		
22.10.1	Repairing of spalled concrete in columns by the following processes as specified and directed by the department complete at all levels:		
	A. Chipping of deteriorated concrete and removal of all loose & friable materials and fully exposing rusted reinforcement followed by proper cleaning and removing of rusts and other foreign materials using sand blasting/ emery cloths/ wire brush etc.	Sqm	88.13
	B. Corroded reinforcement to be replaced by cutting and welding new reinforcement of minimum 25mm dia. and overlapping on both side as specified and directed.(reinforcement bar to be measured and paid separately).	Rm	110.87
	C. Applying two coats of cement based polymer modified anti-corrosive protective coating over exposed reinforcement including proper curing for minimum 24 hours.		405.73
	D. Applying Epoxy based bonding agent Master Bond EP of Choksey /Concresive 1414 of BASF/Dr. Sealkit Bond OTN (Asian Chemicals) after totally saturating the cleaned concrete surface with clean water for proper bonding of old and new concrete.	Sqm	716.02
	E. Concreting the column with 1:1.5:3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade concrete admixed with a normal plasticizer like Master Plast PL-1/Dr. Sealkit Normal Plast (Asian Chemicals) @ 0.2%-0.5% by weight of cement after making the concrete fibre reinforced for reducing the effect of thermal expansion by mixing concrete with Recron 3S @ 1.25kg per bag of cement followed by proper curing for minimum 28days (Formwork to be measured and paid separately).	Cum	7153.94
	F. Plastering with high rich polymer modified mortar 10mm thick with cement sand mortar in prop. 1:4 mixed with 100% acrylic polymer Master Crete M-81 /Dr. Sealkit Sealcrete (Asian Chemicals) / RHEOMIX 131 of BASF @ 15% by weight of cement followed by proper curing as specified and directed by the Department complete.	Sqm	204.48
22.10.2	Repairing of spalled concrete in slabs/ chajjas etc. by the following processes as specified and directed by the department complete at all levels:		
	A. Chipping of deteriorated concrete and removal of all loose & friable materials and fully exposing rusted reinforcement followed by proper cleaning and removing of rusts and other foreign materials using sand blasting/ emery cloths/ wire brush etc.	Sqm	88.13
	B. Applying two coats of cement based polymer modified anti-corrosive protective coating Thoro Stucturite Primer of BASF over exposed reinforcement including proper curing for minimum 24 hours.		405.73
	C. Applying Epoxy based bonding agent Master Bond EP of Choksey /Concresive 1414 of BASF/Dr. Sealkit Bond OTN (Asian Chemicals)/Rheomix 141 of BASF after totally saturating the cleaned concrete surface with clean water for proper bonding of old and new concrete.	Sqm	716.02

	D. Concreting the slab/ chajja etc. with concrete admixed with a normal plasticizer like Master Plast PL-1 /Dr. Sealkit Normal Plast (Asian Chemicals) @ 0.3% by weight of cement after making the concrete fibre reinforced for reducing the effect of thermal expansion by mixing concrete with Recron 3S @ 1.25kg per bag of cement followed by proper curing (Formwork & concrete to be measured and paid separately).	Cum	1978.70
	E. Applying two coats of 2mm thick slurry of 100% pure acrylic polymer like Master Crete M-81/Dr. Sealkit Sealcrete (Asian Chemicals) mixed with ordinary portland cement in prop. 1:2.	Sqm	373.54
	F. Plastering with 20mm thick cement sand mortar in prop.1:4 mixed with integral waterproofing compounds like Master Proof IWP-1 of Choksey Chemicals /Dr. Sealkit No.1 (Asian Chemicals) @ 0.2% by weight of cement conforming to IS:2465 as specified and directed by the Department complete.	Sqm	370.88
22.10.3	Repairing of spalled concrete in columns by the following processes as specified and directed by the department complete at all levels:		
	A. Chipping of deteriorated concrete and removal of all loose & friable materials and fully exposing rusted reinforcement followed by proper cleaning and removing of rusts and other foreign materials using sand blasting/ emery cloths/ wire brush etc.	Sqm	100.51
	B. Corroded reinforcement to be replaced by cutting and welding new reinforcement of minimum 25mm dia. and overlapping on both side as specified and directed.(reinforcement bar to be measured and paid separately).	Rm	110.87
	C. Applying two coats of single component epoxy zinc primer Nito Primer Zinc Rich 90 of Fosroc chemicals with brush over cleaned reinforcement.	Sqm	147.56
	D. Providing and fixing shear connectors of 8mm dia. by drilling 14mm dia. holes on the concrete surface to a depth of 75mm using a rotary cum hammering drilling machine as per specification and thoroughly cleaning the hole using compressed air and water jet and anchoring with polyester resin anchor grout Lokfix P of Fosroc chemicals./ Masterflow 935 of BASF	Each	145.75
	E. Providing and fixing sacrificial Zinc anode, Galvashield XP of Fosroc chemicals with reinforcement bars @ 1 unit per 0.5m run of reinforcement, tightened using a suitable wire-twisting tool restricting free movement and ensuring good electrical continuity.	Rm	1772.38
	F. Supplying, mixing and applying a epoxy resin based bonding agent like Nitobond - EP/ B CONCRETSIVE 1414 of BASF for proper bond of old to new concrete .	Sqm	813.05

	G.(i) Applying structural grade prepacked polymer modified cementitious mortar Renderoc S2/Emaco S88T of BASF for application thickness upto 20mm overhead/ 30mm verticals, after proper mixing with clean water followed by proper curing for minimum 28days (Formwork to be measured and paid separately).	Cum	81852.61
	G (ii). Applying structural grade prepacked polymer modified cementitious mortar Renderoc HB2/Emaco S48CT of BASF for application thickness upto 50mm overhead/ 80mm verticals, after proper mixing with clean water followed by proper curing for minimum 28days (Formwork to be measured and paid separately).	Cum	140144.25
	G.(iii) Applying and placing cementitious micro-concrete Renderoc RG/Emaco S46T of BASF after proper mixing with clean water after erection of formwork followed by proper curing for minimum 28days (Formwork to be measured and paid separately).	Cum	69317.07
22.10.4	Repairing of spalled concrete in slabs/ chajjas etc. by the following processes as specified and directed by the department complete at all levels:		
	A. Chipping of deteriorated concrete and removal of all loose & friable materials and fully exposing rusted reinforcement followed by proper cleaning and removing of rusts and other foreign materials using sand blasting/ emery cloths/ wire brush etc.	Sqm	100.51
	B.(i) Applying two coats of single component epoxy zinc primer Nito Primer Zinc Rich 90 of Fosroc chemicals with brush over cleaned reinforcement.	Sqm	147.56
	B.(ii) Applying two coats of single component epoxy zinc primer Concrecive ZRI of BASF India Ltd, a two component, epoxy zinc rich. It shall be formulated to meet the requirement of BS 4652 Type 2. The primer shall be an active type having zinc content of greater than 90% in dry film state which is capable of negating the generation of incipient anodes in the areas surrounding the repairs. The product shall have mix density not less than 2.25 kg/litre. It shall be fully compatible with EMACO Range of repair mortars. with brush over cleaned reinforcement.	Sqm	266.62
	C. Providing and fixing sacrificial Zinc anode, Galvashield XP of Fosroc chemicals with reinforcement bars @ 1 unit per Sqm of reinforcement, tightened using a suitable wire-twisting tool restricting free movement and ensuring good electrical continuity.	Sqm	895.02
	D. Supplying, mixing and applying a epoxy resin based bonding agent like Nitobond - EP/ CONCRECIVE 1414 of BASF for proper bond of old to new concrete .	Sqm	813.05
	E. (i) Applying cement sand mortar (prop.1:4) admixed with mortar modifier cum bonding agent Nitobond SBR of Fosroc Chemicals/Rheomix 141 of BASF @ 5Ltrs per bag of cement as specified.	Cum	492.33

	E.(ii) Applying and placing cementitious micro-concrete Renderoc RG/Emaco S46T of BASF after proper mixing with clean water after erection of formwork followed by proper curing for minimum 28days (Formwork to be measured and paid separately).	Cum	69317.07
	E.(iii). Concreting the slab/ chajja etc. with concrete 1:1.5:3 (1cement : 1.5coarse sand : 3aggregate) or M-20 grade admixed with Conplast P211 of fosroc chemicals @ 150ml per bag of cement followed by proper curing (Formwork & concrete to be measured and paid separately).	Cum	5598.90
22.10.5	Repairing of spalled concrete in columns by the following processes as specified and directed by the department complete at all levels:		
	A. Chipping of deteriorated concrete and removal of all loose & friable materials and fully exposing rusted reinforcement followed by proper cleaning and removing of rusts and other foreign materials using sand blasting/ emery cloths/ wire brush etc.	Sqm	88.13
	B. Corroded reinforcement to be replaced by cutting and welding new reinforcement of minimum 25mm dia. and overlapping on both side as specified and directed.(reinforcement bar to be measured and paid separately).	Rm	123.25
	C.(i) Applying two coats of cement based polymer modified anti-corrosive protective coating Sika Top Armatec 108 of Sika (India) Ltd. over exposed reinforcement including proper curing for minimum 24 hours.	Sqm	61.00
	C.(ii) . Applying two coats of cement based polymer modified anti-corrosive protective coating Sika Topseal of Sika (India) Ltd. over exposed reinforcement including proper curing for minimum 24 hours.	Sqm	69.52
	D. Applying Epoxy based bonding agent Sikadur 32 LP @ 250gms per sq.m. per coat for proper bonding of old and new concrete.	Sqm	411.71
	E. Applying and placing cementitious micro-concrete Sika Rep by proper mixing with clean water after erection of formwork followed by proper curing for minimum 28days (Formwork to be measured and paid separately).	Cum	59059.65
22.10.6	Repairing of spalled concrete in slabs/ chajjas etc. by the following processes as specified and directed by the department complete at all levels:		
	A. Chipping of deteriorated concrete and removal of all loose & friable materials and fully exposing rusted reinforcement followed by proper cleaning and removing of rusts and other foreign materials using sand blasting/ emery cloths/ wire brush etc.	Sqm	88.13
	B. (i) Applying two coats of cement based polymer modified anti-corrosive protective coating Sika Top Armatec 108 of Sika (India) Ltd./ THORO STRUCTURITE PRIMER of BASF over exposed reinforcement including proper curing for minimum 24 hours.	Sqm	61.00

	B.(ii) Applying two coats of cement based polymer modified anti-corrosive protective coating Sika Topseal of Sika (India) Ltd./Dr. Sealkit Flexibond AP 121 C (Asian Chemicals) over exposed reinforcement including proper curing for minimum 24 hours.	Sqm	69.52
	C. Applying Epoxy based bonding agent Sikadur 32 LP/Dr. Sealkit Bond OTN (Asian Chemicals) @ 250gms per sq.m. per coat for proper bonding of old and new concrete.	Sqm	411.71
	D. Providing, supplying and placing of two component Polymer Modified Mortar Sika Top 122 after erection of formwork as per specification.	Cum	9963.00
22.11	STRENGTHENING OF COLUMNS, BEAMS & SLABS:		
22.11.1	Restrengthening of RCC members by the following processes as specified and directed by the department complete at all levels:		
	A) Doing patch work in depression of concrete surface wherever necessary by laying epoxy putty, Nitocote VF of Fosroc Chemicals as specified & directed by the Department complete after proper surface preparation and proper scraping of concrete protrusion (surface preparation including thorough cleaning of surface to be measured and paid separately).	Sqm	1115.96
	B) Applying of Nitowrap 410 saturant with wet film thickness @ 250 microns over a coat of saturant epoxy primer Nitowrap 30 of fosroc chemicals with brush followed by fixing of Carbon fibre composite system Nitowrap EP(CF) by pressing on the saturant and impregnate the same by applying a final coat of Nitowrap 410 as specified and directed by the department.		
	(i) In Columns	Sqm	3628.21
	(ii) In Slabs & Beams	Sqm	3658.41
	C) Applying of two coats of two component UV resistant coating Nitowrap 512 over fibre system having wet film thickness of 100 micron as specified and directed by the department.		
	(i) In Columns	Sqm	2799.20
	(ii) In Slabs & Beams	sqm	2843.20
22.11.2	Restrengthening of RCC members by the following processes as specified and directed by the department complete at all levels:		
	A) Doing patch work in depression of concrete surface wherever necessary by laying epoxy putty, Concreseive 2200 of BASF India Ltd. as specified & directed by the Department complete after proper surface preparation and proper scraping of concrete protrusion (surface preparation including thorough cleaning of surface to be measured and paid separately).	Sqm	1614.10
	B) Applying of MBACE Fibre strengthening system of BASF India Ltd including laminate and sheet products for increasing the strength of concrete structures. Carbon, Aramid and E-glass can be used to solve problems from change of use to increased load carrying capacity.		

	(i) In Columns : Supply and Application of MBrace CF 230 - Unidirectional Carbon Fiber Reinforcement System of BASF India Ltd having 400Gm/sq.mt weight and having Modulus of elasticity of 240 kN/mm ² at the entire length of the beam with one layer after proper application of Concrecive 2200 at the entire beam area at a thickness of 2mm, which will be followed by the application of Mbrace saturant after proper curing of Concrecive 2200. Mbrace fiber to be applied over Mbrace Saturant at the green condition and after proper wrapping the secod layer of Mbrace Saturant to be applied. Sand to be broadcasted over the Mbrace saturant layer in green condition for application of any top screed or plastering.	Sqm	4707.56
	(ii). In Slabs & Beams: Supply and application of Ready to use Carbon laminates (Fibre Reinforced Polymer) for the reinforcement of concrete elements with MBrace FRP Laminate of 165/2500,50mm wide & 1.4mm thick System at the bottom portion of the beam, after application of Mbrace Laminate Adhesive , which is having minimum 2.5Mpa adhesion strength.	Rm	3451.77
	(iii) Strenghtening With Glass Fibre	Sqm	2653.10
22.12	Providing and application of Insultech safeguard coating for heat reflecting and waterproofing of approved colour as per direction of engineer in charge	sqm	416.96
22.13	Providing applying "XYPEX" Concrete Waterproofing & Durability Enhancement by Crystallization" to be supported by at least two International Test reports showing clear Scanning Electron Microscopic (SEM) photographs of desified and mature crystals forming to plug all the capillary tracks and pores of the concrete permanently and permeability test as per IS 2720 conducted by CBRI. The concrete should not absorb water more than a.5 grams and not more than 1 mm from the concrete surface. The product to be listed in ICC-ES-EVALUATION REPORT INDEX January 2009. The treatment to be carried out on the concrete surfaces as per manufacture's specification	Sqm	1319.77

CH-23: DISMANTLING AND DEMOLISHING

Item No.	Description of works	Unit	Rate
23	DISMANTLING WORKS		
23.1	Demolishing lime concrete without damaging the structure and disposal of debris as directed at all levels	Cum	160.11
23.2	Demolishing plain cement concrete including disposal of debris as directed for all levels		
	(i) Prop (1 : 4 : 8) or leaner mix	Cum	313.80
	(ii) Prop (1 : 3 : 6) or richer mix	Cum	545.40
23.3	Demolishing RCC work including stacking of steel bars and disposal of unserviceable material as directed for all levels	Cum	767.20
23.4	Demolishing RB work including stacking of steel bars and disposal of unserviceable material as directed for all levels	Cum	673.40

23.5	Extra for cutting reinforcement in RCC or RB work at all levels (Payment to be made on cross sectional area of RCC or RB work obtained by multiplying the length of the cut reinforcement and thickness of the member).	Sqm	402.80
23.6	Extra for scraping, cleaning and straightening reinforcement from RCC or RB work at all levels	Kg.	1.90
23.7	Demolishing brickwork including stacking of serviceable materials and disposal of unserviceable materials as directed for all levels	Cum	435.60
23.8	Demolishing stone rubble masonry including stacking of serviceable materials and disposal of unserviceable materials as directed for all levels	Cum	522.90
23.9	Dismantling wood work in frames, trusses, purlins and rafters including stacking the materials as directed by the department for all levels.	Cum	1050.20
23.10	Dismantling steel work in single sections (R.S.Joists, channels, angles, tees and flats etc.) including dismembering and stacking the materials as directed by the Department for all levels.	Kg.	1.10
23.11	Dismantling steel work in built up sections in joists, angles, channels, tees, flats etc. including all gusset plates, bolts, nuts, cutting rivets, welding etc. including dismembering and stacking the materials as directed by the Department for all levels	Kg.	2.20
23.12	Dismantling all types of corrugated/ trapezoidal sheet roofing including ridges, hips, valleys and gutters etc. and stacking the materials as directed by the department for all levels	Sqm	34.20
23.13	Demolishing skirting/ dado in walls/ R.C.C. & P.C.C. members without causing damage to walls/ member upto any height including disposal of debris as directed for all levels	Cum	3849.90
23.14	Demolishing the plastering from walls/ R.C.C. and P.C.C. members without causing any damage to the wall/ member upto any height including disposal of debris as directed for all levels.	Cum	3339.00
23.15	Demolishing CC floors of various thickness including top layer and base course and disposal of debris as directed for all levels.	Sqm	102.60
23.16	Demolishing mosaic floor of various thickness, both mosaic tiles and cast-in-situ type including CC base course and removal of debris as directed for all levels.	Sqm	138.80
23.17	Demolishing R.C.C./ P.C.C. pile head of different diameters carefully without disturbing the remaining portion of the pile including opening out of the pile head to be demolished by excavating the earth upto required depth and clearing the debris as directed.	Cum	707.90
23.18	Dismantling and removing the existing door and window frame (chowkath) both wooden and steel of different sizes without damaging the walls and removing the frame and necessary repairs to the edges of the walls as directed for all levels.		
	(i) Area of door/window up to 3.00 m2	Each	64.65
	(ii) Area of door/window above 3.00 m2	Each	102.47

23.19	Dismantling of existing steel windows/ steel ventilators/ steel windows and steel ventilators without damaging the walls, removing to a safe place and necessary repairs to the edges of walls.	Sqm	101.20
23.20	Dismantling all types of ceiling, partition and wall lining and stacking the materials as directed by the department for all levels	Sqm	22.40

CH- 24: BAMBOO & WIRE-NETTING WORKS

Item No	Description	Unit	Rate
24	BAMBOO AND WIRE NETTING WORKS		
24.1	Spilt bamboo diagonally woven wall tied with necessary bamboo kamis on both sides @300mm apart tied with galvanised wire/cane slit etc complete as directed.	Sq.m	176.07
24.2	Bamboo slit walling in timber frame with bamboo slit fitted in to the grooves of the timber frame and stiffened by means of double bamboo slits 25mm wide and not less than 6mm thick spaced at intervals of not more than 400mm apart one slit attached on each side and tied together by means of cane slit/galvd tying wire including 20mm thick cement plaster in prop 1 cement : 6 fine sand on both sides in two coats with curing etc complete (measurement for timber frame shall be measured and paid separately.)	Sq.m	410.34
24.3	Renewing / providing single tarja enclosure walls made of Muli / jati bamboo with mature Baruah / Bhaluka bamboo (87 mm to 100 mm dia) posts 1300 mm apart and 1800 mm height (approx) above ground and driven 750 mm below ground stiiffened with Muli / Jati bamboo kamis on both the faces 4 lines, including supplying and tying with galvd.wire complete as directed.	Sq.m	134.00
24.4	Renewing / providing Muli / Jati bamboo double tarja enclosure walls with Baruah / Bhaluka bamboo 87 mm to 100 mm dia.1800 mm apart and 1800 mm height (approx) above ground and driven 750 mm below ground and half Muli / Jati bamboo on both the faces 4 lines, including supplying and tying with galvd.wire complete as directed.	Sq.m	242.24
24.5	Renewing / providing double lattice pattern fencing with mature Baruah / Bhaluka bamboo posts 87 mm to 100 mm dia., 950 mm to 1050 mm above ground and minimum 600mm below ground and 4 lines of double kamies with half round Muli / Jati bamboo on top including supplying and tying with galvd.wire complete as directed.	Sq.m	61.57
24.6	Providing jail enclosure wall with matured Jati / Bhaluka / Makhal bamboo 60 mm to 75mm dia.and of height of 3000 mm including taking out and refitting the old timber battens with old and new bolts and nuts, nails etc. and coaltaring (Indoor Coat) two coats on the bamboos and battens and removing and stacking the old materials in proper places as per direction of the deptt.out side the jail compound walls.	Sq.m	480.24
24.7	Providing bamboo sleeping machang with necessary Bhaluka bamboo post etc. complete.	Sq.m	254.49

24.8	Providing, fitting, fixing 20 mmx 50 mm mesh expanded metal wall of 18 gauge (1.2 mm thickness) in sal wood dressed frame not more than 1.20 m centre to centre including sal wood beading of 50 mm x 25 mm size sides chamfered , fitted with necessary wood screws spaced not more than 400 mm apart etc. complete as directed (sal wood frame to be measured and paid separately).	Sq.m	425.65
24.9	Providing, fitting and fixing wire netted wall with 13 mm hexagonal mesh made of 20 gauge (0.95mm) over hollock wood dressed frame not more than 600 mm apart including hollock wood beading of 50 mm x 25 mm size sides chamfered , fitted with necessary wood screws spaced not more than 300 mm apart etc. complete as directed (Hollock wood frame to be measured and paid separately).	Sq.m	513.68
24.10	Providing fly proof M.S. wire netting of I.S.gauge designation 85 with wire of dia 0.05 mm over hollock or other wood dressed frame including 1st class timber beading of 38mmx12 mm size chamfered including fitting, fixing with necessary wood screws etc.complete as directed (timber frame to be measured and paid separately).	Sq.m	471.36
24.11	Providing bird proof hexagonal galvanised wire netting at ceiling level of Assam type building including fitting and fixing with first class local wood frame of size 50mm x 25mm and beads of size 25mm x 10mm (for fitting and fixing the net with wooden frame) including lifting and fitting the same in position etc. complete as directed including two coats kiricide oiling of timber works.	Sq.m	185.55

CH-25: MISCELLANEOUS WORKS

Item No.	Description	Unit	Rate
25	MISCELLANEOUS WORKS		
25.1	Construction of temporary shed with 300mm high plinth and floor , Bhaluka bamboo post driving at least 750mm under ground, height above plinth 2400mm,wattle wall mud plastered and white wash two coats, 100mm thick thatched roof, double mat doors and windows etc. complete as directed as specified.		
	a) With whole Jati bamboo rua frame	Sqm	621.79
	b) With Jati bamboo kami frame	Sqm	557.37
25.2	Supplying fitting and fixing 750mm long F.I.double straps at the bottom of timber posts with 16mm dia. double bolt and nuts and 550 mm length of straps to be embedded in cement concrete pillar in prop 1:3:6 and balance 200 mm length to remain above with bottom 75 mm to be bent at right angles etc. complete as directed (C.C work to be paid separately)		
	(a) With 50mm x 6mm F.I. Double strap.	Set	357.57
	(b) With 45mm x 6mm F.I. Double strap.	Set	221.99
25.3	Jungle clearing including cutting bamboo and trees below 500mm girth and removing the same outside the limit as directed.		
	(a) Light jungle.	Sqm	0.79
	(b) Medium jungle.	Sqm	1.61

25.4	Felling and removing		
	a) Bamboo clumps	Sqm	77.69
	b) Bamboo clumps including uprooting stumps.	Sqm	108.49
	c)Cutting trees including uprooting and removing where necessary.		
	(i) 500 mm to 1000mm girth.	Each	77.58
	(ii) 1000 mm to 1500mm girth.	Each	154.57
	(iii) 1500 mm to 1800mm girth.	Each	306.80
	(iv) 1800mm girth and above	Each	610.69
	N.B.- All girth measurement for trees are to be taken at 1200 mm height above the ground.		
	d) Tea bushes	Each	8.05
	e) Tea bushes including uprooting	Each	17.03
	f) Cutting and clearing bushes consisting of plants and all sorts of bambooes including uprooting and removing as necessary and filling up the holes with earth,dressing,ramming etc. complete.	Sqm	11.08
25.5	Surface dressing of ground including removing vegetation and/ or undulation of ground not exceeding 150mm deep and disposal of rubbish outside the periphery and lift up to 1800mm in all kinds of soil.	Sqm	4.69
25.6	Jungle clearence including uprooting of vegetation, grass, brush-wood, sapling and trees not exceeding 300mm girth and disposal of the same outside the periphery and lift upto 1500mm	Sqm	5.47
25.7	Cutting down trees where the building is to beconstructed of girth exceeding 300mm, grubbing up roots, filling pits and levelling off ground lopping branches, trimming and removeing to distance beyond 30m and stacking serviceable materials and cleaning site etc. as directed.		
	i) Girth exceeding 300mm but not exceeding 1000mm	Each	1044.14
	ii) Girth exceeding 1000mm but not exceeding 1500mm	Each	1810.01
	iii) Girth exceeding 1500mm but not exceeding 2000mm	Each	1927.28
	iv) Girth exceeding 2000mm but not exceeding 2500mm	Each	2383.63
	v) Girth exceeding 2500mm but not exceeding 3000mm	Each	2606.98
25.8	Providing and fixing galvd. / M.S. tube (light grade) outlet (spout) in cement mortar 1:3 including cutting and making good the wall		
	(a) 25mm bore - 300mm long.	Each	57.53
	(b) 40mm bore - 300mm long.	Each	57.53
25.9	Providing 200mm to 450mm long G.I. pipe spout embedded in C.C./ brick works for drainage of water as per direction of the Department including necessary breaking and repair and sealing of joints for all levels.		
	(a) 15mm diameter	Each	144.24
	(b) 20mm diameter	Each	145.70
	(c) 25mm diameter	Each	159.96
	(d) 32mm diameter	Each	171.95
	(e) 40mm diameter	Each	182.45
25.10	Supplying fitting and fixing 200mm x 6mm thick A.C. plain sheet for vertical and horizontal expansion joints fixed to one side of column/ wall by necessary rawal plugs/ screws/ wooden plugs placed not more than 300mm c/c as directed and specified.	Rm	123.66

25.11	Supplying, fitting and fixing in position R.C.C boundary pillars of size 150mm x 150mm x 1500mm long, 750mm above ground and 750mm below ground, including square base of 450mm x 450mm x 150mm with 4 nos. 12mm MS rods main reinforcement and 6mm MS stirrups at 300mm centres tied in position with annealed binding wire, cement concrete proportion 1:2:4 with broken stone aggregate upto 20mm size including cantering moulding the top, curing, 6mm thick cement plastering in proportion 1:3, painted black and white alternately in 230mm strips upto 750mm from the top keeping necessary grooves etc. as per design and direction complete.	Each	784.35
25.12	Providing precast R.C.C slab over drain, septic tank etc. in prop. 1:2:4 reinforced with 10mm M.S. bar @ 150mm centre both ways tying with 20 gauge black annealed wire with necessary shuttering, curing etc. complete including fixing in position as directed.		
	a) 100mm thick slab	Sqm	1410.90
	b) 75mm thick slab	Sqm	1277.11
25.13	Precast concrete block masonry (including quoin blocks, jamb blocks, closure etc.) with hollow cement concrete blocks of approved size conforming to I.S. 2185 - 1965 with 1:3:6 mix (1 cement: 3 coarse sand: 6 graded stone aggregate, 20 mm and down gauge) having solid materials not less than 50% total volume of block including curing complete as directed.		
	(a) 200 mm thick in foundation and substructure up to plinth level in cement mortar (1:6)	Sqm	628.98
	(b) 200 mm thick in super structure in cement mortar (1:6)	Sqm	642.36
	(c) 100mm thick in super structure in cement mortar (1:4)	Sqm	629.15
25.14	Precast concrete block masonry (including quoin blocks, jamb blocks, closure etc.) with hollow cement concrete blocks of approved size conforming to I.S. 2185 - 1965 with 1:6 mix (1 cement: 6 graded stone aggregate, 20 mm and down gauge) 1/4th having solid materials not less than 50% total volume of block including curing complete as directed.		
	(a) 200 mm thick in cement sand mortar (1:6) in foundation and substructure up to plinth level	Sqm	628.98
	(b) 200 mm thick in super structure in cement mortar (1:6)	Sqm	644.45
	(c) 100mm thick in super structure in cement mortar (1:4)	Sqm	490.79
25.15	Labour for fitting and fixing hume pipes in positions with coal tar immediately including caulking and joining pieces with cement mortar of proportion 1:3 as per direction of the Department including all carriage.		
	i) From 300 mm to 600 mm dia.	Rm	39.66
	ii) From 600 mm to 900 mm dia.	Rm	123.33
	iii) From 900 mm to 1200 mm dia.	Rm	184.88
25.16	Labour for fitting and fixing single empty bitumen drum in culvert after cutting to proper sizes and fitting in position including all carriage.	Rm	64.45
25.17	Labour for fitting and fixing double empty bitumen drum in culvert after cutting to proper sizes and fitting in position including all carriage.	Rm	126.36

	Expansion Joint Filler Board		
25.18	Providing and fixing Duraboard HD-100 premoulded compressible filler board at all heights and depths of minimum density 95kg/cum, non-staining having less than 1% water absorption, compression recovery of 93% (minimum) as per manufactured specification with polysulphide sealant to seal the expansion joint including preparation of surface, Polymer Modified Mortar (PMM) (Proportion 1cement: 3 sand:1 acrylic polymer) nozing, carriage of materials, cutting and placing to the required size complete at all levels as per drawings and direction of Engineer-in-Charge.		
	(a) 25mm wide	Sqm	1562.91
	(b) 30mm wide	Sqm	1691.68
	(c) 40mm wide	Sqm	1889.33
	(d) 50mm wide	Sqm	2090.98
	(e) 75mm wide	Sqm	2809.30

CH-26: ANTI-TERMITE TREATMENT

Item No.	Description	Unit	Rate
26.1	PRE CONSTRUCTION ANTI - TERMITE TREATMENT		
26.1.1	Pre construction anti - termite treatment at the bottom of foundation pits and walls on either side up to a height of 300mm from bottom of the excavation made for column plinth beam, wall, trenches etc. by applying the emulsion @ 5 (five) litres per square meter of the surface area before starting construction work (supplying of necessary tools and accessories by the contractor) as per the direction of the Department complete.		
	a) with aqueous emulsion having concentration 1:19 with chloropirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Sqm	77.09
	b) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Sqm	37.88
26.1.2	Pre construction anti - termite treatment in the back fill in immediate contact with the foundation structure after completion, column, wall etc. to a depth of 450mm and width 300mm (when required refill in 200mm thick layers) applying the emulsion @ 5 (five) litres per square meter towards the concrete and masonry walls so as to ensure earth in contact with the structure well treated - with chemical (supplying of necessary tools and accessories by the contractor) as per the direction of the Department complete.		
	a) with aqueous emulsion having concentration 1:19 with chloropirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Sqm	77.09
	b) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Sqm	37.88

26.1.3	Pre construction anti - termite treatment by applying in the entire levelled area after plinth filling and before laying floor by applying the emulsion @ 5 (five) litres per square meter of the surface to be spread with light rodding so as to ensure proper absorption by the filled up earth (supplying of necessary tools and accessories by the contractor) as per the direction of the Department complete.		
	a) with aqueous emulsion having concentration 1:19 with chloropirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Sqm	77.09
	b) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Sqm	37.88
26.1.4	Pre construction anti - termite treatment by applying in the junction of walls, column with the floor before laying floor and after operation of item no. 26.1.3 above making a channel 30mmx30mm all over the walls and making rod holes @ 150mm apart up to ground level below and applying emulsion @ 11 litres per square meter of vertical plinth wall surface inside the building so as to soak the soil with the emulsion up to the ground level and refill the holes and channel with the original materials obtained in making channels (supplying of necessary tools and accessories by the contractor) as per the direction of the Department complete.		
	a) with aqueous emulsion having concentration 1:19 with chloropirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Sqm	168.27
	b) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Sqm	81.99
26.1.5	Pre construction anti - termite treatment by applying the soil along the external perimeter of the building (after construction completed) to a distance close to the plinth wall as specified and as directed by the deptt. up to a depth of 300mm making rod holes along the line @ 150mm apart to a depth of 300mm and applying emulsion @ 4.5 litres per running meter ensure moistening of the soil with emulsion (supplying of necessary tools and accessories by the contractor) as per the direction of the Department complete.		
	a) with aqueous emulsion having concentration 1:19 with chloropirophose solution (trichel or euivalent) and water, i.e. 1% solution.	R.M	69.90
	b) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	R.M	34.60
26.2	POST CONSTRUCTION ANTI - TERMITE TREATMENT		
26.2.1	Diluting and injecting aqueous emulsion for post constructional anti - termite treatment (necessary tools and accessories to be supplied by the contractors).		

a)	Along external wall (where apron is not provided) in the vertical surface of the substructure to a depth of 300mm including excavation channel along the wall and rodding etc. by applying the emulsion at 2.5 litres per square meter complete as directed by the Department.		
	i) with aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Rm	14.63
	ii) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Rm	8.74
b)	Along the external wall below concrete or masonry apron using aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or equivalent) and water @ 1 litre per liner metre including drilling and plugging holes (minimum depth of hole should be 300mm).		
	i) with aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Rm	18.40
	ii) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Rm	10.55
c)	Treatment of soil under existing floors using aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or equivalent) and water @ 0.50 litre per hole, 300mm apart including drilling 12mm diameter holes and plugging with cement mortar 1:2 (1 cement : 2 coarse sand) to match the existing floor. (Cement mortar to be measured and paid separately). (Minimum depth of hole should be 300mm).		
	i) with aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Sqm	44.75
	ii) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Sqm	23.39
d)	Treatment of existing masonry using aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or equivalent) and water @ 0.5 litre per hole, 300mm apart including drilling holes at 45 degree and plugging them with cement mortar 1:2 (1 cement : 2 coarse sand) to the full depth of the hole. (Minimum depth of hole should be 300mm).		
	i) with aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Rm	22.17
	ii) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Rm	12.36
e)	Treatment at points of contact of wood work by aqueous emulsion having concentration 1:19 with chlorpirophose solution (trichel or equivalent) and water @ 0.50 litre/hole by drilling 6mm dia holes at downward angle of 45 degree at 150mm centre to centre and sealing the same.		

	i) with aqueous emulsion having concentration 1:19 with chloropirophose solution (trichel or euivalent) and water, i.e. 1% solution.	Rm	46.42
	ii) with aqueous emulsion having concentration 1:499 with Premise 350 SC (Imidachloprid 30.5% SC) and water, i.e. 0.07% solution.	Rm	24.20

CH-27: REPAIRING WORKS

Item No.	Description	Unit	Rate
27.1	Taking out from position and repairing steel window shutters including steel frames at all levels after cutting out the heavily rusted and damaged portion of the shutters and frames and welded with the new 'Z' section of steel members, corner, hinges, stay etc. including fitting and fixing in position and riveting with new rivets adjusting, cleaning, cutting the holes and making good the damages to metal with existing one. Window grills to be removed during carrying out the work and to be re-fixed in position after completion of work. The glass panes damaged during execution should also be changed.	Sqm	835.30
27.2	Strengthening of beam, columns, lintels, chajjas, slabs by applying water proofing treatment, exposed reinforced treatment, sealant application by Micro concrete CNS 100 + M - 81 up to 15mm complete as directed by the Department at all levels.	Sqm	460.60
27.3	Supplying, fitting and fixing 4mm thick teak plywood sheet after removing the existing damaged teak ply, over timber frame with Fevicol, nails, screws etc. including cutting sheets to proper sizes etc. complete as directed at all levels.	Sqm	1345.00

CH-28: QUALITY CONTROL

Item No.	Description	Unit	Rate
28			
28.1	Collection from the field and testing in the laboratory of undisturbed soil sample for sub-soil investigation (as per relevant I.S. code) by open drive sampler (100 mm dia. Boring) through clayey soil layer/sandy soil layer/sandy clay/sandy silt layer/clayey silt layer,conduction standard penetration test(SPT) and collection of SPT sampler,observation of ground water table,spot level survey to the bore location and carrying out R.L. from nearest B.M.,laboratory test (particle size analysis,hydrometer test,liquid limit determination,plastic limit determination,specific gravity determination,triaxal test,direct shear test,unconfined compression test,consolidation test) including preparation of sub-soil investigation report complete with necessary printing and binding (minimum 5 copies)		
	A) Boring.		

	(i) Construction of 100mm hand wash boring through all types of cycle,sandy clay,silty sand,silty clay soil,excluding transportation charges.	Rm	666.16
	(ii) Construction of 100mm machine boring (Drilling) through all types of cycle,sandy clay,silty sand,silty clay soil,excluding transportation charges.	Rm	849.65
	(iii) Construction of 100mm machine boring (Drilling) through all types of gravel,sandy gravel excluding transportation charges.	Rm	3186.42
	(iv) Construction of 100mm machine boring (Drilling) for rock layer, excluding transportation charges.	Rm	4305.53
	B) Standard Penetration Test (S.P.T.)		
	(i) Conduction of SPT at equal interval of 1.5 m or change of strata whichever is earlier and collection of disturbed soil sample from SPT sampler including proper airtight sealing,labelling.	Rm	88.07
	(ii) Conduction of SPT by DCPT (Method dry) excluding transportation charges.	Rm	346.52
	(iii) Conduction of SPT by DCPT (Method wet) excluding transportation charges.	Rm	397.36
	C) Laboratory test		
	(i) Particle size analusis (dry sieving)	per test	144.46
	(ii) Particle size analusis (wet sieving)	per test	157.70
	(iii) Determination of Hydrometer test.	per test	736.15
	(iv) Determination of liquid limit test.		
	(a) 3 points	per test	230.57
	(b) 2 points	per test	153.71
	(v) Determination of Plastic Limit test.	per test	138.38
	(vi) Determination of Moisture Content test.	per test	88.07
	(vii) Determination of bulk density test.	per test	69.97
	(viii) Determination of Specific Gravity test of fine grained soil..	per test	182.99
	(ix) Determination of Direct shear strength		
	(a) 3 points	per test	474.42
	(b) 2 points	per test	316.27
	(x) Determination of shear strength parameters of a specimen tested in unconsolidated undrained triaxial compression test without measuring pore water pressure by triaxial test.		
	(a) 3 points	per test	723.80
	(b) 2 points	per test	482.54
	(xi) Determination of consolidated drain test determination of the shear strength parameters of a specimen tested in consolidated drain triaxial compression test without measuring pore water pressure,by triaxial test.shear strength parameters of a specimen tested in unconsolidated undrained triaxial compression test without measuring pore water pressure by triaxial test.		
	(a) 3 points	per test	1435.39
	(b) 2 points	per test	956.93
	(xii) Determination of unconfied compressive strength.		
	(a) 3 points	per test	369.50
	(b) 2 points	per test	246.34
	(xiii) Determination of consolidation properties of Soil.		

	(a) 3 points	per test	876.23
	(b) 2 points	per test	584.16
	(xiv) Determination of vane shear strength.	per test	651.61
	(xv) Determination of Permeability of soil by falling head method.	per test	950.65
	(xv) Determination of shear strength parameters of a specimen tested in consolidated undrained triaxial compression test without measuring pore water pressure by triaxial test.		
	(a) 3 points	per test	876.08
	(b) 2 points	per test	584.06
	(xvi) Determination of shear strength parameters of a specimen tested in consolidated undrained undrained test compression test with measuring pore water pressure by triaxial test.		
	(a) 3 points	per test	1329.37
	(b) 2 points	per test	886.25
	(xvii) Determination of Swelling Index.	per test	736.15
	D) Transportation on charges for boring.		
	(i) Hire charge	per day	3811.50
	(ii) Fuel	per km	17.87
28.2	Collection of sample from field and testing in laboratory/field as per relevant I.S. codes of practice and preparation and submission of the reports to the concerned authority in proper format with specified no. of copies (minimum 5 copies)		
	A) Cement		
	(i) Compressive strength.	per test	199.82
	(ii) Setting time..	per test	242.22
	(iii) Fineness test.	per test	526.07
	(iv) Impurities percentage (Bromoform test)	per test	769.08
	(v) Ignition test.	per test	209.35
	(vi) Specific gravity.	per test	508.85
	B) Concrete		
	(i) Compressive strength(including moulding &curing)	per test	262.82
	(ii) Compressive strength(excluding moulding)	per test	127.94
	(iii) (a) Mix design	per test	8458.55
	(b) Subsequent Job Mix Design with same materials.	per test	5686.73
	(iv) Tensile strength of concrete.	per test	766.68
	C) Steel		
	(i) Tensile strength of steel.	per test	507.12
	(ii) Bending & Re-bending of steel.	per test	592.28
	D) Aggregate		
	(i) Aggregate impact value test (in over size)	per test	477.23
	(ii) Aggregate impact value test (in proper size)	per test	301.66
	(iii) Aggregate crushing value test (in over size)	per test	590.18
	(iv) Aggregate crushing value test (in proper size)	per test	497.76
	(v) Los Angeles abrasion test.	per test	559.18
	(vi) Flakiness index test.	per test	382.22
	(vii) Specific gravity test.	per test	273.41
	(viii) Water absorption test.	per test	272.77
	(ix) Soundness test.(5 cycle sodium sulphate	per test	3180.38
	(x) Grading test of coarse aggregate.	per test	428.21

	(xi) Grading test of fine aggregate.(including fineness modulus,gradation,P.C. of silt and clay and silt factor)	per test	477.98
	(xii) Bulk density test of coarse/fine aggregate-Loose compacted.	per test	233.65
	(xiii) Bulkinf factor of fine aggregate by Lab. Method.	per test	327.64
	(xiv) Bulkinf factor of fine aggregate by Field Method.	per test	144.82
	(xv) Aggregate stripping value test.	per test	241.34

CH-29: FIRE-FIGHTING WORKS

Item No.	Description	Unit	Rate
29.1	HYDRANT SYSTEM		
29.1.1	Providing,Fixing, testing &Commissioning of Single Headed Gun Metal ISI Marked oblique pattern Hydrant Landing Valve with 80 mm dia, flanged inlet & 63 mm dia instantaneous type female outlet complete with cap and chain, twist release lug and all accessories as per IS:5290-19833 (Type-'A') and specifications complete as reequred . (MAKE:ESSEL/WINCO/NEWAGE)	each	9028.75
29.1.2	Providing controlled percolating (C.P.) hose ISI Marked (IS:8423) 63 mm dia X 15 M long complete with instantaneous type Gun Metal 63 mm dia ISI Marked male & female couplings (IS:903) bound & riveted to hose pipes with copper rivets and copper wire complete as required. (MAKE: NIRMAL RUBBER/NEWAGE)	each	5602.34
29.1.3	Providing RRL Hose with ISI Marked (IS:636) 63 mm dia X15 M long complete with instantaneous type Gun Metal 3 mm dia ISI Marked male & female couplings (IS:903) bound & riveted to hose pipes with copper rivets and copper wire complete as reqd. (MAKE: NIRMAL RUBBER/NEWAGE)	each	6044.52
29.1.4	Providing & fixing of weather proof hose cabinets fabricated from 16 S.W.G. M.S. Sheet with centre opening full front glass door and locking arrangement, suitable to accommodate one landing valve (Yard Hydrant), 2nos.15M long hoses and 1 no. Branch pipe with nozzle . The cabinet shall be painted with one coat of primer and 2 or more coats of synthetic enamel paint of approved make & shade and shall be suitably mounted on a raised masonry platform complete as reqd.(Make:self Fabricated)		
	a) 900X600X450	each	4855.07
	b) 2000 X 1200 X450	each	9124.18
29.1.5	Providing & Fixing of Gun Metal 63 mm dia Short Branch Pipe with 20 mm dia nozzle, ISI marked (IS:903) complete as required. (Make: ESSEL/WINCO/NEWAGE).	each	2456.70
29.1.6	Providing & Fixing of Gun Metal fire Brigade Connection (inlet collection head) consisting of 63mm instantaneous type male coupling with built in check valves and 150 mm dia flanged outlet complete with bolts, nuts and rubber insertions as per IS:904-1983 complete as reqd. MAKE: ESSEL/WINCO/NEWAGE		
	a) 02 way	each	7979.21
	b) 04 way	each	13186.36

29.1.7	Providing & fixing of Gun Metal fire Brigade connection (withdrawl connecting head) consisting of 63mm instantaneous type male coupling with built in check valves and 150mm dia flanged outlet complete with bolts, nuts and rubber insertions as per IS:904-1983 complete as reqd. MAKE: ESSEL/WINCO/NEWAGE		
	a) 2 way	each	8772.05
	b) 4 way	each	14138.93
29.1.8	Providing & Fixing of Weather Proof M.S. Cabinet (to enclose above Inlet Collection Head Connection) fabricated from 16 S.W.G. M.S. sheet centre opening with 3 mm thick full front glass door and locking arrangement duly painted with one coat of primer and two or more coasts of synthetic enamel paint of appreoved make and shade and suitably mounted. Exact size and shape shall be subject to site conditions and prior approval. (Make:Self Fabricated)	each	4681.95
29.1.9	Providing & fixing of Gun Metal fire Brigade suction hose coupling (draw out connection) with nut for female coupling as per IS:902-1974 complete with 150 mm dia GI suction pipe and foot valve (to be connected to static Tank) complete as reqd.(Make: Self Fabricated)	each	49568.63
29.1.10	Providing & fixing of Weather Proof M.S. cabinet (to enclose above draw out connection) fabricated from 16SWG M.S. sheet centre opening with 3mm thick full front glass door and locking arrangement duly painted with one coat of primer and two or more coats of synthetic enamel paint of appreoved make and shade and sutable mounted . Exact size and shape shall be subject to site conditions and prior approval) (Make: Self Fabricated)	each	4834.42
29.1.11	Providing ,Fixing ,Testing & Commissioning of air Release Valve		
	a) 25 mm dia	each	3905.85
	b) 50 mm dia	each	6589.29
29.1.12	Providing,Fixing,Testing & Commissioning of Mild Steel Black Pipe(IS:1239part-I) Heavy class including cutting, screwing, welding etc. and providing all fittings viz forged steel screwed fitting upto 50mm dia and bolt welded above 50mm dia flanges, bends, tees, reducers, elbow, clamps, hangers etc including cutting holes and chases in brick or RCC walls/ slabs and making good the same to the approval of Engineer-in-charge/ consultant complete including painting with one coat of primer and two or more coats of synthetic enamel paint of approved make and shade including steel work complete as required. (Make:TATA/JINDAL/NEZONE)		
	a) 200 mm dia	rm	2719.77
	b) 150 mm dia	rm	2109.90
	c) 100 mm dia	rm	1597.60
	d) 80 mm dia	rm	1201.03
	e) 65 mm dia	rm	985.00
	f) 50 mm dia	rm	834.58
	g) 40 mm dia	rm	772.95
	h) 32 mm dia	rm	664.30

	i) 25 mm dia	rm	555.71
	j) 20 mm dia	rm	452.03
	k) 15 mm dia	rm	372.75
29.1.13	Providing,Fixing,Testing & Commissioning of Mild Steel Black Pipe(IS:1239part-I) Heavy class including cutting, screwing, welding etc.complete with all fittings (butt welded) viz flanges, bends, tees, elbows, reducers etc including excavation (upto 1.5 m Deep), back filling and removal of surplus soil & providing anti corrosive treatment(coating and wrapping) as per IS:10221 with 4 mm thick tape complete as reqd. (MAKE:TATA/JINDAL/NEZONE)		
	a) 200 mm dia	rm	2940.40
	b) 150 mm dia	rm	2330.53
	c) 100 mm dia	rm	1604.78
	d) 80 mm dia	rm	1409.62
	e) 65 mm dia	rm	1263.25
	f) 40 mm dia	rm	897.33
29.1.14	Supplying & Fixing Cast iron body Non Return valves complete with matching flanges, rubber insertion, nuts, bolts and washer etc. (IS:5312) of following sizes complete as reqd. (Make:Koley/Kalpana/Audco)		
	(a) 150 mm dia	each	22000.07
	(b) 100 mm dia	each	18397.44
	(c) 80 mm dia	each	12485.15
29.1.15	Supplying & Fixing,cast iron body Butterfly Valves with matching flanges, rubber insertion, nuts, bolts and washer etc (IS:13095) of following size complete as reqd. (MAKE:KOLEY/KALPANA/AUDCO)		
	(a) 200 mm dia	each	18584.24
	(b) 150 mm dia	each	17429.39
	(c) 100 mm dia	each	8879.44
	(d) 80 mm dia	each	6543.40
	(e) 65 mm dia	each	5276.51
	(f) 50 mm dia	each	4768.13
	(g) 40 mm dia	each	3330.84
29.1.16	Providing ,Fixing,Testing & Commissioning of G.M. Stop Valve ISI Marked screwed type tested to 30Kg/cm2 pressure as per IS:778-1984 of following sizes complete as reqd.		
	(a) 50 mm dia	each	3712.13
	(b) 25 mm dia	each	2191.67
29.1.17	Providing, Erection, Testing & Commissioning of Cast Iron Sluice Valve/ Gate Valve confirming to IS:14846/2000 complete with matching flanges, rubber insertion, nuts, bolts and washer etc of following size complete as required (MAKE:KOLEY/KALPANA/AUDCO)		
	(a) 200 mm dia	each	15811.38
	(b) 150 mm dia	each	12113.04
	(c) 100 mm dia	each	9100.38
	(d) 80 mm dia	each	6903.65

29.1.18	Providing, erection testing, Commissioning of Cast Iron Foot Valve/Stainer confirming to IS:4038/1538 complete with flanges, nuts, bolts and washer etc as required		
	(a) 200 mm dia	each	24414.72
	(b) 150 mm dia	each	14498.06
29.1.19	Providing, erection testing & commissioning of pressure gauge 100 mm dia (0-10Kg/cm ²)	each	1757.03
29.1.20	Providing ,erection,Testing & Commissioning of Pressure Switch (MAKE:INDFOSS)	each	3796.85
29.1.21	Supplying, Installation,Testing & Commissioning of 15 mm dia quartzoid Bulb Type G.M. Sprinkler Head suitable to operate at 68* C complete as reqd.(UL/FM/LPC listed/Approved) (Make : HD Fire / TYCO)	each	508.41
29.1.22	Supply,installation,testing & commissioning of 150 mm dia installation control valve assembly comprising of vertical wet alarm valve, hydraulic alarm motor & gong, 15 mm dia Test Valve, 50 mm dia , Drain Valve and all necessary trimming connections including 02 nos. pressure gauge as per manufacturers catalogue complete as reqd.(UL/FM/LPC Listed/Approved) (Make : HD FIRE)	each	99356.82
29.1.23	Supplying and laying of following size NP2 class RCC Hume pipe i/c collars jointed with stiff mixture of cement mortar 1:2 (1 cement:2 fine sand) across the road including cutting the road (if reqd) digging the trench, refilling the same after laying pipe, ramming the soil etc as reqd.		
	(a) 400 mm dia	rm	2447.96
	(b) 300 mm dia	rm	1960.06
29.1.24	Supplying, Installation, Testing & Commissioning of First Aid Hose reel drawn with 30 Mtrs. Long rubber hose and G.M shut off nozzle, 25mm dia Gate valve, Union pipe complete set confirming to IS: 884.	each	13206.57
29.2	FIRE EXTINGUISHER		
29.2.1	Supply, Installation, Fixing of DRY CHEMICAL POWDER Fire Extinguisher confirming to IS: 2171 with ISI mark and CO2 cartridge will be confirming to IS: 4947 with ISI mark, Powder will be confirming to IS: 4308 with ISI Mark (5 KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	3543.00
29.2.2	Supply, Installation, Fixing of DRY CHEMICAL POWDER Fire Extinguisher confirming to IS: 2171 with ISI mark and CO2 cartridge will be confirming to IS: 4947 with ISI mark, Powder will be confirming to IS: 4308 with ISI Mark (10 kg) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	4684.34
29.2.3	Supply, Installation, Fixing of DRY CHEMICAL POWDER Fire Extinguisher confirming to IS: 2171 with ISI Mark, and CO2 Cylinder will be confirming to IS: 4947 with ISI Mark, Powder will be confirming to IS: 2878 with ISI Mark (Extinguisher will be TROLLEY MOUNTED)(25 KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	22754.01

29.2.4	Supply, Installation, Fixing of DRY CHEMICAL POWDER Fire Extinguisher confirming to IS: 2171 with ISI mark and CO2 cartridge will be confirming to IS: 4947 with ISI mark, Powder will be confirming to IS: 4308 with ISI Mark (Extinguisher will be TROLLEY MOUNTED).(50 KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	32755.93
29.2.5	Supply, Installation, Fixing of DRY CHEMICAL POWDER Fire Extinguisher confirming to IS: 2171 with ISI mark and CO2 cartridge will be confirming to IS: 4947 with ISI mark, Powder will be confirming to IS: 4308 with ISI Mark (Extinguisher will be TROLLEY MOUNTED)(75 KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	37646.53
29.2.6	Supply, Installation, Fixing of MECHANICAL FOAM (AFFF) Fire Extinguisher confirming to IS: 10204 with ISI Mark and CO2 Cartridge will be confirming to IS: 4947 with ISI mark, (AFFF) will be confirming to IS: 4989 with ISI Mark.(9 LTRS) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	3116.09
29.2.7	Supply, Installation, Fixing of MECHANICAL FOAM (AFFF) Fire Extinguisher confirming to IS: 10204 with ISI Mark and CO2 Cartridge will be confirming to IS: 4947 with ISI mark, (AFFF) will be confirming to IS: 4989 with ISI Mark.(50 LITERS) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	19344.23
29.2.8	Supply, Installation, Fixing of WATER CO2 Fire Extinguisher confirming to IS: 940 with ISI Mark and CO2 Cartridge will be confirming to IS: 4947 with ISI Mark.(09 LITERS) BRAND: FIRESTOP / FIREND / FIRESHIELD	each	2628.19
29.2.9	Supply, Installation, Fixing of CARBON DIOXIDE Fire Extinguisher confirming to IS: 2878 with ISI Mark.(4.5 LITERS) BRAND: FIRESTOP / FIREND / FIRESHIELD	each	9849.09
29.2.10	Supply, Installation, Fixing of CARBON DIOXIDE Fire Extinguisher confirming to IS: 2878 with ISI Mark. (Extinguisher will be TROLLEY MOUNTED)(6.5 KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	11257.90
29.2.11	Supply, Installation, Fixing of CARBON DIOXIDE Fire Extinguisher confirming to IS: 2878 with ISI Mark. (Extinguisher will be TROLLEY MOUNTED)(9 KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	15651.42
29.2.12	Supply, Installation, Fixing of CARBON DIOXIDE Fire Extinguisher confirming to IS: 2878 with ISI Mark, (Extinguisher will be TROLLEY MOUNTED)(2KG) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	6578.99
29.2.13	Supply, Installation, Fixing of ABC Stored Pressure Fire Extinguisher with built – In pressure Gauge to know the Ready ness of the Extinguisher Squeeze Grip Handle As per IS:13849 with ISI mark (05 kg capacity) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	5556.46

29.2.14	Supply, Installation, Fixing of ABC stored Pressure Fire Extinguisher with built – In Pressure Gauge to know the Readiness of the Extinguisher Squeeze Grip Handle as per IS:13849 with ISI mark. (10 kg capacity) BRAND : FIRESTOP / FIREND / FIRE SHIELD	each	11716.18
29.3	FIRE EXTINGUISHER REFILLING & SERVICING		
29.3.1	Refilling & Servicing of DCP Type 05 Kg Capacity Fire Extinguisher with ISI Mark refill etc. complete	each	1450.00
29.3.2	Refilling & Servicing of DCP Type 10 Kg Capacity Fire Extinguisher with ISI Mark refill etc. complete	each	2275.00
29.3.3	Refilling & Servicing of Water CO2 Type 09 Ltrs Capacity Fire Extinguisher with ISI Mark refill etc. complete	each	480.00
29.3.4	Refilling & Servicing of Mechanical Foam AFFF Type 09 Ltrs Capacity Fire Extinguisher with ISI Mark refill etc. complete	each	525.00
29.3.5	Refilling & Servicing of ABC Type 05 Kg Capacity Fire Extinguisher with ISI Mark refill etc. complete	each	1670.00
29.3.6	Refilling & Servicing of CO2 Type Fire Extinguisher with ISI Mark refill etc. complete		
	(a) 02 Kg	each	720.00
	(b) 4.5 Kg	each	1650.00
	(c) 6.5 Kg	each	2325.00
	(d) 09 Kg	each	3630.00

CHAPTER 30. R.C.C APPROACH ROAD

Item No.	Description	Unit	Rate
30.1	Construction of Sub-grade and earthen shoulders with approved materials obtained from borrow pits, with all lifts & leads, transporting to site, spreading, grading to required slope and compacting to meet requirement of table 300-2 (including compensation of earth). (Including cost of testing of materials at site and laboratory as directed by the deptt.)		
	a) from Private land	cum	235.00
	b) from Government land	cum	235.00
	c) Extra lead beyond initial lead of 3.00km	cum/km	8.46
30.2	Construction of Granular sub-base by providing close graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per Clause 401 of MoRTH specifications (with an initial lead of 5 km) (including cost of testing of materials at site and laboratory as directed by the deptt.)		
	(a)Using Grading I material	cum	2040.00
	(b)Using Grading II material	cum	2094.00
	(c)Using Grading III material	cum	1965.00
	(d) Extra lead beyond initial lead of 5.00km	cum/km	8.46

30.3	Supplying, fitting and fixing in position reinforcement bars conforming to relevant I.S. Code for R.C.C. work/ R.B. walling including straightening, cleaning, cutting and bending to proper shapes and length as per details, supplying and binding with 20G annealed black wire and placing in position with proper blocks, supports, chairs, spacers etc. complete		
	a) TMT Bar		
	(i) From Primary Producer	qntl	6902.86
	(ii) Other ISI Marked approved brand	qntl	6026.82
	b) ISI approved - M.S. Rod	qntl	6767.66
30.4	Providing form work of ordinary timber planking so as to give a rough finish including centering, shuttering, strutting and propping etc., and removal of the same for in-situ reinforced concrete and plain concrete work in R.C.C. approach road.		
	(a) Using 38mm thick plank	sqm	276.07
	(b) Using 25mm thick plank	sqm	252.24
30.5	Providing and laying controlled cement concrete pavement of approved grade with 20mm down graded stone aggregate over a prepared sub-base including dewatering if necessary, spread, compacted and finished in a continuous operation including provision of construction and expansion joints, plasticiser and curing compound of Fosroc/ Sika/ Choksey/ Degusa and hardener of Recron/ Degusa/ Fosroc; finishing to lines and grades and curing as specified and directed complete (form work and reinforcement shall be measured and paid separately).		
	(a) M 20 Grade Concrete	cu.m.	7218.26
	(b) M 25 Grade Concrete	cum	7798.14
	(c) M 30Grade Concrete	cum	7848.98
30.6	15 mm thick Cement plaster in single coat with hardener of Recron/ Degusa/ Fosroc on brick/concrete for plastering including arises, internal rounded angles, chamfers and / or rounded angles not exceeding 80mm in girth and finished even and smooth including curing complete as directed.		
	(a) In cement mortar 1:3	sqm	166.34
	(b) In cement mortar 1:4	sqm	146.60
30.7	Providing and laying of inter locking concrete block pavement (ICBP) with cement concrete paver blocks as per IS 15658-2006 of thickness 80mm and 28 days compressive strength not less than 30 N/m over a layer of bedding sand of compacted thickness 30mm, laid either in stretcher/ running/ herringbone pattern of bonds as directed by the engineer and in proper level and grade and compacted with appropriate equipment such as plat vibrator along with spreading of a thin layer of joint filling sand to fill up the gaps between blocks including construction of edge restraint wherever required complete as per drawings and Technical specification clause 1504 of MORD.	sqm	937.68
CH-31: ACOUSTIC TREATMENT			
Item No.	Description	Unit	Rate

31.1	Supplying, fitting and fixing SoundSoak NSS Cord (FSS Cord) wrapped acoustical wall panel system of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	4853.79
	b) 25 MM	Sqm	5149.86
	c) Armstrong OPTRA high acoustical wall panelling with fibre glass substrate 25 MM thick	Sqm	9060.37
	d) Armstrong Wood works acoustical wall panelling	Sqm	9931.33
31.2	Supplying, fitting and fixing SoundSoak FSS Cordura wrapped acoustical wall panel system of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	5900.41
	b) 25 MM	Sqm	6156.16
31.3	Supplying, fitting and fixing SoundSmooth MB Primer (NSS Primer) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	2931.57
	b) 25 MM	Sqm	3298.50
	c) 25 MM	Sqm	3528.30
31.4	Supplying, fitting and fixing SoundSmooth MB White (NSS White) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	3168.78
	b) 25 MM	Sqm	3533.86
	c) 25 MM	Sqm	3728.04
31.5	Supplying, fitting and fixing SoundSmooth MB Trax (NSS Trax) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	3428.23
	b) 25 MM	Sqm	3682.12
31.6	Supplying, fitting and fixing SoundSmooth MB Trax White (NSS Trax) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	3665.44
	b) 25 MM	Sqm	3919.33
31.7	Supplying, fitting and fixing Mat White (Mat White) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM	Sqm	3763.66
	b) 25 MM	Sqm	4130.60
31.8	Supplying, fitting and fixing Mat White Trax (Mat White) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 20 MM		4173.22
31.9	Supplying, fitting and fixing Sound Sonator with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 15 MM	Sqm	3644.44
31.10	Supplying, fitting and fixing Mat Black (Mat Black) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 15 MM	Sqm	3544.56
	b) 20 MM	Sqm	3922.62

31.11	Supplying, fitting and fixing SoundSmooth Ebony (NSS Ebony) with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 15 MM	Sqm	3114.62
	b) 20 MM	Sqm	3689.53
	c) 25 MM	Sqm	3919.33
31.12	Supplying, fitting and fixing Mat BSB with glass wool of Anutone Acoustics Ltd. Including all fixtures and fittings.		
	a) 18mm 900 kg/ M2	Sqm	2794.01
	b) 18mm 1200 kg/ M2	Sqm	3008.86
31.13	Supplying, fitting and fixing SubTex Nubby/Pebble- TEXTURE CEILING of Anutone Acoustics Ltd. Including all fixtures and fittings.		
31.13.1	SubTex		
	a) 15mm Square	Sqm	2137.31
	b) 15mm Steppe	Sqm	2292.98
31.13.2	SubTex Ebony		
	a) 15mm Square	Sqm	2149.25
	b) 15mm Steppe	Sqm	2299.36
31.13.3	FABSTRETCH 25 mm with 50 mm Air Gap		
	a) 25mm	Sqm	5293.39
31.13.4	ACT GLD INSTALLED USING IMPALLING CLIPS		
	a) 25mm	Sqm	10507.32
31.13.5	ACT GLD INSTALLED USING Z Bar		
	a) 25mm	Sqm	11032.00
31.13.6	SOUNDSLATS		
	a) 16 MM PITCH	Sqm	20133.05
	b) 32 MM PITCH	Sqm	17571.91
	c) 16 MM PITCH FR	Sqm	25461.03
	d) 32 MM PITCH FR	Sqm	22899.89
31.14	Armstrong acoustical ceiling		
31.14.1	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with beauty sky (board/tegular) edge tiles with armstrong 24mm exposed grid (600x600x14)mm	Sqm	1785.67
31.14.2	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with beauty sky (board/tegular) edge tiles with armstrong 24mm exposed grid (600x600x15)mm	Sqm	1866.32
31.14.3	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with optra FG (board/tegular) edge tiles with armstrong XL 24mm exposed grid (600x600x15)mm	Sqm	3801.93
31.14.4	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with dune (bev. tegular) edge tiles with armstrong XL 15mm exposed / silhouette grid (600x600x16)mm	Sqm	2511.53
31.14.5	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with dune (board/ tegular) edge tiles with armstrong XL 24mm exposed grid (600x600x16)mm	Sqm	2350.23

31.14.6	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with fine fissured / classic lite (bev. tegular) edge tiles with armstrong XL 15mm exposed / silhouette grid (600x600x16)mm	Sqm	2188.92
31.14.7	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with fine fitted / classic lite (board/ tegular) edge tiles with armstrong 24mm exposed grid (600x600x16)mm	Sqm	2108.27
31.14.8	Providing & fixing Armstrong Mineral fibre Acoustical suspended ceiling system with ANF (bev. tegular) edge tiles with armstrong XL 15mm exposed / silhouette grid (600x600x16)mm	Sqm	1946.97

CH-32 : COMPOSITE ITEMS OF WORKS

Item No.	Description	Unit	Rate
32.1	RCC Work with Shuttering		
32.1.1	Providing and laying plain/reinforced cement concrete works using cement , coarse sand & 20mm down graded stone aggregate with cost of necessary form work including dewatering if necessary, and curing for reinforced cement concrete work complete as directed (Reinforcement will be measured and paid separately)		
	I) Using mixture machine		
	(A) In sub - structure up to plinth level		
	Foundation, footing, columns with base tie and plinth beam, pile cap, base slab, retaining walls, walls of septic tank, inspection pit and the like and other works not less than 100mm thick up to plinth level.		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	6683.67
	(b) M20 or Prop. 1:11/2:3	Cum	7250.67
	(c) M25 grade concrete	Cum	8281.89
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	6904.19
	(b) M20 or Prop. 1:11/2:3	Cum	7516.50
	(c) M25 grade concrete	Cum	8590.77
	B) In super structure up to 1st floor level		
	i) Vertical members like column, pillars, post, struts etc		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	7624.89
	(b) M20 or Prop. 1:11/2:3	Cum	8197.46
	(c) M25 grade concrete	Cum	8742.08
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	7845.41
	(b) M20 or Prop. 1:11/2:3	Cum	8463.29
	(c) M25 grade concrete	Cum	9050.96
	ii) Horizontal members like beam, lintel, girder bressumer etc.		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	7422.09
	(b) M20 or Prop. 1:11/2:3	Cum	7994.66
	(c) M25 grade concrete	Cum	8539.28

	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	7642.61
	(b) M20 or Prop. 1:11/2:3	Cum	8260.49
	(c) M25 grade concrete	Cum	8848.16
	iii) Suspended floor (Slab, balcony, staircase etc)		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	8247.72
	(b) M20 or Prop. 1:11/2:3	Cum	8820.29
	(c) M25 grade concrete	Cum	9364.91
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	8468.24
	(b) M20 or Prop. 1:11/2:3	Cum	9086.12
	(c) M25 grade concrete	Cum	9673.79
	iv) Walls (thickness not less than 100mm) including attached pillastrs, buttresses, plinth and string course etc.		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	7474.02
	(b) M20 or Prop. 1:11/2:3	Cum	8046.59
	(c) M25 grade concrete	Cum	8591.20
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	7694.54
	(b) M20 or Prop. 1:11/2:3	Cum	8312.42
	(c) M25 grade concrete	Cum	8900.08
	v) Vertical and horizontal fins (thickness not more than 100mm) individually or forming box louvre and projected band.		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	12566.67
	(b) M20 or Prop. 1:11/2:3	Cum	13139.19
	(c) M25 grade concrete	Cum	13683.84
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	12787.19
	(b) M20 or Prop. 1:11/2:3	Cum	13405.02
	(c) M25 grade concrete	Cum	13992.72
	vi) On domes, vaults and shell roof		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	9785.54
	(b) M20 or Prop. 1:11/2:3	Cum	10363.66
	(c) M25 grade concrete	Cum	10913.58
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	10006.06
	(b) M20 or Prop. 1:11/2:3	Cum	10629.49
	(c) M25 grade concrete	Cum	11222.46
	(II) Using Batching Plant/ RMC/ Truck mounted batching plant including Transit Mixer and Concrete Pump		
	(A) In sub - structure up to plinth level		
	Foundation, footing, columns with base tie and plinth beam, pile cap, base slab, retaining walls, walls of septic tank, inspection pit and the like and other works not less than 100mm thick up to plinth level.		
	N) Without using admixture, plasticiser		

	(a) M15 or Prop. 1:2:4	Cum	6999.98
	(b) M20 or Prop. 1:11/2:3	Cum	7550.42
	(c) M25 grade concrete	Cum	8073.99
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	7220.50
	(b) M20 or Prop. 1:11/2:3	Cum	7816.25
	(c) M25 grade concrete	Cum	8382.87
	(B) Super-structure from plinth level upto 1st floor level.		
	i) Vertical members (Column, pillars, post, struts etc)		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	7899.48
	(b) M20 or Prop. 1:11/2:3	Cum	8495.23
	(c) M25 grade concrete	Cum	9061.86
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	8120.00
	(b) M20 or Prop. 1:11/2:3	Cum	8761.06
	(c) M25 grade concrete	Cum	9370.74
	ii) Horizontal members (Beam, lintel, girder etc.)		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	7696.68
	(b) M20 or Prop. 1:11/2:3	Cum	8292.43
	(c) M25 grade concrete	Cum	8859.06
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	7917.20
	(b) M20 or Prop. 1:11/2:3	Cum	8558.26
	(c) M25 grade concrete	Cum	9167.94
	iii) Suspended floor (Slab, balcony, staircase etc)		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	8522.31
	(b) M20 or Prop. 1:11/2:3	Cum	9118.06
	(c) M25 grade concrete	Cum	9684.69
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	8742.83
	(b) M20 or Prop. 1:11/2:3	Cum	9383.89
	(c) M25 grade concrete	Cum	9993.57
	iv) Walls (thickness not less than 100mm) including attached pillars, buttresses, plinth and string course etc.		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	7838.68
	(b) M20 or Prop. 1:11/2:3	Cum	8434.43
	(c) M25 grade concrete	Cum	9001.06
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	8059.20
	(b) M20 or Prop. 1:11/2:3	Cum	8700.26
	(c) M25 grade concrete	Cum	9309.94
	v) Vertical and horizontal fins (thickness not more than 100mm) individually or forming box louvre and projected band.		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	12398.08
	(b) M20 or Prop. 1:11/2:3	Cum	12993.84

	(c) M25 grade concrete	Cum	13560.45
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	12618.60
	(b) M20 or Prop. 1:11/2:3	Cum	13259.67
	(c) M25 grade concrete	Cum	13869.33
	v) On domes, vaults and shell roof		
	N) Without using admixture, plasticiser		
	(a) M15 or Prop. 1:2:4	Cum	9490.18
	(b) M20 or Prop. 1:11/2:3	Cum	10085.93
	(c) M25 grade concrete	Cum	10652.56
	P) Using admixture, plasticiser etc		
	(a) M15 or Prop. 1:2:4	Cum	9710.70
	(b) M20 or Prop. 1:11/2:3	Cum	10351.76
	(c) M25 grade concrete	Cum	10961.44
32.1.2	Extra for every subsequent floor above 1st floor on item no 32.1. (l)	Cum	306.19
32.2	Brickwork with Plastering		
32.2.1	Brick work in cement mortar with 1st class brick including cement mortar plastering, racking out joints and dewatering if necessary, and curing complete as directed in sub-structure up to plinth level.		
	(I) One side plastering with 10mm thick		
	i) Using plastering of cement mortar prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	6138.40
	b) Brick work in cement mortar proportion 1:6	Cum	6023.15
	ii) Using plastering of cement mortar prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6099.36
	b) Brick work in cement mortar proportion 1:6	Cum	5984.11
	iii) Using plastering of cement mortar prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6073.32
	b) Brick work in cement mortar proportion 1:6	Cum	5958.07
	(II) One side plastering with 15mm thick		
	i) Using plastering of cement mortar prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	6263.84
	b) Brick work in cement mortar proportion 1:6	Cum	6148.59
	ii) Using plastering of cement mortar prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6217.71
	b) Brick work in cement mortar proportion 1:6	Cum	6102.46
	iii) Using plastering of cement mortar prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6179.66
	b) Brick work in cement mortar proportion 1:6	Cum	6064.41
	(III) One side plastering with 20mm thick		
	i) Using plastering of cement mortar prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	6575.84
	b) Brick work in cement mortar proportion 1:6	Cum	6460.59
	ii) Using plastering of cement mortar prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6513.71
	b) Brick work in cement mortar proportion 1:6	Cum	6398.46
	iii) Using plastering of cement mortar prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6461.62
	b) Brick work in cement mortar proportion 1:6	Cum	6346.37

32.2.2	Brick work in cement mortar with 1st class brick including cement mortar plastering, racking out joints and curing complete as directed in superstructure above plinth level up to 1st floor level.		
	A) One side plastering		
	I) 10mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	6304.91
	b) Brick work in cement mortar proportion 1:6	Cum	6189.65
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6265.87
	b) Brick work in cement mortar proportion 1:6	Cum	6150.61
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6239.83
	b) Brick work in cement mortar proportion 1:6	Cum	6124.57
	II) 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	6430.35
	b) Brick work in cement mortar proportion 1:6	Cum	6315.09
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6384.22
	b) Brick work in cement mortar proportion 1:6	Cum	6268.96
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6346.17
	b) Brick work in cement mortar proportion 1:6	Cum	6230.91
	B) Both side plastering		
	I) With fair side 10mm thick and rough side 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	6945.04
	b) Brick work in cement mortar proportion 1:6	Cum	6829.78
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6859.91
	b) Brick work in cement mortar proportion 1:6	Cum	6744.65
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6795.79
	b) Brick work in cement mortar proportion 1:6	Cum	6680.53
	II) With both side 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	7070.48
	b) Brick work in cement mortar proportion 1:6	Cum	6955.22
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	6978.26
	b) Brick work in cement mortar proportion 1:6	Cum	6863.00
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	6902.13
	b) Brick work in cement mortar proportion 1:6	Cum	6786.87
32.2.3	Extra over item no. 32.2.2 above for brick work in each subsequent floor or part thereof above 1st floor level.	Cum	293.57
32.2.4	Brick work in cement mortar with 1st class brick including cement mortar plastering, racking out joints and dewatering if necessary, and curing complete as directed in sub-structure up to plinth level .		

	I) One side plastering with 10mm thick		
	i) Using plastering of cement mortar prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1411.69
	b) Brick work in cement mortar proportion 1:6	Cum	1384.95
	ii) Using plastering of cement mortar prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1402.71
	b) Brick work in cement mortar proportion 1:6	Cum	1375.97
	iii) Using plastering of cement mortar prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	1396.72
	b) Brick work in cement mortar proportion 1:6	Cum	1369.98
	II) One side plastering with 15mm thick		
	i) Using plastering of cement mortar prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1440.54
	b) Brick work in cement mortar proportion 1:6	Cum	1413.80
	ii) Using plastering of cement mortar prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1429.93
	b) Brick work in cement mortar proportion 1:6	Cum	1403.19
	iii) Using plastering of cement mortar prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	1421.18
	b) Brick work in cement mortar proportion 1:6	Cum	1394.44
	III) One side plastering with 20mm thick		
	i) Using plastering of cement mortar prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1512.30
	b) Brick work in cement mortar proportion 1:6	Cum	1485.56
	ii) Using plastering of cement mortar prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1498.01
	b) Brick work in cement mortar proportion 1:6	Cum	1471.27
	iii) Using plastering of cement mortar prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	1486.03
	b) Brick work in cement mortar proportion 1:6	Cum	1459.29
32.2.5	Brick work in cement mortar with 1st class brick including cement mortar plastering, racking out joints and curing complete as directed in superstructure above plinth level up to 1st floor level.		
	A) One side plastering		
	I) 10mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1450.16
	b) Brick work in cement mortar proportion 1:6	Cum	1423.43
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1441.18
	b) Brick work in cement mortar proportion 1:6	Cum	1414.45
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	1435.19
	b) Brick work in cement mortar proportion 1:6	Cum	1408.46
	II) 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1479.01
	b) Brick work in cement mortar proportion 1:6	Cum	1452.28
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1468.40
	b) Brick work in cement mortar proportion 1:6	Cum	1441.67
	iii) In prop. 1:6		

	a) Brick work in cement mortar proportion 1:5	Cum	1459.65
	b) Brick work in cement mortar proportion 1:6	Cum	1432.92
	B) Both side plastering		
	I) With fair side 10mm thick and rough side 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1597.39
	b) Brick work in cement mortar proportion 1:6	Cum	1570.66
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1577.81
	b) Brick work in cement mortar proportion 1:6	Cum	1551.08
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	1563.06
	b) Brick work in cement mortar proportion 1:6	Cum	1536.33
	II) With both side 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:5	Cum	1626.24
	b) Brick work in cement mortar proportion 1:6	Cum	1599.51
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:5	Cum	1605.03
	b) Brick work in cement mortar proportion 1:6	Cum	1578.30
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:5	Cum	1587.52
	b) Brick work in cement mortar proportion 1:6	Cum	1560.79
32.2.6	Extra over item no. 32.2.2 above for brick work in each subsequent floor or part thereof above 1st floor level.	Cum	67.37
32.2.7	112mm thick 1st class brick nogged wall in cement mortar including racking out joints and curing complete as directed in super structure above plinth up to 1st floor level (protruding M.S rod/Tor steel of column to be embedded in cement mortar and will be measured and paid separately)		
	A) One side plastering		
	I) 10mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:4	Sqm	1507.11
	b) Brick work in cement mortar proportion 1:5	Sqm	1464.70
	c) Brick work in cement mortar proportion 1:6	Sqm	1411.69
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:4	Sqm	1498.13
	b) Brick work in cement mortar proportion 1:5	Sqm	1455.72
	c) Brick work in cement mortar proportion 1:6	Sqm	1402.71
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:4	Sqm	1492.14
	b) Brick work in cement mortar proportion 1:5	Sqm	1449.73
	c) Brick work in cement mortar proportion 1:6	Sqm	1396.72
	II) 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:4	Sqm	1537.49
	b) Brick work in cement mortar proportion 1:5	Sqm	1495.08
	c) Brick work in cement mortar proportion 1:6	Sqm	1442.07
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:4	Sqm	1526.89
	b) Brick work in cement mortar proportion 1:5	Sqm	1484.48

	c) Brick work in cement mortar proportion 1:6	Sqm	1431.47
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:4	Sqm	1518.13
	b) Brick work in cement mortar proportion 1:5	Sqm	1475.72
	c) Brick work in cement mortar proportion 1:6	Sqm	1422.71
	B) Both side plastering		
	I) With fair side 10mm thick and rough side 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:4	Sqm	1654.34
	b) Brick work in cement mortar proportion 1:5	Sqm	1611.93
	c) Brick work in cement mortar proportion 1:6	Sqm	1558.92
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:4	Sqm	1634.76
	b) Brick work in cement mortar proportion 1:5	Sqm	1592.35
	c) Brick work in cement mortar proportion 1:6	Sqm	1539.34
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:4	Sqm	1620.01
	b) Brick work in cement mortar proportion 1:5	Sqm	1577.60
	c) Brick work in cement mortar proportion 1:6	Sqm	1524.59
	II) Both side 15mm thick plastering		
	i) In prop. 1:4		
	a) Brick work in cement mortar proportion 1:4	Sqm	1683.19
	b) Brick work in cement mortar proportion 1:5	Sqm	1640.78
	c) Brick work in cement mortar proportion 1:6	Sqm	1587.77
	ii) In prop. 1:5		
	a) Brick work in cement mortar proportion 1:4	Sqm	1661.98
	b) Brick work in cement mortar proportion 1:5	Sqm	1619.57
	c) Brick work in cement mortar proportion 1:6	Sqm	1566.56
	iii) In prop. 1:6		
	a) Brick work in cement mortar proportion 1:4	Sqm	1644.47
	b) Brick work in cement mortar proportion 1:5	Sqm	1602.06
	c) Brick work in cement mortar proportion 1:6	Sqm	1549.05
32.2.8	Extra over item no 32.2.4 for brick walling including plastering for each subsequent floor or part thereof above the 1st floor level.	Sqm	459.28
32.3	CEMENT CONCRETE FLOORING WITH BRICK SOLING		
32.3.1	40 mm thick cement concrete floor consisting of 25 mm under layer of cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of 12.5 mm and down) and 15 mm thick wearing layer in cement concrete in prop. 1:1:2 (1cement : 1 coarse sand : 2 coarse aggregate of size 10mm down) finished with a floating coat of neat cement finish using cement slurry for bond @ 2.75 kg. per square metre of floor area , wearing layer is to be laid in panels including brick on flat soling, curing etc. complete as directed.	Sqm	764.05

32.3.2	65 mm thick cement concrete floor consisting of 50 mm under layer of cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of 25mm and down) and 15 mm thick wearing layer in cement concrete in prop. 1:1:2 (1cement : 1 coarse sand : 2 coarse aggregate of size 10mm down) laid in panels and finished with a floating coat of neat cement finish (using cement slurry for bond @ 2.75 kg. per square metre of floor area) including brick on flat soling, curing etc. complete as directed.	Sqm	869.62
32.3.3	40 mm thick red oxide floor with under layer of 30 mm thick cement concrete in prop. 1:3:6 (1cement : 3 coarse sand : 6 coarse aggregate of size 20 mm and down) and top layer of 10mm thick plaster of cement red-oxide of iron per 50 kg of cement in prop. 1:3 (1 cement red oxide: 3 coarse sand) finished with a floating coat of neat cement red oxide mix of same proportion including brick on flat soling,	Sqm	800.61
32.3.4	Providing and laying 50mm thick C.C. floor with metallic concrete hardener topping under layer of 35 mm thick cement concrete in prop.1:2:4 (1cement : 2 coarse sand : 4 coarse aggregate of 20 mm and down) and top layer of 15mm thick of metallic cement hardener consisting of mix 1:2 (1 cement hardener mix : 2 stone aggregate 6 mm and down) by volume with which metallic hardening compound of approved quality is mixed in the ratio of 4:1 (4 parts of cement : 1 part of metallic floor hardening compound of approved quality by weight) including finishing brick on flat soling with curing complete as directed. (For Laboratory & Workshop Floor)	Sqm	868.81
32.3.5	Providing and laying 65mm thick C.C. floor with metallic concrete hardener topping under layer of 50 mm thick cement concrete 1:2:4 (1cement:2 coarse sand:4 coarse aggregate of 20 mm and down) and top layer of 15mm thick of metallic cement hardener consisting of mix 1:2 (1 cement hardener mix : 2 stone aggregate 6 mm and down) by volume with which metallic hardening compound of approved quality is mixed in the ratio of 4:1 (4 parts of cement: 1 part of metallic floor hardening compound of approved quality by weight) including brick on flat soling, finishing with curing complete as directed. (For Laboratory & Workshop Floor)	Sqm	979.99
32.4	DOORS / WINDOWS WITH ALL FITTINGS		
32.4.1	Providing fitting and fixing 1/3 panelled 2/3 glazed wooden shutters for doors/windows including oxidised iron butt hinges (3nos X100mm x75 mm x3.55 mm), 300x10 mm Tower bolts-2nos, 150 mm Door handle-2nos, 300x16mm- Sliding Door bolts-2nos and Door stopper with rubber stopper-1nos with necessary screws all complete as directed		
	I) Using Electro Coloured Iron fittings		
	A) With Teak wood / Sissu wood.		
	(i). 40 mm thick.	Sqm	5175.74
	(ii). 35 mm thick.	Sqm	4875.51
	(iii) 30 mm thick.	Sqm	4403.94

	B) With 1st class local wood (Gamari/ Hollock/ Bonsum)		
	(i) 40 mm thick.	Sqm	3832.12
	(ii) 35 mm thick.	Sqm	3666.05
	(iii) 30 mm thick.	Sqm	3408.44
	C) With 1st class local wood(Bola)		
	(i) 40 mm thick.	Sqm	3560.33
	(ii). 35 mm thick.	Sqm	3420.07
	(iii). 30 mm thick.	Sqm	3342.20
	II) Using Aluminium Fittings		
	A) With Teak wood / Sissu wood.		
	(i) 40 mm thick.	Sqm	5330.38
	(ii) 35 mm thick.	Sqm	5030.15
	(iii) 30 mm thick.	Sqm	4558.58
	B) With 1st class local wood (Gamari/ Hollock/ Bonsum)		
	(i) 40 mm thick.	Sqm	3986.76
	(ii) 35 mm thick.	Sqm	3820.69
	(iii) 30 mm thick.	Sqm	3563.08
	C) With 1st class local wood(Bola)		
	(i) 40 mm thick.	Sqm	3714.97
	(ii) 35 mm thick.	Sqm	3574.71
	(iii) 30 mm thick.	Sqm	3347.02
32.4.2	Providing, fitting and fixing full panelled doors/ windows including oxidised M.S butt hinges (100mm x75 mm x3.55mm), 300x10 mm Tower bolts-2nos,150 mm Door handle-2nos, 300x16mm-Sliding Door bolts-2nos and Door stopper with rubber stopper-1nos with necessary screws complete as directed		
	I) Using Electro Coloured Iron fittings		
	A) With Teak wood / Sissu wood.		
	(i). 40 mm thick.	Sqm	5868.63
	(ii). 35 mm thick.	Sqm	5465.68
	(iii) 30 mm thick.	Sqm	4460.18
	B) With 1st class local wood (Gamari/ Hollock/ Bonsum)		
	(i) 40 mm thick.	Sqm	4158.63
	(ii) 35 mm thick.	Sqm	3993.05
	(iii) 30 mm thick.	Sqm	3374.73
	C) With 1st class local wood(Bola)		
	(i) 40 mm thick.	Sqm	4107.39
	(ii). 35 mm thick.	Sqm	3945.75
	(iii). 30 mm thick.	Sqm	3342.20
	II) Using Aluminium Fittings		
	A) With Teak wood / Sissu wood.		
	(i) 40 mm thick.	Sqm	6023.27
	(ii) 35 mm thick.	Sqm	5620.32
	(iii) 30 mm thick.	Sqm	4614.82
	B) With 1st class local wood (Gamari/ Hollock/ Bonsum)		
	(i) 40 mm thick.	Sqm	4313.27
	(ii) 35 mm thick.	Sqm	4147.69
	(iii) 30 mm thick.	Sqm	3529.37
	C) With 1st class local wood(Bola)		
	(i) 40 mm thick.	Sqm	4262.03

	(ii) 35 mm thick.	Sqm	4100.39
	(iii) 30 mm thick.	Sqm	3496.84
32.4.3	Providing, fitting and fixing factory made wooden panelled doors with panel inserts of 75mm to 50mm width and 12mm to 20mm thickness, joined together with continuous tongue and groove joints glued together with Movicol or other approved adhesive leaving a vertical groove of 7.5mm to 15mm between the inserts, the edges of panel inserts being feather tongue into styles and rails including providing three pairs of oxidised M.S butt hinges (100mm x 75 mm x 3.55mm), 300x12 mm Aluminium Tower bolts-2nos, 150 mm Aluminium Door handle-2nos, 300x16mm- Aluminium Sliding Door bolts-2nos and Door stopper with rubber stopper-1nos with necessary screws and furnishing with moulded timber beading of size 25mmx15mm around the panels complete as per drawing		
	I) Using Electro Coloured Iron fittings		
	A) With Teak wood / Sissu wood.		
	(i) 40 mm thick.	Sqm	6219.92
	(ii) 35 mm thick.	Sqm	5435.26
	(iii) 30 mm thick.	Sqm	5232.40
	B) With 1st class local wood (Gamari/ Hollock/ Bonsum)		
	(i) 40 mm thick.	Sqm	4582.51
	(ii) 35 mm thick.	Sqm	4146.06
	(iii) 30 mm thick.	Sqm	4033.22
	C).With 1st class local wood(Bola)		
	(i) 40 mm thick.	Sqm	4534.27
	(ii) 35 mm thick.	Sqm	4108.08
	(iii) 30 mm thick.	Sqm	3997.89
	II) Using Aluminium Fittings		
	A).With Teak wood / Sissu wood.		
	(i) 40 mm thick.	Sqm	6374.56
	(ii) 35 mm thick.	Sqm	5589.90
	(iii) 30 mm thick.	Sqm	5387.04
	B) With 1st class local wood (Hollock/ Bonsum/ Gamari)		
	(i) 40 mm thick.	Sqm	4737.15
	(ii) 35 mm thick.	Sqm	4300.70
	(iii) 30 mm thick.	Sqm	4187.86
	C) With 1st class local wood(Bola)		
	(i) 40 mm thick.	Sqm	4688.91
	(ii) 35 mm thick.	Sqm	4262.72
	(iii) 30 mm thick.	Sqm	4152.53
32.4.4	Providing fitting and fixing 1st class local wood (Gamari/ Bonsum/ Hollock) battened, framed and braced door and window shutters using 25mm thick planks including oxidised iron butt hinges (100mm x 58 mm x 1.90 mm) 6 nos, 250x10 mm Aluminium Tower bolts-2nos, 150 mm Aluminium Door handle-2no, 250x16mm- Aluminium Sliding Door bolts-1nos with necessary screws complete as directed	Sqm	2741.36

32.4.5	Providing, fitting and fixing fully glazed wooden door shutters with necessary butt hinges, 250x10 mm Aluminium Tower bolts-2nos, 150 mm Aluminium Door handle-2nos and Door stopper with rubber stopper-1nos with necessary screws etc. complete as per approved drawing as directed and specified at all levels.(glass panes should be measured and paid separately.)		
	(A) With Gamari Wood.		
	(i) 40mm thick	Sqm	2222.36
	(ii) 35mm thick		2139.59
	(B)With 1st Class Local wood		
	(i) 40mm thick	Sqm	2040.91
	(ii) 35mm thick	Sqm	2040.91
32.4.6	Providing, fitting and fixing fully glazed window with sash bars including oxidised M.S butt hinges (75mmx 60mmx3.15mm) 2nos.on each leaf and 3 nos. on single leaf, 2nos of 150mmx10mm Aluminium tower bolt, 1no of 100mm Aluminium handle & 2 nos of aluminium eye hook complete as directed (Glass panes to be measured and paid separately)		
	(I) With Sash Bars		
	(A) With Gamari/ Bansum Wood.		
	(i) 40mm thick	Sqm	2222.36
	(ii) 35mm thick	Sqm	2139.59
	(ii) 30mm thick	Sqm	1998.72
	(B)With 1st Class Local wood (Bola)		
	(i) 40mm thick	Sqm	2196.36
	(ii) 35mm thick	Sqm	2116.80
	(ii) 30mm thick	Sqm	1979.25
	(II) Without Sash Bars		
	(A) With Gamari/ Bansum Wood.		
	(i) 40mm thick	Sqm	2012.90
	(ii) 35mm thick	Sqm	1956.47
	(ii) 30mm thick	Sqm	1836.90
	(B)With 1st Class Local wood (Bola)		
	(i) 40mm thick	Sqm	1991.49
	(ii) 35mm thick	Sqm	1937.66
	(ii) 30mm thick	Sqm	1820.90
CH-33 : RAIN WATER HARVESTING SYSTEM			
Item No.	Description	Unit	Rate
33.1	RAINWATER HERVESTING SYSTEM		
	For R.C.C Roof		
33.1.1	Providing,Fixing, testing &Commissioning of Rain water Harvesting system for 1000 sq.feet roof area with 110 mm uPVC pipe of Class-3 conforming to IS:4985-200 (with uptodate amendment) with extra UV ray protection by Titanium Dioxide(TiO2) directing the Rainwater to Rainy filter/Pop-up raintap filter (Excluding Civil Works)	Each	87174.84
	For GI sheet/Diana Roof		

33.1.2	Providing,Fixing, testing &Commissioning of Rain water Hervesting system for 1000 sq.feet roof area with 180 mm uPVC Gutter of Class-3 conforming to IS:4985-200 (with uptodate amendment) with extra UV ray protection by Titanium Dioxide(TiO ₂) along the outer periphery of the GCI/Diana roof , 110 mm UPVC pipe directing the Rainwater to Rainy filter/Pop-up raintap filter (Excluding Civil Works)	Each	105895.42
	Storage of Rainwater		
33.1.3	Water storage tank ISI Marked, conforming to IS:12701/1996 (with uptodate amendment). (Excluding Civil Works)	Ltr.	10.77

CH-34 : ORGANIC WASTE DISPOSAL SYSTEM AND STP

Item No.	Description	Unit	Rate
	Organic Waste Disposal System		
34.1	Fully automatic Micro-organism based natural composting system with PLC based control,insulated oil heating chamber,Stainless Steel (SS304) shaft & mixing blade to convert organic waste like kitchen waste , canteen waste etc. to organic manure.		
	a) Capacity 24 Kg	Each	579635.94
	b) Capacity 70 Kg	Each	1048296.88
	c) Capacity 125Kg	Each	1666385.94
	d) Capacity 250 Kg	Each	2243721.88
	Sewage treatment plant		
34.2	Attached growth aerobic Sewage treatment System with FRP treatment chamber with all components including Tertiary filtration system and clorine dosing system to achieve treated water quality suitable for reuse or discharge as per CPCB norms.		
	a) Capacity-10 KLD	Each	2087501.56
	b) Capacity-15 KLD	Each	2223345.31
	c) Capacity-20 KLD	Each	2359189.06
	d) Capacity-30 KLD	Each	2562954.69
	e) Capacity-40 KLD	Each	2834642.19
	f) Capacity-50 KLD	each	2970485.94

CH-35 : THERMAL INSULATION

Item No.	Description	Unit	Rate
35.1	Over Deck (Roof) Insulation: SINGLE LAYER SYSTEM OVER CONCRETE DECK / ACCESSIBLE ROOF (For Hydro & Thermal Insulation with INSUBOARD)		
35.1.1	Day Time Use Building: Providing & Laying of Insuboard over the PCC, 50Mm Thickness.(Make: M/S Supreme or equivalent) (PCC with proper Slope Over RCC)	Sqm	1088.16

35.1.2	24-hours Use Building: Providing & Laying of Insuboard over the PCC, 75 Mm Thickness.(Make: M/S Supreme or equivalent) (PCC with proper Slope Over RCC)	Sqm	1569.83
35.1.3	Providing & laying of Plastic Sheet (120-150GSM)/Silpaulin (45 GSM) as a vapor barrier.	Sqm	59.44
35.1.4	Providing & laying of Geo textile Of 120 GSM as a separating layer over the Poly/Plastic sheet.	Sqm	90.09
35.1.5	Providing & laying of P.C.C. of (1:2:4) 40 mm. Avg. thickness with 40-45 GSM Glass fiber mesh as reinforcement with proper slope.	Sqm	6697.60
35.1.6	Providing & laying of polymer Modified 4mm,Membrane or other water proofing system over the prime slope PCC surface.	Sqm	2456.70
35.1.7	Providing & laying of P.C.C. of (1:2:4) 25 mm. Avg. thickness with 40-45 GSM Glass fiber mesh as reinforcement. The PCC will be done in panel form of 1m x 1 ½ m. panel with Control joints & joints are to be sealed with mastic sealant.	Sqm	6697.60
35.2	Under Deck (Roof) Insulation: SINGLE LAYER SYSTEM UNDER DECK THERMAL INSULATION SYSTEM WITH INSU.BOARD		
35.2.1	Cleaning of the surface by brush	Sqm	11.61
35.2.2	Day Time use Building: Providing & Laying of Insuboard, thickness 50mm of density 32 kgs/m3) over the RCC ceiling (Plastered Surface) from inside. The insuboard will be fixed over the wall surface by the PVC Fastner(side lap) (Make: M/s Supreme or equivalent)	Sqm	1125.44
35.2.3	24- hours use Building: Providing & Laying of Insuboard, thickness 75 mm of density 32 kgs/m3) over the RCC ceiling (Plastered Surface) from inside. The insuboard will be fixed over the wall surface by the PVC Fastner(side lap) (Make: M/s Supreme or equivalent)	Sqm	1607.11
35.2.4	Providing & laying of wall putty 1mm (Max) thick over the Insuboard's bottom surface & then painting with non solvent base paint after levelling the surface. Over the joint self adhesive Glass fibre mesh tape to be provided	Sqm	188.73
35.3	Wall Insulation: Thermal Insulation system of external wall with insuboard and cement plaster finish	Sqm	
35.3.1	Providing & laying of insuboard of thickness 50mm of density 32kgs/m3) over the Brick+ plaster(Plastered surface from outside. The insuboard will be fixed over the wall surface by PVC Fastner(1250X 600MM) shiplap/Sqr Edge profile and the insuboard joints will be sealed with self glass fibre adhesive.	Sqm	1204.66

35.3.2	Providing and laying of plaster (1:4) with GI chicken/PVC Wire Mesh/Glass Fibre Mesh Or Providing or laying of wall putty of 1mm(Max) thick over the insuboard's bottom surface and then painting		
	i) Providing and laying of plaster (1:4) with GI chicken/PVC Wire Mesh/Glass Fibre Mesh	Sqm	291.39
	ii) Providing or laying of wall putty of 1mm(Max) thick over the insuboard's bottom surface and then painting	Sqm	188.73
35.4	Single Layer System inside the cavity wall		
35.4.1	Providing & Laying of Insuboard, thickness 50mm of density 32 kgs/m ³) over the RCC ceiling (Plastered Surface) from inside. The insuboard will be fixed over the wall surface by the PVC Fastener(side lap) (Make: M/s Supreme or equivalent)	Sqm	1088.16

CH-36 : RETROFITTING WORKS

Item No.	Description	Unit	Rate
1	Chipping and breaking /cutting of loose concrete, mortars etc. using electrical chipping hammers to the satisfaction of the Engineer-in-charge for successful execution of the subsequent lifts etc. complete with all tools and machineries required for the job operator applicator, labour, scaffolding for staging and supervision etc.	Sqm	124.66
2	Providing and applying two coats of Nitozinc primer a two component epoxy zinc rich primer / sika top Arimatec-108 of equivalent product on the properly cleaned and prepared steel bars after the required chipping and cleaning of the concrete surface as per the manufacturer's specifications.	Sqm	456.09
3	Before laying of the Micro concrete, the substrate shall be coated with a bonding agent, Nitobond EP Std. / Sikadur -32 LP or equivalent product and the micro concrete shall be poured within 6 hrs/ 4 hrs or as applicable of the application of the bonding agent	Sqm	699.32
4	Providing and placing free flow micro concrete Renderoc RG / sika Micro-Concrete-2 of equivalent product of after fixing slurry tight shuttering. The micro concrete should be mixed mechanically while maintaining water, powder ratio of 0.16 strictly in accordance with the manufacturer's instructions and as per the direction of the Engineer-in-charge. the micro concrete should develop a compressive strength of 10N /mm ² within 1 day/ 40N/mm ² within 7 days and 50N/ mm ² within 28 days the cost is inclusive of materials transportation, labour taxes etc. as complete.	Sqm	66184.80